Network Theory

Detailed Solutions

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Network Theory (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EE	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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SOLUTIONS

1

Basic Concepts of Networks

Topics

1. Introduction to Electrical Quantities
2. Basic Laws and Classification of Network Elements
3. Resistors, Capacitors and Inductors
4. Energy Sources in Electrical Circuits
5. Series-Parallel connections
6. Methods of Analyzing Electrical Circuits - Nodal and Mesh Analysis

Unit Snapshot*

Stream: EE	**Questions:** 42	**MCQ:** 26	**MSQ:** 0	**NAT:** 14
1M: 25	**2M:** 17	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2023)–3M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ | 2026 | 1M | Ans: A

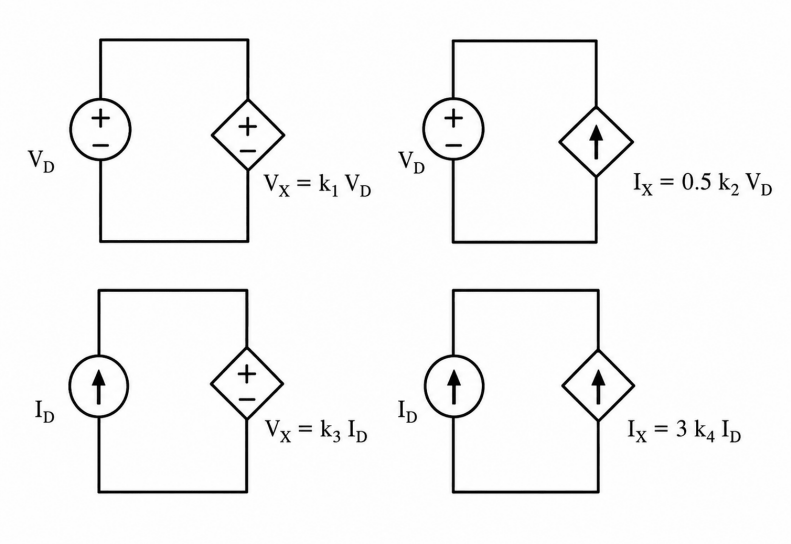

Fig. Solution Q1.26.1

Circuit 1

The independent voltage source is V_D

and the dependent voltage source is $V_x = k_1 V_D$

Since both voltage sources are connected in parallel across the same two nodes,

$$V_x = V_D$$

$$\therefore k_1 V_D = V_D \Rightarrow \boxed{k_1 = 1}$$

Circuit 2

The independent voltage source is V_D

and the dependent current source is $I_x = 0.5 k_2 V_D$

A current source can supply whatever current is required for the specified voltage across it.

Hence, there is no restriction on k_2.

$\boxed{k_2 \text{ can be any value}}$

Circuit 3

The independent current source is I_D

and the dependent voltage source is $V_x = k_3 I_D$

A voltage source can develop whatever voltage is required to support the imposed current.

Hence, there is no restriction on k_3.

$\boxed{k_3 \text{ can be any value}}$

Circuit 4

The independent current source is I_D

and the dependent current source is $I_x = 3 k_4 I_D$

Since the two current sources are connected in series, they must enforce the same current.

Therefore,

$$I_x = -I_D \rightarrow 3 k_4 I_D = -I_D \Rightarrow \boxed{k_4 = -\frac{1}{3}}$$

$\boxed{\textbf{A. } k_1 = 1,\ k_4 = -\frac{1}{3},\ \text{for all values of } k_2 \text{ and } k_3}$

Mistakes to be avoided

- Carefully check the polarity of voltage sources and the direction of current source arrows.

Q1.26.2 MCQ 2026 1M Ans: B ●○○

For an element to be linear, its voltage-current relationship must satisfy superposition and homogeneity.

Resistor

For a resistor, $v = Ri$; which is a linear relationship.

Inductor

For an inductor, $v = L\frac{di}{dt}$; which is also linear.

Capacitor

For a capacitor, $i = C\frac{dv}{dt}$; which is also linear.

Ideal Diode

The v-i characteristic of an ideal diode is

$$\begin{cases} v_D = 0, & i_D > 0 \\ i_D = 0, & v_D < 0 \end{cases}$$

A diode conducts current in one direction only; its voltage-current relationship is not linear.

Therefore, the linear elements are

$R,\ L,$ and C

B. $R,\ L$ and C only

Q1.25.1 MCQ 2025 1M Ans: B ●○○

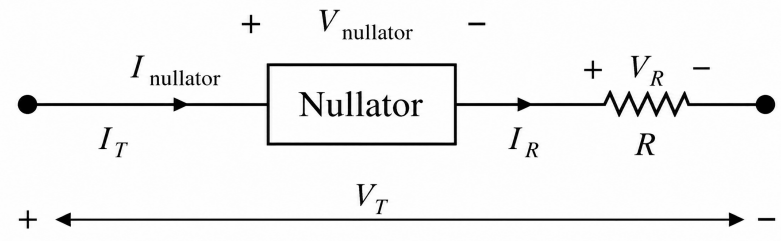

Fig. Solution Q1.25.1

A nullator is a two-terminal element defined by

$$v_N = 0 \quad \text{and} \quad i_N = 0$$

simultaneously.

Consider a nullator connected in series with a resistor R.

Since the elements are in series,

$$i_R = i_N$$

For the nullator,

$$i_N = 0 \quad \text{Hence,} \quad i_R = 0$$

Therefore, the voltage across the resistor is

$$v_R = i_R R = 0$$

Hence, the total voltage across the series combination is

$$v = v_R + v_N = 0$$

Thus, the series combination satisfies

$$v = 0, \qquad i = 0$$

which is exactly the defining characteristic of a nullator.
Therefore,

B. Nullator

Q1.25.2 MCQ 2025 1M Ans: B ● ○ ○

The resistor and current source are connected in parallel.

Current through the resistor

$$I_R = \frac{V_{XY}}{1000}, \qquad \text{flowing from } X \text{ to } Y.$$

Current through the current source

$$I_S = 1 \text{ A}, \qquad \text{from } X \text{ to } Y.$$

Now applying KCL at terminal X,

$$I_{XY} = I_S + I_R = 1 + \frac{V_{XY}}{1000}$$

This is a straight-line relation between I_{XY} and V_{XY}.
The slope is

$$\frac{dI_{XY}}{dV_{XY}} = \frac{1}{1000}, \qquad \text{which is positive.}$$

Hence,

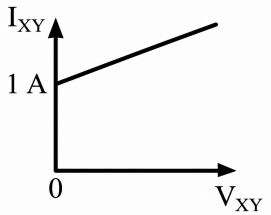

Fig. Solution Q1.25.2

B.

Q1.24.1 MCQ 2024 1M Ans: A ● ○ ○

A junction is a point where three or more branches meet. From the given circuit, the number of such points are circled.

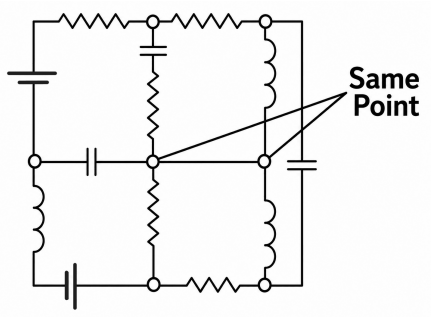

Fig. Solution Q1.24.1

Hence,

A. 6

Mistakes to be avoided

- Count only those points where three or more branches meet. While counting, choose the junctions carefully and ensure that the same junction is not counted more than once.

Q1.24.2 MCQ | 2024 | 1M | Ans: A | ● ○ ○

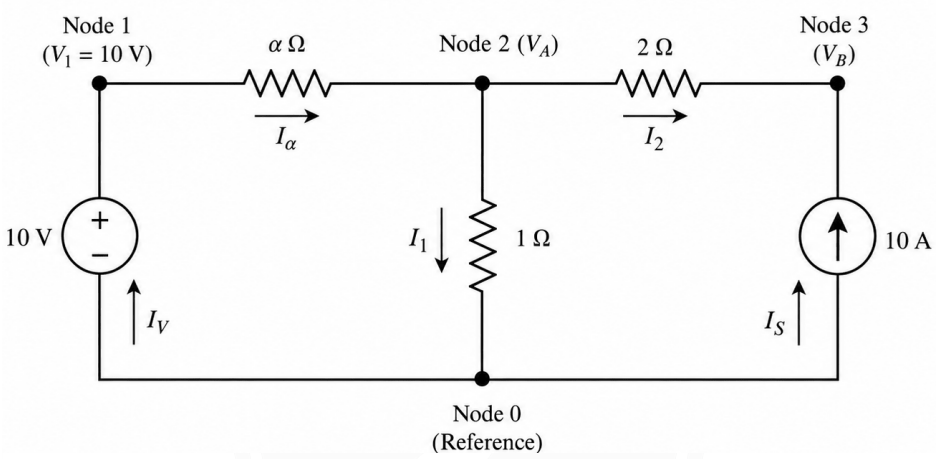

Fig. Solution Q1.24.2

KCL at node A

$$10 = V_A + \frac{V_A - 10}{\alpha}$$

$$10\alpha = \alpha V_A + V_A - 10$$

$$V_A(\alpha + 1) = 10(\alpha + 1) \Rightarrow V_A = 10 \text{ V}$$

Hence,

$$I_\alpha = \frac{V_A - 10}{\alpha} = 0$$

Therefore, no current flows through the $\alpha\Omega$ resistor and the current through the 10 V source is also 0.
Thus, the power delivered by the source is

$$P = VI \Rightarrow P = 0 \text{ W}$$

Hence,

A. 0 W

Q1.23.1 NAT | 2023 | 1M | Ans: 6.0 | ● ○ ○

Method 1

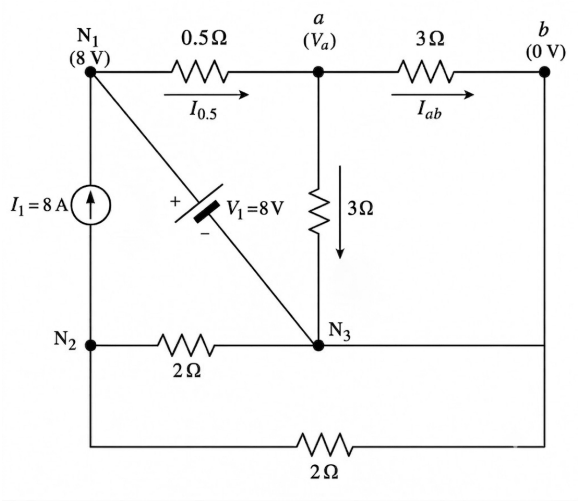

Fig. Solution Q1.23.1

Given,

$$V_1 = 8 \text{ V}$$

Choose node b as the reference node:

$$V_b = 0 \Rightarrow V_{N_3} = 0$$

Let the voltage at node a be V_a.
Then voltage at node N_1 is 8 V ($\because V_{N_3} = 0$).
Applying KCL at node a,

$$\frac{V_a - 8}{0.5} + \frac{V_a}{3} + \frac{V_a}{3} = 0$$

Multiplying throughout by 3,

$$6(V_a - 8) + V_a + V_a = 0$$

$$6V_a - 48 + 2V_a = 0 \Rightarrow V_a = 6 \text{ V}$$

Therefore,

$$V_{ab} = V_a - V_b = 6$$

Method 2

The two 3Ω resistors and the two 2Ω resistors are connected in parallel. Hence,

$$R_{eq_3} = 3 \parallel 3 = \frac{3 \times 3}{3 + 3} = 1.5\Omega, \qquad R_{eq_2} = 2 \parallel 2 = 1\Omega$$

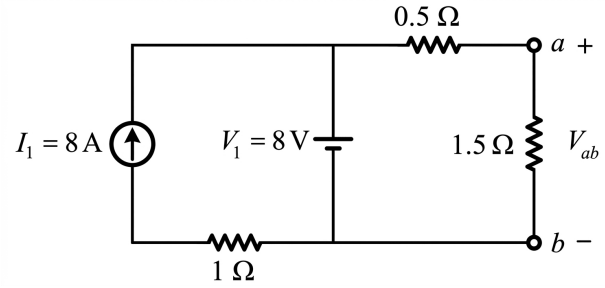

Fig. Solution Q1.23.1

Applying the Voltage Divider Rule,

$$V_{ab} = \frac{1.5}{1.5 + 0.5} \times 8 = 6 \text{ V}$$

Thus,

$$\boxed{6.0}$$

Q1.22.1 NAT | 2022 | 2M | Ans: 100 | ● ○ ○

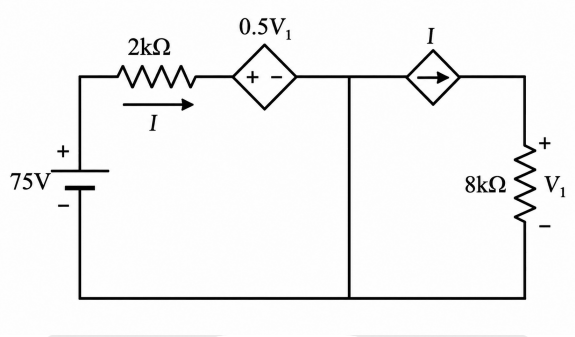

Fig. Solution Q1.22.1

The voltage across the 8 $k\Omega$ resistor is

$$V_1 = (8k)I$$

and,

$$I = \frac{75 - 0.5V_1}{2000}$$

Substituting

$$I = \frac{V_1}{8k}$$

gives

$$75 = 2k\left(\frac{V_1}{8k}\right) + 0.5V_1 \Rightarrow 75 = \frac{V_1}{4} + \frac{V_1}{2}$$

$$V_1 = 100 \text{ V}$$

Hence,

$$\boxed{100}$$

Q1.21.1 NAT | 2021 | 1M | Ans: -3.25 | ● ● ○

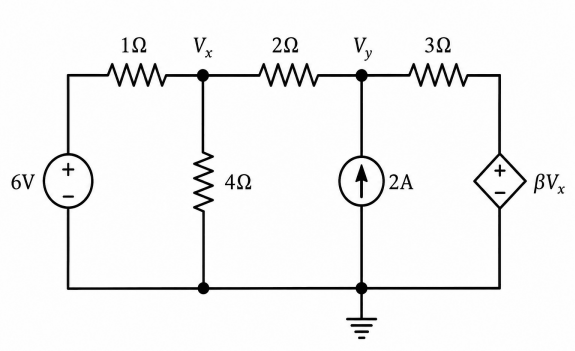

Fig. Solution Q1.21.1

Applying KCL at node V_x,

$$\frac{V_x - 6}{1} + \frac{V_x}{4} + \frac{V_x - V_y}{2} = 0$$

Multiplying throughout by 4,

$$4(V_x - 6) + V_x + 2(V_x - V_y) = 0$$

$$7V_x - 2V_y = 24$$

Since $V_y = 0$,

$$7V_x = 24 \Rightarrow V_x = \frac{24}{7}\text{ V}$$

Applying KCL at node V_y,

$$\frac{V_y - V_x}{2} + \frac{V_y - \beta V_x}{3} = 2$$

Multiplying throughout by 6,

$$3(V_y - V_x) + 2(V_y - \beta V_x) = 12$$

$$5V_y - (3 + 2\beta)V_x = 12$$

Substituting

$$V_y = 0, \qquad V_x = \frac{24}{7}$$

gives

$$-(3 + 2\beta)\frac{24}{7} = 12$$

$$3 + 2\beta = -\frac{12 \times 7}{24} = -\frac{7}{2}$$

$$\beta = -\frac{13}{4} = -3.25$$

Hence,

$$\boxed{-3.25}$$

Mistakes to be avoided

- Be careful with the sign while applying KCL/KVL and solving the equations. For NAT questions, do not forget to include the negative sign in the final answer if the computed value is negative.

Q1.19.1 NAT | 2019 | 1M | Ans: 1.4 | ● ○ ○

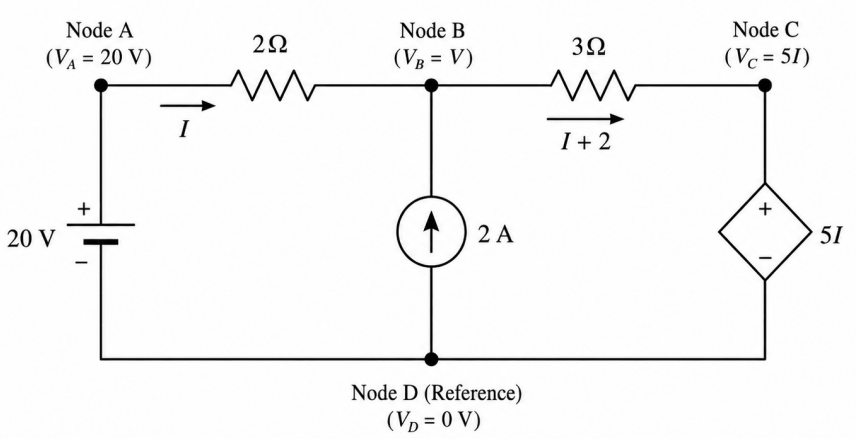

Fig. Solution Q1.19.1

Method KVL

Applying KVL around the outer loop,

$$20 - 2I - 3(I + 2) - 5I = 0$$

$$20 - 2I - 3I - 6 - 5I = 0$$

$$I = 1.4 \text{ A}$$

Method Nodal Analysis

Applying KCL at Node B,

$$\frac{V - 20}{2} + \frac{V - 5I}{3} = 2$$

By KVL on 1st small loop

$$V = 20 - 2I$$

gives

$$\frac{(20 - 2I) - 20}{2} + \frac{(20 - 2I) - 5I}{3} = 2$$

$$-I + \frac{20 - 7I}{3} = 2$$

$$-3I + 20 - 7I = 6 \Rightarrow 20 - 10I = 6$$

$$I = 1.4 \text{ A}$$

Hence,

$$\boxed{1.4}$$

Q1.17.1 NAT | 2017 | 1M | Ans: 250 | ● ○ ○

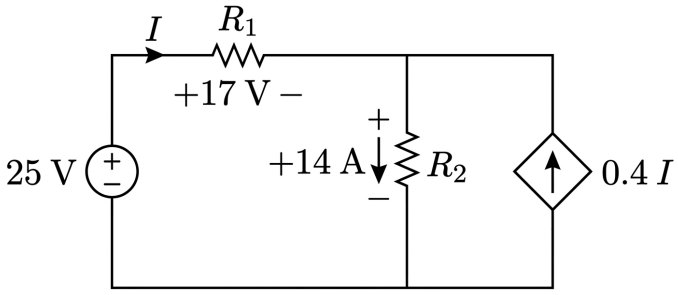

Fig. Solution Q1.17.1

The dependent current source supplies 0.4I
By KCL;

$$I + 0.4I = 14$$

$$I = 10 \text{ A}$$

Therefore, the power supplied by the source is

$$P = VI$$

$$P = 25 \times 10 \Rightarrow P = 250 \text{ W}$$

Hence,

$$\boxed{250}$$

Mistakes to be avoided

- Read the question carefully to determine whether it asks for power *supplied* or power *absorbed*. Apply the passive sign convention correctly and choose the sign of power accordingly.

Q1.16.1 MCQ 2016 1M Ans: D

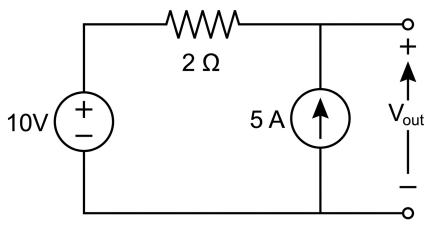

Fig. Solution Q1.16.1

Apply KCL;

$$\frac{V_{out} - 10}{2} = 5$$

$$V_{out} - 10 = 10$$

$$V_{out} = 20 \text{ V}$$

Therefore,

D. 20

Q1.16.2 NAT 2016 1M Ans: 0.5

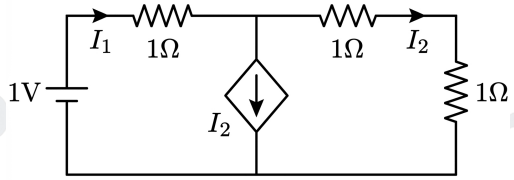

Fig. Solution Q1.16.2

Applying KCL at the middle node

$$I_1 = 2I_2$$

Applying KVL around the outer loop

$$-1 + I_1 + I_2 + I_2 = 0 \Rightarrow I_1 + 2I_2 = 1$$

Substituting

$$I_1 = 2I_2$$

gives

$$2I_2 + 2I_2 = 1 \Rightarrow I_2 = 0.25 \text{ A}$$

Therefore,

$$I_1 = 2I_2 \Rightarrow I_1 = 0.5 \text{ A}$$

Hence,

0.5

Q1.16.3 NAT | 2016 | 1M | Ans: 2 | ● ● ○

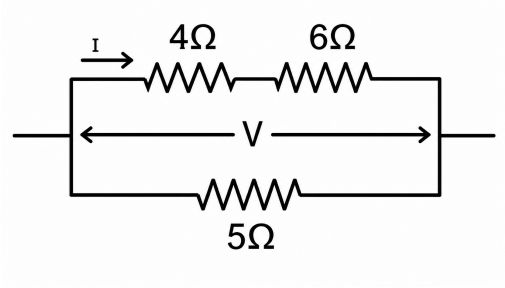

Fig. Solution Q1.16.3

The upper branch consists of 4Ω and 6Ω resistors in series.

$$R_{upper} = 4 + 6 = 10\Omega$$

The lower branch consists of a 5Ω resistor.
Since the two branches are in parallel, the voltage across both branches is the same.
Given that the heat generated in the 5Ω resistor is

$$P_5 = 10 \text{ calories per second.}$$

Using

$$P = \frac{V^2}{R}, \qquad \text{and, } V^2 = 50$$

Current through the upper branch is

$$I = \frac{V}{10}$$

Power dissipated in the 4Ω resistor is

$$P_4 = I^2(4) = \left(\frac{V}{10}\right)^2 (4)$$

$$P_4 = 2 \text{ calories per second.}$$

Therefore,

$$\boxed{2}$$

Q1.16.4 MCQ | 2016 | 1M | Ans: D | ● ○ ○

For a connected electrical network,

$$\text{Number of independent KCL equations} = N - 1$$

$$\text{Number of independent KVL equations} = B - N + 1$$

where

$$B = \text{number of branches=7}, \quad N = \text{number of nodes=5}$$

Therefore,

$$\text{Independent KCL equations} = 5 - 1 = 4$$

and

$$\text{Independent KVL equations} = 7 - 5 + 1 = 3$$

Hence,

$$\boxed{\textbf{D. } 4 \text{ and } 3}$$

Mistakes to be avoided

- Read the question carefully and maintain the required order. Here, the question asks for (independent KCL equations, independent KVL equations), so the answer is $(4, 3)$, not $(3, 4)$.

Q1.15.1 NAT | 2015 | 2M | Ans: 0.5 | ●●○

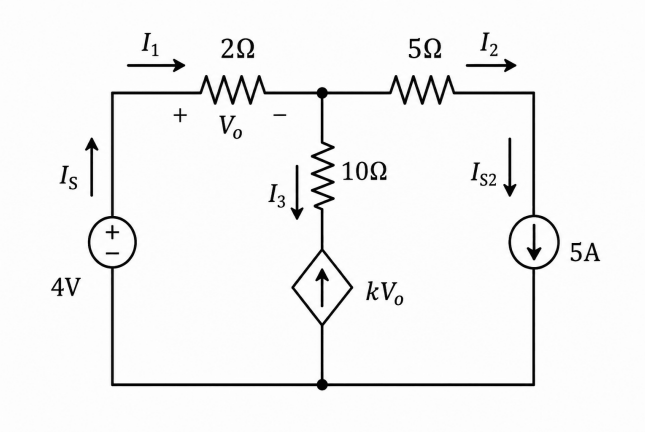

Fig. Solution Q1.15.1

Current through the 2Ω resistor is

$$I_1 = \frac{V_o}{2}$$

Applying KCL;

$$\frac{V_o}{2} + kV_o = 5 \tag{1}$$

Given that the power dissipated in the 2Ω resistor is

$$P_{2\Omega} = 12.5 \text{ W}$$

Using

$$P = I_1^2 R$$

$$12.5 = \left(\frac{V_o}{2}\right)^2 (2)$$

$$V_o^2 = 25 \Rightarrow V_o = \pm 5 \text{ V}$$

Since k is positive,

$$V_o = 5 \text{ V}$$

Substituting into (1),

$$\frac{5}{2} + 5k = 5 \Rightarrow 5k = 2.5$$

$$k = 0.5$$

Hence,

$$\boxed{0.5}$$

Mistakes to be avoided

- From $V_o^2 = 25$, both $V_o = +5$ V and $V_o = -5$ V are possible, leading to two values of k. Read the question carefully and use the given condition ($k > 0$) before selecting the final answer.

Q1.15.2 MCQ | 2015 | 1M | Ans: A

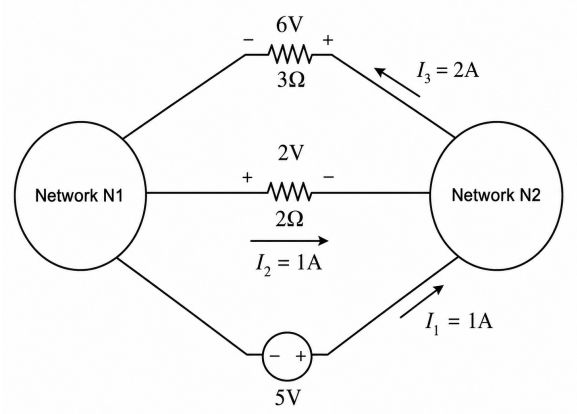

Fig. Solution Q1.15.2

For the 3Ω resistor, $I_3 = \frac{6}{3} = 2$ A

For the 2Ω resistor, $I_2 = \frac{2}{2} = 1$ A

Applying KCL at Network N_2,

Therefore,

$$I_1 + 1 = 2 \Rightarrow I_1 = 1 \text{ A}$$

Since the current leaves the positive terminal of the 5 V source, the source delivers power.

$$P = VI \Rightarrow P = 5 \text{ W}$$

Hence,

A. 5

Q1.15.3 NAT | 2015 | 1M | Ans: 0

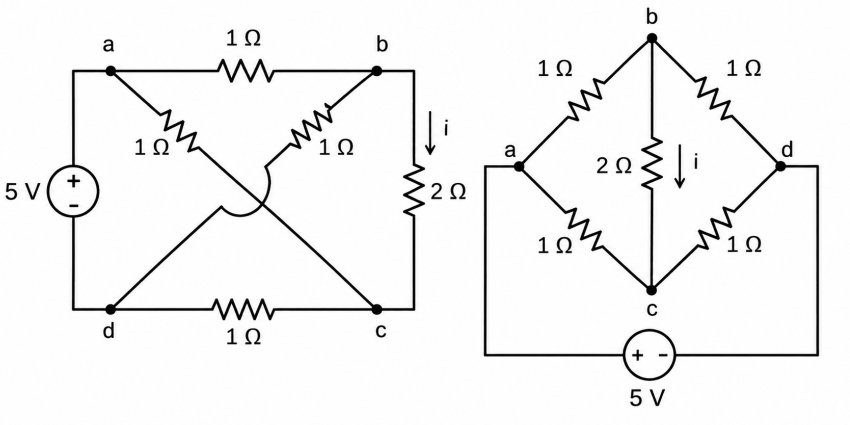

Fig. Solution Q1.15.3

The given network forms a balanced Wheatstone bridge.
Since

$$\frac{R_{ab}}{R_{ac}} = \frac{R_{bd}}{R_{dc}}$$

the bridge is balanced.
Therefore, no current flows through the branch connecting b and c.

$$I = 0$$

Hence,

$$\boxed{0}$$

Q1.14.1 NAT | 2014 | 2M | Ans: 10 | ●○○

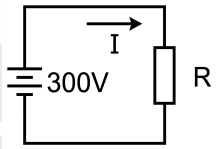

Fig. Solution Q1.14.1

Applying Ohm's law,

$$I = \frac{V}{R} = \frac{300}{25 + \frac{I}{2}}$$

Cross-multiplying,

$$I\left(25 + \frac{I}{2}\right) = 300$$

$$25I + \frac{I^2}{2} = 300$$

Multiplying throughout by 2,

$$I^2 + 50I - 600 = 0$$

Factoring,

$$I^2 + 60I - 10I - 600 = 0$$

$$(I + 60)(I - 10) = 0$$

Therefore,

$$I = -60 \text{ A} \quad \text{or} \quad I = 10 \text{ A}$$

According to the given direction of current,

$$\boxed{I = 10 \text{ A}}$$

Hence,

$$\boxed{10}$$

Q1.14.2 NAT | 2014 | 2M | Ans: 2470 ● ● ○

Given,

$$P = 40\ \text{W}, \quad V = 240\ \text{V}, \quad R_0 = 120\ \Omega$$

$$\alpha = 4.5 \times 10^{-3}/^\circ\text{C}, \quad T_0 = 26^\circ\text{C}$$

Filament resistance in ON state

Using

$$P = \frac{V^2}{R} \qquad \text{(where R = Hot resistance of the filament)}$$

$$R = \frac{V^2}{P} \Rightarrow R = 1440\ \Omega$$

Temperature dependence of resistance

$$R = R_0\left[1 + \alpha(T - T_0)\right]$$

Substituting the given values,

$$1440 = 120\left[1 + 4.5 \times 10^{-3}(T - 26)\right]$$

$$12 = 1 + 4.5 \times 10^{-3}(T - 26)$$

$$11 = 4.5 \times 10^{-3}(T - 26)$$

$$T = 2470.44^\circ\text{C}$$

Therefore,

$$\boxed{2470}$$

Q1.14.3 NAT | 2014 | 1M | Ans: 330 ● ○ ○

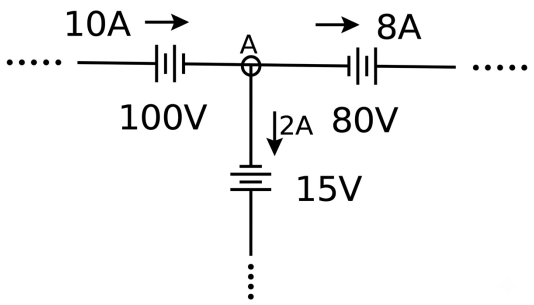

Fig. Solution Q1.14.3

Applying KCL at Node A,

$$10 = 8 + I_{15}$$

$$I_{15} = 2\ \text{A}$$

Power absorbed by the 100 V source

The 10 A current enters the positive terminal.

$$P_{100} = 100 \times 10 = 1000\ \text{W}$$

Power absorbed by the 80 V source

The 8 A current leaves the positive terminal.

$$P_{80} = -(80 \times 8) = -640 \text{ W}$$

Power absorbed by the 15 V source

The 2 A current leaves the positive terminal.

$$P_{15} = -(15 \times 2) = -30 \text{ W}$$

Total power absorbed

$$P_{\text{total}} = P_{100} + P_{80} + P_{15}$$

$$P_{\text{total}} = 1000 - 640 - 30 = 330 \text{ W}$$

Hence,

330

Key Points

- Power sign convention for a voltage source:

Current Direction	Power
Entering $+ve$ terminal	Absorbed
Leaving $+ve$ terminal	Delivered
Entering $-ve$ terminal	Delivered
Leaving $-ve$ terminal	Absorbed

Q1.10.1 MCQ | 2010 | 2M | Ans: B

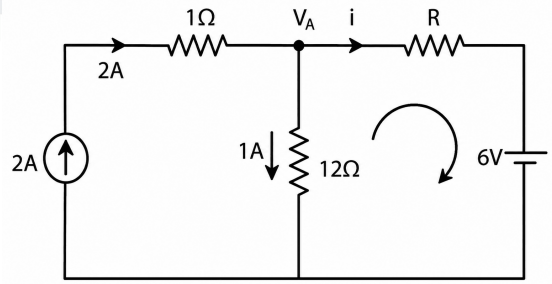

Fig. Solution Q1.10.1

Let the current through resistor R be i

Applying KCL at node A

$$2 = 1 + i \Rightarrow i = 1 \text{ A}$$

Applying KVL in the right loop (Traversing clockwise);

$$-12(1) + iR + 6 = 0$$

$$-12 + R + 6 = 0 \Rightarrow R = 6\ \Omega$$

Therefore,

B. 6 Ω

Q1.10.2 MCQ 2010 1M Ans: B ●○○

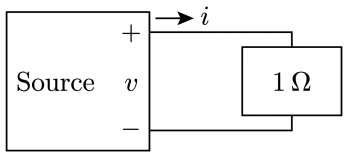

Fig. Solution Q1.10.2

Given, the load-line equation of the source is

$$v + i = 100$$

A 1Ω resistor is connected parallel to the source.
For the resistor,

$$v = iR$$

Since

$$R = 1\Omega \Rightarrow \text{we have, } v = i$$

Substituting into the load-line equation,

$$i + i = 100$$

$$i = 50 \text{ A}$$

Hence,

B. 50 A

Q1.09.1 MCQ 2009 2M Ans: B ●●○

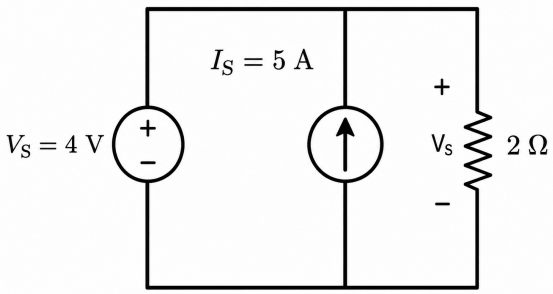

Fig. Solution Q1.09.1

The voltage source, the current source and the 2Ω resistor are connected in parallel.
Therefore, the voltage across the resistor is fixed by the voltage source:

$$V_R = V_S = 4 \text{ V}$$

Hence, the current through the resistor is

$$I_R = \frac{V_R}{R}$$

$$I_R = 2 \text{ A}$$

To double the current, we need to double the Voltage souce;

$$V_S' = 8 \text{ V}$$

Therefore,

B. (2 A; Put $V_s = 8$ V)

Q1.08.1 MCQ | 2008 | 2M | Ans: B

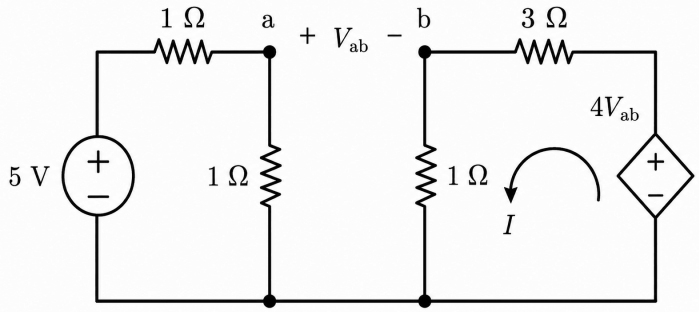

Fig. Solution Q1.08.1

Step 1: Find V_a

The left circuit consists of two 1Ω resistors in series across a 5 V source.

Using voltage division,

$$V_a = \frac{1}{1+1} \times 5$$

$$V_a = 2.5 \text{ V}$$

Step 2: Express V_b in terms of i

The current through the 1Ω resistor in the right loop is i.

Hence,

$$V_b = i$$

Therefore,

$$V_{ab} = V_a - V_b$$

$$V_{ab} = 2.5 - i \quad (1)$$

Step 3: Apply KVL in the right loop (Traversing clockwise);

$$-4V_{ab} + 3i + i = 0$$

$$V_{ab} = i \quad (2)$$

Step 4: Solve for i

Substituting (2) into (1),

$$2.5 - i = i$$

$$2i = 2.5 \Rightarrow i = 1.25 \text{ A}$$

Hence,

B. 1.25 A

Key Points

- **Voltage division rule:** If two resistors R_1 and R_2 are connected in series across a voltage source V, then the voltages across R_1 and R_2 are

$$V_{R_1} = V\frac{R_1}{R_1 + R_2}, \qquad V_{R_2} = V\frac{R_2}{R_1 + R_2}$$

- **Current division rule:** If two resistors R_1 and R_2 are connected in parallel across a current source I, then the currents through R_1 and R_2 are

$$I_{R_1} = I\frac{R_2}{R_1 + R_2}, \qquad I_{R_2} = I\frac{R_1}{R_1 + R_2}$$

Q1.08.2 MCQ 2008 2M Ans: A

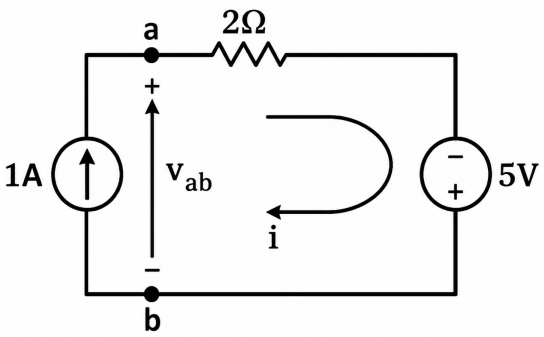

Fig. Solution Q1.08.2

The current through the 2Ω resistor is

$$i = 1 \text{ A}$$

Applying Ohm's law,

$$V_R = iR \Rightarrow V_R = (1)(2)$$

$$V_R = 2 \text{ V}$$

Using KVL;

$$V_{ab} - 2 + 5 = 0$$

$$V_{ab} = -3 \text{ V}$$

Hence,

A. − 3 V

Mistakes to be avoided

- Check the polarity markings carefully while applying KVL and mark the option carefully since both +3V and -3V are present.

Q1.07.1 MCQ 2007 2M Ans: A ● ● ○

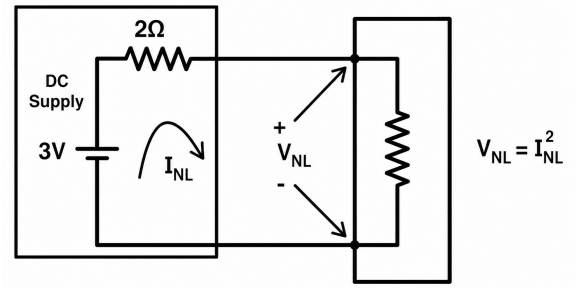

Fig. Solution Q1.07.1

The non-linear resistor characteristic is, $V_{NL} = I^2$

Applying KVL

$$3 = 2I + V_{NL}$$

Substituting $V_{NL} = I^2$;

$$3 = 2I + I^2 \Rightarrow I^2 + 2I - 3 = 0$$

$$(I + 3)(I - 1) = 0$$

$$I = -3 \text{ A} \quad \text{or} \quad I = 1 \text{ A}$$

Rejecting the negative current solution as non -liner resistance is passive,

Power dissipated in the non-linear resistor

Voltage across the non-linear resistor is

$$V_{NL} = I^2 = (1)^2 = 1 \text{ V}$$

$$\Rightarrow P = V_{NL} I = 1 \text{ W}$$

Therefore,

A. 1.0 W

Key Points

- A **passive element** cannot generate energy on its own; it can only absorb or store energy. Examples: resistor, inductor and capacitor.
- Since $V_{NL} = I^2$, the absorbed power is; $P = V_{NL} I = I^3$
- For a passive resistor, $P \geq 0$ (it cannot deliver power). Hence, $I \geq 0$, and the negative-current solution is rejected.

Q1.05.1 MCQ 2005 1M Ans: C ● ○ ○

The two 10Ω resistors are connected in parallel ⇒ their equivalent is 5Ω.

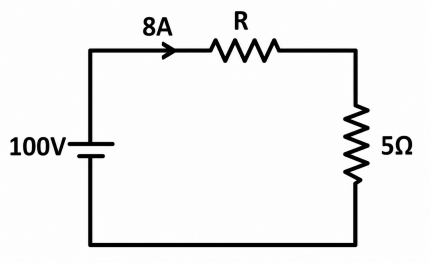

Fig. Solution Q1.05.1

Therefore, the total equivalent resistance is

$$R_{eq} = \frac{V}{I}$$

$$R_{eq} = \frac{100}{8} = 12.5\Omega$$

Since R is in series with 5Ω;

$$R + 5 = 12.5 \Rightarrow R = 7.5\Omega$$

Therefore,

C. 7.5 Ω

Q1.04.1 MCQ | 2004 | 2M | Ans: C

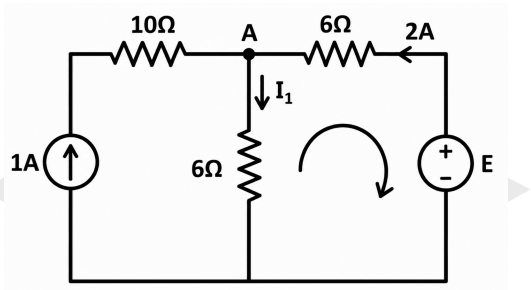

Fig. Solution Q1.04.1

Applying KCL at Node A

$$1 + 2 - I_1 = 0$$

$$I_1 = 3 \text{ A}$$

Applying KVL in the right loop (Traversing clockwise);

$$-E + (2)(6) + (I_1)(6) = 0$$

gives

$$-E + 12 + 18 = 0 \Rightarrow E = 30 \text{ V}$$

Therefore,

C. 30 V

Q1.03.1 MCQ | 2003 | 1M | Ans: C

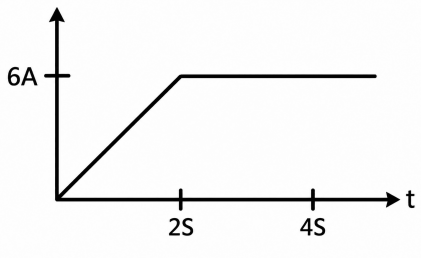

Fig. Solution Q1.03.1

The current waveform is

$$i(t) = \begin{cases} 3t, & 0 \le t \le 2 \text{ s} \\ 6 \text{ A}, & 2 \le t \le 4 \text{ s} \end{cases}$$

Energy absorbed by the resistor

$$W_R = \int_0^4 i^2 R \, dt$$

$$W_R = \int_0^2 (3t)^2 (1) \, dt + \int_2^4 6^2 (1) \, dt = 9 \int_0^2 t^2 \, dt + 36 \int_2^4 dt$$

$$W_R = 9 \left[\frac{t^3}{3} \right]_0^2 + 36(4 - 2) = 9 \left(\frac{8}{3} \right) + 72$$

$$W_R = 24 + 72 = 96 \text{ J}$$

Energy absorbed by the inductor

$$W_L = \frac{1}{2} L i^2$$

At $t = 4$ s, i=6 A

$$W_L = \frac{1}{2}(2)(6)^2 = 36 \text{ J}$$

Total energy absorbed

$$W = W_R + W_L = 132 \text{ J}$$

Hence,

C. 132 J

Key Points

- Energy stored in an inductor can be calculated using

$$W_L = \frac{1}{2} L i^2$$

where i is the current through the inductor at the instant for which the stored energy is required.

- Alternatively, the energy stored can be obtained from the power integral:

$$W_L = \int_0^t vi \, dt = \int_0^t \left(L \frac{di}{dt} i \right) dt$$

Q1.01.1 MCQ | 2001 | 2M | Ans: B | ● ○ ○

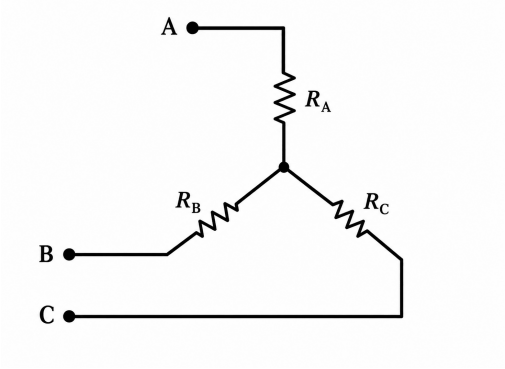

Fig. Solution Q1.01.1

Given,

$$R_{AB} = 6\Omega, \quad \text{with terminal C open.}$$

Hence,

$$R_A + R_B = 6 \tag{1}$$

Similarly,

$$R_{BC} = 11\Omega, \quad \text{with terminal A open.}$$

$$R_B + R_C = 11 \tag{2}$$

Also,

$$R_{CA} = 9\Omega, \quad \text{with terminal B open.}$$

$$R_C + R_A = 9 \tag{3}$$

Adding (1), (2) and (3),

$$2(R_A + R_B + R_C) = 6 + 11 + 9$$

$$2(R_A + R_B + R_C) = 26$$

$$R_A + R_B + R_C = 13 \tag{4}$$

From (4) and (2),

$$R_A = 13 - 11 = 2\Omega$$

From (1) and (2),

$$R_B = 6 - 2 = 4\Omega, \qquad R_C = 11 - 4 = 7\Omega$$

Hence,

B. $R_A = 2\Omega,\ R_B = 4\Omega,\ R_C = 7\Omega$

Q1.99.1 NAT 1999 2M Ans: *

Use the clockwise mesh currents shown in the original circuit. The 2 A source between meshes 1 and 3 gives $I_3 - I_1 = 2$. The supermesh and upper mesh equations are

$$4I_1 - 10I_2 + 6I_3 = 2,$$
$$-4I_1 + 12I_2 - 6I_3 = -2,$$
$$I_3 - I_1 = 2.$$

Adding the first two equations gives $2I_2 = 0$. Substituting $I_3 = I_1 + 2$ in the first gives $10I_1 + 12 = 2$. Therefore

$$\boxed{I_1 = -1 \text{ A}}, \qquad \boxed{I_2 = 0}, \qquad \boxed{I_3 = 1 \text{ A}}.$$

The 1 A current enters the positive terminal of the 10 V source. It absorbs 10 W, so the power *developed* by that source is $\boxed{-10 \text{ W}}$.

Key Points

- A **supermesh** is formed when a current source lies on the common branch between two adjacent meshes. Since the voltage across a current source is unknown, KVL cannot be applied to the individual meshes.
- To identify a supermesh, look for a current source shared by two neighboring meshes. Combine those meshes into a single larger loop excluding the current-source branch, apply KVL around the outer perimeter, and use the current-source relation as an additional constraint equation.
- A **supernode** is formed when a voltage source lies between two non-reference nodes. Since the current through a voltage source is unknown, KCL cannot be applied directly at the individual nodes.
- To identify a supernode, look for a voltage source connecting two non-reference nodes. Treat the two nodes and the voltage source as a single extended node, apply KCL to the entire supernode, and use the voltage-source relation as an additional constraint equation.

Q1.99.2 MCQ | 1999 | 2M | Ans: B

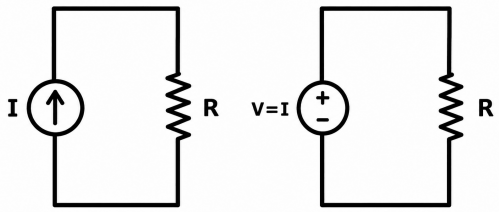

Fig. Solution Q1.99.2

When resistor R is connected to the current source,

$$P = I^2R = 18 \tag{1}$$

When the same resistor is connected to a voltage source having the same numerical value as the current source,

$$V = I$$

Hence,

$$P = \frac{V^2}{R}$$

$$\frac{V^2}{R} = 4.5 \tag{2}$$

Multiplying (1) and (2),

$$(I^2R)\left(\frac{I^2}{R}\right) = 18 \times 4.5$$

$$I^4 = 81 \Rightarrow I = 3 \text{ A}$$

Substituting into (1),

$$R = 2\Omega$$

Hence,

B. 3 A and 2 Ω

Q1.99.3 MCQ 1999 2M Ans: C ● ○ ○

Standard Resistor Color Code Table

Color	Digit	Multiplier	Tolerance	Temp. Coeff. (ppm/°C)
Black	0	10^0	–	250
Brown	1	10^1	±1%	100
Red	2	10^2	±2%	50
Orange	3	10^3	–	15
Yellow	4	10^4	–	25
Green	5	10^5	±0.5%	20
Blue	6	10^6	±0.25%	10
Violet	7	10^7	±0.1%	5
Grey	8	10^8	±0.05%	1
White	9	10^9	–	–
Gold	–	10^{-1}	±5%	–
Silver	–	10^{-2}	±10%	–

For a 3-band resistor,

$$\text{Value} = (\text{1st digit})(\text{2nd digit}) \times 10^{(\text{Multiplier})}$$

Given,

$$1\ \text{k}\Omega = 1000\ \Omega = 10 \times 10^2$$

Band	Value	Color
1st Digit	1	Brown
2nd Digit	0	Black
Multiplier	10^2	Red

Hence,

C. Brown, Black, Red

Q1.97.1 Descriptive 1997 2M Ans: Inductor, 2H ● ● ○

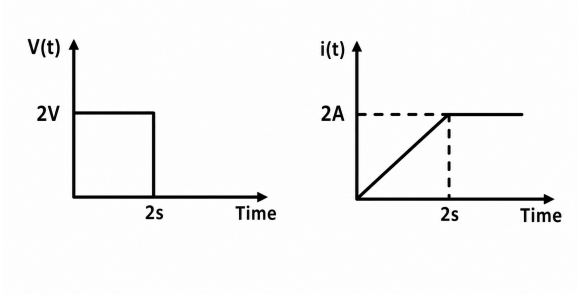

Fig. Solution Q1.97.1

Given,

$$v(t) = \begin{cases} 2\ \text{V}, & 0 \le t \le 2\ \text{s} \\ 0, & t > 2\ \text{s} \end{cases}$$

The current increases linearly from 0 to 2 A during the first 2 s.

A resistor cannot satisfy the given waveforms because for a resistor,

$$v = Ri$$

and the voltage and current must have the same waveform.
A capacitor cannot satisfy the given waveforms because

$$i = C\frac{dv}{dt}$$

and a constant voltage would imply zero current.
Now Inductor,

$$v = L\frac{di}{dt}$$

The slope of the current waveform is

$$\frac{di}{dt} = \frac{2-0}{2-0} = 1 \text{ A/s} = \text{constant}$$

i.e. from 0 to 2 second v(t) is constant.
Substituting the given values,

$$2 = L(1) \Rightarrow L = 2 \text{ H}$$

After $t = 2$ s,

$$v = 0$$

and the current remains constant at 2 A, which is consistent with the behavior of an ideal inductor since

$$v = L\frac{di}{dt} = 0 \quad \Rightarrow \quad \frac{di}{dt} = 0$$

Hence,

Inductor, 2 H

Q1.97.2 Descriptive | 1997 | 2M | Ans: 31V, 13A

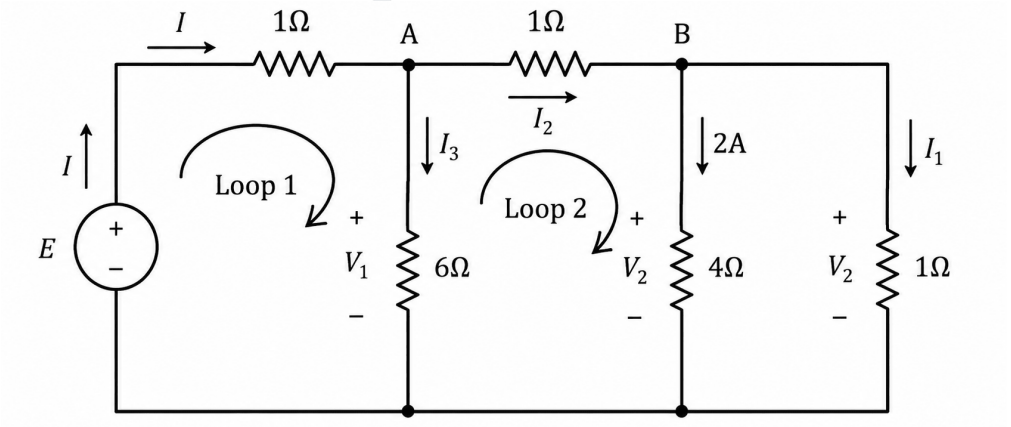

Fig. Solution Q1.97.2

Voltage across the 4Ω resistor is

$$V_2 = 4 \times 2 = 8 \text{ V}$$

$$\Rightarrow I_1 = \frac{8}{1} = 8 \text{ A}$$

Applying KCL at Node B,

$$I_2 = I_1 + 2$$

$$I_2 = 8 + 2 = 10 \text{ A}$$

Applying KVL (Loop 2),

$$V_1 = V_2 + I_2(1)$$

$$V_1 = 8 + 10 = 18 \text{ V}$$

Current through the 6Ω resistor is

$$I_3 = \frac{V_1}{6}$$

$$I_3 = \frac{18}{6} = 3 \text{ A}$$

Applying KCL at Node A,

$$I = I_2 + I_3$$

$$I = 10 + 3 = 13 \text{ A}$$

Finally, applying KVL (Loop 1),

$$E = V_1 + I(1)$$

$$E = 18 + 13 = 31 \text{ V}$$

Therefore,

$$\boxed{31 \text{ V}, 13 \text{ A}}$$

Q1.97.3 NAT | 1997 | 2M | Ans: 5 | ● ○ ○

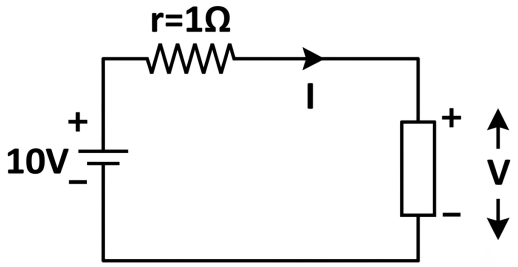

Fig. Solution Q1.97.3

The terminal voltage across the nonlinear load is

$$V = E - Ir$$

$$V = 10 - I \quad (1)$$

The load characteristic is

$$7I = V^2 + 2V \quad (2)$$

Substituting (1) into (2),

$$7I = (10 - I)^2 + 2(10 - I)$$

$$7I = 120 - 22I + I^2$$

$$I^2 - 29I + 120 = 0$$

$$(I - 24)(I - 5) = 0$$

Therefore,

$$I = 24 \text{ A} \qquad \text{or} \qquad I = 5 \text{ A}$$

For

$$I = 24 \text{ A} \quad \Rightarrow \quad V = 10 - 24 = -14 \text{ V}$$

which is not admissible for the given nonlinear load as load is delivering power to battery in this case.
and for,

$$I = 5 \text{ A} \quad \Rightarrow \quad V = 10 - 5 = 5 \text{ V}$$

Hence, the valid operating point is

$$\boxed{I = 5 \text{ A}}$$

Therefore,

$$\boxed{5}$$

Q1.97.4 MCQ 1997 1M Ans: C ● ○ ○

For a capacitor, the energy stored at any instant is

$$W = \frac{1}{2}CV^2$$

When connected to an AC source, the capacitor stores energy during part of the cycle and returns the same energy back to the source during the remaining part.
Hence,

Energy absorbed in one cycle = Energy returned in one cycle

Therefore, the net energy stored over one complete cycle is 0.
Equivalently, the average power absorbed by an ideal capacitor is

$$P_{avg} = V_{rms} I_{rms} \cos \phi$$

For a capacitor current leads voltage by, $\phi = 90°$
Hence,

$$P_{avg} = V_{rms} I_{rms} \cos 90° = 0$$

Therefore,

$$\boxed{\textbf{C.}\ \text{Zero}}$$

Q1.97.5 MCQ 1997 1M Ans: B ● ○ ○

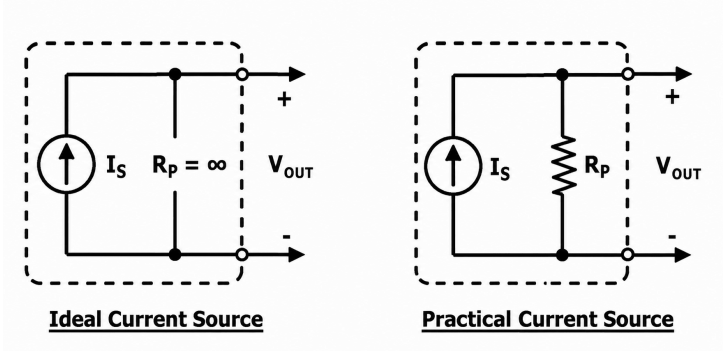

Fig. Solution Q1.97.5

The practical models of ideal sources are:

Source	Practical Model
Voltage Source	Ideal voltage source + series resistance
Current Source	Ideal current source + parallel resistance

Therefore, a practical current source is represented by an ideal current source in parallel with a resistance.
Hence,

B. A resistance in parallel with an ideal current source

Q1.97.6 MCQ 1997 1M Ans: C

For a capacitor,

$$i = C\frac{dv}{dt}$$

When an ideal capacitor is connected directly across an ideal voltage source, the capacitor voltage must immediately become equal to the source voltage.
The charging time constant is $\tau = RC$
Since both the voltage source and capacitor are ideal, $R = 0$
Therefore,

$$\tau = RC = 0$$

A zero time constant implies that the capacitor reaches its final voltage instantaneously.
Equivalently,

$$v_C(0^+) = V \qquad \text{which requires, } \frac{dv}{dt} \to \infty$$

and hence

$$i = C\frac{dv}{dt} \to \infty$$

An ideal voltage source can supply this impulse current. Therefore, the capacitor charges instantaneously.
Hence,

C. Instantaneously

Q1.96.1 MCQ 1996 1M Ans: C

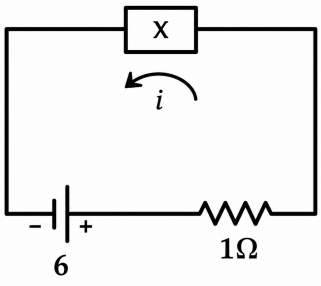

Fig. Solution Q1.96.1

Applying KVL,

$$V_X = 6 - i$$

Since element X always absorbs power,

$$P_X = V_X i \;\Rightarrow\; P_X = i(6 - i)$$

For the given operating condition, i=1A

$$\Rightarrow P_X = 1(6-1) = 5\ \text{W}$$

Let another current i produce the same absorbed power.

$$i(6-i) = 5 \Rightarrow i^2 - 6i + 5 = 0$$

$$(i-1)(i-5) = 0$$

$$i = 1\ \text{A} \quad \text{or} \quad i = 5\ \text{A}$$

Since the question asks for another current,

$$i = 5\ \text{A}$$

Therefore,

C. 5 A

Q1.92.1 MCQ | 1992 | 1M | Ans: C | ● ○ ○

Switch Open

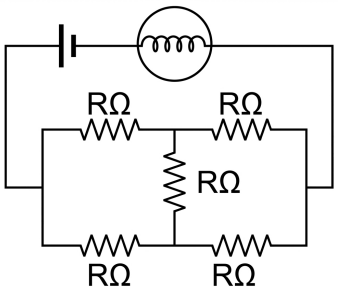

Fig. Solution Q1.92.1

The bridge is balanced since $\frac{R}{R} = \frac{R}{R}$

Hence, no current flows through the central resistor.

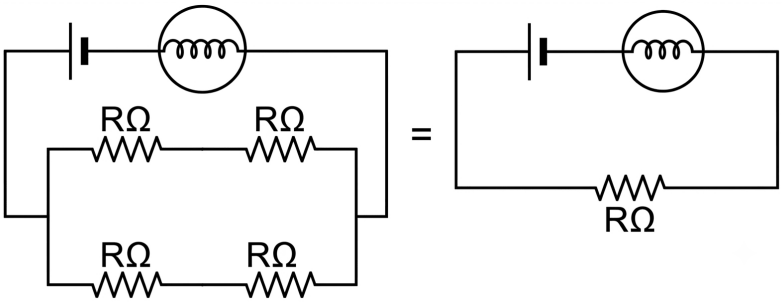

Fig. Solution Q1.92.1

$$R_{\text{eq, open}} = R$$

Switch Closed

When the switch is closed, the top and bottom nodes of the bridge become shorted.

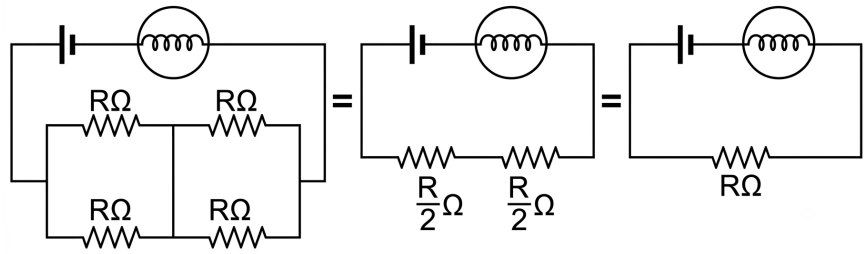

Fig. Solution Q1.92.1

$$R_{\text{eq, closed}} = \frac{R}{2} + \frac{R}{2}$$

$$R_{\text{eq, closed}} = R$$

Therefore,

$$R_{\text{eq, open}} = R_{\text{eq, closed}} = R$$

Hence, the total circuit resistance does not change when the switch is closed.
Therefore, the current through the lamp remains unchanged and so does its power (intensity).

Lamp intensity remains same

Hence,

C. Remains same

SOLUTIONS

11

Transient Analysis

Topics

1. Introduction to Transient Analysis
2. Basics of Signals for Transient Analysis
3. RC Circuits - Introduction and Examples
4. Impulse Current in Capacitor and Basics of Laplace in Circuit Analysis
5. Slope at t=0+, Switched Capacitor and Impulse Response of RC Circuits
6. RC Circuits with Pulse Wave Input
7. Initial and Final Value Theorems
8. RL Circuits

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 50	**MCQ:** 37	**MSQ:** 0	**NAT:** 13
1M: 20	**2M:** 26	**Descriptive:** 4	**Avg Weightage** ~2M	**Range:** 1M(2024)–4M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ | 2026 | 2M | Ans: B ● ● ○

For $t < 0$, the current source is OFF

From -∞ to 0, the capacitor would have been completely charged and behaves as open circuit.

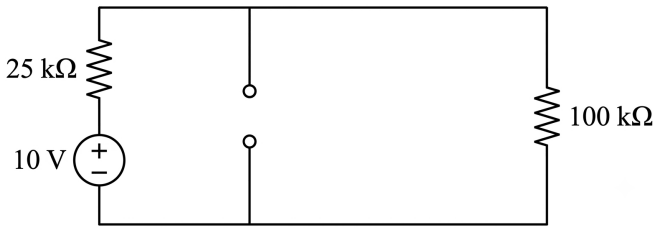

Fig. Solution Q2.26.1

Hence, the circuit reduces to a voltage divider:

$$v_C(0^-) = 10\left(\frac{100k}{25k + 100k}\right)$$

$$v_C(0^-) = 8 \text{ V}$$

Since capacitor voltage cannot change instantaneously,

$$\boxed{v_C(0^+) = v_C(0^-) = 8 \text{ V}}$$

Final value $v_C(\infty)$

At $t = \infty$ (steady state), the capacitor again behaves as an open circuit as it is fully charged.

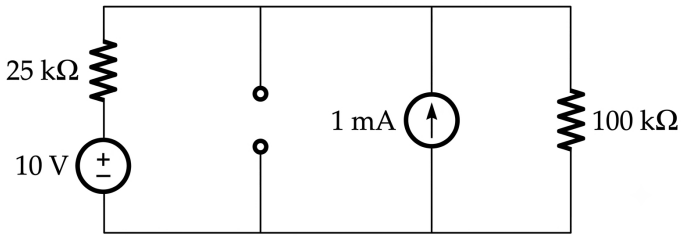

Fig. Solution Q2.26.1

Applying KCL at the capacitor node,

$$\frac{v_C - 10}{25k} + \frac{v_C}{100k} = 1 \text{ mA}$$

Multiplying by $100k$,

$$4(v_C - 10) + v_C = 100$$

$$5v_C = 140$$

$$\boxed{v_C(\infty) = 28 \text{ V}}$$

Time constant

$$\tau = R_{\text{th}}C$$

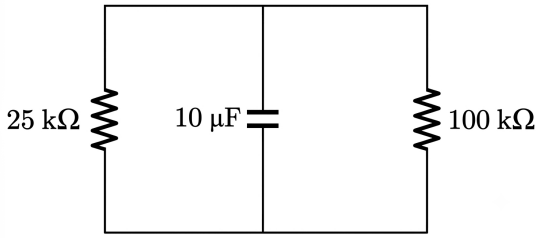

Fig. Solution Q2.26.1

where

$$R_{th} = 25k \parallel 100k = 20k\Omega$$

Therefore,

$$\tau = (20k)(10\mu F)$$

$$\tau = 0.2 \text{ s}$$

Transient response

$$v_C(t) = v_C(\infty) + \left[v_C(0^+) - v_C(\infty)\right]e^{-t/\tau}$$

$$v_C(t) = 28 + (8 - 28)e^{-t/0.2}$$

At t=0.5s;

$$v_C(0.5) = 28 - 20e^{-2.5} = 28 - 20(0.0821)$$

$$v_C(0.5) \approx 26.36 \text{ V}$$

Hence,

B. 8.00 V, 26.36 V, 28.00 V

Key Points

Condition	Capacitor	Inductor
DC Steady State ($t = \infty$)	Open Circuit	Short Circuit
Immediately After Switching ($t = 0^+$)	Short Circuit	Open Circuit

Q2.25.1 MCQ | 2025 | 2M | Ans: C | ● ● ○

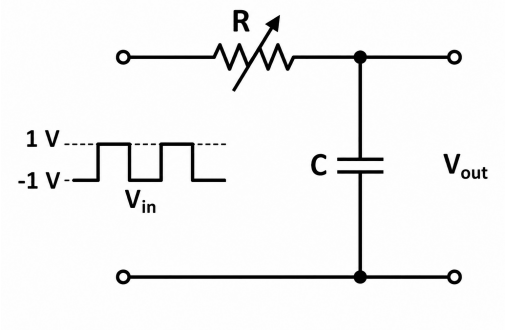

Fig. Solution Q2.25.1

Method Frequency-Domain (Filter) Approach

The circuit is an RC low-pass filter with output taken across the capacitor.
The cut-off frequency of the filter is

$$f_c = \frac{1}{2\pi RC}$$

Since R and C are very large,

$$f_c = \frac{1}{2\pi RC}, \quad \text{is small.}$$

The input square wave can be represented by its Fourier series as

$$v_{in}(t) = \frac{4}{\pi}\left(\sin\omega t + \frac{1}{3}\sin 3\omega t + \frac{1}{5}\sin 5\omega t + \cdots\right)$$

A low-pass filter passes frequency components satisfying

$$f < f_c$$

and attenuates components satisfying

$$f > f_c$$

Since the input square wave is of very high frequency, all of its harmonic components are heavily attenuated.
Only a small portion of the lower-frequency components can appear at the output.
Therefore, the output resembles a sinusoidal waveform rather than a square wave. Since multiple sinusoidal components are still present, the waveform is not perfectly sinusoidal but slightly distorted.
Hence, the output is best represented by

C

Method Time-Domain (Charging/Discharging) Approach

The time constant is

$$\tau = RC$$

Since R and C are very large,

$$\tau \gg \frac{T}{2}$$

where T is the period of the square wave.
Thus, during each half-cycle, the capacitor does not have sufficient time to charge or discharge completely.
During the positive half-cycle,

$$v_C(t) = V_f + (V_i - V_f)e^{-t/\tau}$$

where $V_f = +1$ V.
During the negative half-cycle,

$$v_C(t) = V_f + (V_i - V_f)e^{-t/\tau}$$

where $V_f = -1$ V.
So the capacitor charges and dischares exponentially.

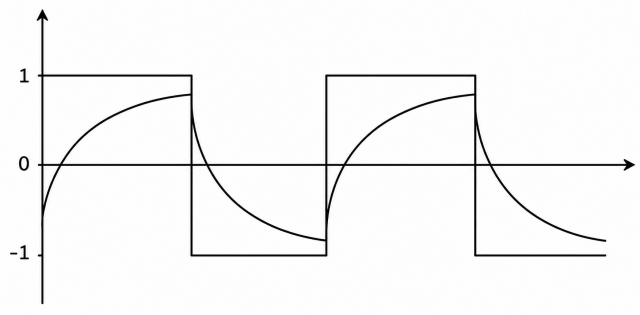

Fig. Solution Q2.25.1

Hence, the waveform is best represented by

C

Q2.25.2 NAT 2025 2M Ans: 4.1 ●●○

Before $t = 0$, the switch is open.

At DC steady state ($t=0^-$), the capacitor behaves as an open circuit.

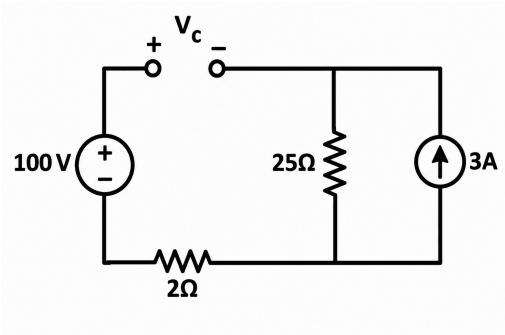

Fig. Solution Q2.25.2

Hence, no current flows through the 2Ω resistor

The 3 A current source drives current through the 25Ω resistor.

$$V_{25} = IR = 3 \times 25$$

$$V_{25} = 75 \text{ V}$$

Therefore, the right plate of the capacitor is at 75V

Thus,

$$v_C(0^-) = 100 - 75 = 25 \text{ V}$$

Since capacitor voltage cannot change instantaneously,

$$\boxed{v_C(0^+) = 25 \text{ V}}$$

At $t = 0$, the switch is closed.

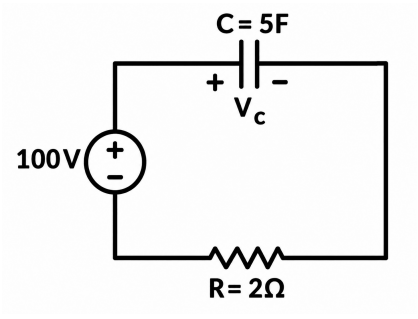

Fig. Solution Q2.25.2

At $t = \infty$,

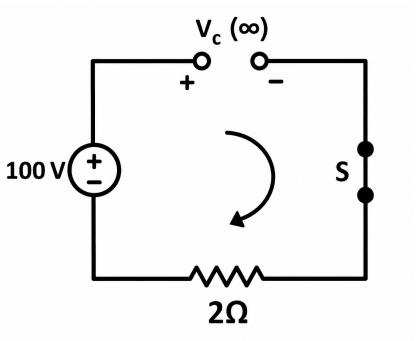

Fig. Solution Q2.25.2

Therefore, no current flows through the 2Ω resistor and the left plate remains at 100V

Thus,

$$\boxed{v_C(\infty) = 100 \text{ V}}$$

Time Constant

To find the time constant, suppress all independent sources.

$$100\text{ V source} \rightarrow \text{short circuit}, \quad 3\text{ A source} \rightarrow \text{open circuit}$$

Since the switch is closed, the capacitor sees only

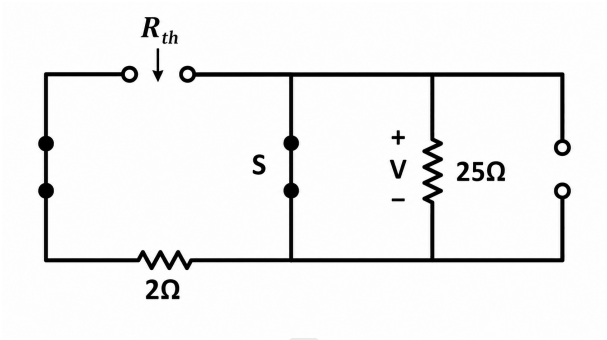

Fig. Solution Q2.25.2

$$R_{th} = 2\Omega$$

Hence,

$$\tau = R_{th}C = 2 \times 5 = 10\text{ s}$$

Therefore,

$$v_C(t) = v_C(\infty) + \left[v_C(0^+) - v_C(\infty)\right]e^{-t/\tau}$$

$$v_C(t) = 100 + (25 - 100)e^{-t/10} \Rightarrow v_C(t) = 100 - 75e^{-t/10}$$

The capacitor reaches 50 V when

$$50 = 100 - 75e^{-t/10}$$

$$e^{-t/10} = \frac{2}{3} \Rightarrow t \approx 4.05\text{ s}$$

Hence,

4.1

Q2.24.1 MCQ 2024 1M Ans: A ● ○ ○

After the switch is closed, the time constant is

$$\tau = \frac{L_{eq}}{R_{eq}}$$

To find the time constant, suppress all independent sources.

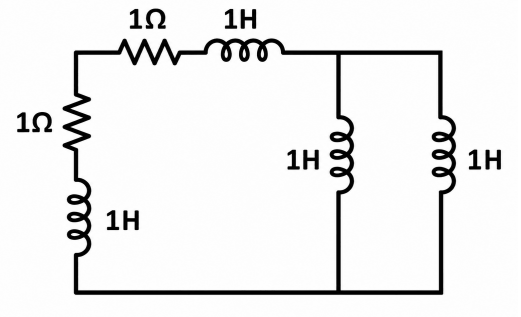

Fig. Solution Q2.24.1

Equivalent Resistance

The two 1Ω resistors are in series.

$$R_{eq} = 1 + 1 = 2\Omega$$

Equivalent Inductance

$$L_s = 1 + 1 = 2 \text{ H}$$

The two right-side inductors are in parallel.

$$L_p = 1 \parallel 1 = \frac{1 \times 1}{1 + 1} = 0.5 \text{ H}$$

Hence,

$$L_{eq} = L_s + L_p = 2 + 0.5 = 2.5 \text{ H}$$

Therefore,

$$\tau = \frac{L_{eq}}{R_{eq}} = \frac{2.5}{2} = 1.25 \text{ s}$$

A. 1.25

Q2.23.1 NAT 2023 1M Ans: 8.0

For $t < 0$, the switch is closed and the circuit is in steady state.

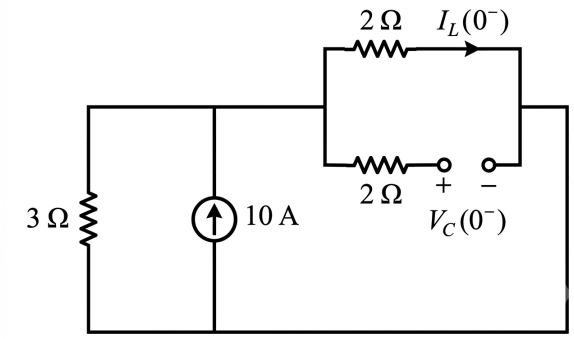

Fig. Solution Q2.23.1

Hence,

Inductor → short circuit, Capacitor → open circuit

Therefore, the capacitor branch carries no current.

The remaining circuit consists of $R_1 = 2\Omega$ in parallel with $R_3 = 3\Omega$, fed by the 10 A current source.

$$R_{eq} = R_1 \parallel R_3 = \frac{2 \times 3}{2 + 3} = 1.2\Omega$$

Hence, the voltage across the parallel network is

$$V_2 = V_3 = 10 \times 1.2 = 12 \text{ V}$$

Therefore,

$$i_L(0^-) = \frac{12}{2} = 6 \text{ A}$$

and

$$v_C(0^-) = 12 \text{ V}$$

Using continuity of inductor current and capacitor voltage,

$$i_L(0^+) = i_L(0^-) = 6 \text{ A}, \qquad v_C(0^+) = v_C(0^-) = 12 \text{ V}$$

After the switch is opened, R_3 is disconnected.
At $t = 0^+$,
the inductor behaves as a 6 A current source and the capacitor behaves as a 12 V voltage source.

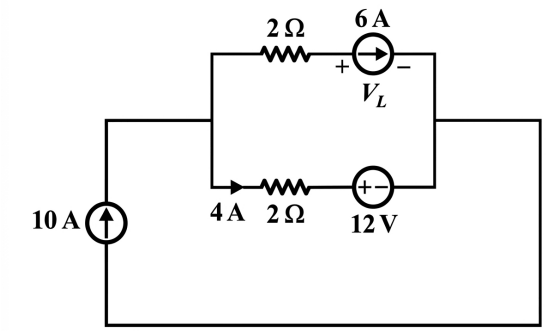

Fig. Solution Q2.23.1

Current through R_1 is, $i_1 = 6$ A
Hence, the voltage drop across R_1 is

$$V_{R_1} = i_1 R_1 = 6 \times 2 = 12 \text{ V}$$

Let V be the potential across 10A source.
The current through R_2 is

$$i_2 = \frac{V - 12}{2}$$

Also,

$$i_1 + i_2 = 10$$

$$6 + \frac{V - 12}{2} = 10$$

$$V = 20 \text{ V}$$

Applying KVL,

$$V = V_{R_1} + V_L$$

$$20 = 12 + V_L \Rightarrow V_L = 8 \text{ V}$$

Hence,

$$\boxed{8.0}$$

Q2.22.1 NAT 2022 1M Ans: 5.00 ● ○ ○

At steady state, the capacitor behaves as an open circuit for DC.
Therefore,

$$R_{eq} = \frac{6 \times 3}{6 + 3} = 2\Omega$$

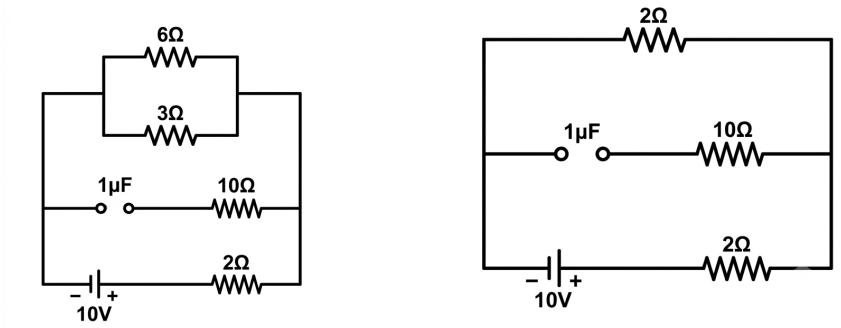

Fig. Solution Q2.22.1

Applying voltage division,

$$V_{2\Omega} = 10\left(\frac{2}{2+2}\right) = 5\text{ V}$$

Since the 6Ω and 3Ω are in parallel

$$V_{6\Omega} = 5\text{ V}$$

Hence,

$$\boxed{5.00}$$

Q2.21.1 NAT 2021 2M Ans: 12.46 ● ● ●

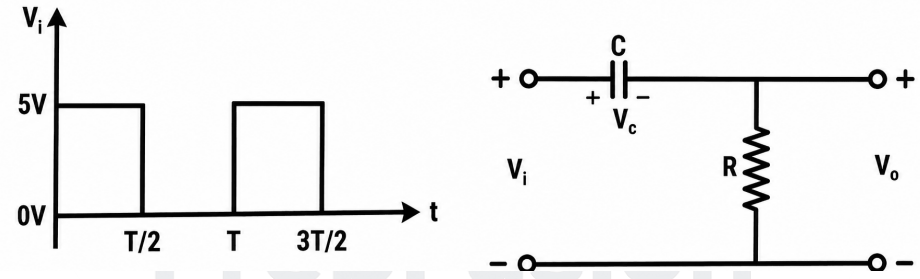

Fig. Solution Q2.21.1

$$T = \frac{1}{f} = \frac{1}{100} = 0.01\text{ s} \quad \Rightarrow \quad \frac{T}{2} = 5\text{ ms}$$

During

$$0 < t < \frac{T}{2}$$

the input is 5 V, and the capacitor charges towards 5 V. At the end of this interval, the capacitor voltage reaches its maximum value $V_{c,\max}$.

Similarly, during

$$\frac{T}{2} < t < T$$

the input is 0 V, and the capacitor discharges towards 0 V. At the end of this interval, the capacitor voltage reaches its minimum value $V_{c,\min}$.

Since

$$V_o = V_i - V_c$$

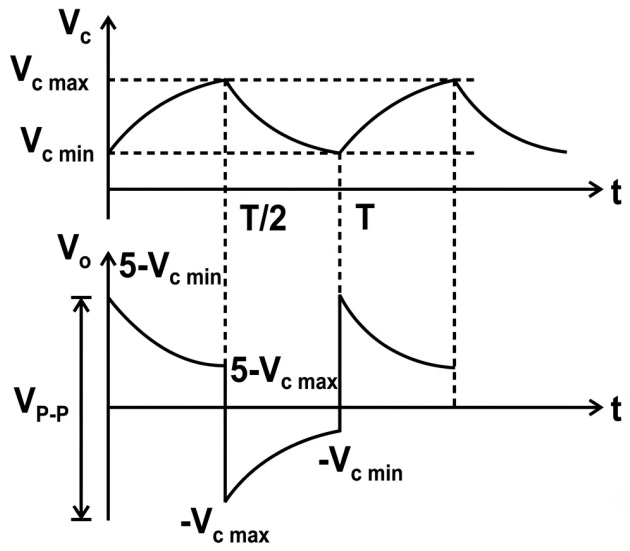

Fig. Solution Q2.21.1

the peak-to-peak output voltage is

$$V_{o(pp)} = (5 - V_{c,\min}) - (-V_{c,\max})$$

Given,

$$V_{o(pp)} = 6.2 \text{ V}$$

Therefore,

$$6.2 = 5 + V_{c,\max} - V_{c,\min}$$

$$V_{c,\max} - V_{c,\min} = 1.2 \quad (1)$$

For $0 < t < T/2$

The input is 5 V, so the capacitor charges towards 5 V.

$$V_c(t) = 5 + (V_{c,\min} - 5)\, e^{-t/\tau}$$

At $t = \dfrac{T}{2}$;

$$V_{c,\max} = 5 + (V_{c,\min} - 5)\, e^{-T/(2\tau)} \quad (2)$$

For $T/2 < t < T$

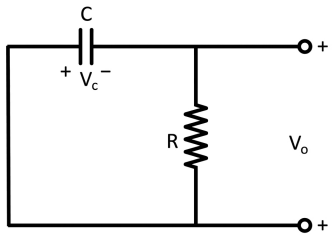

Fig. Solution Q2.21.1

The input becomes 0 V, so the capacitor discharges towards 0 V.

$$V_c(t) = V_{c,\max} e^{-(t-\frac{T}{2})/\tau}$$

At t=T ;

$$V_{c,\min} = V_{c,\max} e^{-T/(2\tau)} \tag{3}$$

Substituting (3) into (2),

$$V_{c,\max} = 5 + \left(V_{c,\max} e^{-T/(2\tau)} - 5\right) e^{-T/(2\tau)}$$

Using

$$V_{c,\max} - V_{c,\min} = 1.2$$

$$5 - 6.2\, e^{-T/(2\tau)} = 1.2 \;\Rightarrow\; e^{-T/(2\tau)} = \frac{3.8}{6.2} = 0.6129$$

$$-\frac{T}{2\tau} = \ln(0.6129) = -0.4895$$

$$\frac{T}{2\tau} = 0.4895$$

Using

$$T = 0.01 \text{ s}$$

$$\tau = \frac{0.01}{2(0.4895)} = 0.01021 \text{ s}$$

Since

$$\tau = RC$$

$$0.01021 = 820\, C \;\Rightarrow\; C = 12.46 \times 10^{-6} \text{F}$$

$$\boxed{12.46}$$

Q2.21.2 MCQ 2021 2M Ans: C ●●○

Step 1: Find $I_L(0^-)$

The switch has been closed for a long time.

At DC steady state,

Inductor → short circuit

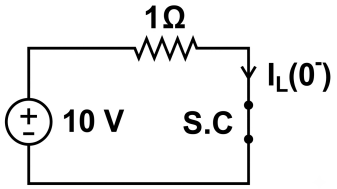

Fig. Solution Q2.21.2

Hence, the circuit reduces to a 10 V source in series with a 1Ω resistor.

$$I_L(0^-) = \frac{10}{1} = 10 \text{ A}$$

Using continuity of inductor current,

$$I_L(0^+) = I_L(0^-) = 10 \text{ A}$$

Step 2: Find $I_L(\infty)$

After the switch is opened,

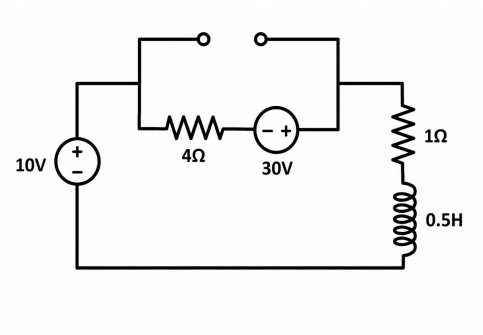

Fig. Solution Q2.21.2

At steady state,

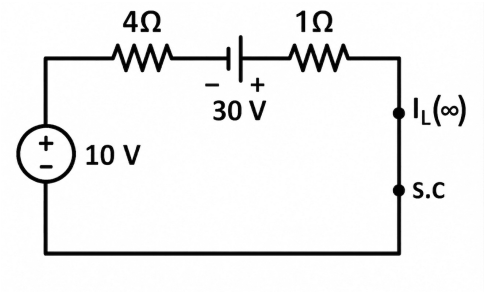

Fig. Solution Q2.21.2

Applying KVL,

$$10 + 30 - I(4 + 1) = 0$$

$$40 - 5I = 0 \Rightarrow I_L(\infty) = 8 \text{ A}$$

Step 3: Find the time constant

To find R_{eq}, suppress all independent voltage sources.

10 V source $\rightarrow$ short circuit, 30 V source $\rightarrow$ short circuit

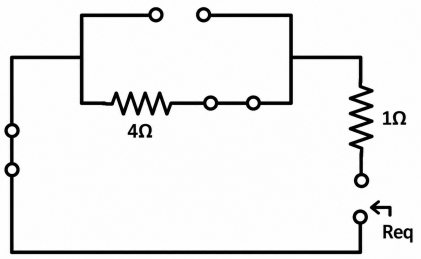

Fig. Solution Q2.21.2

The inductor sees

$$R_{eq} = 4 + 1 = 5\Omega$$

Hence,

$$\tau = \frac{L}{R_{eq}} = \frac{0.5}{5} = 0.1 \text{ s}$$

Step 4: Write the complete response

For a first-order RL circuit,

$$I_L(t) = I_L(\infty) + \left[I_L(0^+) - I_L(\infty)\right]e^{-t/\tau}$$

Substituting the values,

$$I_L(t) = 8 + (10 - 8)e^{-t/0.1}$$

$$I_L(t) = 8 + 2e^{-10t}$$

Hence,

$$\boxed{\textbf{C. } 8 + 2e^{-10t}}$$

Q2.20.1 NAT | 2020 | 2M | Ans: 54.99 | ● ● ○

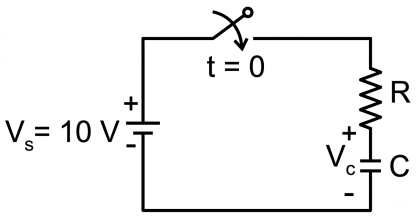

Fig. Solution Q2.20.1

For a first-order RC circuit,

$$v_C(t) = V_f + (V_i - V_f)e^{-t/\tau}$$

where V_i is the initial capacitor voltage at $t = 0^+$, V_f is the final (steady-state) capacitor voltage as $t \to \infty$, and $\tau = RC$ is the time constant of the circuit.
Here,

$$V_f = 10 \text{ V}$$

Given that the capacitor voltage crosses zero at

$$t = 0.4\tau$$

Therefore,

$$0 = 10 + (V_i - 10)e^{-0.4} \Rightarrow V_i - 10 = -10e^{0.4}$$

$$V_i = 10(1 - e^{0.4}) \Rightarrow V_i = -4.918 \text{ V}$$

Hence, the magnitude of the initial capacitor voltage is

$$|V_i| = 4.918 \text{ V}$$

At $t = 0.2\tau$;

$$v_C(0.2\tau) = 10 + (V_i - 10)e^{-0.2}$$

Substituting

$$V_i - 10 = -10e^{0.4}$$

gives

$$v_C(0.2\tau) = 10 - 10e^{0.2}$$

$$v_C(0.2\tau) = -2.214 \text{ V}$$

Thus, the magnitude of the capacitor voltage at $t = 0.2\tau$ is

$$|v_C(0.2\tau)| = 2.214 \text{ V}$$

Therefore, the percentage change required in the capacitor voltage magnitude is

$$\frac{|V_i| - |v_C(0.2\tau)|}{|V_i|} \times 100 = \frac{4.918 - 2.214}{4.918} \times 100$$

$$= 54.99\%$$

Hence,

$$\boxed{54.99}$$

Q2.19.1 NAT | 2019 | 2M | Ans: 0.46 | ● ○ ○

The capacitor is initially charged to $V_i = 100$ V

For a discharging RC circuit,

$$v_C(t) = V_i e^{-t/\tau}$$

Given,

$$R = 1\ \text{k}\Omega = 1000\ \Omega, \quad C = 0.1\ \mu\text{F} = 0.1 \times 10^{-6}\text{F}$$

Hence,

$$\tau = RC = 1000 \times 0.1 \times 10^{-6} = 10^{-4}\ \text{s}$$

The capacitor voltage falls to 1 V

$$\Rightarrow 1 = 100e^{-t/\tau}$$

$$e^{-t/\tau} = 0.01$$

Taking natural logarithm,

$$-\frac{t}{\tau} = \ln(0.01) = -4.6052$$

$$t = 4.6052\,\tau = 0.4605\ \text{ms}$$

Hence,

$$\boxed{0.46}$$

Key Points

- The **time constant** (τ) is the time required for a first-order response to complete 63.2% of its total change (or reduce to 36.8% of its initial value).

Q2.17.1 NAT | 2017 | 1M | Ans: 100.0 | ● ● ○

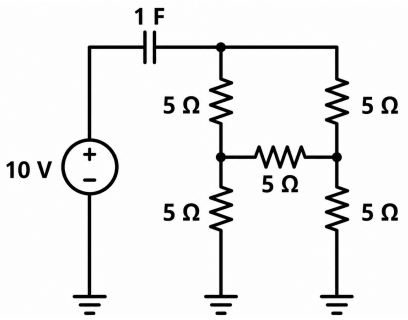

Fig. Solution Q2.17.1

The bridge is balanced since

$$\frac{5}{5} = \frac{5}{5}$$

Hence, no current flows through the middle 5Ω resistor.

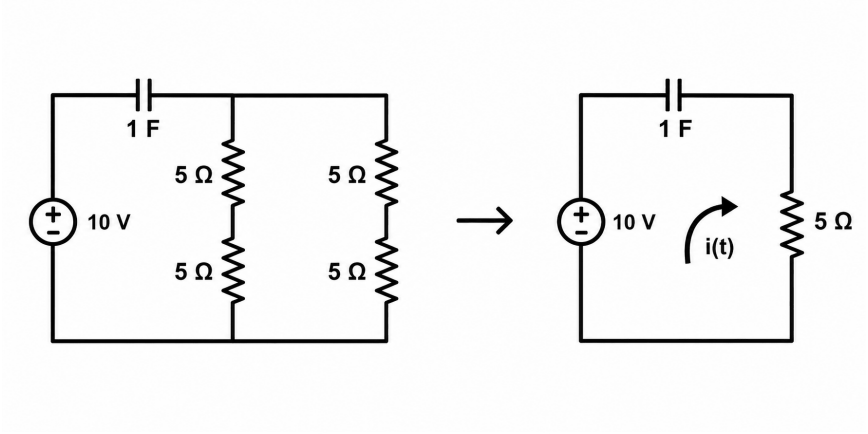

Fig. Solution Q2.17.1

$$R_{eq} = 10 \parallel 10 = 5\Omega$$

Method 1

The charging current is

$$i(t) = \frac{V}{R} e^{-t/RC}$$

$$i(t) = \frac{10}{5} e^{-t/5} = 2e^{-0.2t}$$

The energy supplied by the source is

$$E = \int_0^\infty v_s \, i(t) \, dt$$

$$E = \int_0^\infty 10 \left(2e^{-0.2t}\right) dt = 20 \int_0^\infty e^{-0.2t} \, dt$$

$$= 20 \left[-\frac{e^{-0.2t}}{0.2} \right]_0^\infty = 20 \left(0 + \frac{1}{0.2}\right)$$

$$= 100 \text{ J}$$

Method 2

For an initially uncharged capacitor charged from a DC source,

$$E_{\text{stored}} = \frac{1}{2} CV^2$$

and an equal amount of energy is dissipated in the resistor.
Therefore, the total energy supplied by the source is

$$E_{\text{source}} = CV^2$$

Substituting,

$$E_{\text{source}} = 1 \times 10^2 = 100 \text{ J}$$

Hence,

$$\boxed{100.0}$$

Mistakes to be avoided

- Do not directly use $\frac{1}{2}CV^2$ for the energy supplied by the source. It gives only the energy finally stored in the capacitor.

Q2.17.2 MCQ 2017 2M Ans: A

For $t < 0$

The switch has been closed for a long time. Hence, the inductor is in steady state and behaves as a short circuit.

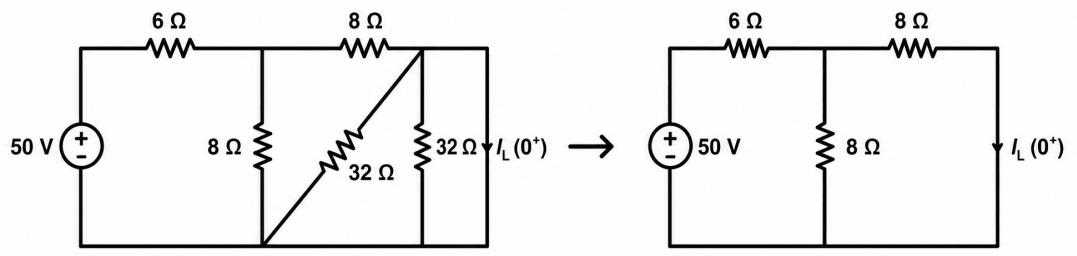

Fig. Solution Q2.17.2

The circuit reduces to

$$R_{eq} = 6 + (8 \parallel 8)$$

$$R_{eq} = 6 + 4 = 10\Omega$$

Therefore, the source current is

$$I = \frac{50}{10} = 5 \text{ A}$$

Since the two 8Ω branches are identical, the current divides equally.
Hence, the current through the branch containing the inductor is

$$i_L(0^-) = \frac{5}{2} = 2.5 \text{ A}$$

Using continuity of inductor current,

$$i_L(0^+) = i_L(0^-) = 2.5 \text{ A}$$

For $t > 0$

The switch is opened and the source is disconnected.
The inductor discharges through the resistive network connected across its terminals.

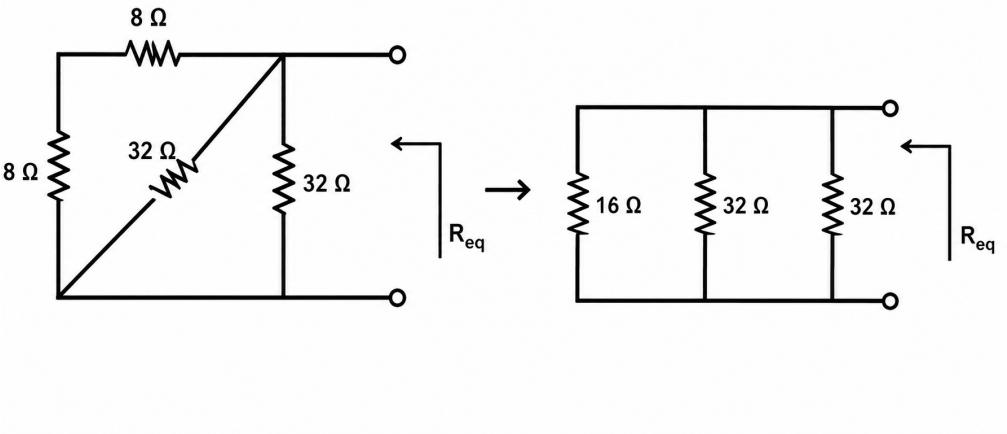

Fig. Solution Q2.17.2

The resistance seen by the inductor is

$$R_{eq} = 16 \parallel 32 \parallel 32 = 8\Omega$$

The time constant is

$$\tau = \frac{L}{R_{eq}} = \frac{2}{8} = 0.25 \text{ s}$$

For a source-free RL circuit,

$$i_L(t) = i_L(0^+)e^{-t/\tau} = 2.5e^{-t/0.25}$$

$$\boxed{i_L(t) = 2.5e^{-4t} \text{ A}}$$

Hence,

$$\boxed{\textbf{A. } 2.5e^{-4t}}$$

Q2.16.1 NAT | 2016 | 2M | Ans: 6.99 | ● ○ ○

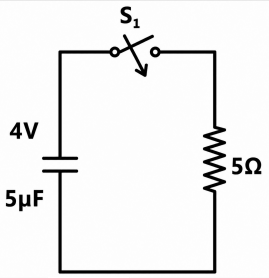

Fig. Solution Q2.16.1

The circuit is a source-free RC discharging circuit.
The initial capacitor voltage is

$$V_0 = 4 \text{ V}$$

The time constant is

$$\tau = RC = (5)(5 \times 10^{-6})$$

$$\tau = 25\ \mu s$$

For a discharging capacitor,

$$v_C(t) = V_0 e^{-t/\tau}$$

Hence,

$$v_C(25\mu s) = 4e^{-25/25} = 4e^{-1}$$

and

$$v_C(100\mu s) = 4e^{-100/25} = 4e^{-4}$$

The charge on the capacitor is

$$q(t) = Cv_C(t)$$

Therefore, the charge lost by the capacitor from

$$t = 25\ \mu s \quad \text{to} \quad t = 100\ \mu s$$

is

$$\Delta q = C\left[v_C(25\mu s) - v_C(100\mu s)\right]$$

$$= 5 \times 10^{-6}\left(4e^{-1} - 4e^{-4}\right) = 20 \times 10^{-6}\left(e^{-1} - e^{-4}\right)$$

$$= 20 \times 10^{-6}(0.3679 - 0.0183) = 6.99 \times 10^{-6}\ \text{C}$$

$$\boxed{6.99}$$

Mistakes to be avoided

- Do not write a negative answer. The question asks for **charge lost**, which is a positive quantity. If your calculation gives a negative value due to the order of subtraction, report only its magnitude.

Q2.16.2 NAT 2016 2M Ans: 2 ●●○

For $t < 0$, S_2 is closed and S_1 is open.

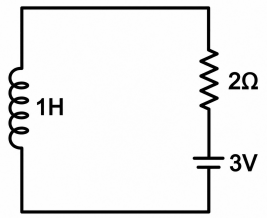

Fig. Solution Q2.16.2

Since the circuit is in steady state,

$$\text{Inductor} \rightarrow \text{short circuit}$$

$$\Rightarrow i_L(0^-) = \frac{3}{2} = 1.5\ \text{A}$$

Using continuity of inductor current,

$$i_L(0^+) = i_L(0^-) = 1.5\ \text{A}$$

Thus, at $t = 0^+$, the inductor can be replaced by a 1.5 A current source.
Let

$$V = V_L(0^+)$$

be the voltage across the inductor.

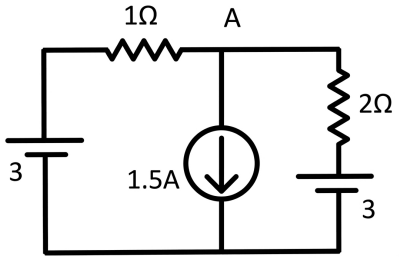

Fig. Solution Q2.16.2

Applying KCL at node A,

$$\frac{V-3}{1} + \frac{V-3}{2} + 1.5 = 0$$

Multiplying by 2,

$$2(V-3)+(V-3)+3=0 \Rightarrow 3V-6=0$$

$$V=2\text{ V} \Rightarrow V_L(0^+)=2\text{ V}$$

Using

$$V_L = L\frac{di_L}{dt}$$

$$2=1\times\frac{di_L}{dt} \Rightarrow \left.\frac{di_L}{dt}\right|_{t=0^+}=2\text{ A/s}$$

$$\boxed{2}$$

Q2.15.1 MCQ | 2015 | 1M | Ans: D

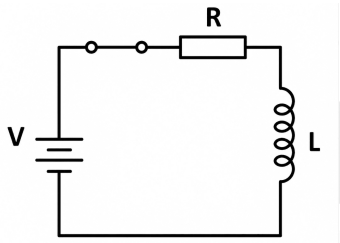

Fig. Solution Q2.15.1

Applying KVL,

$$V = Ri + L\frac{di}{dt}$$

Differentiating both sides,

$$0 = R\frac{di}{dt} + L\frac{d^2i}{dt^2}$$

Hence,

$$\frac{d^2i}{dt^2} = -\frac{R}{L}\frac{di}{dt}$$

At $t = 0^+$, the inductor current cannot change instantaneously, so $\Rightarrow i(0^+) = 0$
Substituting into the KVL equation,

$$V = L\left.\frac{di}{dt}\right|_{0^+} \Rightarrow \left.\frac{di}{dt}\right|_{0^+} = \frac{V}{L}$$

Therefore,

$$\left.\frac{d^2i}{dt^2}\right|_{0^+} = -\frac{R}{L}\left(\frac{V}{L}\right) = -\frac{RV}{L^2}$$

Therefore,

$$\boxed{\textbf{D. } \frac{-RV}{L^2}}$$

Mistakes to be avoided

- $i_L(0) = 0$ does *not* imply

$$\left.\frac{di_L}{dt}\right|_0 = 0 \quad \text{or} \quad \left.\frac{d^2i_L}{dt^2}\right|_0 = 0.$$

A function can be zero at a point while its derivatives are non-zero (e.g., $f(t) = t^2$).

Q2.14.1 MCQ | 2014 | 1M | Ans: A ● ○ ○

Convert the circuit into the s-domain.

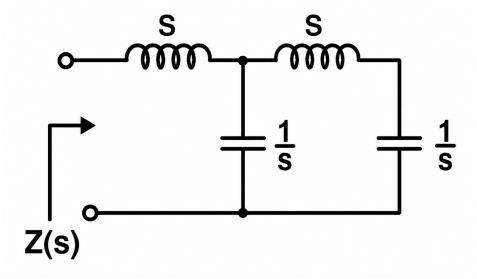

Fig. Solution Q2.14.1

Solving,

$$Z_2 = \frac{1}{s} \parallel \left(s + \frac{1}{s}\right) = \frac{\frac{1}{s}\left(s + \frac{1}{s}\right)}{\frac{1}{s} + \left(s + \frac{1}{s}\right)}$$

$$Z_2 = \frac{s^2 + 1}{s(s^2 + 2)}$$

The driving-point impedance is

$$Z(s) = s + Z_2$$

$$Z(s) = s + \frac{s^2 + 1}{s(s^2 + 2)} = \frac{s^2(s^2 + 2) + s^2 + 1}{s(s^2 + 2)}$$

Hence,

$$Z(s) = \frac{s^4 + 3s^2 + 1}{s^3 + 2s}$$

Therefore,

$$\boxed{\textbf{A.}\ \frac{s^4 + 3s^2 + 1}{s^3 + 2s}}$$

Q2.14.2 MCQ | 2014 | 1M | Ans: MTA ● ● ○

At position 1 for a long time the capacitor is open at steady state, so $v_C(0^-) = V_{01}$. Its voltage is continuous: $v_C(0^+) = V_{01}$.

After moving to position 2, the two resistors are in series, giving $\tau = 2RC$. The polarity of the left source is reversed in the actual figure, so $v_C(\infty) = -V_{02}$. Therefore

$$v_C(t) = -V_{02} + (V_{01} + V_{02})e^{-t/(2RC)}.$$

This has the required initial value V_{01} and final value $-V_{02}$. None of the printed options matches both limits.

Source/answer-checking warning: The official archive GATE 2014 EE Set 2 Q7 reproduces this opposite source polarity but lists D in its answer key. D instead tends to $+V_{02}$. The printed source polarity and the archived key therefore conflict.

Q2.14.3 MCQ | 2014 | 1M | Ans: C ● ● ○

At $t = 0$, both switches are operated.

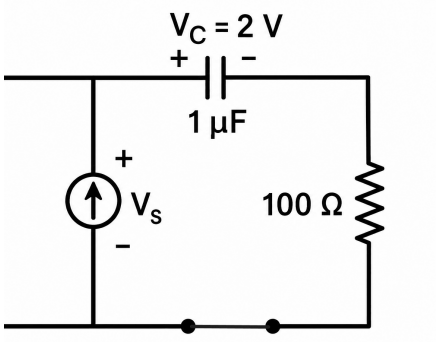

Fig. Solution Q2.14.3

Method Time-Domain Approach

The switch across the current source is opened, forcing the source current

$$I_s = 20 \text{ mA}$$

to flow through the resistor-capacitor series circuit.
Since capacitor voltage cannot change instantaneously,

$$V_C(0^+) = V_C(0^-) = -2 \text{ V}$$

The resistor voltage at $t = 0^+$ is

$$V_R(0^+) = I_s R = 20 \times 10^{-3} \times 100 = 2 \text{ V}$$

Hence,

$$V_s(0^+) = V_C(0^+) + V_R(0^+)$$
$$= -2 + 2 = 0$$

For $t > 0$, the current through the capacitor remains constant:

$$i_C = I_s$$

Using

$$i_C = C\frac{dV_C}{dt}$$

Integrating,

$$V_C(t) = V_C(0^+) + \frac{I_s}{C}t$$

$$V_C(t) = -2 + \frac{I_s}{C}t$$

Therefore,

$$V_s(t) = V_C(t) + I_s R$$

$$V_s(t) = \left(-2 + \frac{I_s}{C}t\right) + I_s R$$

Since

$$I_s R = 2 \text{ V}$$

$$V_s(t) = \frac{I_s}{C}t$$

Thus, $V_s(t)$ is a straight line passing through the origin with positive slope.

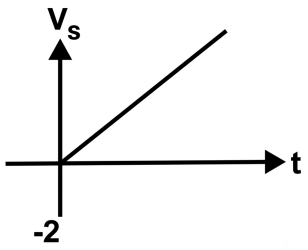

Fig. Solution Q2.14.3

Hence,

C

Method Laplace Transform Approach

Using Laplace transforms,

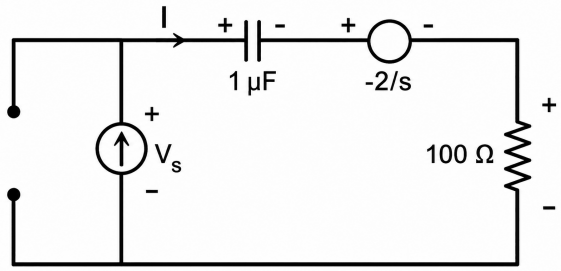

Fig. Solution Q2.14.3

$$V_s(s) = \left(\frac{I}{s}\right)\left(\frac{1}{sC}\right) + \left(\frac{I}{s}\right)R + \left(\frac{-2}{s}\right)$$

$$V_s(s) = \frac{1}{s}\left(IR - 2 + \frac{I}{sC}\right)$$

Substituting

$$I = 20 \times 10^{-3}\ \text{A}, \qquad R = 100\Omega, \qquad C = 1\mu\text{F}$$

$$V_s(s) = \frac{1}{s}\left(20 \times 10^{-3} \times 100 - 2 + \frac{20 \times 10^{-3}}{s \times 10^{-6}}\right)$$

$$V_s(s) = \frac{1}{s}\left(2 - 2 + \frac{20 \times 10^{3}}{s}\right)$$

$$V_s(s) = \frac{20 \times 10^3}{s^2} \Rightarrow V_s(s) = \frac{2 \times 10^4}{s^2}$$

Taking inverse Laplace transform, $\Rightarrow V_s(t) = 2 \times 10^4\, t\, u(t)$
which is the equation of a straight line passing through the origin.
Therefore, the correct waveform is

C

Key Points

- **Common Laplace Transform pairs** (These Laplace formulas are consistent only if the function is considered for $t > 0$ and 0 for $t < 0$ (i.e. causal signals)):

Function	Laplace	Remark
$A\,u(t)$	$\frac{A}{s}$	Constant
$At\,u(t)$	$\frac{A}{s^2}$	Ramp
$Ae^{-at}u(t)$	$\frac{A}{s+a}$	Exponential

Function	Laplace	Remark
$\sin(\omega t)\,u(t)$	$\frac{\omega}{s^2+\omega^2}$	Sine
$\cos(\omega t)\,u(t)$	$\frac{s}{s^2+\omega^2}$	Cosine
$\delta(t)$	1	Impulse

- A capacitor with initial voltage V_0 can be represented as a capacitor in series with a voltage source V_0.
- In the Laplace domain, $Z_C = \frac{1}{sC}$ and the initial voltage source becomes $\frac{V_0}{s}$. Thus, a capacitor with initial voltage V_0 is represented by an impedance $\frac{1}{sC}$ in series with a source $\frac{V_0}{s}$.

Q2.12.1 MCQ | 2012 | 1M | Ans: D

Method Time-Domain (Physical Interpretation)

Initially,

$$V_{C_1}(0^-) = 12\text{ V}, \qquad \text{and } C_2 \text{ is uncharged.}$$

When the switch is closed, two ideal capacitors having different voltages are connected directly together.
Since there is no resistor or inductor in the circuit, there is no mechanism to limit the current.
The capacitor voltages must equalize instantaneously.
An instantaneous transfer of a finite amount of charge requires an infinite current acting for zero time.
Hence, the current waveform is an impulse:

$$i(t) = Q\,\delta(t), \qquad \text{where } Q \text{ is the charge transferred.}$$

$\Rightarrow$ Current is an impulse function

Method Laplace-Domain Approach

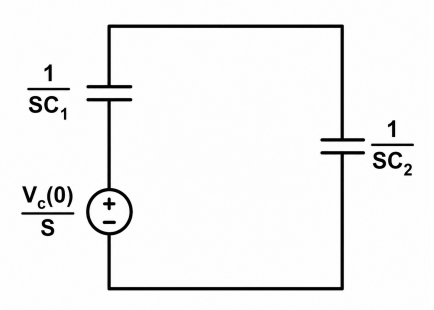

Fig. Solution Q2.12.1

Let the initial voltage on C_1 be, $V_C(0) = 12$ V
In the Laplace domain,

$$I(s) = \frac{\frac{V_C(0)}{s}}{\frac{1}{C_1 s} + \frac{1}{C_2 s}}$$

$$= \frac{V_C(0)}{\frac{1}{C_1} + \frac{1}{C_2}} = V_C(0)\left(\frac{C_1 C_2}{C_1 + C_2}\right)$$

Let

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$$

$$\Rightarrow I(s) = 12\, C_{eq}$$

$$\because \mathcal{L}\{\delta(t)\} = 1$$

taking inverse Laplace transform,

$$i(t) = 12\, C_{eq}\, \delta(t), \qquad \text{which is an impulse current.}$$

Hence,

D. Impulse function

Q2.10.1 MCQ 2010 1M Ans: B

For $t < 0$

The switch has been closed for a long time.
Hence, the capacitor is in steady state and behaves as an open circuit.

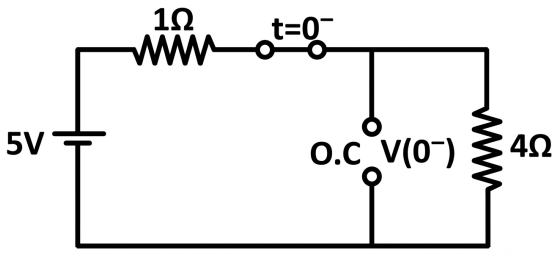

Fig. Solution Q2.10.1

Therefore,

$$V_C(0^-) = 5\left(\frac{4}{1+4}\right) = 4 \text{ V}$$

Since capacitor voltage cannot change instantaneously,

$$V_C(0^+) = V_C(0^-) = 4 \text{ V}$$

For $t = 0^+$

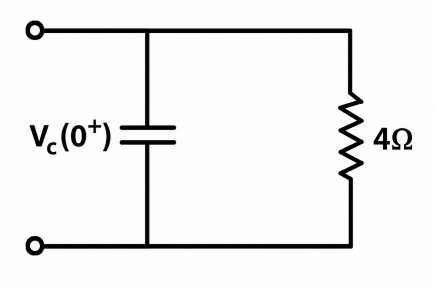

Fig. Solution Q2.10.1

At the instant $t = 0^+$, the capacitor behaves as a 4 V voltage source.
Hence, the current through the 4Ω resistor is

$$i(0^+) = \frac{V_C(0^+)}{4} = \frac{4}{4} = 1 \text{ A}$$

The capacitor current has the same magnitude at that instant.
Therefore,

$$\boxed{i_C(0^+) = 1 \text{ A}}$$

Hence,

B. 1 A

Q2.09.1 MCQ 2009 2M Ans: A ●●○

For $t < 0$

S_1 is closed and S_2 is open.
Hence, capacitor C_1 is connected across the 3 V source and reaches steady state.

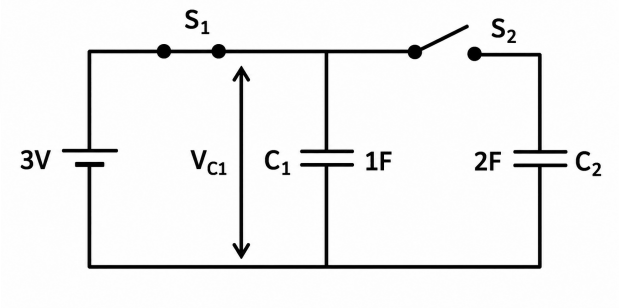

Fig. Solution Q2.09.1

$$\Rightarrow V_{C_1}(0^-) = 3 \text{ V}, \quad V_{C_2}(0^-) = 0 \text{ V}$$

Since capacitor voltage cannot change instantaneously,

$$V_{C_1}(0^+) = 3 \text{ V} \quad \text{and} \quad V_{C_2}(0^+) = 0 \text{ V}$$

At $t = 0$

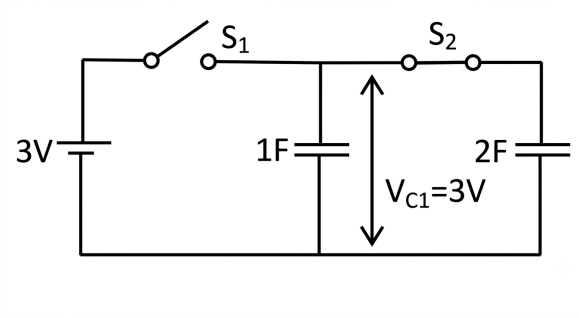

Fig. Solution Q2.09.1

Method 1

Since there is no resistor in the circuit, total charge is conserved.
Initial charge stored in C_1 is

$$Q_{\text{initial}} = C_1 V_{C_1}(0^-) = 1 \times 3 = 3 \text{ C}$$

Initial charge on C_2 is 0
Hence,

$$Q_{\text{total}} = 3 \text{ C}$$

After connection, both capacitors attain the same voltage V.
Therefore,

$$Q_{\text{total}} = (C_1 + C_2)V \implies 3 = (1 + 2)V$$
$$V = 1 \text{ V}$$

Thus, the voltage across the capacitor combination at $t = 0^+$ is 1V

Method 2

When two capacitors having initial voltages V_1 and V_2 are connected, then by charge distribution,

$$V_f = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$ (where V_f is the final voltage across cap. after charge sharing)

Substituting,

$$V_f = \frac{(1)(3) + (2)(0)}{1 + 2}$$
$$V_f = 1 \text{ V}$$

Hence,

A. 1 V

Q2.09.2 MCQ 2009 1M Ans: A ●●○

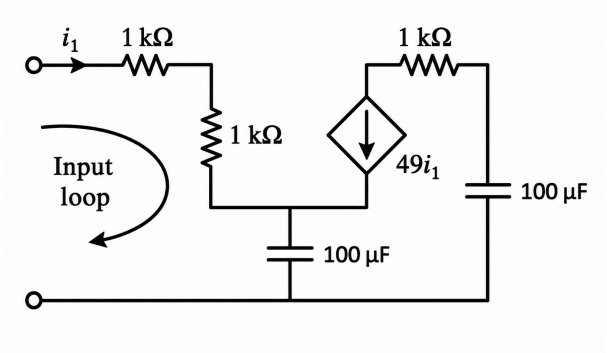

Fig. Solution Q2.09.2

Method 1

For a capacitor,

$$Q = CV, \qquad \text{For constant voltage; } Q \propto C$$

and

$$i = \frac{dQ}{dt} \Rightarrow C \propto i$$

The current through middle 100μF is,

$$i_s = i_1 + 49i_1 = 50i_1$$

Since capacitance is proportional to current for constant voltage,

$$\frac{C_{eq}}{100\mu F} = \frac{i_1}{50i_1}$$

$$C_{eq} = \frac{100}{50}\mu F$$

$$C_{eq} = 2\mu F$$

Method 2

Let

$$R = 1\ k\Omega, \qquad C = 100\ \mu F$$

By KVL;

$$V_{in} = 2Ri_1 + \frac{50i_1}{j\omega C}$$

Representing the entire network by an equivalent capacitance C_{eq},

$$V_{in} = 2Ri_1 + \frac{i_1}{j\omega C_{eq}}$$

Comparing the capacitive terms,

$$\frac{i_1}{j\omega C_{eq}} = \frac{50i_1}{j\omega C}$$

$$C_{eq} = \frac{C}{50}$$

$$C_{eq} = \frac{100}{50}\mu F = 2\mu F$$

Hence,

$$\boxed{C_{eq} = 2\ \mu F}$$

Therefore,

$$\boxed{\textbf{A. } 2\ \mu F}$$

Key Points

- The main idea in solving such problems is to find the input impedance

$$Z_{in} = \frac{V_{in}}{I_{in}}$$

and compare the resulting expression with the impedance of the required equivalent circuit to determine the unknown quantity.

Q2.08.1 MCQ | 2008 | 2M | Ans: C ● ● ○

To find the time constant, deactivate all independent sources.

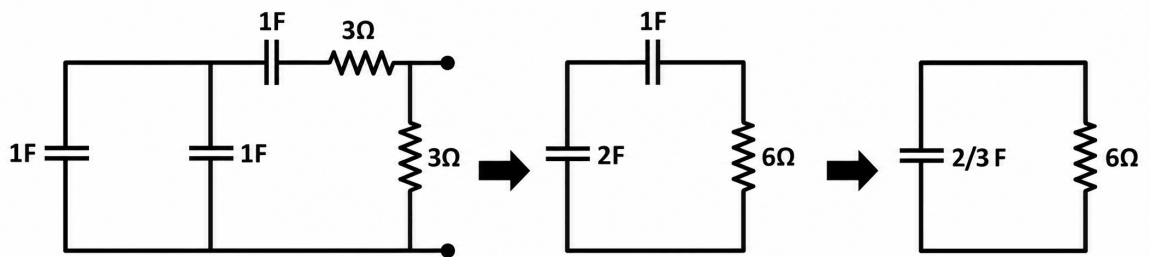

Fig. Solution Q2.08.1

Since the circuit contains an independent current source,

Current source → Open Circuit

Step 1: Find the equivalent capacitance

$$C_{eq} = \frac{(2)(1)}{2+1} = \frac{2}{3}\text{ F}$$

Step 2: Find the equivalent resistance

$$R_{eq} = 3 + 3 = 6\Omega$$

Step 3: Compute the time constant

$$\tau = R_{eq}C_{eq}$$

$$\tau = 6 \times \frac{2}{3}$$

Therefore,

C. 4 s

Q2.08.2 MCQ | 2008 | 2M | Ans: C ● ○ ○

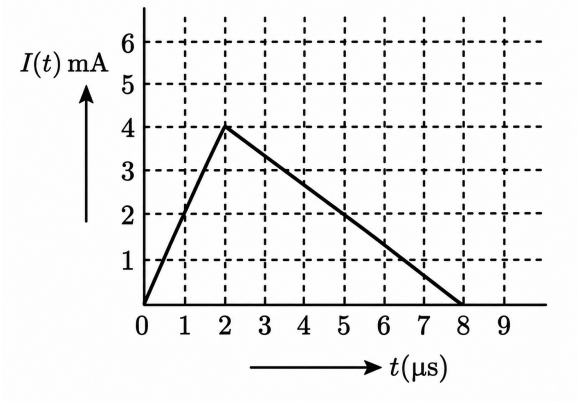

Fig. Solution Q2.08.2

The charge stored in the capacitor at $t = 5\,\mu s$ is equal to the area under the current waveform till $5\,\mu s$.

$$Q = \int_0^{5\mu s} i(t)\,dt$$

Area from 0 **to** $2\,\mu s$

$$A_1 = \frac{1}{2}(2)(4) = 4 \text{ nC}$$

Area from 2 **to** $5\,\mu s$

$$A_2 = (3)(2) + \frac{1}{2}(3)(2)$$

$$A_2 = 6 + 3 = 9 \text{ nC}$$

Hence,

$$Q = A_1 + A_2 = 4 + 9 = 13 \text{ nC}$$

Therefore,

C. 13 nC

Q2.08.3 MCQ 2008 1M Ans: D ● ● ○

From Q2.08.2,

$$Q_0 = 13 \text{ nC}, \qquad C = 0.3 \text{ nF}$$

Hence,

$$V_0 = \frac{Q_0}{C} = \frac{13}{0.3} = 43.3 \text{ V}$$

Initially,

$$V_C(0^-) = 43.3 \text{ V}, \qquad i_L(0^-) = 0$$

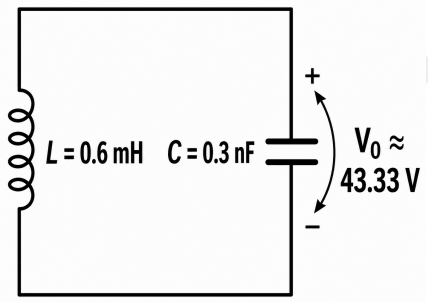

Fig. Solution Q2.08.3

For the capacitor,

$$i_C(t) = C\frac{dv_C}{dt}$$

Taking Laplace transform,

$$I(s) = CsV_C(s) - CV_C(0^-)$$

Therefore,

$$V_C(s) = \frac{I(s)}{Cs} + \frac{V_0}{s}$$

For the inductor,

$$v_L(t) = L\frac{di_L}{dt}$$

Taking Laplace transform,

$$V_L(s) = LsI(s) - Li_L(0^-) \Rightarrow V_L(s) = LsI(s)$$

By KVL,

$$V_C(s) + V_L(s) = 0$$

Substituting,

$$\frac{I(s)}{Cs} + \frac{V_0}{s} + LsI(s) = 0$$

Multiplying by s,

$$\frac{I(s)}{C} + V_0 + Ls^2 I(s) = 0$$

$$I(s) = -\frac{CV_0}{1 + LCs^2}$$

$$\Rightarrow V_C(s) = \frac{I(s)}{Cs} + \frac{V_0}{s} = V_0 \frac{s}{s^2 + \frac{1}{LC}}$$

Let, $\omega_0 = \frac{1}{\sqrt{LC}}$ $\Rightarrow V_C(s) = V_0 \frac{s}{s^2+\omega_0^2}$

Using

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + \omega_0^2}\right\} = \cos(\omega_0 t)$$

$$V_C(t) = V_0 \cos(\omega_0 t)$$

Now,

$$\omega_0 = \frac{1}{\sqrt{(0.3 \times 10^{-9})(0.6 \times 10^{-3})}} = 2.35 \times 10^6 \text{ rad/s}$$

At $t = 1\ \mu s$;

$$V_C = 43.3 \cos\left(2.35 \times 10^6 \times 10^{-6}\right) = 43.3 \cos(2.35)$$

$$= 43.3(-0.706) = -30.6 \text{ V}$$

Therefore,

D. -30.6 V

Q2.07.1 MCQ 2007 2M Ans: D ●●○

Before switching,

$$i_L(0^-) = 10 \text{ A}, \qquad v_C(0^-) = 10 \text{ V}$$

Since inductor current and capacitor voltage cannot change instantaneously,

$$i_L(0^+) = i_L(0^-) = 10 \text{ A}$$

$$v_C(0^+) = v_C(0^-) = 10 \text{ V}$$

Thus, at $t = 0^+$, the inductor can be represented by a 10 A current source.
Circuit at $t = 0^+$

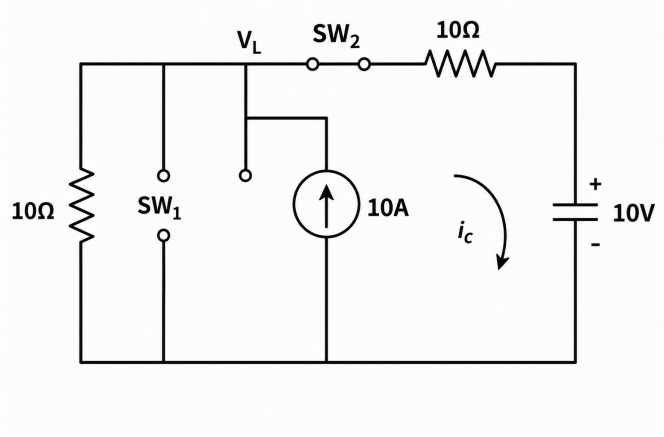

Fig. Solution Q2.07.1

Applying KCL at node V_L,

$$\frac{V_L}{10} + \frac{V_L - 10}{10} - 10 = 0$$

Multiplying throughout by 10,

$$V_L + V_L - 10 - 100 = 0$$

$$2V_L = 110 \Rightarrow V_L = 55 \text{ V}$$

Hence,

$$v_L(0^+) = 55 \text{ V}$$

Now,

$$i_C(0^+) = \frac{V_L - 10}{10}$$

$$i_C(0^+) = \frac{55 - 10}{10} = \frac{45}{10} = 4.5 \text{ A}$$

Hence,

D. 4.5 A, 55 V

Q2.06.1 MCQ 2006 2M Ans: D ● ○ ○

Since no switching action takes place, the circuit is in steady state.

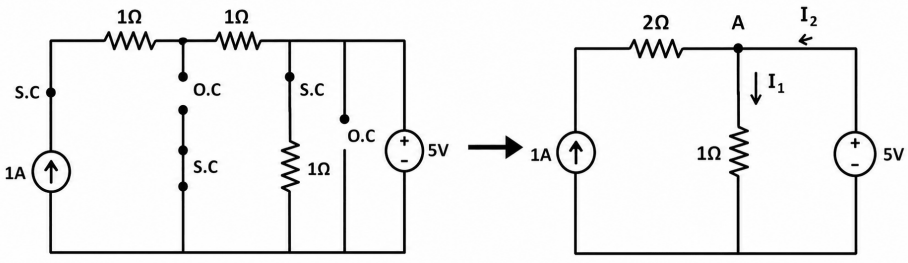

Fig. Solution Q2.06.1

Applying KCL at node A,

$$1 + I_2 = I_1 \tag{1}$$

Since the 5 V source is directly connected across R_3,

$$V_{R_3} = 5 \text{ V}$$

Using Ohm's law,

$$I_1 = \frac{V_{R_3}}{R_3}$$

$$I_1 = \frac{5}{1} = 5 \text{ A}$$

Substituting $I_1 = 5$ A into (1),

$$1 + I_2 = 5$$

$$I_2 = 4 \text{ A}$$

Therefore,

D. 5, 4

Q2.05.1 MCQ 2005 1M Ans: B

The switch is closed at $t = 0$. We are required to find the *final steady-state* voltage across the capacitor.

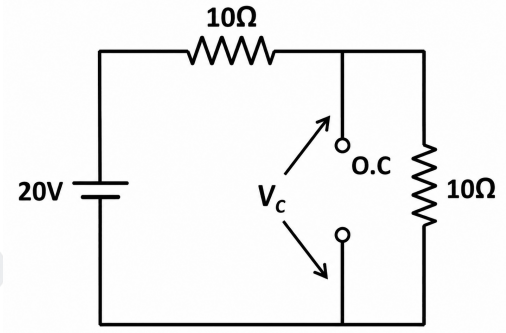

Fig. Solution Q2.05.1

At steady state ($t \to \infty$), the capacitor behaves as an open circuit.
The voltage across the capacitor is the same as the voltage across the right-hand 10Ω resistor.
Using the voltage divider rule,

$$V_C = 20\left(\frac{10}{10+10}\right) = 20\left(\frac{10}{20}\right)$$

$$V_C = 10 \text{ V}$$

Hence, the correct answer is

B. 10 V

Q2.05.2 MCQ | 2005 | 2M | Ans: C

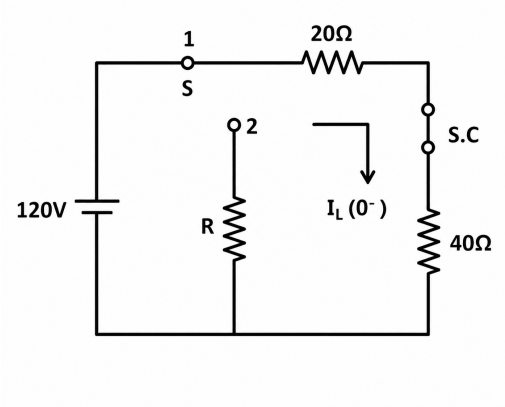

Fig. Solution Q2.05.2

For $t < 0$, the switch is connected to position 1 for a long time.
Hence, the circuit is in steady state and the inductor behaves as a short circuit.
Therefore, the current through the inductor just before switching is

$$i_L(0^-) = \frac{120}{20 + 40} = 2 \text{ A}$$

Since the current through an inductor cannot change instantaneously,

$$i_L(0^+) = i_L(0^-) = 2 \text{ A}$$

At $t = 0^+$, the source is disconnected and the inductor behaves as a constant current source of value $i_L(0^+) = 2$ A

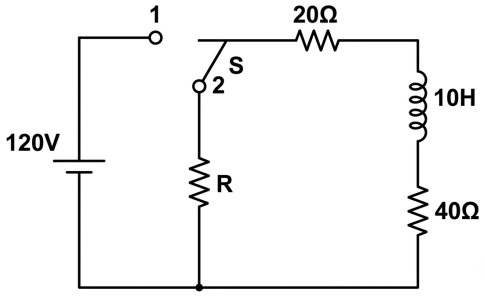

Fig. Solution Q2.05.2

The voltage across the coil is equal to the voltage drop across the external resistors R and 20Ω,

$$V_L = i_L(0^+)(R + 20)$$

Given that

$$V_L = 120 \text{ V}$$

$$120 = 2(R + 20)$$

$$R + 20 = 60 = 40 \ \Omega$$

Hence, the correct answer is

C. 40 Ω

Mistakes to be avoided

- Do not treat 40ohm as external resistance, the drop across coil means the total drop across L and its internal resistance.

Q2.05.3 MCQ | 2005 | 2M | Ans: B ●●○

From Q.2.05.2,

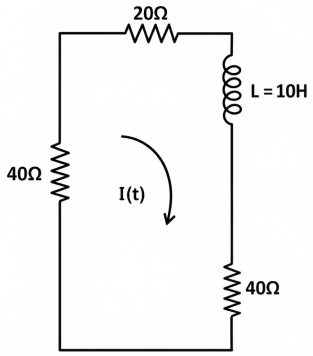

Fig. Solution Q2.05.3

For $t > 0$, the inductor discharges through the resistive network.
The equivalent resistance seen by the inductor is

$$R_{eq} = 40 + 40 + 20 = 100\ \Omega$$

The time constant is

$$\tau = \frac{L}{R_{eq}} = \frac{10}{100} = 0.1\ \text{s}$$

For a source-free RL circuit,

$$i_L(t) = i_L(0^+)e^{-t/\tau}$$

Substituting

$$i_L(0^+) = i_L(0^-) = 2\ \text{A}$$

$$i_L(t) = 2e^{-10t}$$

The initial energy stored in the inductor is

$$W_0 = \frac{1}{2}Li_L^2(0^-) = \frac{1}{2} \times 10 \times (2)^2$$

$$W_0 = 20\ \text{J}$$

If 95% of the energy is dissipated, then only 5% remains:

$$W = 0.05W_0 = 0.05 \times 20$$

$$W = 1\ \text{J}$$

Using

$$W = \frac{1}{2}Li_L^2$$

(where i_L is the current in the inductor after 95% dissipation of energy)

$$\Rightarrow \frac{1}{2} \times 10 \times i_L^2 = 1 \Rightarrow i_L^2 = \frac{1}{5}$$

$$i_L = \frac{1}{\sqrt{5}} = 0.4472 \text{ A}$$

Substituting into the current expression,

$$2e^{-10t} = 0.4472$$

$$e^{-10t} = 0.2236 \Rightarrow -10t = \ln(0.2236)$$

$$t = 0.15 \text{ s}$$

Hence, the correct answer is

B. 0.15 s

Q2.05.4 MCQ 2005 2M Ans: C ● ○ ○

Before $t = 0$, the switch is open and the circuit has been in steady state for a long time.

In steady state, an ideal inductor behaves as a short circuit.

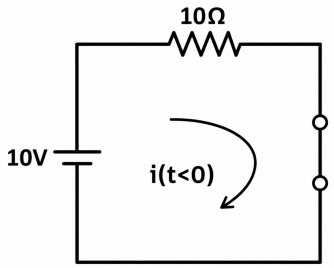

Fig. Solution Q2.05.4

Hence, the current through the inductor just before switching is

$$i_L(0^-) = \frac{10}{10} = 1 \text{ A}$$

Since the current through an inductor cannot change instantaneously,

$$i_L(0^+) = i_L(0^-)$$

$$i_L(0^+) = 1 \text{ A}$$

Hence, the correct answer is

C. 1 A

Q2.04.1 MCQ 2004 2M Ans: A ● ○ ○

Given,

$$Q(0^-) = 10 \text{ C}, \qquad C = 0.5 \text{ F}$$

Therefore, the initial capacitor voltage is

$$V_C(0^-) = \frac{Q(0^-)}{C} = \frac{10}{0.5} = 20 \text{ V}$$

Since the voltage across a capacitor cannot change instantaneously,

$$V_C(0^+) = V_C(0^-) = 20 \text{ V}$$

At steady state ($t \to \infty$), the capacitor behaves as an open circuit.

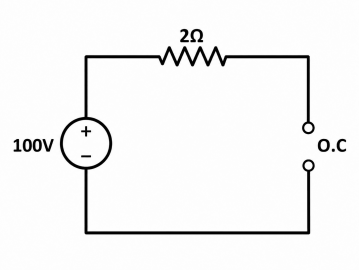

Fig. Solution Q2.04.1

$$V_C(\infty) = 100 \text{ V}$$

For an RC charging circuit,

$$V_C(t) = V_C(\infty) + \left[V_C(0^+) - V_C(\infty)\right] e^{-t/RC}$$

Substituting,

$$R = 2\ \Omega, \qquad C = 0.5 \text{ F}$$

$$RC = 2 \times 0.5 = 1 \text{ s}$$

Therefore,

$$V_C(t) = 100 + (20 - 100)e^{-t}$$

$$V_C(t) = 100 - 80e^{-t}$$

The capacitor current is

$$i_C(t) = C\frac{dV_C(t)}{dt} = 0.5\frac{d}{dt}\left(100 - 80e^{-t}\right)$$

$$i_C(t) = 0.5\left(80e^{-t}\right) = 40e^{-t} \text{ A}$$

At $t = 1$ s,

$$i_C(1) = 40e^{-1}$$

$$i_C(1) \approx 14.71 \text{ A}$$

Therefore, the correct answer is

A. 14.7 A

Q2.03.1 MCQ 2003 2M Ans: B ●●○

Before the switch is closed, the circuit is unenergized. Therefore,

$$i_L(0^-) = 0, \qquad v_C(0^-) = 0$$

Since the current through an inductor and the voltage across a capacitor cannot change instantaneously,

$$i_L(0^+) = i_L(0^-) = 0, \qquad v_C(0^+) = v_C(0^-) = 0$$

Hence, at $t = 0^+$,

- the inductor behaves as an open circuit,
- the capacitor behaves as a short circuit.

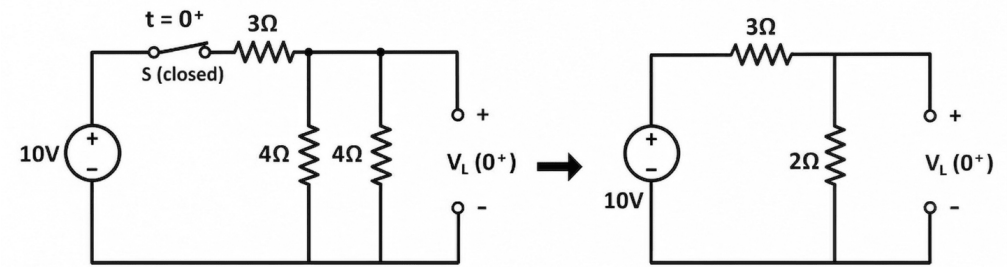

Fig. Solution Q2.03.1

The equivalent parallel resistance is

$$R_p = 4 \parallel 4 = \frac{4 \times 4}{4 + 4} = 2\ \Omega$$

Hence, the source current is

$$I = \frac{10}{3 + 2} = 2\ \text{A}$$

The voltage across the parallel combination is

$$V_L(0^+) = IR_p$$

$$V_L(0^+) = 2 \times 2 = 4\ \text{V}$$

Since the inductor is connected across the same two nodes,

$$\boxed{V_L(0^+) = 4\ \text{V}}$$

Therefore, the correct answer is

$$\boxed{\textbf{B.}\ 4\ \text{V}}$$

Q2.02.1 NAT 2002 5M Ans: * ●●○

Method 1

When the switch S is closed for a long time, the switch provides a short-circuit path across the network. Hence, the entire source current flows through the switch and no current flows through the inductor. Therefore,

$$i_L(0^-) = 0$$

Since the current through an inductor cannot change instantaneously,

$$i_L(0^+) = i_L(0^-) = 0$$

After the switch is opened, the source current divides between the resistor and the inductor.

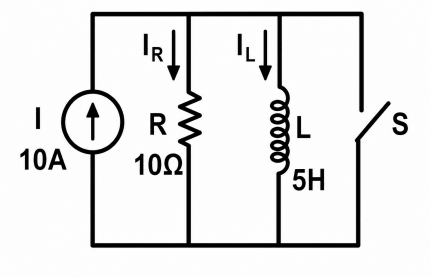

Fig. Solution Q2.02.1

Applying KCL at the top node,

$$i_L + i_R = 10$$

$$i_R = 10 - i_L \tag{1}$$

Since the resistor and inductor are connected in parallel,

$$v_R = v_L$$

Using the element relations,

$$Ri_R = L\frac{di_L}{dt}$$

Substituting

$$R = 10\Omega, \qquad L = 5\text{ H}$$

$$10\, i_R = 5\frac{di_L}{dt}$$

Using (1),

$$10(10 - i_L) = 5\frac{di_L}{dt}$$

$$100 - 10i_L = 5\frac{di_L}{dt}$$

$$\frac{di_L}{dt} + 2i_L = 20$$

Hence,

$$\tau = \frac{L}{R} = \frac{5}{10} = 0.5\text{ s}$$

The steady-state current is

$$i_L(\infty) = 10\text{ A}$$

Using the standard first-order response,

$$i_L(t) = i_L(\infty) + \left[i_L(0^+) - i_L(\infty)\right]e^{-t/\tau}$$

Substituting $i_L(0^+) = 0$, $i_L(\infty) = 10$ A and $\tau = 0.5$ s,

$$i_L(t) = 10 + (0 - 10)e^{-t/0.5}$$

$$\boxed{i_L(t) = 10\left(1 - e^{-2t}\right)\text{ A}}$$

At steady state ($t \to \infty$),

$$i_L(\infty) = 10\text{ A}$$

The energy stored in the inductor is

$$W_L = \frac{1}{2}Li_L^2$$

Hence,

$$W_L = \frac{1}{2}(5)(10)^2$$

$$\boxed{W_L = 250\text{ J}}$$

Therefore,

$$\boxed{\text{Energy stored in the inductor after a long time } = 250\text{ J}}$$

Method 2 (Using Source Transformation):

Transform the current source and parallel resistance into their Thevenin equivalent:

$$V_{th} = IR = (10)(10) = 100 \text{ V}$$

$$R_{th} = 10\ \Omega$$

Thus, the circuit becomes,

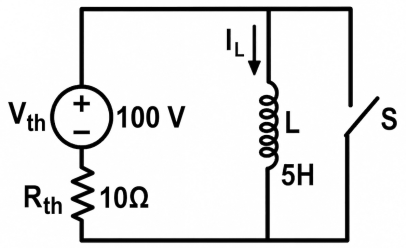

Fig. Solution Q2.02.1

Since the switch was closed for a long time before $t = 0$, the inductor was short-circuited and hence

$$i_L(0^-) = 0$$

Current through an inductor cannot change instantaneously, therefore

$$i_L(0^+) = i_L(0^-) = 0$$

Applying KVL to the equivalent series RL circuit,

$$V_{th} = R_{th} i_L + L\frac{di_L}{dt}$$

Substituting the values,

$$100 = 10i_L + 5\frac{di_L}{dt}$$

$$\frac{di_L}{dt} + 2i_L = 20$$

Rest solution is same as Method 1.

Q2.02.2 MCQ 2002 2M Ans: C ● ○ ○

At steady state,

Inductor → Short Circuit, Capacitor → Open Circuit

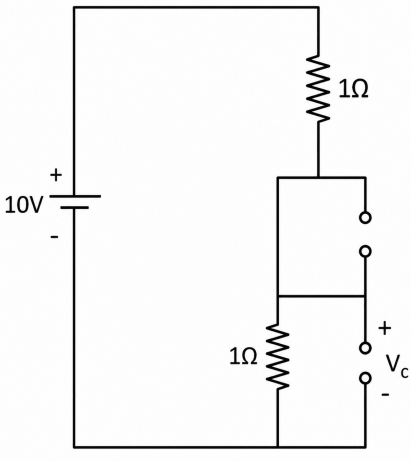

Fig. Solution Q2.02.2

The voltage across the capacitor is the same as the voltage across the lower 1 Ω resistor.
Using the voltage divider rule,

$$V_C = 10\left(\frac{1}{1+1}\right)$$

$$V_C = 10\left(\frac{1}{2}\right)$$

$$V_C = 5\ \text{V}$$

Hence, the correct answer is

C. 5 V

Q2.02.3 MCQ 2002 2M Ans: C ●●○

Before solving, compare the pulse width with the circuit time constant.

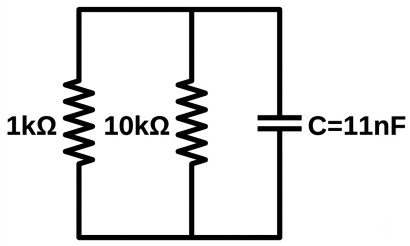

Fig. Solution Q2.02.3

The equivalent resistance seen by the capacitor is

$$R_{\text{eq}} = 1\,\text{k}\Omega \parallel 10\,\text{k}\Omega = \frac{1\times 10}{1+10}\,\text{k}\Omega = \frac{10}{11}\,\text{k}\Omega$$

Hence,

$$\tau = R_{\text{eq}}C = \left(\frac{10}{11}\times 10^{3}\right)(11\times 10^{-9})$$

$$\tau = 10\ \mu\text{s}$$

The applied pulse width is

$$t_p = 10\ \mu\text{s}$$

Since

$$t_p = \tau$$

Method 1

The capacitor does *not* get sufficient time to charge completely. Therefore, the peak capacitor voltage occurs at the end of the pulse ($t = 10\ \mu$s).
Initially,

$$V_C(0) = 0$$

At steady state, the capacitor behaves as an open circuit. Hence, using the voltage divider rule,

$$V_C(\infty) = 11\left(\frac{10}{10+1}\right) = 10\ \text{V}$$

The capacitor charging equation is

$$V_C(t) = V_C(\infty) + \left[V_C(0) - V_C(\infty)\right]e^{-t/\tau}$$

Substituting the values,

$$V_C(t) = 10 + (0 - 10)e^{-t/(10\mu s)}$$

$$V_C(t) = 10\left(1 - e^{-t/(10\mu s)}\right)$$

The peak voltage occurs at

$$t = 10\ \mu s$$

Therefore,

$$V_{peak} = 10\left(1 - e^{-1}\right)$$

$$V_{peak} = 10(1 - 0.3679)$$

$$V_{peak} = 6.32\ V$$

Therefore, the correct answer is

C. 6.32 V

Method 2

Initially,

$$V_C(0) = 0$$

At steady state, the capacitor behaves as an open circuit. Hence, using the voltage divider rule,

$$V_C(\infty) = 11\left(\frac{10}{10+1}\right) = 10\ V$$

Since the capacitor is subjected to the input for exactly one time constant, by definition of time constant, the capacitor voltage reaches 63.2% of its total change.
Therefore,

$$V_C(\tau) = 63.2(\frac{10}{100})$$

$$V_C(\tau) = 6.32\ V$$

Hence, the correct answer is

C. 6.32 V

Mistakes to be avoided

- In the question it is mentioned that $10\mu s$ is the duration of 11V pulse. Do not assume $10\mu s$ as the time period of pulse.

Q2.02.4 MCQ 2002 1M Ans: D ●○○

For a capacitor,

$$i_C(t) = C\frac{dv_C(t)}{dt}$$

Given,

$$i_C(t) = 5\delta(t)$$

Hence,

$$C\frac{dv_C(t)}{dt} = 5\delta(t)$$

$$\frac{dv_C(t)}{dt} = \frac{5}{C}\delta(t)$$

Integrating both sides,

$$v_C(t) = \frac{5}{C}\int \delta(t)\,dt$$

Using

$$\int \delta(t)\,dt = u(t)$$

$$v_C(t) = \frac{5u(t)}{C}$$

Hence, the correct answer is

D. $\frac{5u(t)}{C}$

Q2.01.1 MCQ 2001 1M Ans: B

For a series RL circuit excited by a unit-step voltage and having zero initial conditions,

$$i(0^-) = 0$$

Since the current through an inductor cannot change instantaneously,

$$i(0^+) = i(0^-) = 0$$

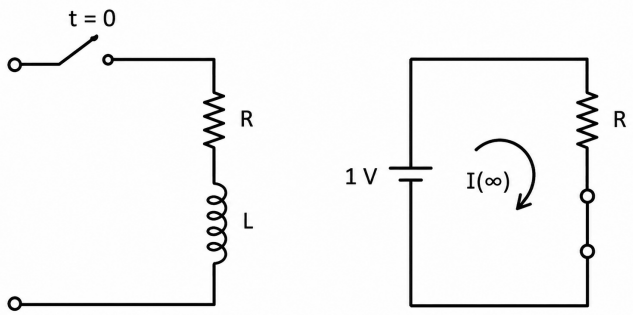

Fig. Solution Q2.01.1

The steady-state current is

$$i(\infty) = \frac{V}{R}$$

and the current response is

$$i(t) = i(\infty) + \left[i(0^+) - i(\infty)\right]e^{-t/\tau}$$

where

$$\tau = \frac{L}{R}$$

Substituting $i(0^+) = 0$,

$$i(t) = \frac{V}{R}\left(1 - e^{-Rt/L}\right)$$

Option A: *It is possible for the current to be oscillatory.*

Incorrect. A series RL circuit contains only one energy-storage element and is therefore a first-order circuit. Its response is exponential and cannot be oscillatory.

Option B: *The voltage across the resistor at $t = 0^+$ is zero.*
Correct. Since

$$i(0^+) = 0$$

the resistor voltage is

$$v_R(0^+) = i(0^+)R = 0$$

Option C: *The energy stored in the inductor in steady state is zero.*
Incorrect. At steady state,

$$i(\infty) = \frac{V}{R} \neq 0$$

Hence, the energy stored in the inductor is

$$W_L = \frac{1}{2}Li^2(\infty) = \frac{1}{2}L\left(\frac{V}{R}\right)^2$$

which is non-zero.

Option D: *The resistor current eventually falls to zero.*
Incorrect. Since the resistor and inductor are in series, both carry the same current. As $t \to \infty$,

$$i(\infty) = \frac{V}{R}$$

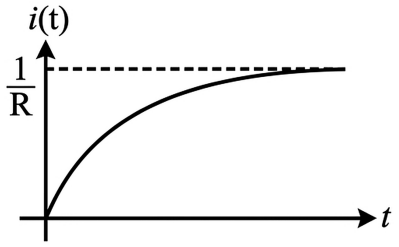

Fig. Solution Q2.01.1

which is a non-zero constant. Therefore, the resistor current does not fall to zero.

Hence, the correct answer is

B. The voltage across the resistor at $t = 0^+$ is zero.

Q2.00.1 MCQ 2000 1M Ans: D ●○○

For an inductor,

$$v(t) = L\frac{di(t)}{dt}$$

Given,

$$v(t) = 12t^2, \qquad L = 1\ \text{H}$$

Therefore,

$$12t^2 = \frac{di(t)}{dt}$$

Integrating both sides,

$$i(t) = \int 12t^2\,dt = 12\frac{t^3}{3} + C$$

$$i(t) = 4t^3 + C$$

Given that the initial current through the inductor is zero,

$$i(0) = 0$$

Substituting $t = 0$,

$$0 = 4(0)^3 + C$$

$$C = 0$$

Hence,

$$\boxed{i(t) = 4t^3}$$

Therefore, the correct answer is

D. $4t^3$

Q2.99.1 NAT | 1999 | 5M | Ans: *

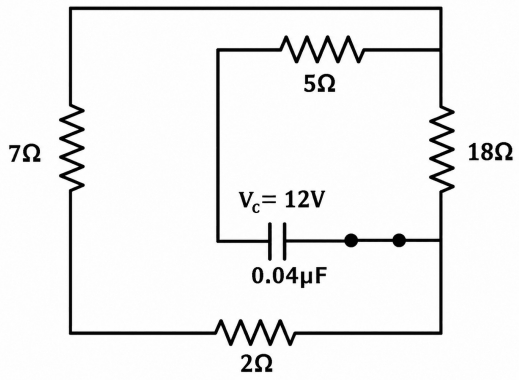

Fig. Solution Q2.99.1

$$V_C(0) = 12\ \text{V}$$

Since the circuit contains no independent source, the capacitor eventually discharges completely.

$$V_C(\infty) = 0$$

To determine the time constant, find the equivalent resistance seen by the capacitor.
The capacitor terminals are opened and the equivalent resistance is calculated across its terminals.

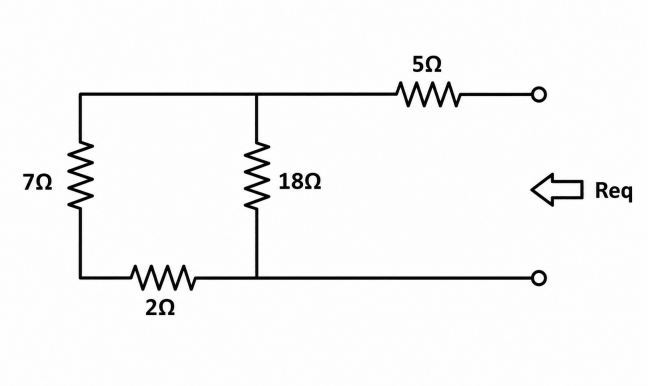

Fig. Solution Q2.99.1

$$R_{eq} = (7 + 2) \parallel 18 + 5$$

$$R_{eq} = 9 \parallel 18 + 5 = 11\ \Omega$$

Hence,

$$\tau = R_{eq}C = 11 \times 0.04 \times 10^{-6} = 0.44 \times 10^{-6}\ \text{s}$$

Using the capacitor voltage response,

$$V_C(t) = V_C(\infty) + \left[V_C(0) - V_C(\infty)\right]e^{-t/\tau}$$

Substituting $V_C(0) = 12$ V, $V_C(\infty) = 0$ and $\tau = 0.44 \times 10^{-6}$ s,

$$\boxed{V_C(t) = 12e^{-t/(0.44\times10^{-6})}\ \text{V}}$$

Q2.99.2 MCQ | 1999 | 2M | Ans: D | ● ● ○

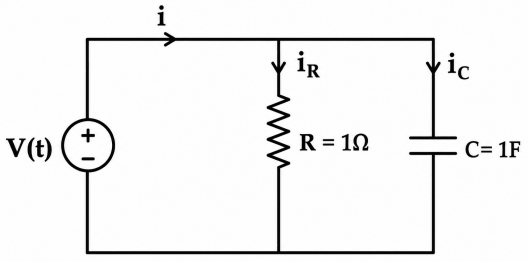

Fig. Solution Q2.99.2

$$i = i_R + i_C$$

Using the element relations,

$$i_R = \frac{v(t)}{R}, \qquad i_C = C\frac{dv(t)}{dt}$$

Given,

$$R = 1\ \Omega, \qquad C = 1\ \text{F}$$

Hence,

$$i = v(t) + \frac{dv(t)}{dt}$$

The applied voltage is a triangular wave of peak value 1 V and frequency 0.5 Hz.

The resistor current is

$$i_R = v(t)$$

Therefore, i_R is also a triangular waveform varying between 0 and 1 A.

For a triangular waveform, the slope is constant in each interval. Therefore,

$$\frac{dv(t)}{dt}$$

is a square wave.
Since the waveform rises from 0 to 1 V in 1 s and falls from 1 to 0 V in 1 s,

$$\frac{dv(t)}{dt} = +1$$

during the rising interval and

$$\frac{dv(t)}{dt} = -1$$

during the falling interval.
Thus,

$$i_C = \frac{dv(t)}{dt}$$

is a square wave alternating between +1 A and −1 A.

The source current is

$$i = i_R + i_C$$

During the rising interval,

$$i = i_R + 1$$

Hence, the triangular waveform is shifted upward by 1 A.
During the falling interval,

$$i = i_R - 1$$

Hence, the triangular waveform is shifted downward by 1 A.
Therefore, the resulting waveform consists of triangular segments with sudden jumps at the instants where the slope of the input voltage changes.

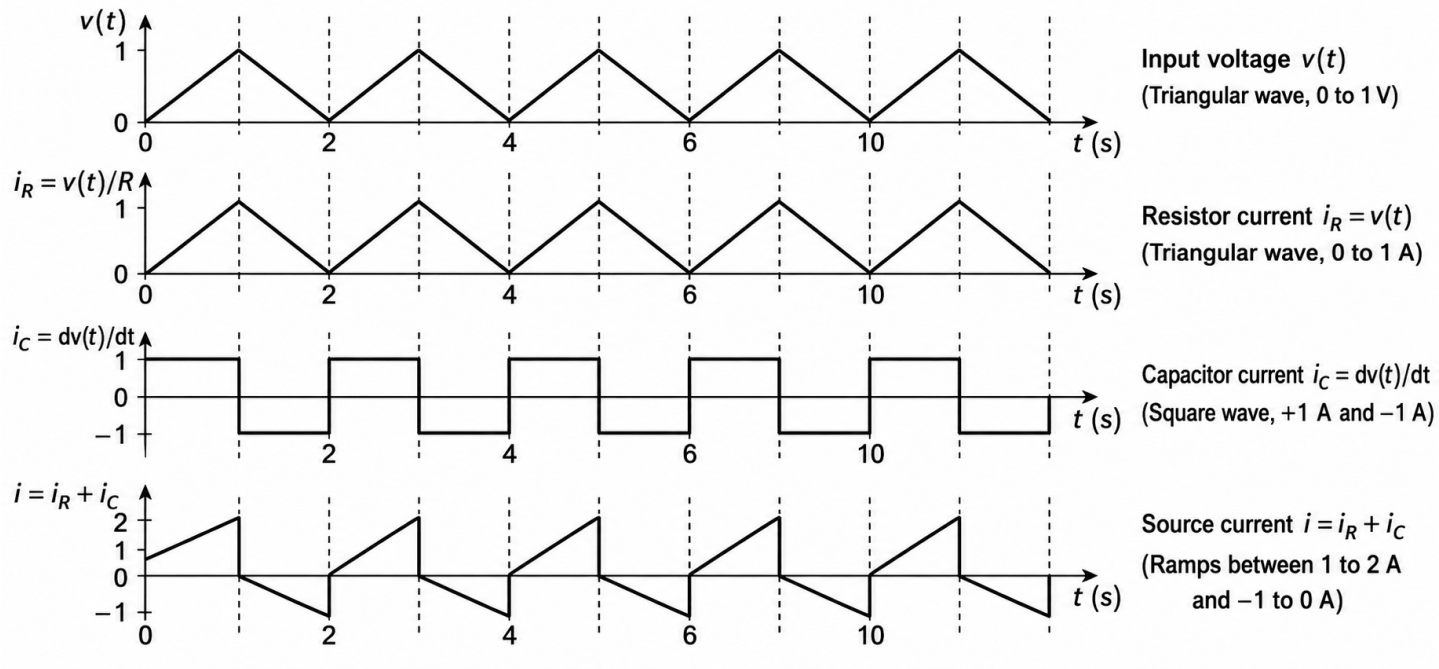

Fig. Solution Q2.99.2

This corresponds to

D

Mistakes to be avoided

- Mark the options carefully.

Q2.99.3 MCQ | 1999 | 2M | Ans: A | ● ○ ○

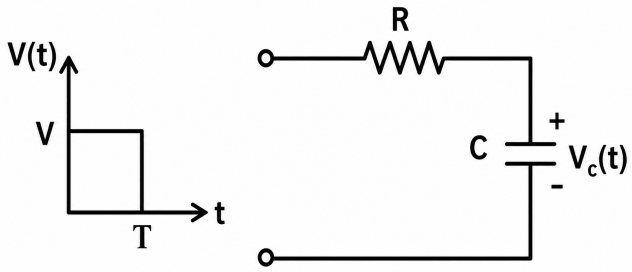

Fig. Solution Q2.99.3

A rectangular pulse of magnitude V and duration T can be represented as

$$v_{in}(t) = V\left[u(t) - u(t-T)\right]$$

For $0 \le t \le T$, the circuit behaves as a standard charging RC circuit.
Hence, the capacitor voltage is

$$v_C(t) = V\left(1 - e^{-t/RC}\right)$$

At $t = T$,

$$v_C(T) = V\left(1 - e^{-T/RC}\right)$$

For $t > T$, the input becomes zero and the capacitor starts discharging. Therefore,

$$v_C(t) = v_C(T)e^{-(t-T)/RC}$$

Since the capacitor voltage decreases exponentially for $t > T$, the maximum capacitor voltage occurs at the instant the pulse ends
Thus,

$$V_{C,\max} = v_C(T) = V\left(1 - e^{-T/RC}\right)$$

Hence, the correct answer is

A. $V\left[1 - e^{-T/RC}\right]$

Q2.98.1 MCQ 1998 1M Ans: C ●○○

For an ideal inductor,

$$v(t) = L\frac{di(t)}{dt}$$

A constant direct current can be represented as a step current:

$$i(t) = I\,u(t)$$

where $u(t)$ is the unit-step function.
Differentiating,

$$\frac{di(t)}{dt} = I\frac{du(t)}{dt}$$

Using

$$\frac{du(t)}{dt} = \delta(t)$$

we get

$$\frac{di(t)}{dt} = I\delta(t)$$

Hence,

$$v(t) = L\frac{di(t)}{dt} = LI\delta(t)$$

Since $\delta(t)$ is an impulse function, the applied voltage must be an ideal impulse.
Therefore,

C. Ideal impulse

Key Points

- To establish a constant current instantaneously in an ideal inductor, the current must jump from 0 to I. Since an ideal inductor opposes any instantaneous change in current, an impulse voltage is required.

Q2.98.2 NAT | 1998 | 5M | Ans: * | ● ● ●

Given,

$$L = 2\text{ H}, \qquad R_1 = 10\ \Omega, \qquad R_2 = 2\ \Omega, \qquad C = 0.25\ \mu\text{F}$$

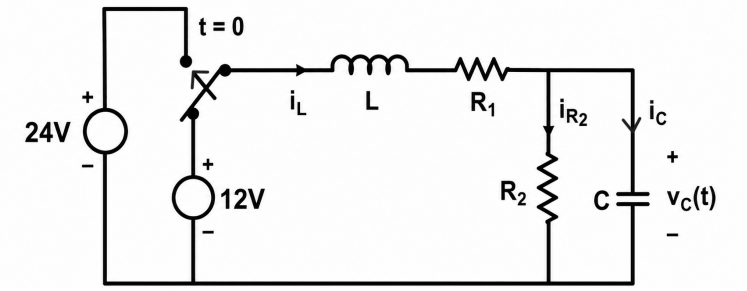

Fig. Solution Q2.98.2

(a) Determine $i_L(0)$ and $V_C(0)$

Before switching, the circuit has been connected to the 12 V source for a long time.

At steady state,

$$\text{Inductor} \rightarrow \text{Short Circuit}, \qquad \text{Capacitor} \rightarrow \text{Open Circuit}$$

Therefore, the equivalent circuit is

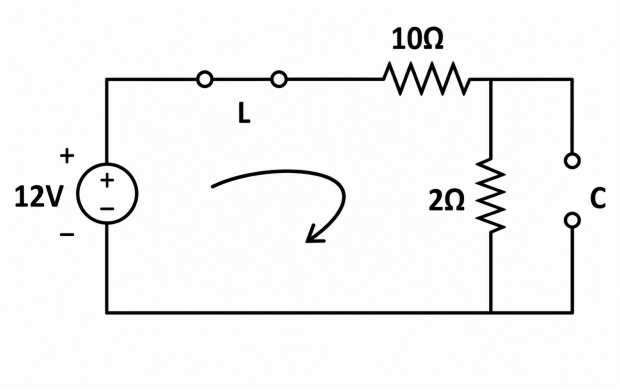

Fig. Solution Q2.98.2

$$I = \frac{12}{R_1 + R_2} = \frac{12}{10 + 2} = 1\text{ A}$$

Hence,

$$\boxed{i_L(0^-) = 1\text{ A}}$$

Since the current through an inductor cannot change instantaneously,

$$\boxed{i_L(0^+) = i_L(0^-) = 1\text{ A}}$$

The capacitor voltage is the voltage across R_2,

$$V_C(0^-) = R_2 I = 2 \times 1 = 2\text{ V}$$

Since the voltage across a capacitor cannot change instantaneously,

$$\boxed{V_C(0^+) = V_C(0^-) = 2\text{ V}}$$

(b) Differential equation governing $V_C(t)$ for $t > 0$

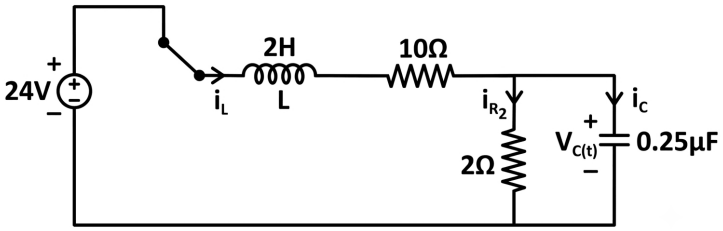

Fig. Solution Q2.98.2

For $t > 0$, the switch is connected to the 24 V source.
The capacitor current is

$$i_C = C\frac{dV_C(t)}{dt}$$

The current through R_2 is

$$i_{R_2} = \frac{V_C(t)}{R_2} = \frac{V_C(t)}{2}$$

Applying KCL at the capacitor node,

$$i_L = i_C + i_{R_2}$$

$$i_L = C\frac{dV_C(t)}{dt} + \frac{V_C(t)}{2} \quad (1)$$

Applying KVL,

$$L\frac{di_L}{dt} + R_1 i_L + V_C(t) - 24 = 0$$

Substituting $L = 2$ H and $R_1 = 10\ \Omega$,

$$2\frac{di_L}{dt} + 10i_L + V_C(t) - 24 = 0 \quad (2)$$

Differentiating (1),

$$\frac{di_L}{dt} = C\frac{d^2V_C(t)}{dt^2} + \frac{1}{2}\frac{dV_C(t)}{dt}$$

Substituting i_L and di_L/dt into (2),

$$2\left(C\frac{d^2V_C(t)}{dt^2} + \frac{1}{2}\frac{dV_C(t)}{dt}\right) + 10\left(C\frac{dV_C(t)}{dt} + \frac{V_C(t)}{2}\right) + V_C(t) = 24$$

$$2C\frac{d^2V_C(t)}{dt^2} + (1 + 10C)\frac{dV_C(t)}{dt} + 6V_C(t) = 24$$

Substituting

$$C = 0.25 \times 10^{-6}\ \text{F}$$

$$0.5 \times 10^{-6}\frac{d^2V_C(t)}{dt^2} + (1 + 2.5 \times 10^{-6})\frac{dV_C(t)}{dt} + 6V_C(t) = 24$$

Multiplying throughout by 2×10^6,

$$\boxed{\frac{d^2V_C(t)}{dt^2} + 2 \times 10^6\frac{dV_C(t)}{dt} + 12 \times 10^6 V_C(t) = 48 \times 10^6}$$

(c) Steady-state value of $V_C(t)$

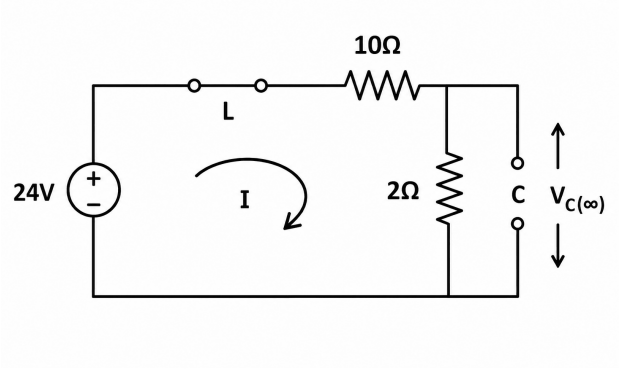

Fig. Solution Q2.98.2

$$I = \frac{24}{10+2} = 2 \text{ A}$$

The capacitor voltage is equal to the voltage across R_2,

$$V_C(\infty) = R_2 I = 2 \times 2 = 4 \text{ V}$$

Therefore,

$$\boxed{V_C(\infty) = 4 \text{ V}}$$

Q2.97.1 NAT 1997 5M Ans: *

Given,

$$e_g(t) = 2.5t \text{ V}, \qquad R = 2\ \Omega, \qquad L = 4 \text{ H}$$

Taking Laplace transform,

$$E_g(s) = \frac{2.5}{s^2}$$

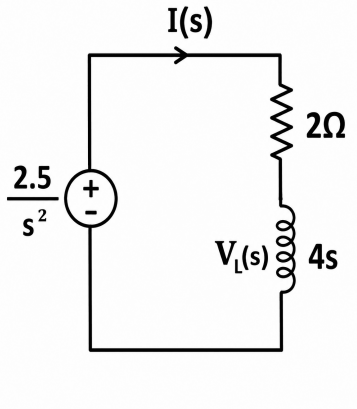

Fig. Solution Q2.97.1

Applying KVL in the s-domain,

$$\frac{2.5}{s^2} = (2+4s)I(s)$$

$$I(s) = \frac{2.5}{s^2(2+4s)} = \frac{0.625}{s^2(s+0.5)}$$

Using partial fractions,

$$I(s) = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{s+0.5}$$

Multiplying by $s^2(s+0.5)$,

$$0.625 = A(s+0.5) + Bs(s+0.5) + Cs^2$$

Putting $s = 0$,

$$A = 1.25$$

Putting $s = -0.5$,

$$C = 2.5$$

Also,

$$B + C = 0 \Rightarrow B = -2.5$$

Hence,

$$I(s) = \frac{1.25}{s^2} - \frac{2.5}{s} + \frac{2.5}{s+0.5}$$

Taking inverse Laplace transform,

$$i(t) = 1.25t - 2.5 + 2.5e^{-0.5t}$$

$$\boxed{i(t) = \frac{5}{4}t - \frac{5}{2} + \frac{5}{2}e^{-t/2}}$$

At $t = 4$ s,

$$i(4) = 1.25(4) - 2.5 + 2.5e^{-2} = 2.838 \text{ A}$$

$$\boxed{i(4) = 2.84 \text{ A}}$$

Further,

$$V_L(s) = LsI(s) = 4sI(s) = \frac{5}{s} - \frac{10}{2s+1}$$

Therefore,

$$v_L(t) = 5 - 5e^{-t/2} = 5\left(1 - e^{-t/2}\right)$$

At $t = 4$ s,

$$v_L(4) = 5(1 - e^{-2}) = 4.323 \text{ V}$$

$$\boxed{v_L(4) = 4.32 \text{ V}}$$

Q2.96.1 MCQ | 1996 | 1M | Ans: D | ● ○ ○

Before switching,

$$V_C(0^-) = 0$$

Since the voltage across a capacitor cannot change instantaneously,

$$V_C(0^+) = V_C(0^-) = 0$$

The capacitor current is

$$i_C = C\frac{dV_C}{dt}$$

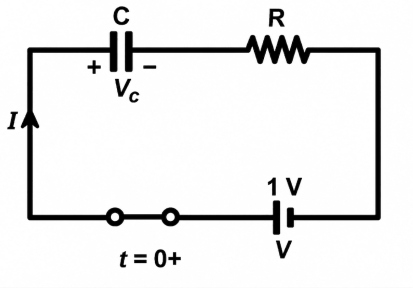

Fig. Solution Q2.96.1

Applying KVL at $t = 0^+$,

$$-1 + V_C(0^+) + Ri_C(0^+) = 0$$

Substituting $V_C(0^+) = 0$ and $i_C = C\dfrac{dV_C(0^+)}{dt}$,

$$-1 + RC\frac{dV_C(0^+)}{dt} = 0$$

Hence,

$$RC\frac{dV_C(0^+)}{dt} = 1$$

$$\boxed{\left.\frac{dV_C}{dt}\right|_{t=0^+} = \frac{1}{RC}}$$

Therefore, the correct answer is

$$\boxed{\textbf{D. } \frac{1}{RC}}$$

Q2.96.2 MCQ 1996 1M Ans: A ●○○

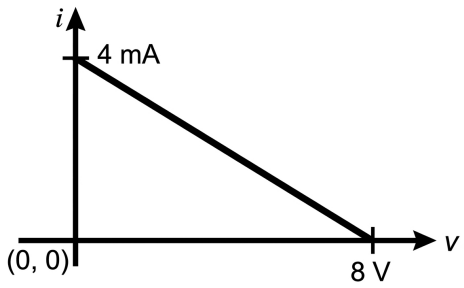

Fig. Solution Q2.96.2

Equivalent resistance of the network can be found out from slope of the given curve;

$$R_{th} = \frac{V}{I} = \frac{8}{4 \times 10^{-3}} = 2000\ \Omega$$

$$R_{th} = 2\ \text{k}\Omega$$

Hence, the network can be replaced by its Thevenin equivalent.

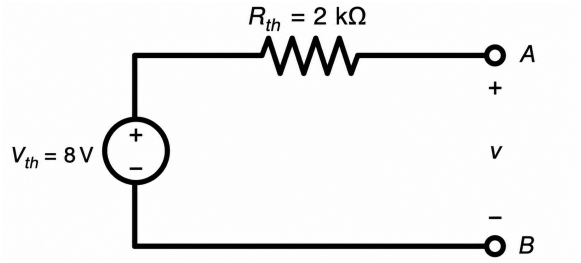

Fig. Solution Q2.96.2

When a 6 mH inductor is connected across terminals A and B, the time constant is

$$\tau = \frac{L}{R_{th}}$$

$$\tau = \frac{6 \times 10^{-3}}{2 \times 10^{3}} = 3 \times 10^{-6} \text{ s}$$

Hence, the correct answer is

A. 3 μs

Q2.92.1 MCQ 1992 1M Ans: D

For an RC circuit,

$$\tau = R_{eq}C$$

where R_{eq} is the resistance seen by the capacitor.
To find R_{eq}, deactivate the independent voltage source.

Voltage source $\rightarrow$ Short Circuit

The equivalent circuit seen from the capacitor terminals is shown below.

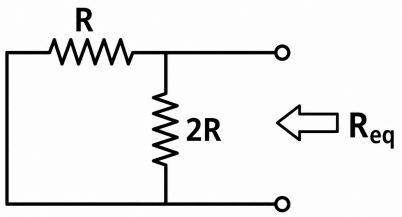

Fig. Solution Q2.92.1

Hence,

$$R_{eq} = R \parallel 2R$$

$$R_{eq} = \frac{R(2R)}{R + 2R} = \frac{2R}{3}$$

Therefore,

$$\tau = R_{eq}C = \frac{2R}{3}C$$

Hence, the correct answer is

D. $\frac{2RC}{3}$

Q2.91.1 MCQ | 1991 | 2M | Ans: D | ● ● ●

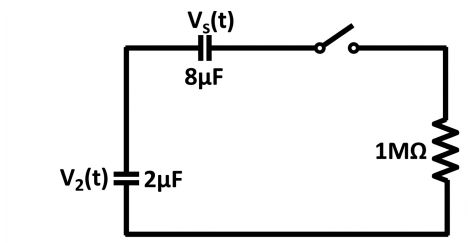

Fig. Solution Q2.91.1

The capacitor polarities during charge redistribution are shown below.

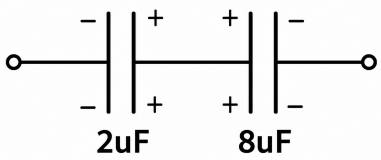

Fig. Solution Q2.91.1

Observe that the inner plates of both capacitors are positively charged. Therefore, the voltages across the capacitors oppose each other when viewed across the resistor terminals.
Hence, the voltage across the resistor is

$$v_R = v_{2\mu F} - v_{8\mu F}$$

Initially,

$$v_{2\mu F}(0) = 10\text{ V}, \qquad v_{8\mu F}(0) = 0$$
$$\Rightarrow v_R(0) = 10 - 0 = 10\text{ V}$$

and current flows through the resistor.
As charge is transferred, the 8 μF capacitor charges up while the 2 μF capacitor discharges. Consequently, $v_{8\mu F}$ increases and $v_{2\mu F}$ decreases, causing the resistor voltage

$$v_R = v_{2\mu F} - v_{8\mu F}$$

to gradually reduce.
At steady state,

$$i(\infty) = 0 \qquad \therefore v_R(\infty) = 0$$

Hence,

$$v_{2\mu F}(\infty) - v_{8\mu F}(\infty) = 0$$
$$\Rightarrow v_{2\mu F}(\infty) = v_{8\mu F}(\infty).$$

Let this common voltage be x.
At steady state,

$$i(\infty) = 0$$

Hence, the voltage across the resistor is; $v_R(\infty) = 0$
Therefore, both capacitors are connected across the same two nodes and must have the same voltage:

$$v_2(\infty) = v_s(\infty) = x$$

The equivalent capacitance seen by the resistor is

$$C_{eq} = \frac{8 \times 2}{8 + 2} = 1.6\ \mu F$$

Hence,

$$\tau = RC_{eq} = (1\ M\Omega)(1.6\ \mu F) = 1.6\ s$$

Let $i(t)$ be the loop current.
For the 8 μF capacitor,

$$i = 8\mu F \frac{dv_s}{dt}$$

Integrating from 0 to ∞,

$$\int_0^\infty i\,dt = 8\mu F\,[x - 0] = 8x\ \mu C \tag{1}$$

For the 2 μF capacitor,

$$i = -2\mu F \frac{dv_2}{dt}$$

Integrating,

$$\int_0^\infty i\,dt = 2\mu F(10 - x) = (20 - 2x)\mu C \tag{2}$$

Since the same current flows through both capacitors,

$$8x = 20 - 2x$$

$$10x = 20 \Rightarrow x = 2\ V$$

Therefore,

$$\boxed{v_2(\infty) = v_s(\infty) = 2\ V}$$

Hence, the correct answer is

$$\boxed{\textbf{D. } v_2(\infty) = v_s(\infty) = 2\ V}$$

Mistakes to be avoided

- Do not directly assume that due to resistance final voltage will be zero. Analyze the circuit carefully.

Q2.91.2 MCQ 1991 2M Ans: B ●●○

Let the two coils have inductances L_1 and L_2, and resistances R_1 and R_2.

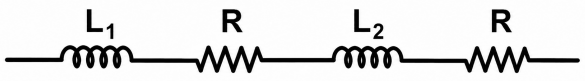

Fig. Solution Q2.91.2

The individual time constants are

$$\tau_1 = \frac{L_1}{R_1}, \qquad \tau_2 = \frac{L_2}{R_2}$$

When the coils are connected in series,

$$L_{eq} = L_1 + L_2, \qquad R_{eq} = R_1 + R_2$$

Hence, the time constant of the series combination is

$$\tau = \frac{L_1 + L_2}{R_1 + R_2}$$

Given,

$$R_1 = R_2 = R$$

Therefore,

$$\tau = \frac{L_1 + L_2}{2R}$$

$$\tau = \frac{1}{2}\left(\frac{L_1}{R} + \frac{L_2}{R}\right)$$

Using

$$\tau_1 = \frac{L_1}{R}, \qquad \tau_2 = \frac{L_2}{R}$$

we get

$$\tau = \frac{\tau_1 + \tau_2}{2}$$

Thus, the time constant of the series combination is the arithmetic mean of the individual time constants.
Hence, the correct answer is

B. Average of the time constant of the individual coils

PrepFusion

SOLUTIONS

III

Network Theorems For DC & Miscellaneous Topics

Topics

1. Source Transformation and Superposition Theorem
2. Thevenin's Theorem
3. Norton's Theorem
4. Concept of Linearity
5. Reciprocity Theorem
6. Maximum Power Transfer Theorem
7. Star-Delta Connections and conversions
8. Finding Equivalent Resistances
9. Equivalent Resistance of a Cube Across Different Nodes
10. Equivalent Resistance in an Infinite Ladder
11. Bulb-Based Problems

Unit Snapshot*

Stream: EE	**Questions:** 36	**MCQ:** 24	**MSQ:** 0	**NAT:** 10
1M: 16	**2M:** 17	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2021)–4M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ | 2026 | 1M | Ans: B

Finding V_{TH}:

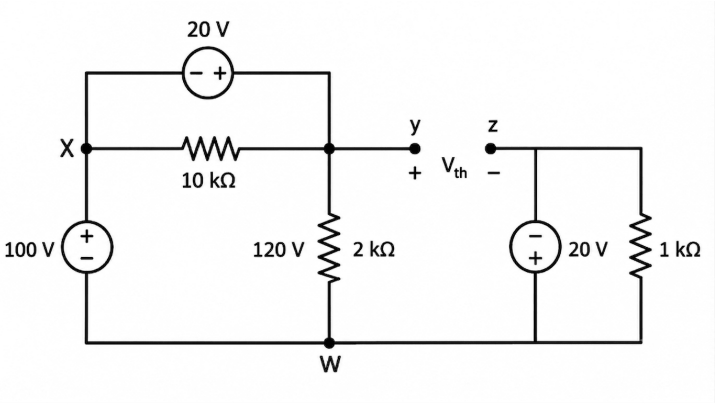

Fig. Solution Q3.26.1

Take node W as the reference node.

$$V_X = 100 \text{ V} \qquad \text{and} \qquad V_Y - V_X = 20$$

Hence,

$$V_Y = 100 + 20 = 120 \text{ V}$$

Similarly,

$$V_Z = -20 \text{ V}$$

Therefore,

$$V_{TH} = V_Y - V_Z = 120 + 20 = 140 \text{ V}$$

Finding R_{TH}:

Deactivate all independent voltage sources.

$$100 \text{ V} \rightarrow \text{Short Circuit}, \quad 20 \text{ V} \rightarrow \text{Short Circuit}, \quad 20 \text{ V} \rightarrow \text{Short Circuit}$$

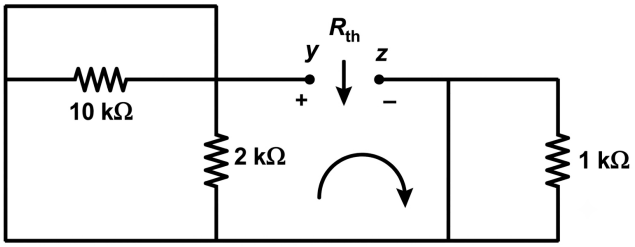

Fig. Solution Q3.26.1

Thus, terminals Y and Z are directly connected by an ideal short circuit.
Therefore,

$$\boxed{R_{TH} = 0 \ \Omega}$$

The resistors 10 kΩ, 2 kΩ and 1 kΩ are bypassed by the short circuit and do not affect R_{TH}.
Hence,

$$\boxed{V_{TH} = 140 \text{ V}, \qquad R_{TH} = 0 \ \Omega}$$

Therefore, the correct answer is

$$\boxed{\textbf{B. } V_{TH} = 140 \text{ V}, \ R_{TH} = 0 \ k\Omega}$$

Q3.26.2 MCQ | 2026 | 2M | Ans: C

Let the Thevenin equivalent of the network be

V_{TH} in series with R_{TH}.

Using the data point

$$v_t = 30 \text{ V}, \qquad i_x = 0.5 \text{ A},$$

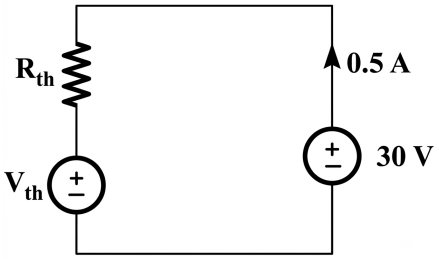

Fig. Solution Q3.26.2

$$0.5 = \frac{30 - V_{TH}}{R_{TH}}$$

$$0.5R_{TH} = 30 - V_{TH} \tag{1}$$

Using the data point

$$v_t = 36 \text{ V}, \qquad i_x = 1 \text{ A},$$

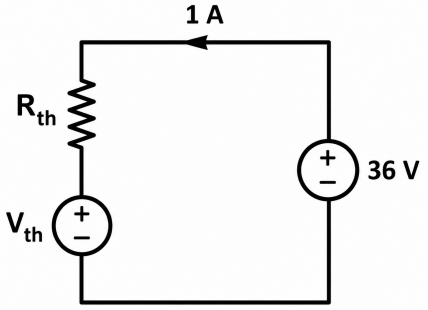

Fig. Solution Q3.26.2

$$1 = \frac{36 - V_{TH}}{R_{TH}}$$

$$R_{TH} = 36 - V_{TH} \tag{2}$$

Substituting (2) into (1),

$$0.5(36 - V_{TH}) = 30 - V_{TH}$$

$$18 - 0.5V_{TH} = 30 - V_{TH}$$

$$0.5V_{TH} = 12 \Rightarrow V_{TH} = 24 \text{ V}$$

From (2),

$$R_{TH} = 36 - 24 = 12\ \Omega$$

The Norton current is

$$I_N = \frac{V_{TH}}{R_{TH}} = \frac{24}{12} = 2 \text{ A}$$

and

$$R_N = R_{TH} = 12\ \Omega.$$

Therefore,

$$\boxed{I_N = 2 \text{ A}, \qquad R_N = 12\ \Omega}$$

Hence, the correct answer is

$$\boxed{\textbf{C. } I_N = 2 \text{ A}, \ R_N = 12\ \Omega}$$

Q3.26.3 NAT | 2026 | 1M | Ans: 2.86 | ● ○ ○

For maximum power transfer,

$$R_L = R_{th}$$

To find R_{th}, deactivate the independent voltage source. The resulting circuit is shown below;

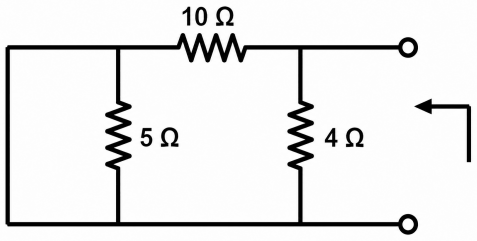

Fig. Solution Q3.26.3

The 5 Ω resistor is short-circuited and hence does not affect the equivalent resistance.
Therefore, looking into the load terminals,

$$R_{th} = 10 \parallel 4$$

$$R_{th} = \frac{10 \times 4}{10 + 4}$$

$$R_{th} = \frac{40}{14} = 2.857\ \Omega$$

Therefore,

$$\boxed{2.86}$$

Q3.21.1 MCQ | 2021 | 1M | Ans: B | ● ● ○

Finding V_{TH}:

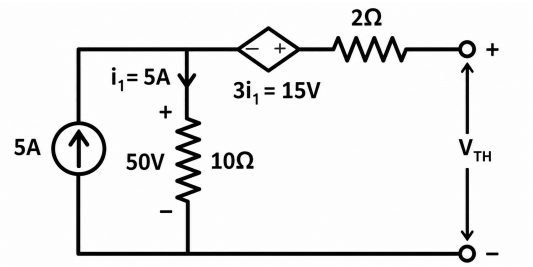

Fig. Solution Q3.21.1

For open-circuit condition,

$$i_a = 0$$

Hence, no current flows through the 2 Ω resistor.
The 5 A source current therefore flows entirely through the 10 Ω resistor.

$$i_1 = 5 \text{ A}$$

Voltage across the 10 Ω resistor is

$$V = 10 \times 5 = 50 \text{ V}$$

The dependent voltage source is

$$3i_1 = 3 \times 5 = 15 \text{ V}$$

Therefore,

$$V_{TH} = 50 + 15 = 65 \text{ V}$$

$$\boxed{V_{TH} = 65 \text{ V}}$$

Finding R_{TH} using V_{oc}/I_{sc}:
Short-circuit terminals a and b.

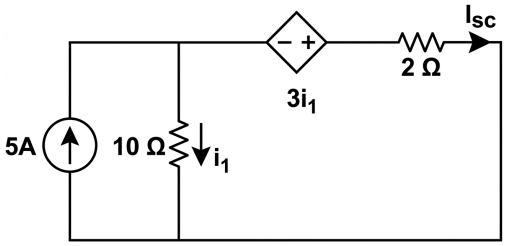

Fig. Solution Q3.21.1

Let the short-circuit current be I_{sc}.
Applying KCL at the top node,

$$I_{sc} + i_1 = 5 \quad (1)$$

Applying KVL,

$$10i_1 + 3i_1 - 2I_{sc} = 0$$

$$I_{sc} = 6.5\, i_1$$

Using KCL,

$$I_{sc} + i_1 = 5 \Rightarrow 6.5i_1 + i_1 = 5$$

$$i_1 = \frac{2}{3} \text{ A}$$

$$I_{sc} = 6.5\left(\frac{2}{3}\right) = \frac{13}{3} \text{ A}$$

Hence,

$$R_{TH} = \frac{V_{oc}}{I_{sc}} = \frac{65}{13/3} = 15 \ \Omega$$

$$\boxed{V_{TH} = 65 \text{ V}, \qquad R_{TH} = 15 \ \Omega}$$

Therefore, the correct answer is

$$\boxed{\textbf{B.}\ 65\text{V in series with } 15\text{ohms}}$$

Q3.20.1 MCQ | 2020 | 2M | Ans: C ● ● ○

Method 1 (Using V-I Characteristics)

The supply behaves as:

- An ideal 4 A current source for terminal voltages below 10 V.
- An ideal 10 V voltage source for load currents between 0 and 4 A.

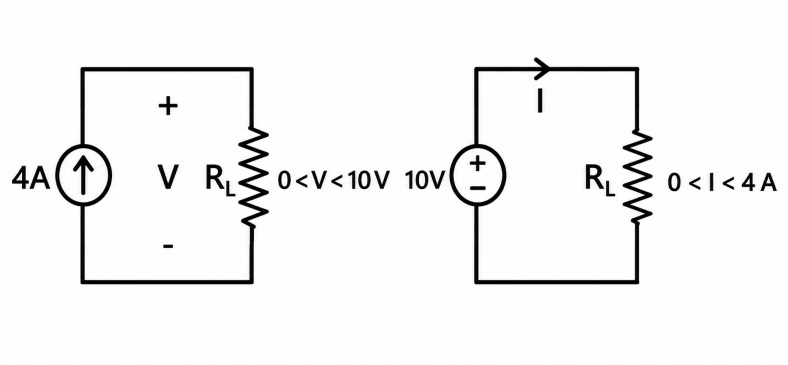

Fig. Solution Q3.20.1

Hence, the operating point is limited by

$$V = 10 \text{ V}, \qquad I = 4 \text{ A}$$

and the maximum power transfer occurs at the transition point.

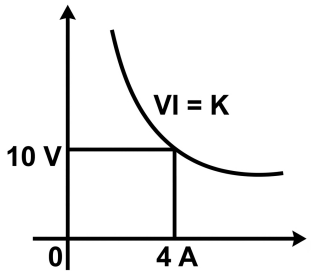

Fig. Solution Q3.20.1

Therefore,

$$P_{\max} = VI = (10)(4) = 40 \text{ W}$$

Since

$$P = I^2 R_L$$

$$40 = 4^2 R_L$$

$$R_L = \frac{40}{16} = 2.5\ \Omega$$

Thus,

$$\boxed{R_L = 2.5\ \Omega, \quad I_L = 4 \text{ A}, \quad V_L = 10 \text{ V}}$$

Method 2 (Direct Analysis)

When the supply operates as a 4 A current source,

$$V_L = 4R_L$$

The current-source mode is valid only while

$$V_L < 10 \text{ V} \Rightarrow 4R_L < 10$$

$$R_L < 2.5\ \Omega$$

The load power is

$$P_L = I^2 R_L = 16R_L$$

which increases with R_L. Hence, the maximum power in current-source mode occurs at

$$R_L = 2.5\ \Omega$$

giving

$$P_{\max} = 16(2.5) = 40 \text{ W}$$

For

$$R_L > 2.5\ \Omega$$

the supply behaves as a 10 V voltage source.
Then,

$$I_L = \frac{10}{R_L}$$

and

$$P_L = \frac{10^2}{R_L}$$

which decreases as R_L increases.
Thus, the maximum power in voltage-source mode also occurs at

$$R_L = 2.5\ \Omega$$

Hence,

$$I_L = \frac{10}{2.5} = 4 \text{ A}$$

and

$$V_L = 10 \text{ V}$$

Hence, the correct answer is

C. $R_L = 2.5\ \Omega, \quad I_L = 4 \text{ A}, \quad V_L = 10 \text{ V}$

Q3.20.2 NAT 2020 1M Ans: 14 ●○○

Let the voltage at node (1) be V_1 and the open-circuit voltage be V_{TH}.

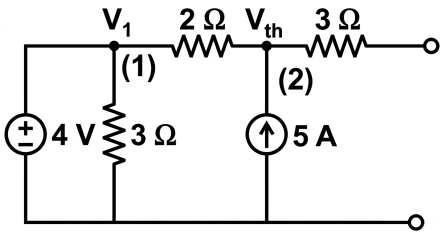

Fig. Solution Q3.20.2

Since the 4 V source is directly connected between node (1) and ground,

$$V_1 = 4 \text{ V}$$

Since the output terminal is open-circuited, no current flows through the 3 Ω resistor connected to the output. Hence, the voltage at the output terminal is equal to the voltage at node (2), i.e.,

$$V_{TH} = V_2$$

Applying KCL at node (2),

$$\frac{V_{TH} - V_1}{2} = 5$$

Substituting $V_1 = 4$ V,

$$\frac{V_{TH} - 4}{2} = 5$$

$$V_{TH} - 4 = 10 \Rightarrow V_{TH} = 14 \text{ V}$$

Therefore,

$$\boxed{14.00}$$

Q3.19.1 NAT 2019 2M Ans: 0 ●●○

Method 1 (Nodal Analysis)

Let the voltage across every parallel branch be V.

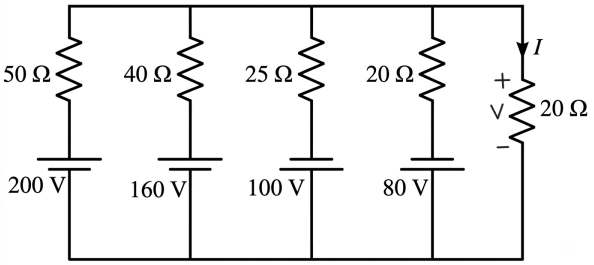

Fig. Solution Q3.19.1

Applying KCL at the top node,

$$\frac{V - 200}{50} + \frac{V - 160}{40} + \frac{V + 100}{25} + \frac{V + 80}{20} + \frac{V}{20} = 0$$

Multiplying throughout by 200,

$$4V - 800 + 5V - 800 + 8V + 800 + 10V + 800 + 10V = 0$$

$$37V = 0$$

$$V = 0$$

Hence,

$$I = \frac{V}{20} = 0$$

$$\boxed{I = 0 \text{ A}}$$

Method 2 (Source Transformation)

Transform each practical voltage source into an equivalent current source using

$$I_s = \frac{V_s}{R_s}$$

with the same resistance R_s connected in parallel.

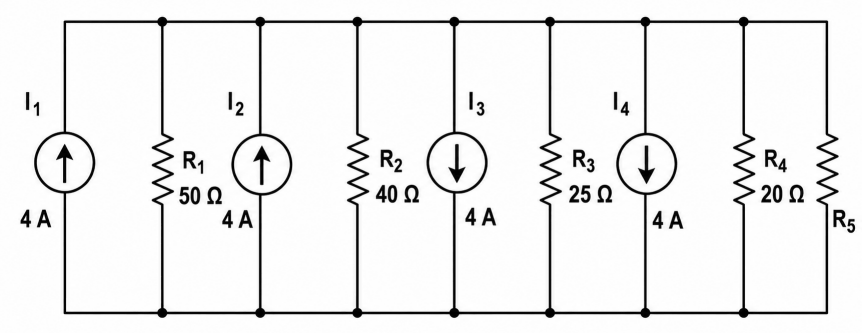

Fig. Solution Q3.19.1

After source transformation, two 4A current sources inject current into the node while the other two 4A current sources draw current from the node.
Hence, the net source current is

$$I_{net} = 4 + 4 - 4 - 4 = 0$$

$$\boxed{0}$$

Q3.17.1 MCQ | 2017 | 2M | Ans: A | ● ● ○

Using source transformation,

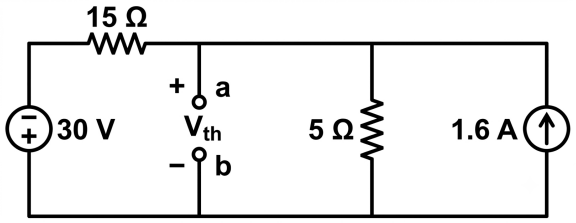

Fig. Solution Q3.17.1

Transforming once again,

$$1.6\text{ A} \parallel 5\Omega \implies 8\text{ V in series with } 5\Omega$$

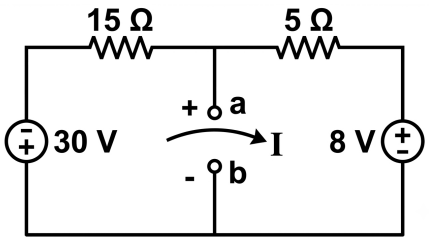

Fig. Solution Q3.17.1

Applying KVL,

$$30 + 15I + 5I + 8 = 0$$

$$20I = -38$$

$$I = -1.9 \text{ A}$$

Applying KVL in the left loop,

$$30 + 15I + V_{ab} = 0$$

$$V_{ab} = -30 - 15I \Rightarrow V_{ab} = -30 - 15(-1.9)$$

$$V_{ab} = -1.5 \text{ V}$$

Hence, the correct answer is

A. $-1.5V$

Q3.17.2 NAT 2017 2M Ans: 3.025 ●●○

For maximum power transfer, first find the Thevenin equivalent seen by R.

Thevenin Voltage V_{th}:

Remove R and find the open-circuit voltage across its terminals.

Transform the 2 A current source in parallel with 5Ω into an equivalent voltage source:

$$V = IR = (2)(5) = 10 \text{ V}$$

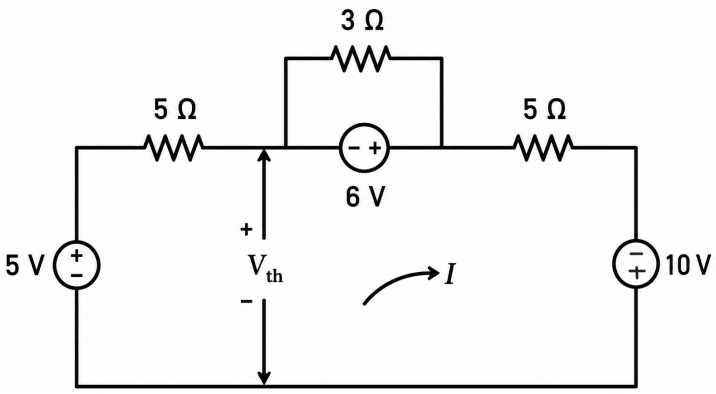

Fig. Solution Q3.17.2

Applying KVL,

$$-5 + 5I - 6 + 5I - 10 = 0$$

$$10I = 21$$

$$I = 2.1 \text{ A}$$

Also,

$$V_{th} = 5 - 5I$$

$$V_{th} = 5 - 5(2.1) = -5.5 \text{ V}$$

$$|V_{th}| = 5.5 \text{ V}$$

Thevenin Resistance R_{th}:

Deactivate all independent sources:

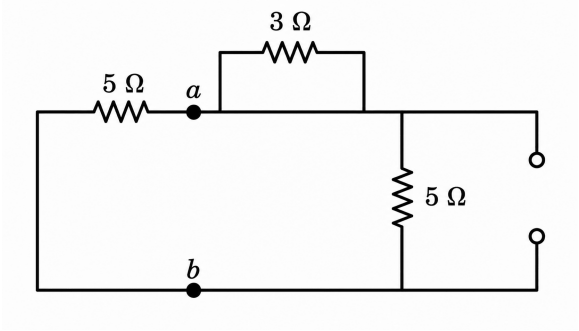

Fig. Solution Q3.17.2

The 3Ω resistor gets short-circuited by the deactivated 6 V source.
Hence,

$$R_{th} = 5 \parallel 5$$

$$R_{th} = \frac{5 \times 5}{5 + 5} = 2.5\ \Omega$$

Therefore, the maximum power transferred to R is

$$P_{\max} = \frac{V_{th}^2}{4R_{th}}$$

$$P_{\max} = \frac{(5.5)^2}{4(2.5)} = 3.025\ \text{W}$$

$$\boxed{3.025}$$

Key Points

- **Source Transformation:**

  A current source I_s in parallel with a resistance R is equivalent to a voltage source $V_s = I_s R$ in series with the same resistance R. The **positive** terminal of the voltage source is at the terminal from which the current source arrow originates.

Q3.17.3 NAT | 2017 | 1M | Ans: 3 | ●●○

Redrawing the circuit;

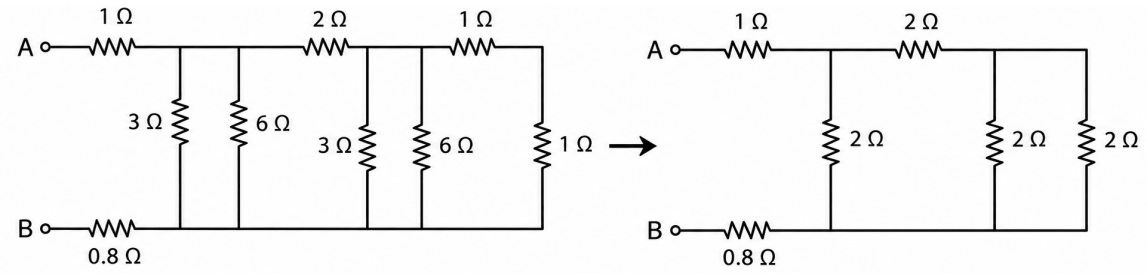

Fig. Solution Q3.17.3

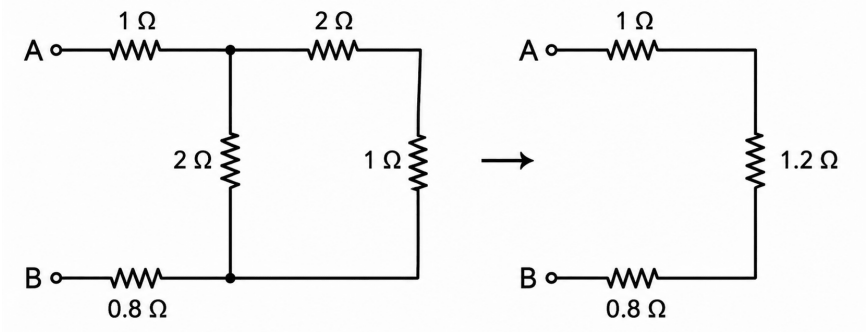

Fig. Solution Q3.17.3

Therefore, the equivalent circuit reduces to three series resistors:

$$R_{AB} = 1 + 1.2 + 0.8 = 3\ \Omega$$

3

Q3.16.1 MCQ 2016 1M Ans: D

For R_A:

Since the network extends infinitely, the portion to the right of the first 2Ω resistor is again R_A.

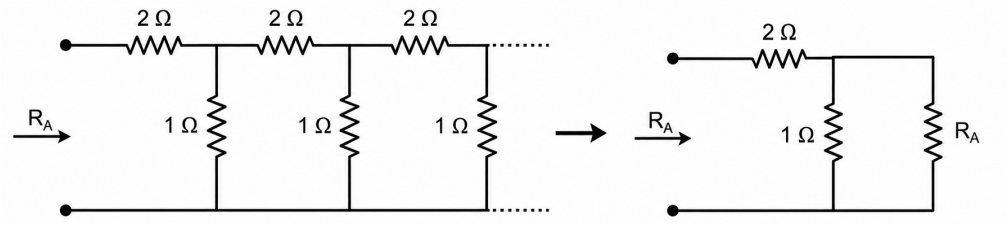

Fig. Solution Q3.16.1

Thus,

$$R_A = 2 + (1 \parallel R_A) = 2 + \frac{R_A}{1 + R_A}$$

$$R_A^2 - 2R_A - 2 = 0$$

Solving we get (negative root rejected);

$$R_A = 1 + \sqrt{3}$$

For R_B:

Method 1

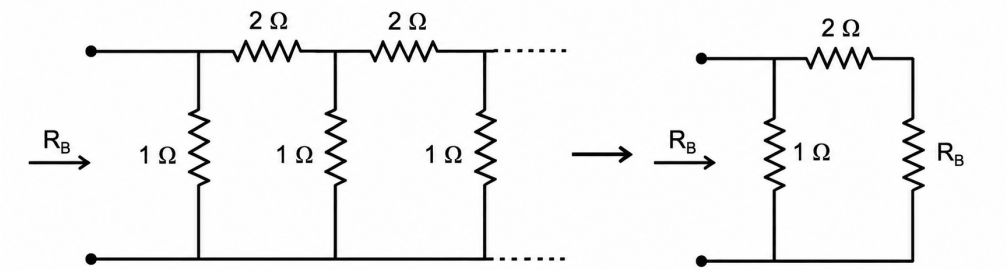

Fig. Solution Q3.16.1

$$R_B = 1 \parallel (2 + R_B)$$

$$R_B = \frac{R_B + 2}{R_B + 3}$$

$$R_B^2 + 2R_B - 2 = 0$$

Solving we get (negative root rejected);

$$R_B = -1 + \sqrt{3}$$

Option A, B and C are all wrong.
Checking option D,

$$\frac{R_A}{1 + R_A} = \frac{1 + \sqrt{3}}{2 + \sqrt{3}}$$

Rationalizing,

$$\frac{1 + \sqrt{3}}{2 + \sqrt{3}} \cdot \frac{2 - \sqrt{3}}{2 - \sqrt{3}} = (1 + \sqrt{3})(2 - \sqrt{3})$$

$$= 2 - \sqrt{3} + 2\sqrt{3} - 3$$

$$= \sqrt{3} - 1$$

Also,

$$R_B = \sqrt{3} - 1$$

Hence,

$$\boxed{R_B = \frac{R_A}{1 + R_A}}$$

Method 2

We can draw R_B as;

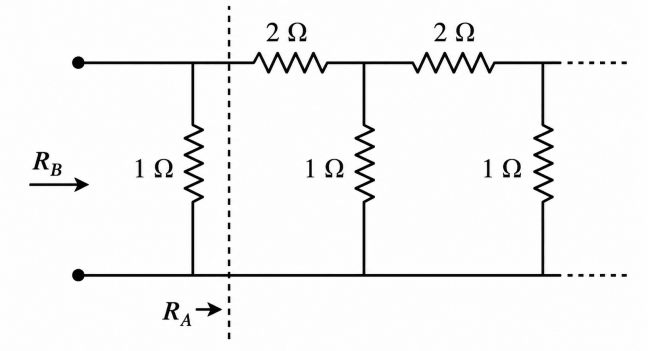

Fig. Solution Q3.16.1

$$\Rightarrow R_B = 1 \parallel R_A$$

$$R_B = \frac{1 \times R_A}{1 + R_A}$$

We also get another relation:

$$R_A = 2 + R_B \qquad (\because 2 \text{ is in series with } R_B \text{[refer to 1st image]})$$

which is consistent with the above result. However, since only $R_B = \dfrac{R_A}{1 + R_A}$ appears in the options.

Therefore, the correct answer is

$$\boxed{\textbf{D. } R_B = \frac{R_A}{1 + R_A}}$$

Q3.16.2 NAT | 2016 | 2M | Ans: 11.43 | ● ○ ○

Convert the 5 A source in parallel with 5Ω into its equivalent voltage source:

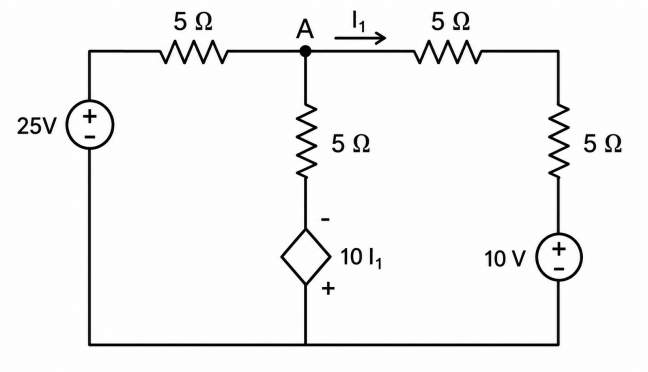

Fig. Solution Q3.16.2

Applying KCL at node A,

$$\frac{V_A - 25}{5} + \frac{V_A - 10}{10} + \frac{V_A + 10I_1}{5} = 0$$

Multiplying by 10,

$$2(V_A - 25) + (V_A - 10) + 2(V_A + 10I_1) = 0$$

$$5V_A + 20I_1 = 60$$

Substituting

$$I_1 = \frac{V_A - 10}{10}$$

gives

$$5V_A + 2(V_A - 10) = 60$$

$$7V_A = 80 \Rightarrow V_A = 11.43 \text{ V}$$

$$\boxed{11.43}$$

Q3.15.1 NAT | 2015 | 1M | Ans: 3.36 | ● ○ ○

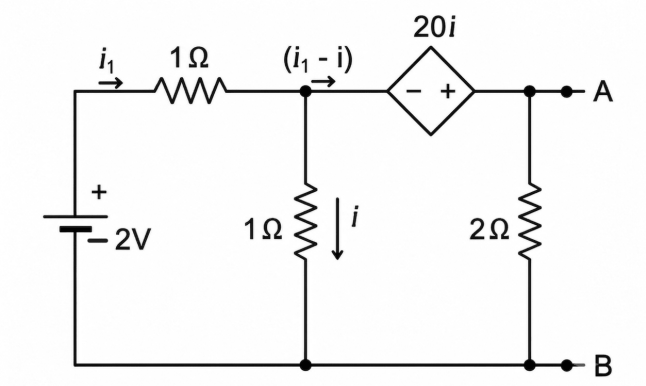

Fig. Solution Q3.15.1

Applying KVL in the left loop,

$$2 = i_1 + i$$

$$2i_1 + 2i = 4 \quad (1)$$

For the right loop,

$$i + 20i - 2(i_1 - i) = 0$$

$$21i - 2i_1 + 2i = 0$$

$$23i = 2i_1 \quad (2)$$

From (2),

$$i_1 = 11.5\,i$$

Substituting into (1),

$$2(11.5i) + 2i = 4 \Rightarrow 25i = 4$$

$$i = 0.16 \text{ A} \Rightarrow i_1 = 1.84 \text{ A}$$

The Thevenin voltage is

$$V_{TH} = V_{AB}$$

$$V_{AB} = 2(i_1 - i)$$

$$= 2(1.84 - 0.16) = 3.36 \text{ V}$$

$$\boxed{3.36}$$

Q3.14.1 NAT | 2014 | 2M | Ans: 3 | ● ○ ○

Method 1 (Nodal Analysis)

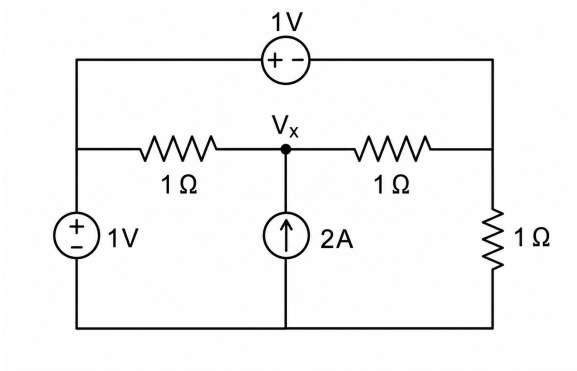

Fig. Solution Q3.14.1

Applying KCL at node V_x,

$$\frac{1 - V_x}{1} + 2 = \frac{V_x}{1}$$

$$(1 - V_x) + 2 = V_x$$

$$2V_x = 3 \Rightarrow V_x = 1.5 \text{ V}$$

Power delivered by the 2 A source is

$$P = V_x I = (1.5)(2) = 3W$$

$$\boxed{3}$$

Method 2 (Superposition)

Case 1: Only the left 1 V source active

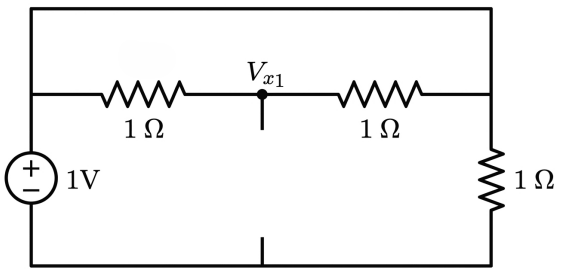

Fig. Solution Q3.14.1

Applying KCL at V_{x1},

$$\frac{1 - V_{x1}}{1} = \frac{V_{x1} - 1}{1}$$

$$V_{x1} = 1 \text{ V}$$

Case 2: Only the top 1 V source active

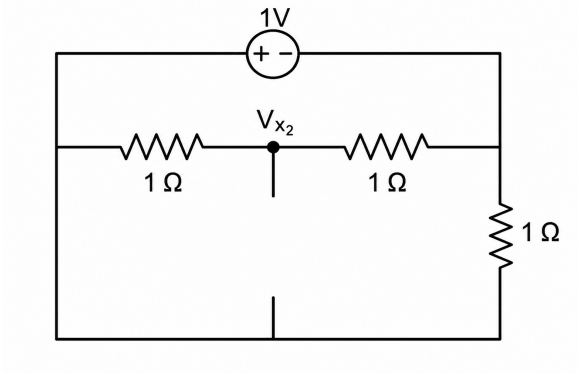

Fig. Solution Q3.14.1

Applying KCL at V_{x2},

$$\frac{0 - V_{x2}}{1} = \frac{V_{x2} + 1}{1}$$

$$V_{x2} = -0.5 \text{ V}$$

Case 3: Only the 2 A source active

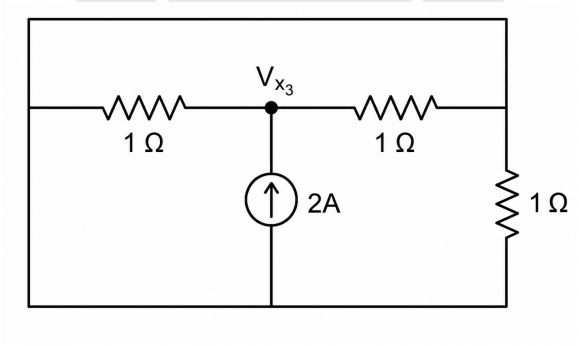

Fig. Solution Q3.14.1

Applying KCL at V_{x3},

$$2 = \frac{V_{x3}}{1} + \frac{V_{x3}}{1}$$

$$2 = 2V_{x3} \Rightarrow V_{x3} = 1 \text{ V}$$

By superposition,

$$V_x = V_{x1} + V_{x2} + V_{x3}$$

$$V_x = 1 - 0.5 + 1 = 1.5 \text{ V}$$

$$\Rightarrow P = V_x I = (1.5)(2) = 3W$$

$$\boxed{3}$$

Q3.14.2 NAT | 2014 | 2M | Ans: 1 | ● ● ○

Method 1

To find Norton's current, short-circuit terminals X and Y.

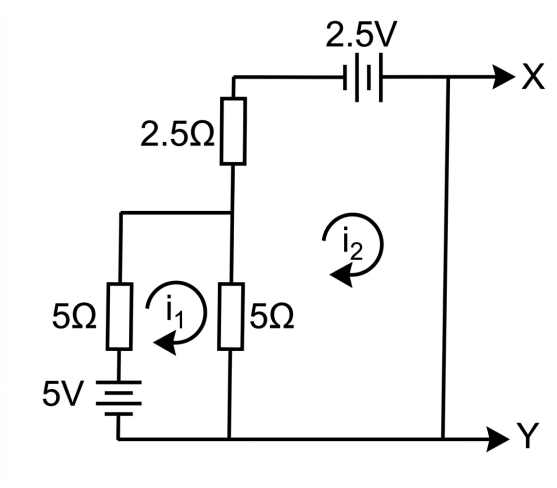

Fig. Solution Q3.14.2

Let mesh currents be i_1 and i_2.

$$5 = 10i_1 - 5i_2$$

$$2.5 = 5i_1 - 2.5i_2 \quad (1)$$

For the second mesh,

$$2.5 = 7.5i_2 - 5i_1 \quad (2)$$

Adding (1) and (2),

$$5 = 5i_2$$

$$i_2 = 1 \text{ A}$$

Since the short-circuit current equals the current through the shorted branch,

$$\boxed{1}$$

Method 2

Transform the 5 V source in series with 5Ω into its equivalent current source;

$$I = \frac{V}{R} = \frac{5}{5} = 1 \text{ A}$$

in parallel with 5Ω.

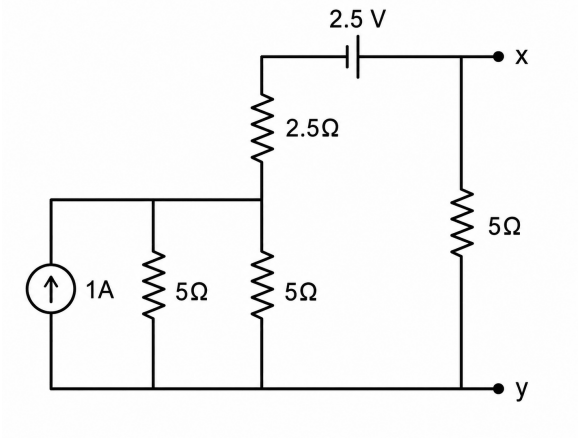

Fig. Solution Q3.14.2

The two parallel 5Ω resistors combine to give

$$5 \parallel 5 = \frac{5 \times 5}{5 + 5} = 2.5\Omega$$

Now transform the 1 A source in parallel with 2.5Ω back into equivalent Voltage source:

$$V = IR = (1)(2.5) = 2.5 \text{ V}$$

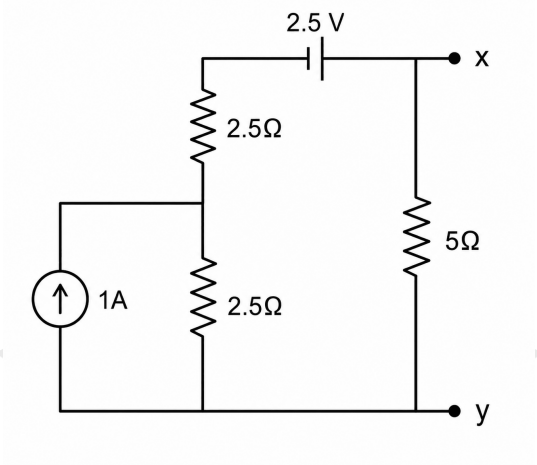

Fig. Solution Q3.14.2

Thus the left network becomes a 2.5 V source in series with two 2.5Ω resistors.

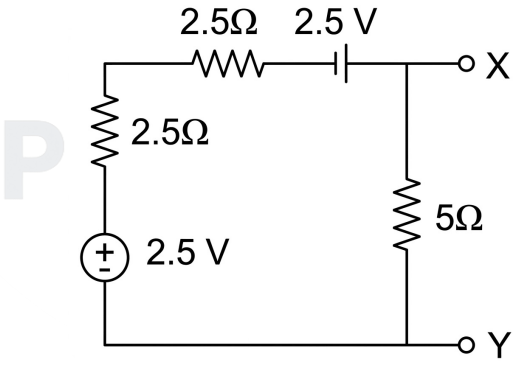

Fig. Solution Q3.14.2

Hence,

$$R_{\text{series}} = 2.5 + 2.5 = 5\Omega$$

and

$$V_{\text{series}} = 2.5 + 2.5 = 5 \text{ V}$$

Now short-circuit terminals X and Y.

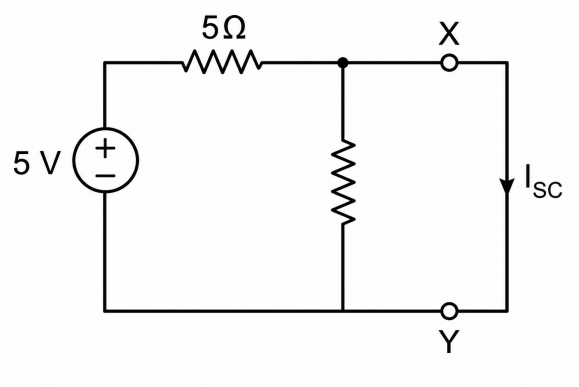

Fig. Solution Q3.14.2

$$I_{SC} = \frac{5}{5} = 1 \text{ A}$$

Hence,

1

Q3.13.1 MCQ | 2013 | 1M | Ans: B | ● ○ ○

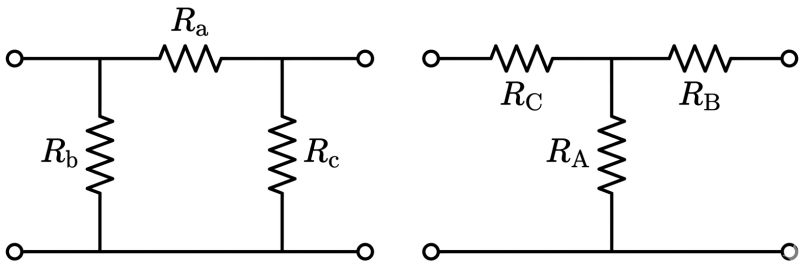

Fig. Solution Q3.13.1

For a delta-to-star transformation,

$$R_A = \frac{R_b R_c}{R_a + R_b + R_c}, \quad R_B = \frac{R_c R_a}{R_a + R_b + R_c}, \quad R_C = \frac{R_a R_b}{R_a + R_b + R_c}$$

If all delta resistances are scaled by a factor k,

$$R'_a = kR_a, \qquad R'_b = kR_b, \qquad R'_c = kR_c$$

Then,

$$R'_A = \frac{R'_b R'_c}{R'_a + R'_b + R'_c} = \frac{(kR_b)(kR_c)}{k(R_a + R_b + R_c)}$$

$$R'_A = k\frac{R_b R_c}{R_a + R_b + R_c} = kR_A$$

Similarly,

$$R'_B = kR_B, \qquad R'_C = kR_C$$

Therefore,

B. k

Q3.12.1 MCQ 2012 2M Ans: A ●●○

For a one-port network, the current entering one terminal is equal to the current leaving the other terminal.

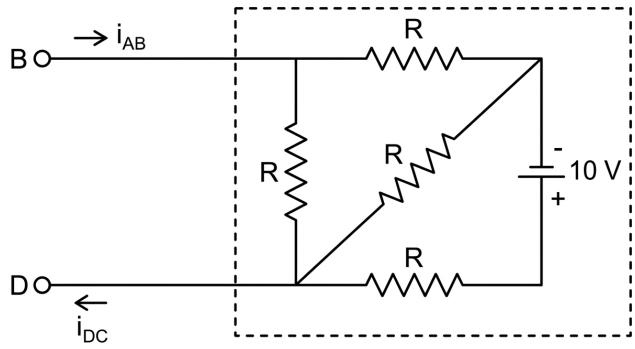

Fig. Solution Q3.12.1

Hence,

$$i_{AB} = i_{DC}$$

Therefore, if a current i flows from V_A to V_B, the same current i must flow through the branch connecting V_C and V_D.

Method 1 (Using Source Transformation)

Transform the 2 A current source in parallel with 1Ω into its equivalent Voltage source:

$$V = IR = (2)(1) = 2 \text{ V}$$

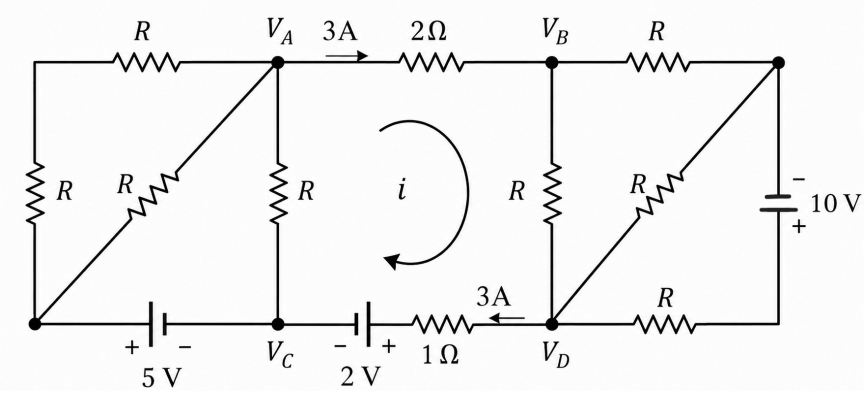

Fig. Solution Q3.12.1

Given,

$$V_A - V_B = 6 \text{ V}$$

Current through the 2Ω resistor is

$$i = \frac{V_A - V_B}{2} = \frac{6}{2} = 3 \text{ A}$$

By the above one-port property, the same current flows through the branch between C and D.

$$i_{DC} = 3 \text{ A}$$

in series with 1Ω.
Hence,

$$V_{DC} = (3)(1) + 2 = 5 \text{ V}$$

Therefore,

$$V_{CD} = V_C - V_D = -5 \text{ V}$$

Method 2 (Without Source Transformation)

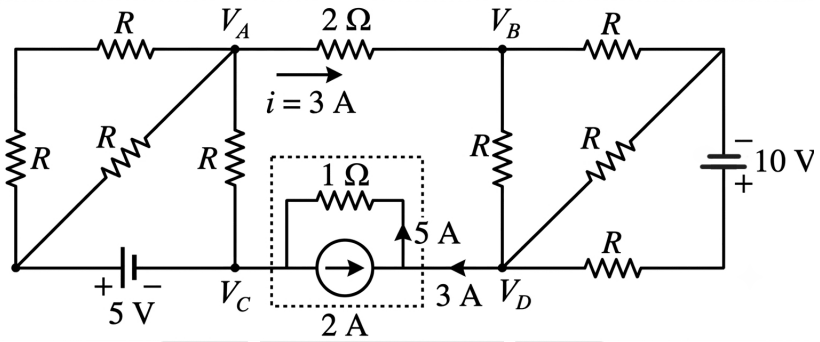

Fig. Solution Q3.12.1

Given,

$$V_A - V_B = 6 \text{ V}$$

$$i = \frac{V_A - V_B}{2} = \frac{6}{2} = 3 \text{ A}$$

The current source supplies

$$2 \text{ A}$$

from C to D. Hence the current through the parallel 1Ω resistor is

$$i_R = 3 + 2 = 5 \text{ A}$$

Thus,

$$V_{DC} = (5)(1) = 5 \text{ V}$$

Therefore,

$$V_{CD} = V_C - V_D = -5 \text{ V}$$

Hence, the correct answer is

A. − 5 V

Q3.12.2 MCQ 2012 1M Ans: A ● ● ○

Apply a test voltage source V across terminals 1 and 2.
Let the current drawn from the source be I.

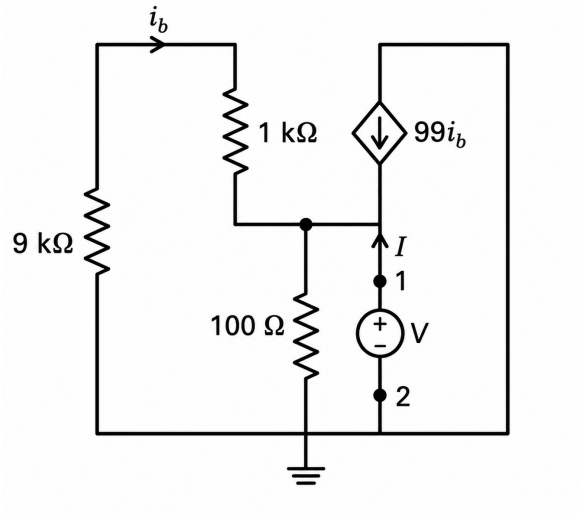

Fig. Solution Q3.12.2

Then,

$$Z_{12} = \frac{V}{I}$$

The 1 kΩ and 9 kΩ resistors are in series, hence

$$i_b = \frac{0 - V}{1\text{k} + 9\text{k}}$$

$$i_b = -\frac{V}{10^4} \tag{1}$$

Applying KCL at node 1,

$$-i_b + \frac{V}{100} - 99i_b - I = 0$$

$$V = 10^4 i_b + 100I \tag{2}$$

Substituting (1) into (2),

$$V = 10^4\left(-\frac{V}{10^4}\right) + 100I$$

$$2V = 100I \Rightarrow \frac{V}{I} = 50\ \Omega$$

Therefore,

$$\boxed{Z_{12} = 50\ \Omega}$$

Hence, the correct answer is

$$\boxed{\textbf{A. } 50\ \Omega}$$

Q3.11.1 MCQ | 2011 | 1M | Ans: A | ●●○

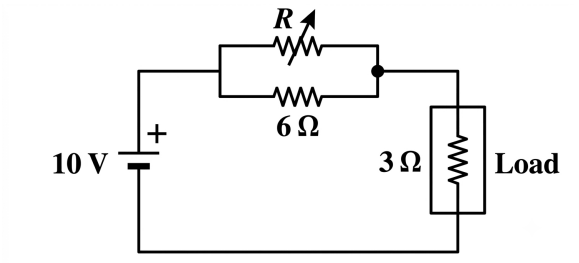

Fig. Solution Q3.11.1

The resistance seen in series with the load is

$$R_s = R \parallel 6 = \frac{6R}{R+6}$$

Hence the circuit current is

$$I = \frac{10}{3 + R_s}$$

Power delivered to the fixed 3Ω load is

$$P_L = I^2(3)$$

$$P_L = 3\left(\frac{10}{3+R_s}\right)^2$$

Since P_L decreases as R_s increases, maximum load power is obtained when R_s is minimum.
Now,

$$R_s = \frac{6R}{R+6}$$

and for $R \geq 0$,

$$R_s \geq 0$$

The minimum value of R_s occurs at

$$R = 0$$

Thus the entire source voltage appears across the load and the load current becomes maximum.
Therefore,

$$\boxed{R = 0\ \Omega}$$

Hence, the correct answer is

$$\boxed{\textbf{A. Zero}}$$

Mistakes to be avoided

- Do not apply Maximum Power Transfer Theorem.
  Maximum Power Transfer Theorem ($R_L = R_{th}$) is applicable when the *load resistance is variable*. Here the load is fixed at 3Ω, while R is varied on the source side. Therefore, the load power is maximized by minimizing the source-side resistance, i.e., $R = 0\Omega$.

Q3.09.1 MCQ 2009 2M Ans: B ●●○

To find R_{TH}, first calculate the open-circuit voltage $V_{OC} = V_{TH}$.

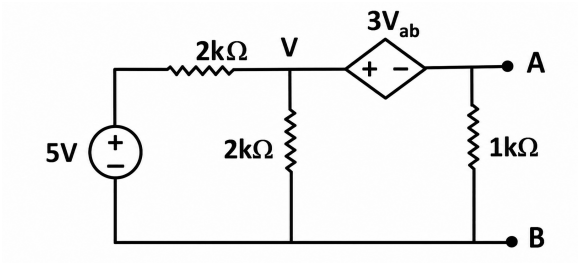

Fig. Solution Q3.09.1

Applying KCL at node V,

$$\frac{V-5}{2k} + \frac{V}{2k} + \frac{V-3V_{AB}}{1k} = 0 \qquad (1)$$

From the dependent source,

$$V_{AB} + 3V_{AB} = V$$

$$V = 4V_{AB} \tag{2}$$

Substituting (2) into (1),

$$\frac{4V_{AB} - 5}{2k} + \frac{4V_{AB}}{2k} + \frac{V_{AB}}{1k} = 0$$

$$10V_{AB} - 5 = 0$$

$$V_{AB} = 0.5 \text{ V}$$

Hence,

$$V_{OC} = 0.5 \text{ V}$$

Now short-circuit terminals A and B.

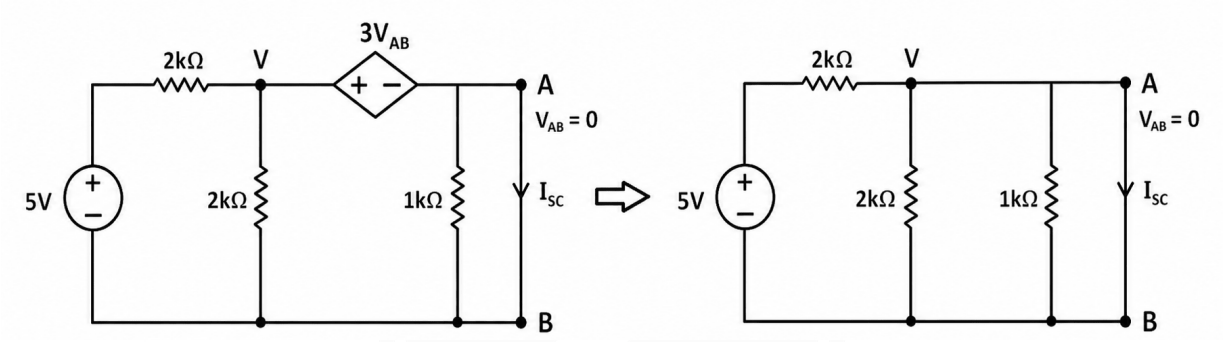

Fig. Solution Q3.09.1

Since A and B are shorted,

$$V_{AB} = 0$$

Therefore the dependent source voltage is zero and acts as a short circuit.

$$\Rightarrow I_{SC} = \frac{5}{2k} = \frac{2.5}{1000} = 2.5 \text{ mA}$$

Hence,

$$R_{TH} = \frac{V_{OC}}{I_{SC}} = \frac{0.5}{2.5 \times 10^{-3}} = 200\ \Omega = 0.2\ \text{k}\Omega$$

Therefore,

B. 0.2 kΩ

Q3.09.2 MCQ 2009 2M Ans: D ●●○

Taking result from Q3.09.1

$$V_{TH} = 0.5 \text{ V}$$

Hence, the correct answer is

D. 0.5 V

Q3.09.3 MCQ 2009 1M Ans: D ●○○

For a lamp rated P watts at voltage V, $R = \dfrac{V^2}{P}$

For a 100 W, 220 V lamp,

$$R_{100} = \frac{220^2}{100} = 484\ \Omega$$

Hence the power consumed at 220 V is

$$P_{100} = \frac{220^2}{484} = 100\ \text{W}$$

For a 200 W, 220 V lamp,

$$R_{200} = \frac{220^2}{200} = 242\ \Omega$$

Let n such lamps be connected in series. $\Rightarrow R_{eq} = 242n$

Power consumed by the series combination is

$$P = \frac{220^2}{242n} = \frac{200}{n}$$

For equal power consumption,

$$\frac{200}{n} = 100$$

$$n = 2$$

Hence, the correct answer is

D. 2

Q3.05.1 MCQ 2005 2M Ans: A ●○○

To find R_{TH}:

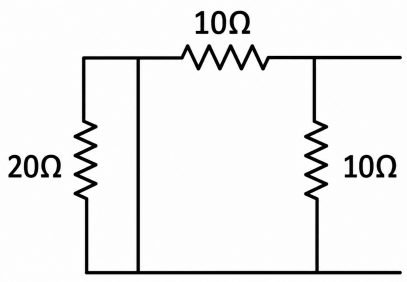

Fig. Solution Q3.05.1

The 20Ω resistor is bypassed by the short circuit.

Hence, the resistance seen from terminals P and Q is

$$R_{TH} = 10\Omega \parallel 10\Omega$$

$$R_{TH} = \frac{10 \times 10}{10 + 10} = 5\Omega$$

To find V_{TH}:

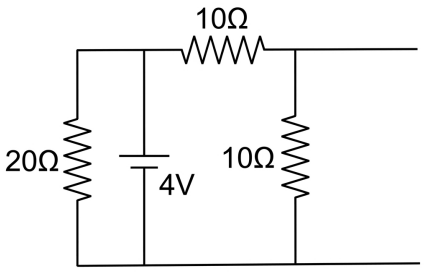

Fig. Solution Q3.05.1

Open-circuit the terminals P and Q.
The 4 V source is applied across two series 10Ω resistors.
Using voltage division,

$$V_{TH} = \frac{10}{10+10} \times 4$$

$$V_{TH} = 2 \text{ V}$$

Therefore,

$$(V_{TH}, R_{TH}) = (2 \text{ V}, 5\ \Omega)$$

Hence, the correct answer is

A. 2 V, 5 Ω

Q3.04.1 MCQ 2004 2M Ans: A

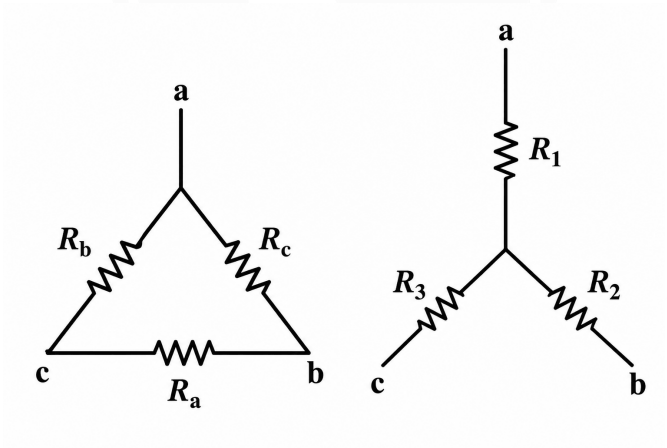

Fig. Solution Q3.04.1

For a delta-to-star transformation,

$$R_1 = \frac{R_b R_c}{R_a + R_b + R_c}, \quad R_2 = \frac{R_c R_a}{R_a + R_b + R_c}, \quad R_3 = \frac{R_a R_b}{R_a + R_b + R_c}$$

Given,

$$R_a = 20\Omega, \quad R_b = 10\Omega, \quad R_c = 10\Omega$$

Hence,

$$R_1 = \frac{10 \times 10}{20 + 10 + 10} = \frac{100}{40} = 2.5\Omega$$

$$R_2 = \frac{10 \times 20}{20 + 10 + 10} = \frac{200}{40} = 5\Omega$$

$$R_3 = \frac{20 \times 10}{20 + 10 + 10} = \frac{200}{40} = 5\Omega$$

Therefore,

$$\boxed{R_1 = 2.5\Omega, \quad R_2 = 5\Omega, \quad R_3 = 5\Omega}$$

Hence, the correct answer is

$$\boxed{\textbf{A. } 2.5,\ 5,\ 5}$$

Q3.03.1 MCQ | 2003 | 2M | Ans: C | ● ○ ○

Method 1 (Nodal Analysis)

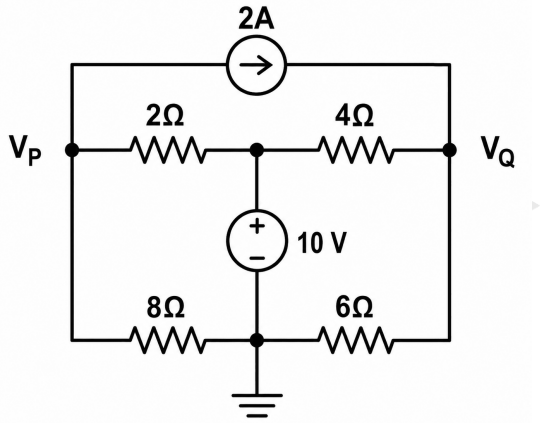

Fig. Solution Q3.03.1

The middle node is fixed at

$$V = 10 \text{ V}$$

due to the ideal voltage source.
Applying KCL at node P,

$$-2 = \frac{V_P}{8} + \frac{V_P - 10}{2}$$

$$-16 = V_P + 4(V_P - 10)$$

$$\boxed{V_P = 4.8 \text{ V}}$$

Applying KCL at node Q,

$$2 = \frac{V_Q}{6} + \frac{V_Q - 10}{4}$$

$$24 = 2V_Q + 3(V_Q - 10)$$

$$\boxed{V_Q = 10.8 \text{ V}}$$

Hence,

$$V_{PQ} = V_P - V_Q \Rightarrow V_{PQ} = 4.8 - 10.8$$

$$\boxed{V_{PQ} = -6 \text{ V}}$$

Method 2 (Superposition)

Case 1: 10 V source acting alone

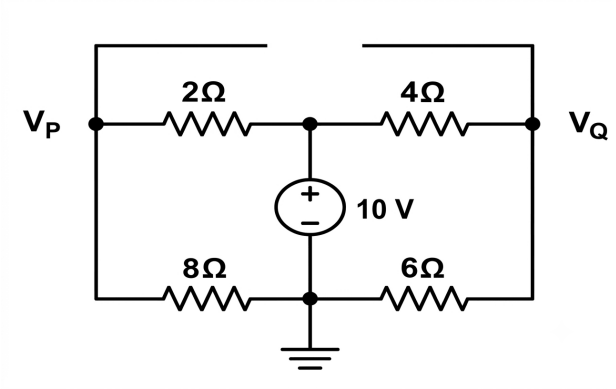

Fig. Solution Q3.03.1

At node P,

$$\frac{V_P}{8} + \frac{V_P - 10}{2} = 0$$

$$V_P = 8 \text{ V}$$

At node Q,

$$\frac{V_Q}{6} + \frac{V_Q - 10}{4} = 0$$

$$V_Q = 6 \text{ V}$$

Therefore,

$$V_{PQ}^{(10V)} = 8 - 6 = 2 \text{ V}$$

Case 2: 2 A source acting alone

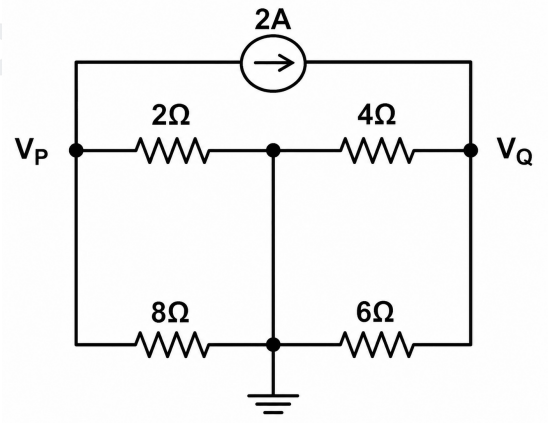

Fig. Solution Q3.03.1

Short-circuit the 10 V source.
Then the middle node becomes ground.
For node P,

$$-2 = \frac{V_P}{8} + \frac{V_P}{2}$$

$$V_P = -3.2 \text{ V}$$

For node Q,

$$2 = \frac{V_Q}{6} + \frac{V_Q}{4}$$

$$V_Q = 4.8 \text{ V}$$

Hence,

$$V_{PQ}^{(2A)} = -3.2 - 4.8 = -8 \text{ V}$$

Adding both contributions,

$$V_{PQ} = V_{PQ}^{(10V)} + V_{PQ}^{(2A)}$$

$$V_{PQ} = 2 + (-8) = -6 \text{ V}$$

Hence, the correct answer is

C. -6 V

Q3.02.1 MCQ 2002 5M Ans: *

Convert the delta network into equivalent star.

For a delta with all sides equal to 1Ω,

$$R_{\text{star}} = \frac{1 \times 1}{1+1+1} = \frac{1}{3}\Omega$$

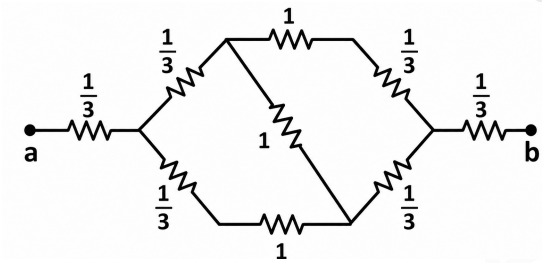

Fig. Solution Q3.02.1

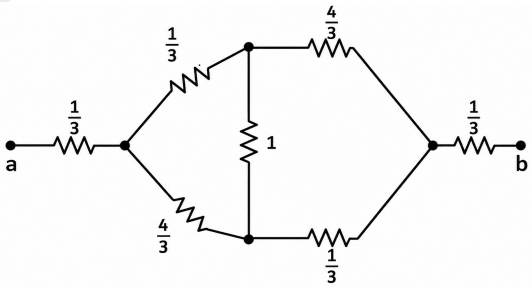

Fig. Solution Q3.02.1

Again Converting this delta into a star,

$$R_1 = \frac{\frac{1}{3}\cdot\frac{4}{3}}{\frac{1}{3}+\frac{4}{3}+1} = \frac{\frac{4}{9}}{\frac{8}{3}} = \frac{1}{6}\Omega, \qquad R_2 = \frac{1\cdot\frac{4}{3}}{\frac{1}{3}+\frac{4}{3}+1} = \frac{\frac{4}{3}}{\frac{8}{3}} = \frac{1}{2}\Omega$$

$$R_3 = \frac{1\cdot\frac{1}{3}}{\frac{1}{3}+\frac{4}{3}+1} = \frac{\frac{1}{3}}{\frac{8}{3}} = \frac{1}{8}\Omega$$

The network reduces to

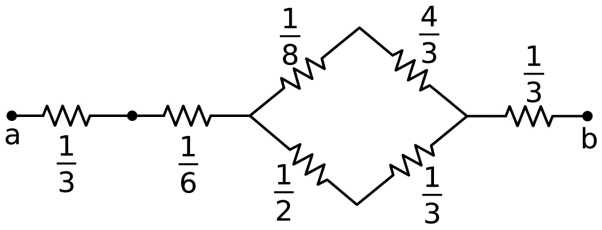

Fig. Solution Q3.02.1

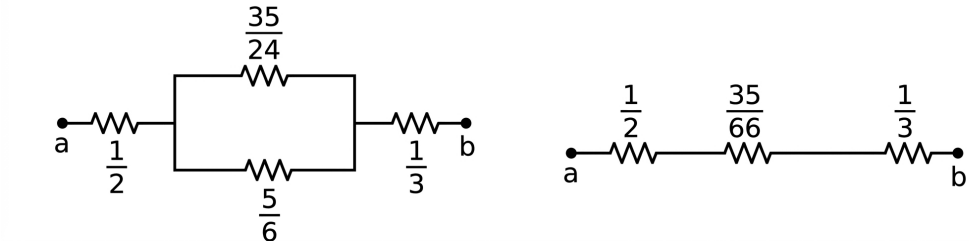

Fig. Solution Q3.02.1

Hence,

$$R_{ab} = \frac{1}{2} + \frac{35}{66} + \frac{1}{3} = \frac{15}{11}\Omega$$

Since $I = 1$ A,

$$V_{ab} = IR_{ab} = 1 \times \frac{15}{11}$$

$$\boxed{V_{ab} = \frac{15}{11}\text{ V}}$$

Mistakes to be avoided

- Do not apply the equipotential-node concept because the network is not symmetric.
  Although the resistor network appears symmetric, the excitation between terminals a and b is not symmetric with respect to the internal nodes under consideration. Hence those nodes are not guaranteed to be at the same potential and cannot be merged.

Q3.01.1 MCQ | 2001 | 1M | Ans: D | ● ○ ○

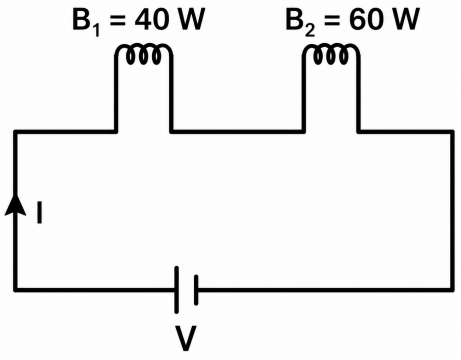

Fig. Solution Q3.01.1

For a bulb rated at voltage V and power P,

$$R = \frac{V^2}{P}$$

Hence,

$$R_{40} = \frac{V^2}{40}, \qquad R_{60} = \frac{V^2}{60}$$

Therefore,

$$R_{40} > R_{60}$$

Since the bulbs are connected in series, the same current flows through both bulbs.

Power dissipated in each bulb is

$$P = I^2R$$

Thus,

$$P_{40} = I^2R_{40}, \qquad P_{60} = I^2R_{60}$$

Since

$$R_{40} > R_{60} \Rightarrow P_{40} > P_{60}$$

Hence the 40 W bulb dissipates more power and glows brighter.
Hence, the correct answer is

D. the 40 W bulb glows brighter

Key Points

- The bulbs are no longer operating at their rated voltages, so their powers are not 40 W and 60 W. Therefore, the total power consumed is less than 100 W, making options A and B incorrect.

Q3.00.1 MCQ | 2000 | 2M | Ans: B

To find the equivalent resistance across the terminals, connect a test current source of value I A and let the terminal voltage be V.

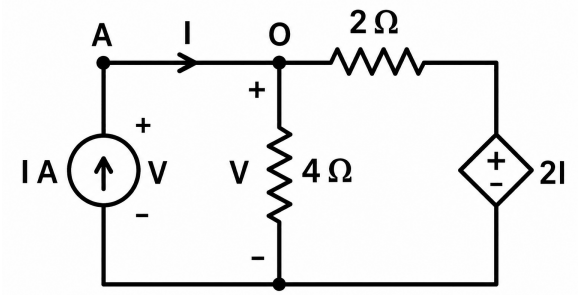

Fig. Solution Q3.00.1

Then,

$$R_{eq} = \frac{V}{I}$$

Applying KCL at O,

$$\frac{V}{4} + \frac{V - 2I}{2} = I$$

Multiplying throughout by 4,

$$V + 2(V - 2I) = 4I$$

$$3V - 4I = 4I$$

$$3V = 8I \Rightarrow V = \frac{8}{3}I$$

Hence,

$$R_{eq} = \frac{V}{I} = \frac{8}{3}\Omega$$

Hence, the correct answer is

B. $\frac{8}{3}\Omega$

Q3.99.1 MCQ 1999 1M Ans: C ● ○ ○

Since node R is directly connected to ground, the circuit can be redrawn as:

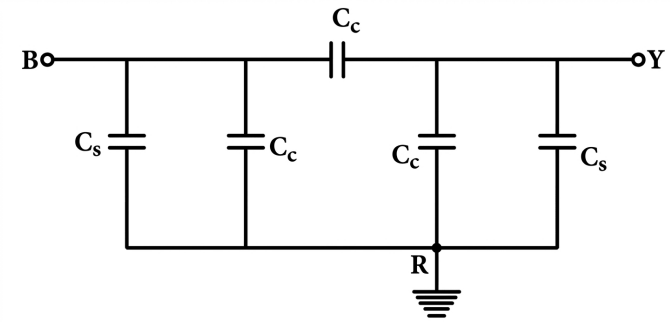

Fig. Solution Q3.99.1

The capacitors connected from node B to ground are in parallel, giving

$$C_B = C_s + C_c.$$

Similarly, the capacitors connected from node Y to ground are also in parallel, giving

$$C_Y = C_s + C_c.$$

The series equivalent through ground is

$$C_{\text{series}} = \frac{C_B C_Y}{C_B + C_Y} = \frac{(C_s + C_c)^2}{2(C_s + C_c)} = \frac{C_s + C_c}{2}.$$

Since this branch is in parallel with the capacitor C_c,

$$C_{BY} = C_c + C_{\text{series}} = C_c + \frac{C_s + C_c}{2}.$$

Therefore,

$$C_{BY} = \frac{2C_c + C_s + C_c}{2} = \frac{C_s + 3C_c}{2}.$$

Hence,

$$\boxed{\text{C. } \frac{C_s + 3C_c}{2}}.$$

Q3.98.1 MCQ 1998 2M Ans: B ● ○ ○

Thevenin resistance:

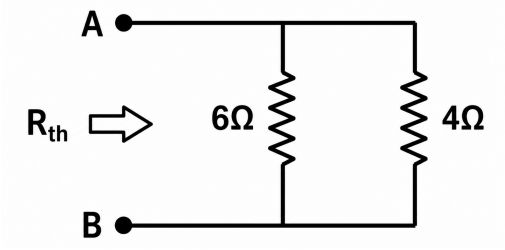

Fig. Solution Q3.98.1

$$R_{th} = 4 \parallel 6 = \frac{4 \times 6}{4 + 6} = 2.4\Omega$$

Thevenin voltage:

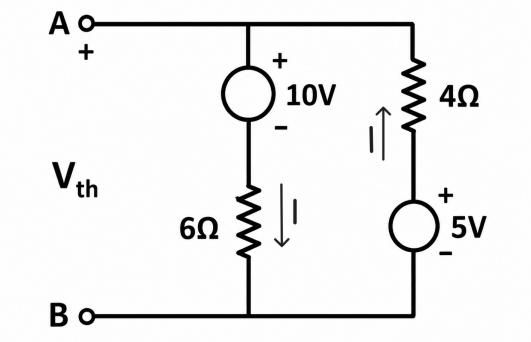

Fig. Solution Q3.98.1

Applying KVL,

$$5 - 4I - 10 - 6I = 0$$

$$10I = -5 \Rightarrow I = -0.5 \text{ A}$$

Now,

$$V_{th} = 10 + 6I$$

$$V_{th} = 10 + 6(-0.5) = 7 \text{ V}$$

Hence,

$$\boxed{V_{th} = 7 \text{ V}, \qquad R_{th} = 2.4\ \Omega}$$

The correct answer is

B. 7 volt source in series with 2.4 Ω resistor

Q3.97.1 Descriptive | 1997 | 5M | Ans: * | ●●○

To find V_{th}:

Open-circuit terminals A and B.

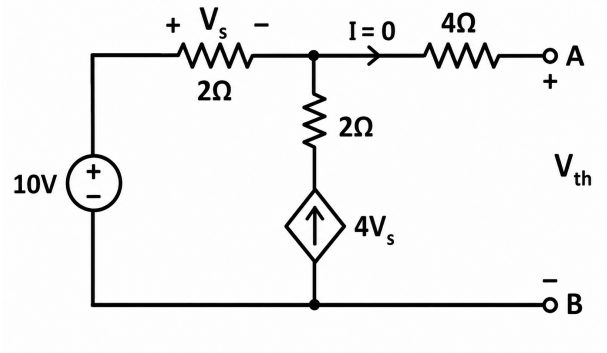

Fig. Solution Q3.97.1

Since the terminals are open-circuited, the current through the 4Ω resistor is zero.

Applying KVL,

$$-10 + V_s + V_{th} = 0 \Rightarrow V_{th} = 10 - V_s \qquad (1)$$

By KCL,

$$\frac{V_{th} - 10}{2} = 4V_s \Rightarrow V_{th} = 8V_s + 10 \quad (2)$$

From (1) and (2),

$$10 - V_s = 8V_s + 10 \Rightarrow V_s = 0$$

Hence,

$$\boxed{V_{th} = 10 \text{ V}}$$

To find R_{th}:

Deactivate the independent 10 V source and apply a 1 V test source across terminals A and B.

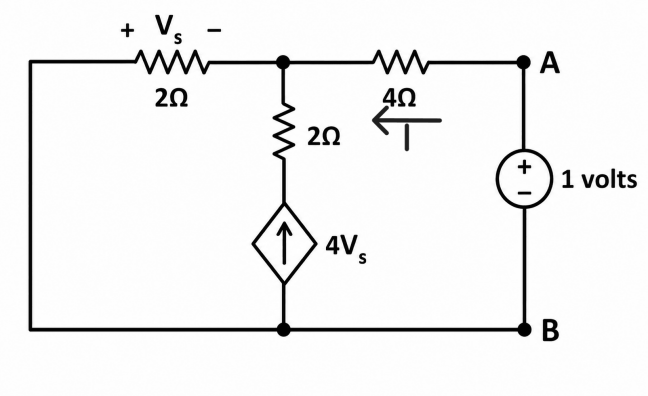

Fig. Solution Q3.97.1

Let the current supplied by the test source be I.

Applying KCL,

$$\frac{-V_s}{2} - 4V_s + \frac{-V_s - 1}{4} = 0$$

$$-2V_s - 16V_s - V_s - 1 = 0 \Rightarrow 19V_s = -1$$

$$V_s = -\frac{1}{19} \text{ V}$$

The test-source current is

$$I = \frac{1 - (-V_s)}{4} = \frac{1 - \frac{1}{19}}{4} = \frac{9}{38} \text{ A}$$

Therefore,

$$R_{th} = \frac{1}{I} = \frac{38}{9} = 4.22\ \Omega$$

$$\boxed{V_{th} = 10 \text{ V}, \qquad R_{th} = 4.22\ \Omega}$$

Q3.97.2 Descriptive | 1997 | 2M | Ans: 2, 4.5 | ●●○

To find Norton resistance R_N:

Deactivate the independent sources

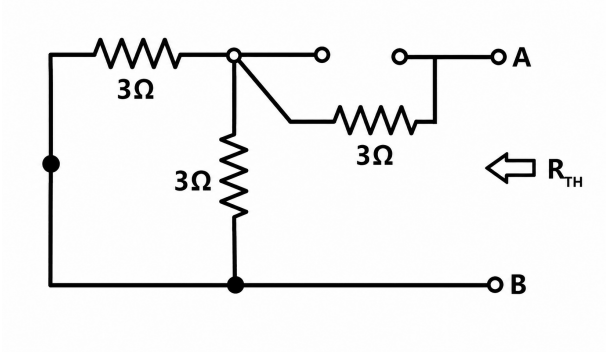

Fig. Solution Q3.97.2

The two 3Ω resistors connected between the middle node and terminal B are in parallel.

$$R_N = 3 + (3 \parallel 3) = 3 + \frac{3 \times 3}{3 + 3}$$

$$R_N = 3 + 1.5 = 4.5\Omega$$

To find Norton current I_N:
Short-circuit terminals A and B.
Transform the 2 A source in parallel with 3Ω into an equivalent voltage source:

$$V = IR = 2 \times 3 = 6 \text{ V}$$

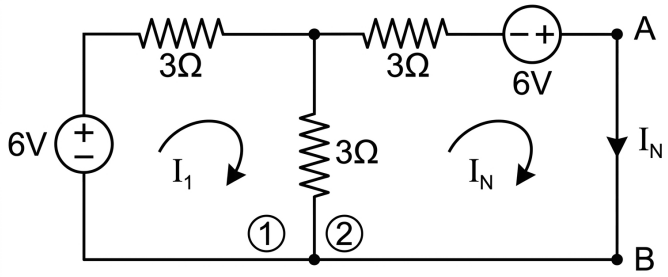

Fig. Solution Q3.97.2

Applying KVL in Loop 1,

$$-6 + 3I_1 + 3(I_1 - I_N) = 0 \Rightarrow 6I_1 - 3I_N = 6 \quad (1)$$

Applying KVL in Loop 2,

$$3(I_N - I_1) + 3I_N - 6 = 0$$

$$6I_N - 3I_1 = 6 \quad (2)$$

Solving (1) and (2),

$$I_1 = I_N = 2 \text{ A}$$

Therefore,

$$\boxed{I_N = 2 \text{ A}, \qquad R_N = 4.5\Omega}$$

Q3.95.1 NAT | 1995 | 5M | Ans: * | ● ● ●

To find the current through the 10Ω resistor connected between terminals a and b, first determine the Thevenin equivalent seen from terminals a and b.

Step 1: Find R_{th}

Deactivate all independent voltage sources by replacing them with short circuits.

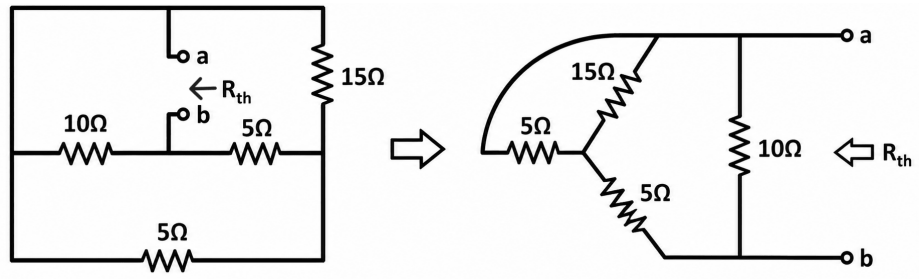

Fig. Solution Q3.95.1

The 15Ω and 5Ω resistors are in parallel:

$$15 \parallel 5 = \frac{15 \times 5}{15 + 5} = 3.75\Omega$$

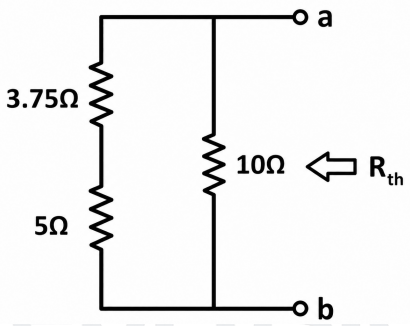

Fig. Solution Q3.95.1

This 3.75Ω is in series with the remaining 5Ω:

$$R = 3.75 + 5 = 8.75\Omega$$

Hence,

$$R_{th} = 10 \parallel 8.75 = \frac{10 \times 8.75}{10 + 8.75} = 4.67\Omega$$

Step 2: Find V_{th}

Open-circuit the 10Ω resistor between terminals a and b.

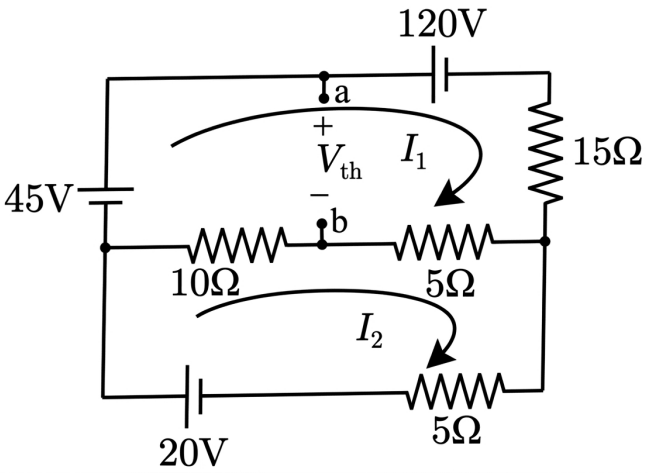

Fig. Solution Q3.95.1

Let clockwise mesh currents in the upper and lower loops be I_1 and I_2, respectively.
Applying KVL in the upper loop,

$$-45 + 120 + 15I_1 + (10 + 5)(I_1 - I_2) = 0$$

$$30I_1 - 15I_2 = -75 \quad (1)$$

Applying KVL in the lower loop,

$$(10 + 5)(I_2 - I_1) + 5I_2 - 20 = 0$$

$$-15I_1 + 20I_2 = 20 \quad (2)$$

Solving (1) and (2),

$$I_1 = -3.2 \text{ A}, \qquad I_2 = -1.4 \text{ A}$$

Now applying KVL between terminals a and b,

$$-45 + V_{th} + 10(I_1 - I_2) = 0$$

$$V_{th} = 63 \text{ V}$$

Step 3: Connect the load
The Thevenin equivalent is a 63 V source in series with $R_{th} = 4.67\Omega$, supplying the load $R = 10\Omega$.

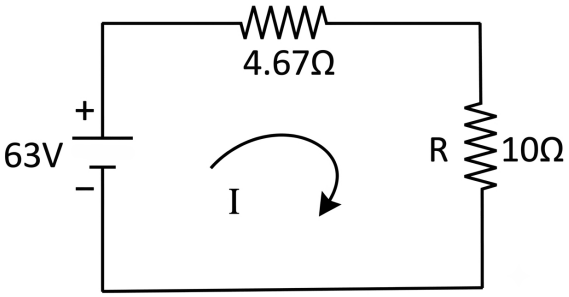

Fig. Solution Q3.95.1

$$I = \frac{V_{th}}{R_{th} + R} = \frac{63}{4.67 + 10} = 4.29 \text{ A}$$

$$\boxed{I = 4.29 \text{ A}}$$

Q3.94.1 MCQ | 1994 | 1M | Ans: B | ● ○ ○

The statement is **false**. Superposition theorem is applicable to all *linear* circuits. The elements may be time-invariant or time-varying; this does not violate superposition as long as the circuit remains linear. Superposition fails only when the circuit contains *nonlinear* elements such as diodes operating in their nonlinear region, transistors in nonlinear operation, saturating magnetic cores, etc. Therefore, the presence of time-varying resistors does not by itself invalidate superposition. Hence, the correct answer is

B. FALSE

Q3.92.1 MCQ | 1992 | 1M | Ans: D | ● ○ ○

All resistors are 1Ω.

Starting from the rightmost side,

$$1 \parallel 1 = \frac{1 \times 1}{1+1} = \frac{1}{2}\Omega$$

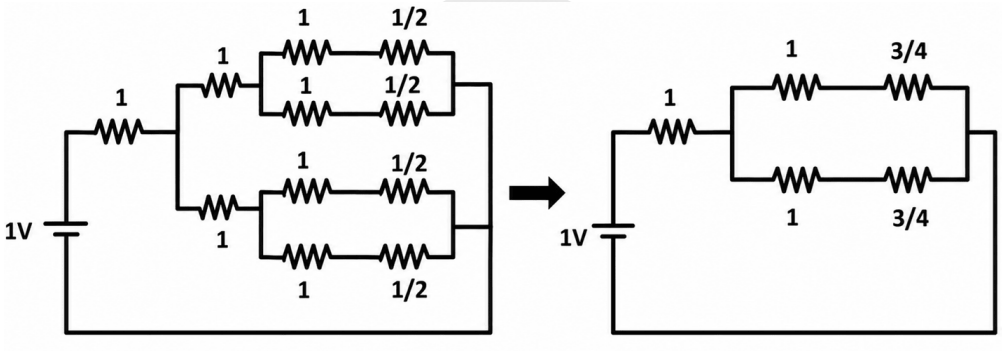

Fig. Solution Q3.92.1

The network reduces to;

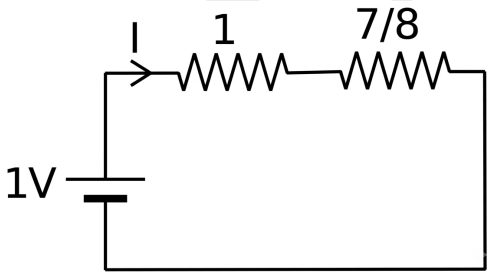

Fig. Solution Q3.92.1

Therefore the total equivalent resistance is

$$R_{eq} = 1 + \frac{7}{8} = \frac{15}{8}\Omega$$

The source voltage is 1 V, so

$$I = \frac{V}{R_{eq}} = \frac{1}{15/8} = \frac{8}{15}\text{ A}$$

Hence, the correct answer is

D. $\frac{8}{15}$ A

Q3.91.1 MCQ | 1991 | 1M | Ans: C

For a linear network, the V-I characteristic can be represented by its Thevenin equivalent.

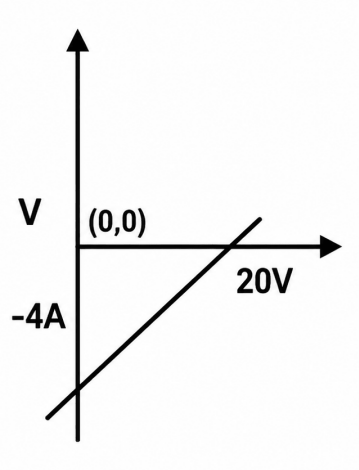

Fig. Solution Q3.91.1

From the graph,

$$V_{OC} = 20 \text{ V}, \qquad I_{SC} = 4 \text{ A}$$

Therefore,

$$R_{TH} = \frac{V_{OC}}{I_{SC}} = \frac{20}{4} = 5\Omega$$

and

$$V_{TH} = V_{OC} = 20 \text{ V}$$

So the Thevenin equivalent circuit is;

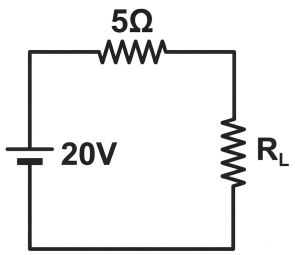

Fig. Solution Q3.91.1

For maximum power transfer,

$$R_L = R_{TH} = 5\Omega$$

Hence the load current is

$$I = \frac{V_{TH}}{R_{TH} + R_L} = \frac{20}{5+5} = 2 \text{ A}$$

Therefore,

$$P_{\max} = I^2 R_L = (2)^2(5) = 20 \text{ W}$$

Or,

$$P_{\max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{20^2}{4 \times 5} = 20 \text{ W}$$

Hence, the correct answer is

C. 20 W

PrepFusion

SOLUTIONS

IV

2nd Order Dynamics and Complex Impedance

Topics

1. Source-Free 2nd Order RLC Circuits
2. Initial Conditions and Sourced RLC Circuits
3. LC Oscillators
4. Equivalent Impedance of AC Circuits
5. AC transients

Unit Snapshot*

Stream: EE	**Questions:** 7	**MCQ:** 2	**MSQ:** 0	**NAT:** 5
1M: 1	**2M:** 6	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–4M(2018)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 NAT | 2026 | 1M | Ans: 32.3 to 32.5 | ● ● ○

Given data for a single-phase half-wave rectifier with an RL load and a freewheeling/circuit diode:

$$V_s = 325\sin(100\pi t)$$

$$L = 500\,\mu\text{H}$$

$$R = 10\,\Omega$$

The angular frequency of the source is:

$$\omega = 100\pi \text{ rad/s}$$

The inductive reactance is calculated as:

$$X_L = \omega L = 100\pi \times 500 \times 10^{-6} \approx 0.157\,\Omega$$

Since $X_L \ll R$, the impedance magnitude is approximated as:

$$|Z| = \sqrt{R^2 + (\omega L)^2} \approx R = 10\,\Omega$$

The impedance angle θ is given by:

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) = \tan^{-1}\left(\frac{100\pi \times 500 \times 10^{-6}}{10}\right) \approx 0.9^\circ$$

The expression for the transient current $i(t)$ through the diode during conduction is:

$$i(t) = ce^{-t/\tau} + \frac{V_m}{|Z|}\sin(\omega t - \theta)$$

where $V_m = 325$ V and $\tau = \frac{L}{R} = \frac{500\times10^{-6}}{10} = 50\,\mu$s. Since L is very small we take |Z|=R. Substituting the values:

$$i(t) = ce^{-t/50\,\mu\text{s}} + \frac{325}{10}\sin(\omega t - 0.9^\circ)$$

Using the initial condition at $t = 0$, $i(0) = 0$:

$$0 = c + 32.5\sin(-0.9^\circ)$$

$$c = -32.5\sin(-0.9^\circ) \approx 0.51\text{ A}$$

Thus, the complete current equation is:

$$i(t) = 0.51e^{-t/50\,\mu\text{s}} + 32.5\sin(\omega t - 0.9^\circ)$$

Since the transient component coefficient (0.51 A) is extremely small compared to the steady-state amplitude (32.5 A), the maximum current occurs very close to the peak of the sinusoidal component, i.e., when $\omega t - 0.9^\circ \approx 90^\circ$: Therefore, the maximum current through the diode is dominated by the peak steady-state current value:

$$i_{\max} \approx \frac{V_m}{R} = \frac{325}{10} = 32.5\text{ A}$$

Key Points

- The total response in an RL circuit connected to an AC source consists of a transient (exponential decay) and a steady-state (sinusoidal) component.
- When the load resistance R is significantly larger than the inductive reactance X_L, the circuit behaves almost purely

resistively, making the impedance phase angle $\theta \approx 0°$.

- The time constant $\tau = \frac{L}{R}$ determines how fast the transient component dies out.

Mistakes to be avoided

- Ensure correct unit of angle (degree or radian) when computing

32.5

Q4.23.1 NAT | 2023 | 2M | Ans: 1.57 | ● ● ○

Before $t = t_1$, switch K is open and switch $\overline{K}$ is closed. The circuit has reached steady state under the excitation of the AC current source and the DC voltage source.

The capacitive reactance is calculated as:

$$X_c = \frac{1}{\omega C} = \frac{1}{1000 \times 100 \times 10^{-6}} = 10\,\Omega$$

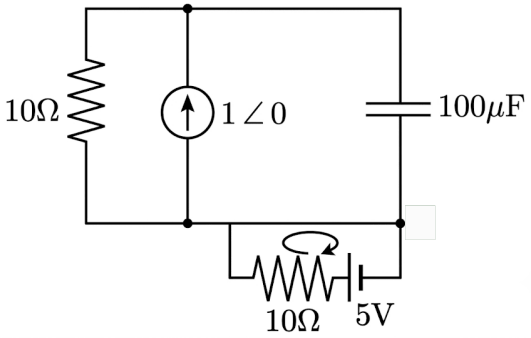

Fig. Solution Q4.23.1

In this state, the current from the 5 V DC source keeps circulating in its own local loop through $\overline{K}$ and does not affect the capacitor. Thus, the steady-state voltage V_c across the capacitor due to the AC source ($1\angle 0°$ A at $\omega = 1000$ rad/s) is:

$$V_c = 1\angle 0° \cdot [10 \parallel (-j10)] = 1\angle 0° \cdot \left(\frac{-j100}{10 - j10}\right) = \frac{10}{\sqrt{2}}\angle -45° \text{ V}$$

Converting this to the time-domain expression for the capacitor voltage gives:

$$v_c(t) = \frac{10}{\sqrt{2}} \sin(1000t - 45°)$$

At $t = t_1$, the initial voltage across the capacitor is:

$$v_c(t_1) = \frac{10}{\sqrt{2}} \sin(1000t_1 - 45°)$$

After $t = t_1$, switch K is closed (shorting the AC current source and the 10 Ω parallel resistor) and switch $\bar{K}$ is opened. This configures a series RC circuit connected to the 5 V DC source.

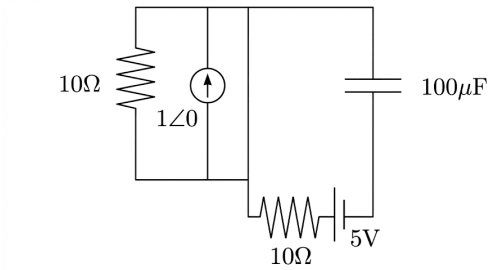

Fig. Solution Q4.23.1

The parameters for this new configuration are:

$$\text{Steady-state capacitor voltage, } V_{ss} = 5\,\text{V}$$

$$\text{Time constant, } \tau = RC = 10\,\Omega \times 100\,\mu\text{F} = 1000\,\mu\text{s} = 1\,\text{ms}$$

The complete voltage response across the capacitor for $t \geq t_1$ is given by:

$$v_c(t) = V_{ss} + [v_c(t_1) - V_{ss}]e^{-(t-t_1)/\tau}$$

$$v_c(t) = 5 + \left[\frac{10}{\sqrt{2}}\sin(1000t_1 - 45^\circ) - 5\right] e^{-(t-t_1)/\tau}$$

The second term in the expression represents the transient response. For a transient-free response, the coefficient of the exponential term must equal zero:

$$\frac{10}{\sqrt{2}}\sin(1000t_1 - 45^\circ) - 5 = 0$$

$$\sin\left(1000t_1 - \frac{\pi}{4}\right) = \frac{5}{\frac{10}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$$

Taking the appropriate principal angle to solve for the timing condition:

$$1000t_1 - \frac{\pi}{4} = \frac{\pi}{4}$$

$$1000t_1 = \frac{\pi}{2}$$

$$t_1 = \frac{\pi}{2000}\,\text{s} = \frac{\pi}{2}\,\text{ms} \approx 1.57\,\text{ms}$$

$$\boxed{1.57}$$

Q4.18.1 MCQ 2018 2M Ans: A ●●○

Based on the provided infinite ladder network diagram, we aim to find the input equivalent impedance Z looking into the circuit terminals. Since the ladder network is semi-infinite and repeats periodically, the remaining network down the line after the first stage maintains the exact same equivalent input impedance Z.

From the circuit topology, the total input impedance Z is equal to the series combination of an inductive reactance $j9\,\Omega$ and the parallel combination of another inductive reactance $j4\,\Omega$ with the repeating network impedance Z:

$$Z = j9 + (j4 \parallel Z)$$

Writing the parallel branch equation explicitly:

$$Z = j9 + \frac{Z(j4)}{Z + j4}$$

Finding a common denominator to simplify the algebraic expression:

$$Z = \frac{j9(Z + j4) + j4\,Z}{Z + j4}$$

$$Z(Z + j4) = j9\,Z + (j9)(j4) + j4\,Z$$

$$Z^2 + j4\,Z = j9\,Z - 36 + j4\,Z$$

Subtracting $j4\,Z$ from both sides:

$$Z^2 = j9\,Z - 36$$

$$Z^2 - j9\,Z + 36 = 0$$

To solve this quadratic equation, we can split the middle term into a combination whose sum is $-j9$ and whose product is 36:

$$Z^2 - (j12 + (-j3))Z + 36 = 0$$

$$Z^2 - j12\,Z + j3\,Z + 36 = 0$$

$$Z(Z - j12) + j3(Z - j12) = 0$$

$$(Z - j12)(Z + j3) = 0$$

This yields two possible mathematical roots:

$$Z = j12\,\Omega \quad \text{or} \quad Z = -j3\,\Omega$$

To determine the physically correct root, we can check the convergence by analyzing a finite truncation of the ladder:
1-stage approximation: $Z = j9 + j4 = j13\,\Omega$
2-stage approximation: $Z = j9 + (j4 \parallel j13) = j9 + j3.06 = j12.06\,\Omega$
As more stages are added, the value clearly converges toward $j12\,\Omega$.
Thus, the final equivalent input impedance of the network is:

$$Z = j12\,\Omega$$

$$\boxed{(A)\,j12\Omega}$$

PrepFusion

Q4.18.2 NAT | 2018 | 2M | Ans: 50.0 to 50.0 | ● ○ ○

When the inductor current is in phase with the voltage v(t), it implies that the driving point voltage and current are in phase with each other and the circuit is in resonance. Thus, we investigate the problem further based on the conditions of resonance.

First, let us find the capacitive reactance X_c of the capacitor:

$$X_c = \frac{1}{\omega C} = \frac{1}{100 \times 10^{-4}} = 100\,\Omega$$

The parallel combination impedance Z_p of the resistor $R = 100\,\Omega$ and the capacitive reactance $-jX_c = -j100\,\Omega$ is calculated as:

$$Z_p = \frac{100(-j100)}{100 - j100}$$

$$Z_p = \frac{-j10000(100 + j100)}{100^2 + 100^2} = \frac{1000000 - j1000000}{20000} = (50 - j50)\,\Omega$$

The total input impedance Z_{in} of the network, including the series inductor, is expressed as:

$$Z_{in} = j\omega L + Z_p$$

$$Z_{in} = j\omega L + 50 - j50$$

$$Z_{in} = 50 + j(\omega L - 50)$$

Resonance implies that the total input impedance is purely real, and its imaginary part must equal zero:

$$\text{Im}(Z_{in}) = 0$$

$$\omega L - 50 = 0$$

Substituting $\omega L = 50$ back into the total impedance equation yields:

$$Z_{in} = 50\,\Omega$$

$$\boxed{50}$$

Q4.16.1 NAT | 2016 | 2M | Ans: 1.0 to 1.0 | ● ○ ○

The given AC network features an operating frequency of $\omega = 1000\,\text{rad/s}$ controlled by the source voltage $v_s(t) = 10\sin(1000t)$ V.

The network components in first LC parallel branch are a capacitor $C_A = 250\,\mu\text{F}$ and an inductor $L_A = 4\,\text{mH}$. The capacitive reactance magnitude is:

$$X_{CA} = \frac{1}{\omega C_A} = \frac{1}{1000 \times 250 \times 10^{-6}} = 4\,\Omega$$

The complex capacitive impedance is $Z_{CA} = -j4\,\Omega$.

The inductive reactance magnitude is:

$$X_{LA} = \omega L_A = 1000 \times 4 \times 10^{-3} = 4\,\Omega$$

The complex inductive impedance is $Z_{LA} = j4\,\Omega$.

The total equivalent impedance of the parallel tank circuit branch A is calculated as:

$$Z_A = \frac{Z_{LA} \times Z_{CA}}{Z_{LA} + Z_{CA}} = \frac{j4 \times (-j4)}{j4 - j4} = \frac{16}{0} = \infty\,\Omega$$

Since the total parallel impedance Z_A is infinite, branch A acts as an ideal open circuit at this operating frequency.

The network components in second parallel LC branch are a capacitor $C_B = 2\,\mu\text{F}$ and an inductor $L_B = 500\,\text{mH}$. The capacitive reactance magnitude is:

$$X_{CB} = \frac{1}{\omega C_B} = \frac{1}{1000 \times 2 \times 10^{-6}} = 500\,\Omega$$

The complex capacitive impedance is $Z_{CB} = -j500\,\Omega$.

The inductive reactance magnitude is:

$$X_{LB} = \omega L_B = 1000 \times 500 \times 10^{-3} = 500\,\Omega$$

The complex inductive impedance is $Z_{LB} = j500\,\Omega$.

The total equivalent impedance of the parallel tank circuit branch B is given by:

$$Z_B = \frac{Z_{LB} \times Z_{CB}}{Z_{LB} + Z_{CB}} = \frac{j500 \times (-j500)}{j500 - j500} = \frac{250000}{0} = \infty\,\Omega$$

Consequently, branch B also acts as an ideal open circuit at this operating frequency.

By substituting the open circuits for the two LC parallel branches into the main configuration, the active path reduces to a single-loop closed circuit containing only the independent voltage source and the two series resistors. The total series resistance seen by the voltage source is:

$$R_{total} = 4\,\Omega + 5\,\Omega + 1\,\Omega = 10\,\Omega$$

The instantaneous loop current tracking through the network is derived via Ohm's law:

$$i(t) = \frac{10\sin(1000t)}{R_{total}} = \frac{10\sin(1000t)}{10} = \sin(1000t)\,\text{A}$$

Hence, the peak magnitude of the steady-state current flowing through the 1 Ω resistor is:

$$I_{max} = 1\,\text{A}$$

[1]

Q4.15.1 NAT | 2015 | 2M | Ans: 10 | ● ○ ○

The total root-mean-square (RMS) response of a multi-frequency network containing independent linear AC and DC excitations can be calculated using the principle of superposition.

Case 1: Under the action of the 8 V DC source acting alone For a pure DC excitation, the steady-state operating frequency is $\omega_1 = 0$ rad/s. At this zero frequency condition, the reactive response of any network inductance transitions to zero:

$$X_{L1} = \omega_1 L = 0 \times 1 = 0\,\Omega$$

Hence, the inductor acts as an ideal short circuit. Assuming a basic loop path configuration with a series resistance of 1 Ω, the resulting DC current component is given by:

$$I_{dc} = \frac{8}{1} = 8\,\text{A}$$

Case 2: Under the action of the $12\sin(t)$ V sinusoidal source acting alone The operating frequency for this independent excitation is $\omega_2 = 1$ rad/s. The inductive reactance at this specific frequency is calculated as:

$$X_{L2} = \omega_2 L = 1 \times 1 = 1\,\Omega$$

The complex impedance of the series RL branch layout is:

$$Z = R + jX_{L2} = 1 + j1\,\Omega$$

Expressing the total impedance in polar form yields:

$$Z = \sqrt{1^2 + 1^2}\angle\tan^{-1}\left(\frac{1}{1}\right) = \sqrt{2}\angle 45^\circ\,\Omega$$

The corresponding AC phasor current component is evaluated using Ohm's law:

$$I_{ac} = \frac{12\angle 0^\circ}{1 + j1} = \frac{12\angle 0^\circ}{\sqrt{2}\angle 45^\circ} = \frac{12}{\sqrt{2}}\angle -45^\circ\,\text{A}$$

Converting the phasor response back into its time-domain equivalent form:

$$i_{ac}(t) = \frac{12}{\sqrt{2}}\sin(t - 45^\circ)\,\text{A}$$

The RMS value of this specific sinusoidal current component is:

$$I_{rms,ac} = \frac{I_{max}}{\sqrt{2}} = \frac{12/\sqrt{2}}{\sqrt{2}} = \frac{12}{2} = 6\,\text{A}$$

Total Effective Current Evaluation: Since the components originate from independent sources operating at completely distinct frequencies (0 rad/s and 1 rad/s), their individual effective root-mean-square values are combined orthogonally:

$$I_{rms} = \sqrt{I_{dc}^2 + I_{rms,ac}^2}$$

$$I_{rms} = \sqrt{8^2 + \left(\frac{12/\sqrt{2}}{\sqrt{2}}\right)^2} = \sqrt{64 + 6^2} = \sqrt{64 + 36} = \sqrt{100} = 10\,\text{A}$$

$$\boxed{10}$$

Key Points

- **DC Inductor Behavior:** Under stable DC excitation conditions ($\omega = 0$), the inductive reactance drops to zero ($X_L = 0$), reducing the component model down to a basic short circuit path.
- **Multi-Frequency RMS Combination:** Total effective current for distinct orthogonal harmonic/DC frequencies cannot be algebraically summed up. The calculation demands a root-sum-square calculation profile: $I_{rms} = \sqrt{I_0^2 + I_{rms1}^2 + I_{rms2}^2 + \ldots}$.

Mistakes to be avoided

- Remember to divide the maximum peak amplitude of the sinusoidal component ($I_{max} = 12/\sqrt{2}$ A) by $\sqrt{2}$ to convert it accurately into its standalone I_{rms} value before squaring it inside the final radical expression.

Q4.02.1 MCQ 2002 2M Ans: B ●○○

When the switch is closed at $t = t_0$, a certain voltage equal to $\sin(\omega t_0)$ appears across the RL circuit at once, giving rise to transients.The transients eventually die out as the response reaches sinusoidal steady state.

The expression for circuit current would be

$$i(t) = ce^{-t/\tau} + \frac{V_m}{|Z|}sin(\omega t - \theta)$$

where $V_m = 1$ and |Z| is the magnitude of circuit impedance. To determine the constant c, we apply the initial condition at the switching instant $t = t_{sub|0}$. Assuming the circuit is initially unenergized, the initial current at this exact moment is zero:

$$i(t_0) = 0$$

Substituting $t = t_0$ into the current equation yields:

$$0 = ce^{-t_0/\tau} + \frac{V_m}{|Z|}\sin(\omega t_0 - \theta)$$

By rearranging this equation, the arbitrary coefficient c is explicitly determined by the initial switching conditions:

$$c = -\frac{V_m}{|Z|}\sin(\omega t_0 - \theta)e^{t_0/\tau}$$

To achieve a completely transient-free operation, we must entirely eliminate the decaying exponential transient term from the current expression. This requires the coefficient to vanish completely ($c = 0$).Therefore, the only way to satisfy $c = 0$ is for the sinusoidal component to be zero:

$$\sin(\omega t_0 - \theta) = 0$$

Hence, the condition for transient free switching requires:

$$\omega t_0 = \theta$$

Where θ is the power factor impedance angle:

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

Given parameters from the problem: Frequency, $f = 50\,\text{Hz}$ Angular frequency, $\omega = 2\pi \times 50\,\text{rad/s}$ Inductance, $L = 0.01\,\text{H}$ Resistance, $R = 5\,\Omega$

Substituting these values into the condition equation:

$$2\pi \times 50 t_0 = \tan^{-1}\left(\frac{2\pi \times 50 \times 0.01}{5}\right)$$

Simplifying the term inside the inverse tangent function:

$$2\pi \times 50 \times t_0 = \tan^{-1}\left(\frac{\pi}{5}\right)$$

$$100\pi \times t_0 = 0.561\,\text{rad}$$

Solving for the switching time t_0:

$$t_0 = \frac{0.561}{100\pi}$$

$$t_0 = 1.786 \times 10^{-3}\,\text{s}$$

$$t_0 = 1.78\,\text{ms}$$

$$\boxed{(B) 1.78\ ms}$$

PrepFusion

SOLUTIONS

V

AC Circuit Analysis

Topics

1. Basics of Phasor Diagrams
2. Phasor Diagram for Series RL, RC and RLC Circuits
3. Phasor Diagram for Parallel RL, RC and RLC Circuits
4. Locus Diagram of Current, Admittance and Impedance
5. Application of Network Theorems in AC Circuits
6. Maximum Power Transfer in AC Circuits

Unit Snapshot*

Stream: EE	**Questions:** 40	**MCQ:** 28	**MSQ:** 0	**NAT:** 11
1M: 17	**2M:** 20	**Descriptive:** 3	**Avg Weightage** ~2M	**Range:** 1M(2020)–6M(2011)

**See preface for how Avg Weightage and Range are calculated.*

Q5.25.1 NAT | 2025 | 2M | Ans: 78.0 to 81.0 | ● ● ○

Since the voltmeters are ideal, i.e., they have ∞ resistance, and have no current flowing through them, the source, 2Ω resistance and RL load are in series (have the same current flowing through them). Taking the current I as reference, we can draw the phasor diagram as:

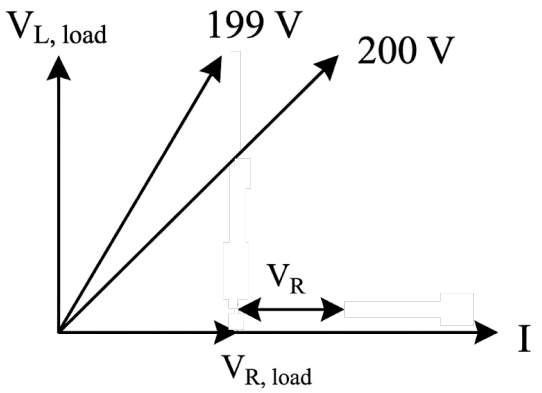

Fig. Solution Q5.25.1

Given parameters : * Voltage across the series resistor: $V_R = 2I = 9\,\text{V}$ Load voltage: $V_{\text{load}} = 199\,\text{V}$ Source voltage: $V_s = 200\,\text{V}$ From the resistor voltage we can calculate current I:

$$I = \frac{9}{2} = 4.5\,\text{A}$$

The voltage across the load is given by:

$$V_{\text{load}} = I\sqrt{R^2 + X_L^2} = 199\,\text{V}$$

$$R^2 + X_L^2 = \frac{199^2}{4.5^2}$$

The total circuit voltage expression considering the series resistance of 2 Ω is:

$$(R+2)^2 + X_L^2 = \frac{200^2}{4.5^2}$$

Expanding and subtracting the two equations:

$$\left(R^2 + 4R + 4 + X_L^2\right) - \left(R^2 + X_L^2\right) = \frac{200^2 - 199^2}{4.5^2}$$

$$4R + 4 = \frac{200^2 - 199^2}{4.5^2}$$

$$4R + 4 = \frac{399}{20.25} \approx 19.7037$$

$$4R = 15.7037 \implies R \approx 3.926\,\Omega$$

The active power consumed by the load is:

$$P_{\text{load}} = I^2R = (4.5)^2 \times 3.926 = 20.25 \times 3.926 = 79.5\,\text{W}$$

78.0 *to* 81.0

Key Points

- The total voltage in an R-L series circuit is calculated using the phasor sum of the in-phase (resistive) and quadrature (inductive) components.
- Active load power is solely dissipated by the resistive component of the load, defined by $P = I^2R$.
- To draw the phasor diagram for a series circuit, current is taken as reference while for a parallel circuit, voltage is taken as reference.

Q5.23.1 NAT | 2023 | 2M | Ans: -11.98 | ●●○

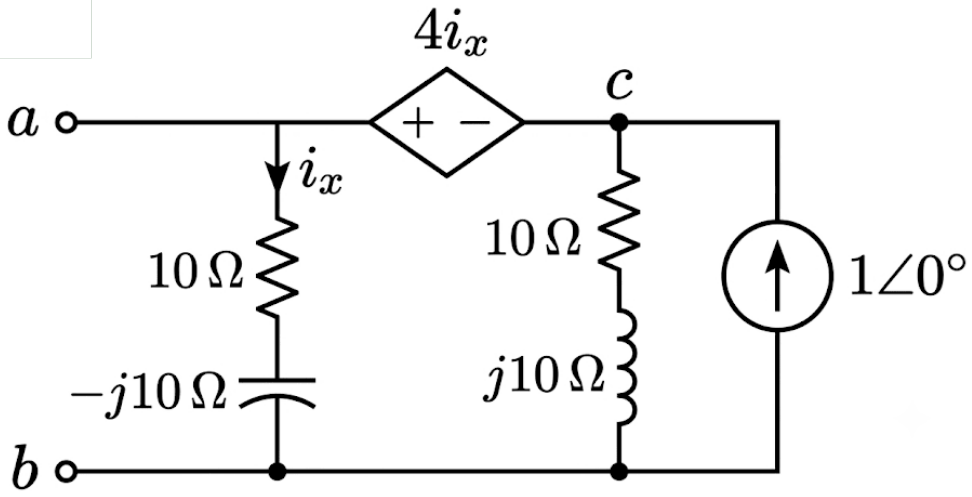

Fig. Solution Q5.23.1

Let node b be selected as the reference ground node ($V_b = 0\,\text{V}$). Thus, the node voltages can be defined relative to node b as V_{ab} and V_{cb}.

From the circuit diagram, the relationship across the dependent voltage source is:

$$V_{ab} = V_{cb} + 4\,i_x$$

The current i_x flowing through the left branch is given by:

$$i_x = \frac{V_{ab}}{10 - j10}$$

Substituting V_{ab} into the equation for i_x:

$$i_x(10 - j10) = V_{cb} + 4\,i_x$$

$$i_x(6 - j10) = V_{cb} \implies i_x = \frac{V_{cb}}{6 - j10}$$

Writing the nodal analysis equation at node c (supernode structure combined with the independent current source entering the node):

$$-1\angle 0^\circ + \frac{V_{cb}}{10 + j10} + \frac{V_{ab}}{10 - j10} = 0$$

Substituting $V_{ab} = V_{cb} + 4\,i_x$ into the nodal equation:

$$\frac{V_{cb}}{10 + j10} + \frac{V_{cb} + 4\,i_x}{10 - j10} = 1$$

$$V_{cb}\left(\frac{1}{10+j10}+\frac{1}{10-j10}\right)+\frac{4}{10-j10}\left(\frac{V_{cb}}{6-j10}\right)=1$$

$$V_{cb}\left(\frac{1}{10+j10}+\frac{1}{10-j10}+\frac{4}{(10-j10)(6-j10)}\right)=1$$

Solving this complex algebraic equation yields:

$$V_{cb}=10-j2.5\,\text{V}$$

Using the relationship found for i_x:

$$i_x=\frac{V_{cb}}{6-j10}=0.625+j0.625\,\text{A}$$

Now, calculating the terminal voltage V_{ab}:

$$V_{ab}=V_{cb}+4\,i_x=12.5\angle 0^\circ\,\text{V}=12.5\sin(\omega t)\,\text{V}$$

Evaluating the instantaneous voltage at $t=5\,\text{ms}$ as shown in the derivation:

$$V_{ab}=-11.98\,\text{V}$$

$$\boxed{-11.98}$$

Mistakes to be avoided

- Do not treat the dependent voltage source as an independent node; its constraint equation ($V_{ab}-V_{cb}=4\,i_x$) must be explicitly maintained.

Q5.21.1 NAT 2021 1M Ans: 20.00 to 20.00 ● ○ ○

The total network current I becomes zero if the parallel network impedance blocking the path approaches infinity (∞), which physically represents an open-circuit condition at that frequency.

Let the capacitive reactance of the capacitor be denoted as X_c. The network consists of a parallel combination of an inductive impedance component $j5$ and a series branch containing another inductive component $j5$ and the capacitive reactance $-jX_c$.

The equivalent impedance Z_{eq} of this parallel combination is given by:

$$Z_{\text{eq}}=j5\parallel(j5-jX_c)=\frac{j5\cdot(j5-jX_c)}{j5+(j5-jX_c)}$$

$$Z_{\text{eq}}=\frac{j5\cdot(j5-jX_c)}{j10-jX_c}$$

For the current I to be zero, this equivalent parallel impedance configuration must become infinitely large ($Z_{\text{eq}}\to\infty$). This condition is met when the denominator of the impedance transfer equation equals zero:

$$j10-jX_c=0$$

$$X_c=10\,\Omega$$

The expression for capacitive reactance is:

$$X_c=\frac{1}{\omega C}=10\,\Omega$$

Given that the operating frequency parameter from the system context is $\omega=5000\,\text{rad/s}$:

$$\frac{1}{5000\times C}=10$$

$$C=\frac{1}{50000}=20\times10^{-6}\,\text{F}=20\,\mu\text{F}$$

$$\boxed{20}$$

Q5.21.2 NAT | 2021 | 1M | Ans: 2.30 to 2.50 | ● ○ ○

Based on the provided low-pass RC filter network, we aim to find the required capacitance value C to satisfy the input-output voltage conditions at the operating frequency. The circuit looks like the one shown below.

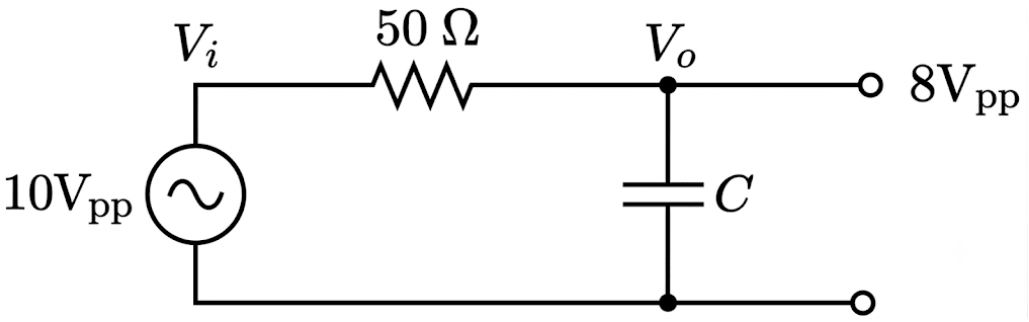

Fig. Solution Q5.21.2

Given parameters from the circuit schematic: Input voltage magnitude: $V_i = 10\,\text{V}_{\text{pp}}$ Output voltage magnitude: $V_o = 8\,\text{V}_{\text{pp}}$ Series resistance: $R = 50\,\Omega$ Signal frequency parameter: $\omega = 2\pi \times 1000\,\text{rad/s}$

The transfer function $H(s)$ of the first-order low-pass filter system is given by:

$$H(s) = \frac{V_o}{V_i}(s) = \frac{1}{sCR + 1}$$

Substituting $s = j\omega$, the frequency response relation is:

$$H(\omega) = \frac{1}{j\omega CR + 1}$$

The steady-state output magnitude of an LTI system is equal to the magnitude of the input multiplied by the magnitude of the frequency response at the operating frequency:

$$|V_o(\omega)| = |V_i(\omega)| \cdot |H(\omega)|$$

Given that $|V_o(\omega)| = 8\,\text{V}$ and $|V_i(\omega)| = 10\,\text{V}$:

$$|H(\omega)| = \frac{8}{10} = 0.8$$

Expressing the magnitude of the transfer function explicitly:

$$\frac{1}{\sqrt{1 + \omega^2 C^2 R^2}} = 0.8$$

Squaring both sides and solving for capacitance C:

$$1 + \omega^2 C^2 R^2 = \frac{1}{0.8^2}$$

$$\omega^2 C^2 R^2 = \frac{1}{0.8^2} - 1$$

$$C = \sqrt{\frac{\frac{1}{0.8^2} - 1}{\omega^2 R^2}}$$

Substituting the numerical values $\omega = 2\pi \times 1000\,\text{rad/s}$ and $R = 50\,\Omega$:

$$C = \sqrt{\frac{1.5625 - 1}{(2\pi \times 1000)^2 \times 50^2}}$$

$$C = \frac{\sqrt{0.5625}}{(2\pi \times 1000) \times 50} = \frac{0.75}{100000\pi} \approx 2.38 \times 10^{-6}\,\text{F} = 2.38\,\mu\text{F}$$

$$\boxed{2.30\ to\ 2.50}$$

Q5.21.3 NAT | 2021 | 1M | Ans: 1.40 to 1.42 | ●○○

According to the Maximum Power Transfer Theorem (MPT) for a purely resistive load connected to an AC network, the optimal load resistance R_L is equal to the magnitude of the Thevenin equivalent impedance Z_{th} looking back into the network from the load terminals:

$$R_L = |Z_{th}|$$

First, let us calculate the individual reactances of the passive components given the operating parameters from the system network ($\omega = 1$ krad/s $= 1000$ rad/s):

1. For the inductive component ($L = 4$ mH):

$$X_L = j\omega L = j(1000)(4 \times 10^{-3}) = j4\,\Omega$$

2. For the capacitive component ($C = 0.5$ mF):

$$X_C = \frac{-j}{\omega C} = \frac{-j}{(1000)(0.5 \times 10^{-3})} = -j2\,\Omega$$

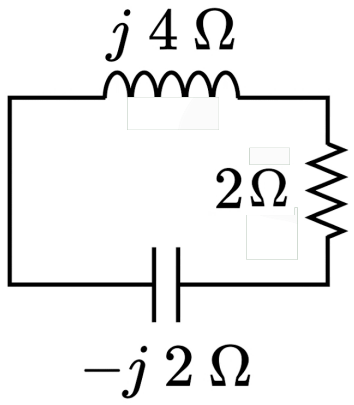

Fig. Solution Q5.21.3

Looking into the open terminals to find the internal impedance Z_{th}, the series combination of the reactive elements forms a branch in parallel with the $2\,\Omega$ resistor. The net series reactance of the loop connected across the resistor is:

$$X_{net} = X_L + X_C = j4 - j2 = j2\,\Omega$$

Thus, the Thevenin impedance Z_{th} is the parallel combination of the $2\,\Omega$ resistor and the net $j2\,\Omega$ reactive impedance branch:

$$Z_{th} = j2 \parallel 2 = \frac{2 \times j2}{2 + j2}$$

$$Z_{th} = \frac{j4}{2 + j2} = \frac{j2}{1 + j1} = \frac{j2(1 - j1)}{2} = 1 + j1\,\Omega$$

To find the optimal load resistance R_L, we take the absolute magnitude of Z_{th}:

$$|Z_{th}| = \sqrt{1^2 + 1^2} = \sqrt{2} \approx 1.414\,\Omega$$

Therefore, the value of the load resistance for maximum power transfer is:

$$R_L = 1.41\,\Omega$$

Mistakes to be avoided

- Do not equate R_L simply to the real part R_{th} of the Thevenin impedance (1 Ω) when the network contains uncompensated reactive components.

$$\boxed{1.40 \textit{ to } 1.42}$$

Q5.20.1 NAT | 2020 | 1M | Ans: 1.700 to 1.750 | ● ○ ○

The reading of ammeter A1 would be the magnitude of the sum of currents flowing through ammeters A2 and A3.

It is given that currents through ammeters A2 and A3 are $1\angle 10°$ and $1\angle 70°$ A.

$$\therefore I_{A1} = 1\angle 10° + 1\angle 70° = \sqrt{3}\angle 40° A$$

Reading of ammeter A1 would be $\sqrt{3}A = 1.732A$.

$$\boxed{1.700 \textit{ to } 1.750}$$

Q5.18.1 MCQ | 2018 | 2M | Ans: B | ● ● ○

The circuit can be drawn in Laplace domain as :

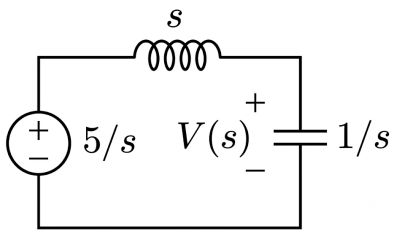

Fig. Solution Q5.18.1

The total loop impedance in the s-domain is:

$$Z(s) = sL + \frac{1}{sC}$$

Given the component values in the s-domain from the schematic ($L = 1$ H, $C = 1$ F):

$$Z(s) = s + \frac{1}{s}$$

The loop current $I(s)$ is given by:

$$I(s) = \frac{V_{in}(s)}{Z(s)} = \frac{\frac{5}{s}}{s + \frac{1}{s}} = \frac{5}{s^2 + 1}$$

Taking the inverse Laplace transform to find the time-domain current $i(t)$:

$$i(t) = 5\sin(t)$$

Next, the capacitor voltage $V(s)$ in the s-domain is given by:

$$V(s) = I(s) \cdot \frac{1}{s} = \frac{5}{s(s^2 + 1)}$$

Using partial fraction expansion:

$$\frac{5}{s(s^2+1)} = \frac{A}{s} + \frac{Bs+C}{s^2+1}$$

$$5 = A(s^2+1) + (Bs+C)s$$

$$5 = (A+B)s^2 + Cs + A$$

Equating the coefficients on both sides:

$$A = 5$$

$$C = 0$$

$$A + B = 0 \implies B = -5$$

Substituting the coefficients back into the $V(s)$ expression:

$$V(s) = \frac{5}{s} - \frac{5s}{s^2+1}$$

Taking the inverse Laplace transform to determine the time-domain capacitor voltage $v(t)$:

$$v(t) = 5 - 5\cos(t)$$

To find the state-space locus or mathematical path of the system, we square and add the state variable expressions. First, rearrange the voltage expression:

$$v(t) - 5 = -5\cos(t)$$

Squaring both sides of $i(t)$ and the modified $v(t)$ equations:

$$(i(t))^2 = 25\sin^2(t)$$

$$(v(t)-5)^2 = 25\cos^2(t)$$

Adding the two squared equations together:

$$(v(t)-5)^2 + (i(t))^2 = 25(\sin^2(t) + \cos^2(t))$$

$$(v(t)-5)^2 + (i(t))^2 = 5^2$$

This is the standard equation of a circle in the state-space plane with axes $v(t)$ and $i(t)$, which matches the form:

$$(x - x_o)^2 + (y - y_o)^2 = R^2$$

Therefore, the coordinates of the center and the radius of the circle are:

$$\text{Center} = (5, 0), \text{Radius} = 5 \text{ units}$$

$\boxed{(b)}$

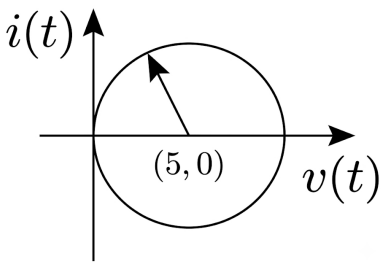

Fig. Solution Q5.18.1

Q5.17.2 MCQ | 2017 | 2M | Ans: D

According to the Maximum Power Transfer Theorem for AC circuits, maximum power is transferred from the source to the load when the load impedance Z_l is the complex conjugate of the source internal impedance Z_s:

$$Z_l = Z_s^*$$

Given that the resistive part of the source impedance is $R_s = 0.5\,\Omega$, the real part of the load impedance must satisfy:

$$\mathrm{Re}(Z_l) = R_s = 0.5\,\Omega$$

The system operating angular frequency is given as:

$$\omega = 100\,\text{rad/s}$$

The inductive reactance X_l is given by:

$$X_l = j\omega L = j100 \times 5 \times 10^{-3} = j0.5\,\Omega$$

Let the capacitive reactance magnitude be X_c, so the complex impedance of the capacitor is $-jX_c$. The load network consists of an inductor in series with a parallel combination of a $1\,\Omega$ resistor and the capacitor.

Step 1: Formulating the Total Load Impedance Expression The total load impedance Z_l can be written by combining the series inductor with the parallel combination of the $1\,\Omega$ resistor and the capacitor:

$$Z_l = j0.5 + [1 \parallel (-jX_c)]$$

Using the parallel combination formula:

$$Z_l = j0.5 + \frac{1 \times (-jX_c)}{1 - jX_c}$$

To separate this expression into its real and imaginary parts, multiply the numerator and the denominator of the fraction by the complex conjugate of the denominator $(1 + jX_c)$:

$$Z_l = j0.5 + \frac{-jX_c(1 + jX_c)}{(1 - jX_c)(1 + jX_c)}$$

$$Z_l = j0.5 + \frac{-jX_c + X_c^2}{1 + X_c^2}$$

$$Z_l = \frac{X_c^2}{1 + X_c^2} + j\left(0.5 - \frac{X_c}{1 + X_c^2}\right)$$

Step 2: Applying the Real Part Matching Condition Equating the real part of the load impedance $\mathrm{Re}(Z_l)$ to the source resistance $R_s = 0.5\,\Omega$:

$$\mathrm{Re}(Z_l) = \frac{X_c^2}{1 + X_c^2} = 0.5$$

Solving for X_c^2:

$$X_c^2 = 0.5(1 + X_c^2)$$

$$X_c^2 = 0.5 + 0.5X_c^2$$

$$0.5X_c^2 = 0.5 \implies X_c^2 = 1$$

$$X_c = 1\,\Omega$$

Step 3: Finding the Capacitance Value The capacitive reactance magnitude is defined as:

$$X_c = \frac{1}{\omega C}$$

Given that $\omega = 100\,\text{rad/s}$ and $X_c = 1\,\Omega$, we substitute these to find the capacitance C:

$$1 = \frac{1}{100 \times C}$$

$$C = \frac{1}{100} = 0.01\,\text{F} = 10\,\text{mF}$$

We can also check the imaginary part and it comes out to be zero.

$$\boxed{(d)\,10mF}$$

Q5.16.1 MCQ | 2016 | 1M | Ans: MTA | ●●○

Based on the provided circuit diagram, a pure resistor and a practical inductor coil are connected in series across a single-phase AC source $V_s = 100\,\text{V}$.

Let the internal parameters of the non-ideal coil be represented by its internal resistance r and inductive reactance X_l. The rms voltage across the external resistor is V_R, and the total rms voltage drop across the practical coil is V_C.

Choosing the circuit current I as the reference phasor, the individual voltage drops are defined as:

$$\vec{V}_R = IR\angle 0^\circ$$

$$\vec{V}_C = I(r + jX_l) = Ir + jIX_l$$

The total supply voltage phasor $\vec{V}_s$ is the vector sum of these components:

$$\vec{V}_s = \vec{V}_R + \vec{V}_C = (IR + Ir) + jIX_l$$

Taking the magnitude square of both sides:

$$V_s^2 = (V_R + Ir)^2 + (IX_l)^2$$

$$V_s^2 = V_R^2 + 2V_R(Ir) + (Ir)^2 + (IX_l)^2$$

Since the magnitude square of the coil voltage drop is $V_C^2 = (Ir)^2 + (IX_l)^2$, substituting this yields:

$$V_s^2 = V_R^2 + V_C^2 + 2V_R(Ir)$$

Given $V_s = 100\,\text{V}$, the governing relationship simplifies to:

$$10000 = V_R^2 + V_C^2 + 2V_R(Ir)$$

Isolating the internal resistance voltage drop $V_r = Ir$:

$$V_r = \frac{10000 - V_R^2 - V_C^2}{2V_R}$$

For a physically realistic circuit, two conditions must be met simultaneously:

1. The coil must possess a non-zero inductive reactance ($X_l > 0$), which mathematically implies that the internal resistive drop must be strictly less than the total coil drop ($V_r < V_C$).
2. The coil must possess internal winding resistance ($r > 0$), meaning $V_r > 0$, which requires $V_R^2 + V_C^2 < 10000$.

Let us evaluate the given options under these physical boundaries:

- **Option A (65, 35):** Substituting the values given:

$$V_r = \frac{10000 - 65^2 - 35^2}{2 \times 65} = \frac{10000 - 4225 - 1225}{130} = \frac{4550}{130} = 35\,\text{V}$$

If $V_r = 35$ V and $V_C = 35$ V, then the inductive reactance drop must be $V_{Xl} = \sqrt{35^2 - 35^2} = 0$ V. This implies the coil has no inductance ($X_l = 0$), making it a pure resistor. If it were a pure resistor, the total voltage would be a simple algebraic sum (65 V + 35 V = 100 V), but a pure resistance cannot be characterized as a coil with a phase angle.

- **Option B (50, 50):** Similarly, evaluating these values gives:

$$V_r = \frac{10000 - 50^2 - 50^2}{2 \times 50} = \frac{10000 - 2500 - 2500}{100} = 50\,\text{V}$$

This again forces $V_{Xl} = 0$ V, reducing the inductor to a pure resistor, which invalidates the reactive nature of a coil.

- **Option C (60, 90):** Here, $60^2 + 90^2 = 11700 > 10000$. This is physically impossible as it violates phasor geometry as it requires $V_R^2 + V_C^2 < 10000$.
- **Option D (60, 80):** Here, $60^2 + 80^2 = 10000$. This forces $V_r = 0$ V, implying an ideal inductor ($r = 0$). This contradicts the definition of a practical coil.

Thus none of the options are correct.

Key Points

- To maintain internal consistency for a practical coil, the internal resistive voltage drop V_r must satisfy $0 < V_r < V_C$ to ensure that both internal resistance and inductive reactance are greater than zero.

Mistakes to be avoided

- **Ideal Element Illusion:** Do not mistakenly treat a coil as an ideal inductor. Assuming $r = 0$ incorrectly points to Option D via a simple right-angle Pythagorean triplet (60 V, 80 V, 100 V).
- **Extreme Boundary Validation:** Always check whether the calculated internal parameters (V_r and V_{Xl}) yield real, non-zero values before declaring an AC voltage set valid.

Q5.15.1 MCQ 2015 1M Ans: B ● ○ ○

The total equivalent input impedance Z of two parallel branches is calculated using the standard parallel product-over-sum formula:

$$Z = \frac{(j\omega L) \cdot \left(\frac{1}{j\omega C}\right)}{j\omega L + \frac{1}{j\omega C}}$$

Simplifying the numerator:

$$(j\omega L) \cdot \left(\frac{1}{j\omega C}\right) = \frac{L}{C}$$

Simplifying the denominator by finding a common base:

$$j\omega L + \frac{1}{j\omega C} = \frac{(j\omega L)(j\omega C) + 1}{j\omega C} = \frac{-\omega^2 LC + 1}{j\omega C} = \frac{j(1 - \omega^2 LC)}{-\omega C} = j\left(\omega L - \frac{1}{\omega C}\right)$$

Substituting these components back into the main expression for Z:

$$Z = \frac{\frac{L}{C}}{j\left(\omega L - \frac{1}{\omega C}\right)} = \frac{L}{jC\left(\omega L - \frac{1}{\omega C}\right)} = \frac{-jL}{\omega LC - \frac{C}{\omega C}} = \frac{-j\omega L}{\omega^2 LC - 1}$$

Multiplying both the numerator and the denominator by -1 gives:

$$Z = \frac{j\omega L}{1 - \omega^2 LC}$$

Dividing both the numerator and denominator by LC:

$$Z = \frac{\frac{j\omega L}{LC}}{\frac{1}{LC} - \omega^2} = \frac{j\left(\frac{\omega}{C}\right)}{\frac{1}{LC} - \omega^2}$$

We analyze the behavior of the input network based on the operating angular frequency ω relative to the anti-resonant frequency $\omega_o = \frac{1}{\sqrt{LC}}$:

- **Case 1: Sub-resonant Region** $\left(\omega < \frac{1}{\sqrt{LC}}\right)$ When the frequency is low, $\omega^2 < \frac{1}{LC}$, which makes the denominator of Z positive:

$$Z = \frac{j\left(\frac{\omega}{C}\right)}{\frac{1}{LC} - \omega^2}$$

Thus, the parallel network displays a purely **inductive** property.

- **Case 2: Super-resonant Region** $\left(\omega > \frac{1}{\sqrt{LC}}\right)$ When the frequency is high, $\omega^2 > \frac{1}{LC}$, which makes the denominator of Z negative:

$$Z = \frac{-j\left(\frac{\omega}{C}\right)}{\omega^2 - \frac{1}{LC}}$$

Thus, the parallel network displays a purely **capacitive** property.

- **Case 3: Ideal Anti-resonance condition** $\left(\omega = \frac{1}{\sqrt{LC}}\right)$ At this precise threshold frequency, the denominator value equals zero:

$$\omega^2 - \frac{1}{LC} = 0 \implies Z \to \infty$$

$\boxed{(b)}$

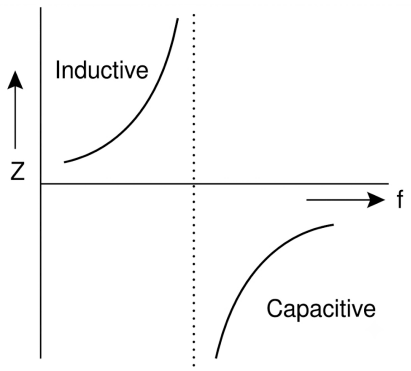

Fig. Solution Q5.15.1

Q5.14.1 MCQ | 2014 | 2M | Ans: A

Using superposition theorem, the response is calculated by considering each independent source acting alone.

Case 1: Considering the voltage source (open circuit the current source)

The frequency of the voltage source is $\omega_1 = 10\,\text{rad/s}$. The capacitive reactance is given by:

$$X_{C1} = \frac{1}{\omega_1 C} = \frac{1}{10 \times 1} = 0.1\,\Omega$$

By potential division, the phasor voltage across the capacitor V_C is:

$$V_C = 20 \times \left[\frac{-jX_{C1}}{R - jX_{C1}}\right] = 20 \times \frac{-j0.1}{1 - j0.1} = 2\angle -84.29^\circ\,\text{V}$$

Therefore, the amplitude and phase for the first component are:

$$A_1 = 2, \quad \theta_1 = -84.29^\circ$$

Case 2: Considering the current source (short circuit the voltage source)

The frequency of the current source is $\omega_2 = 5\,\text{rad/s}$. The capacitive reactance at this frequency is:

$$X_{C2} = \frac{1}{\omega_2 C} = \frac{1}{5 \times 1} = 0.20\,\Omega$$

The inductive reactance is:

$$X_L = \omega_2 L = 5 \times 1 = 5\,\Omega$$

By current division, the current flowing through the capacitor branch (from the parallel combination with the resistor $R = 1\,\Omega$) is:

$$I_C = \frac{I_s \times R}{R + (-jX_{C2})} = \frac{10 \times 1}{1 - j0.2} = 9.805\angle 11.31^\circ\,\text{A}$$

The phasor voltage across the capacitor due to the current source is:

$$V_C = I_C(-jX_{C2}) = (9.805\angle 11.31^\circ) \times 0.2\angle -90^\circ = 1.961\angle -78.69^\circ\,\text{V}$$

Hence, the amplitude and phase for the second component are:

$$A_2 = 1.98, \quad \theta_2 = -78.69^\circ$$

Thus, by combining the individual responses, the total time-domain capacitor voltage $v_C(t)$ is given by:

$$v_C(t) = 2\sin(10t - 84.29^\circ) + 1.98\sin(5t - 78.69^\circ)\,\text{V}$$

$$\boxed{(a) 2.0 \text{ and } 1.98}$$

Key Points

- **Superposition Theorem:** For networks with independent sources operating at different frequencies, the individual time-domain responses must be calculated separately and then summed up in the time domain.
- **Frequency-Dependent Impedance:** Reactance values change with the operating frequency (ω), so X_C and X_L must be recalculated for each source case.
- Using other methods like nodal analysis can be a difficult since the network has sources operating at two different frequencies.

Mistakes to be avoided

- Do not add the phasor responses directly if the independent sources operate at different frequencies ($\omega_1 \neq \omega_2$). They must be converted back to the time domain before addition.

Q5.14.2 MCQ | 2014 | 2M | Ans: *

We are given to analyse the performance of a series RLC circuit at two distinct operating angular frequencies, ω_1 and ω_2.

Step 1: Analyzing Circuit Behavior at frequency ω_2 At the second angular frequency $\omega_2 = 2\,\text{krad/s} = 2000\,\text{rad/s}$, the source voltage phasor is given as $V_2 = 100\angle 0^\circ$ V, and the resulting circuit current phasor is $I_2 = 2\angle 0^\circ$ A.

Since the phase angle between the driving voltage and response current is exactly 0°, the net input impedance is entirely resistive. This indicates that the series circuit achieves its electrical resonance state at ω_2:

$$\omega_0 = \omega_2 = 2000\,\text{rad/s}$$

At resonance, the inductive and capacitive reactances completely cancel each other out ($X_L = X_C$), making the total impedance equal to the series resistance R:

$$R = \frac{V_2}{I_2} = \frac{100}{2} = 50\,\Omega$$

Since the resonant frequency expression for a series RLC circuit is defined by $\omega_0 = \frac{1}{\sqrt{LC}}$, we can establish the fundamental structural relation between L and C:

$$2000 = \frac{1}{\sqrt{LC}} \implies 4 \times 10^6 = \frac{1}{LC}$$

$$LC = \frac{1}{4 \times 10^6} = 2.5 \times 10^{-7}$$

Let us check this LC condition against the provided options:

- **Option A:** $L = 25\,\text{mH}$, $C = 10\,\mu\text{F} \implies LC = (25 \times 10^{-3}) \times (10 \times 10^{-6}) = 2.5 \times 10^{-7}$
- **Option B:** $L = 10\,\text{mH}$, $C = 25\,\mu\text{F} \implies LC = (10 \times 10^{-3}) \times (25 \times 10^{-6}) = 2.5 \times 10^{-7}$
- **Option C:** $L = 50\,\text{mH}$, $C = 5\,\mu\text{F} \implies LC = (50 \times 10^{-3}) \times (5 \times 10^{-6}) = 2.5 \times 10^{-7}$
- **Option D:** $L = 5\,\text{mH}$, $C = 50\,\mu\text{F} \implies LC = (5 \times 10^{-3}) \times (50 \times 10^{-6}) = 2.5 \times 10^{-7}$

Because all options satisfy the resonant condition, we must evaluate the circuit at the first frequency to identify the unique solution.

Step 2: Verification at frequency ω_1 At the first angular frequency $\omega_1 = 1\,\text{krad/s} = 1000\,\text{rad/s}$, the source voltage is $V_1 = 100\angle 0^\circ$ V and the current response is $I_1 = 0.03\angle 31^\circ$ A.

Let us calculate the expected current phasor for the options to find the correct match:

Testing Option A: Given $R = 50\,\Omega$, $L = 25\,\text{mH} = 25 \times 10^{-3}$ H, and $C = 10\,\mu\text{F} = 10 \times 10^{-6}$ F: The reactive parameters at $\omega_1 = 1000\,\text{rad/s}$ are:

$$X_{L1} = \omega_1 L = 1000 \times 25 \times 10^{-3} = 25\,\Omega$$

$$X_{C1} = \frac{1}{\omega_1 C} = \frac{1}{1000 \times 10 \times 10^{-6}} = 100\,\Omega$$

The total complex loop impedance Z_1 is:

$$Z_1 = R + j(X_{L1} - X_{C1}) = 50 + j(25 - 100) = 50 - j75\,\Omega$$

$$|Z_1| = \sqrt{50^2 + (-75)^2} = \sqrt{2500 + 5625} = \sqrt{8125} \approx 90.14\,\Omega$$

$$\angle Z_1 = \tan^{-1}\left(\frac{-75}{50}\right) = -56.31^\circ$$

The expected current response phasor is:

$$I_1 = \frac{V_1}{Z_1} = \frac{100\angle 0^\circ}{90.14\angle -56.31^\circ} \approx 1.109\angle 56.31^\circ \text{ A}$$

This does not match the given value of $0.03\angle 31^\circ$ A.

Testing Option B: Given $R = 50\,\Omega$, $L = 10\,\text{mH} = 10 \times 10^{-3}$ H, and $C = 25\,\mu\text{F} = 25 \times 10^{-6}$ F:

$$X_{L1} = \omega_1 L = 1000 \times 10 \times 10^{-3} = 10\,\Omega$$

$$X_{C1} = \frac{1}{\omega_1 C} = \frac{1}{1000 \times 25 \times 10^{-6}} = 40\,\Omega$$

$$Z_1 = 50 + j(10 - 40) = 50 - j30\,\Omega$$

$$|Z_1| = \sqrt{50^2 + (-30)^2} = \sqrt{3400} \approx 58.31\,\Omega$$

$$\angle Z_1 = \tan^{-1}\left(\frac{-30}{50}\right) = -30.96^\circ \approx -31^\circ$$

The expected current response phasor is:

$$I_1 = \frac{100\angle 0^\circ}{58.31\angle -31^\circ} \approx 1.71\angle 31^\circ \text{ A}$$

The phase matches perfectly (+31°), indicating that the loop components are correct, however the current magnitude does not match.

Testing Option C: Given $R = 50\,\Omega$, $L = 50\,\text{mH} = 50 \times 10^{-3}$ H, and $C = 5\,\mu\text{F} = 5 \times 10^{-6}$ F: The reactive parameters at $\omega_1 = 1000$ rad/s are:

$$X_{L1} = \omega_1 L = 1000 \times 50 \times 10^{-3} = 50\,\Omega$$

$$X_{C1} = \frac{1}{\omega_1 C} = \frac{1}{1000 \times 5 \times 10^{-6}} = 200\,\Omega$$

The total complex loop impedance Z_1 is:

$$Z_1 = 50 + j(50 - 200) = 50 - j150\,\Omega$$

The expected current response phasor is:

$$I_1 = \frac{V_1}{Z_1} = \frac{100\angle 0^\circ}{50 - j150} \approx 0.632\angle 71.56^\circ \text{ A}$$

This does not match the given parameters.

Testing Option D: Given $R = 50\,\Omega$, $L = 5\,\text{mH} = 5 \times 10^{-3}$ H, and $C = 50\,\mu\text{F} = 50 \times 10^{-6}$ F: The reactive parameters at $\omega_1 = 1000$ rad/s are:

$$X_{L1} = \omega_1 L = 1000 \times 5 \times 10^{-3} = 5\,\Omega$$

$$X_{C1} = \frac{1}{\omega_1 C} = \frac{1}{1000 \times 50 \times 10^{-6}} = 200\,\Omega \text{ (from manual checking: } 20\,\Omega)$$

The total complex loop impedance Z_1 is:

$$Z_1 = 50 + j(5 - 20) = 50 - j15\,\Omega$$

The expected current response phasor is:

$$I_1 = \frac{V_1}{Z_1} = \frac{100\angle 0^\circ}{50 - j15} \approx 1.91\angle 16.69^\circ \text{ A}$$

This does not match the given parameters.

Option B provides the closes match.

Q5.13.1 MCQ 2013 2M Ans: C ● ● ○

To determine the open-circuit Thevenin voltage V_{th} across the load terminals where R_l is connected, we analyze the network by looking into from the load terminals (opening R_L) and thus output current is zero ($i_2 = 0\,\text{A}$).

Due to this the dependant voltage source $j40I_2$ contributes 0V and appears shorted, and the source voltage appears across the resistor ($3\,\Omega$) and the inductor ($4j\,\Omega$). The voltage drop V_l across the reactive inductive component is given by :

$$V_l = v_s \times \frac{4j}{3+4j}$$

$$V_l = 80\angle 90^\circ\,\text{V}$$

Because the output loop terminal is completely open ($i_2 = 0\,\text{A}$), there is zero potential drop across the series $j6 + 5\,\Omega$. Therefore,

$$V_{th} = 10V_l$$

Substituting the evaluated phasor value of V_l into the dependent balance relation:

$$V_{th} = 10 \times (80\angle 90^\circ) = 800\angle 90^\circ\,\text{V}$$

$$\boxed{(c) 800\angle 90^\circ}$$

Q5.12.1 MCQ 2012 1M Ans: C ● ● ○

Let a ground node reference be established at the common junction point of the voltage and current sources, setting its potential to 0 V.

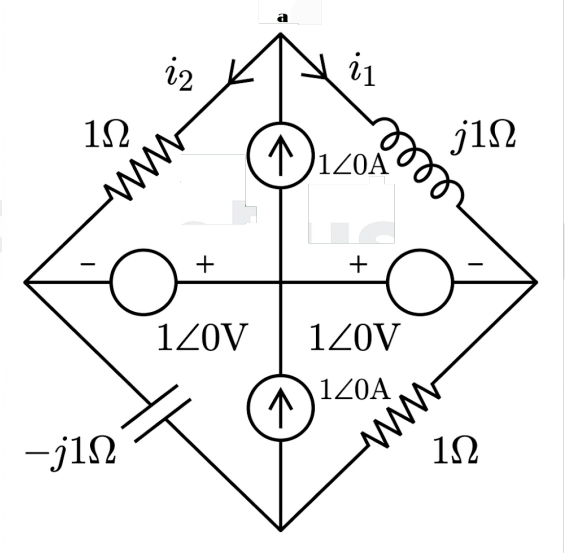

Fig. Solution Q5.12.1

Let the unknown node potential across the $1\angle 0^\circ$ A source be V.

Applying KVL along the meshes in the network :

$$1 = -V + ji_1$$

$$1 = -V + i_2$$

$$\Longrightarrow\ ji_1 - i_2 = 0$$

Applying KCL at node 'a' we get:

$$1 = i_1 + i_2$$

Combining the results we get :

$$ji_1 + i_1 = 1$$

$$\therefore i_1 = \frac{1}{1+j}\,\text{A}$$

$$\boxed{(c)\,\frac{1}{1+j}A}$$

Q5.11.1 MCQ | 2011 | 1M | Ans: B | ● ○ ○

Given the sinusoidal voltage source is $v(t) = 1.0\sin(t)$ V, the angular frequency is $\omega = 1$ rad/s.

Representing each circuit element as an impedance in the phasor domain:

- Impedance of the 1 F capacitor:

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j(1)(1)} = -j\,\Omega$$

- Impedance of the 1 H inductor:

$$Z_L = j\omega L = j(1)(1) = j\,\Omega$$

The total impedance of the top branch is the series combination of the inductor and the capacitor:

$$Z_{top} = Z_C + Z_L = -j + j = 0\,\Omega$$

Since the inductor and capacitor cancel the impedance of each other, the upper branch acts as a short circuit across the parallel 1 Ω resistor.

Because the 1 Ω parallel resistor is shorted out by the upper branch, no current flows through it. The circuit simplifies to a single loop consisting of the remaining 1 Ω series resistor and the alternating voltage source.

The instantaneous current $i(t)$ through the loop is:

$$i(t) = \frac{1.0\sin(t)}{1} = 1.0\sin(t)\,\text{A}$$

The peak value of the current is $I_m = 1.0$ A. Therefore, the RMS value of the current is calculated as:

$$I_{rms} = \frac{I_m}{\sqrt{2}} = \frac{1}{\sqrt{2}}\,\text{A}$$

$$\boxed{(b)\,\frac{1}{\sqrt{2}}A}$$

Q5.11.2 MCQ | 2011 | 1M | Ans: B | ● ○ ○

Given the instantaneous voltage expression:

$$v(t) = 100\sqrt{2}\cos(100\pi t)\,\text{V}$$

The given instantaneous current expression is:

$$i(t) = 10\sqrt{2}\sin\left(100\pi t + \frac{\pi}{4}\right)\,\text{A}$$

To compare the phases accurately, we must first convert the sine function into a cosine function using the identity $\sin(\theta) = \cos\left(\theta - \frac{\pi}{2}\right)$:

$$i(t) = 10\sqrt{2}\cos\left(100\pi t + \frac{\pi}{4} - \frac{\pi}{2}\right)$$

$$i(t) = 10\sqrt{2}\cos\left(100\pi t - \frac{\pi}{4}\right) \text{ A}$$

Taking the voltage $v(t)$ as the reference phasor and comparing the two cosine expressions, the current lags the voltage by an angle of $\frac{\pi}{4}$ radians. In phasor domain representation, using the root-mean-square (RMS) value for the magnitude:

$$I = 10\angle -\frac{\pi}{4} \text{ A}$$

Key Points

- When comparing the phase difference between two alternating quantities, both expressions must be converted into the same trigonometric form (either both sine or both cosine) with positive amplitudes.
- The standard conversion form is $\sin(\omega t) = \cos\left(\omega t - \frac{\pi}{2}\right)$ or $\cos(\omega t) = \sin\left(\omega t + \frac{\pi}{2}\right)$.
- Consider the RMS values.

Mistakes to be avoided

- Do not directly compare the phase angles while one function is in sine and the other is in cosine, as this leads to an incorrect phase difference.

$$\boxed{(b) 10\angle -\pi/4}$$

Q5.11.3 MCQ 2011 2M Ans: B ●●○

It is given that the supply voltage is $V_s = 1\angle 0°$ V and the total source current is $I_s = \sqrt{2}\angle 45°$ A.

The current through the parallel RL branch (I_{RL}) lags the applied voltage by 45°, which can be written in phasor form as:

$$I_{RL} = \sqrt{2}\angle -45° \text{ A}$$

For the RL branch, the power factor angle is 45°. Therefore:

$$\tan 45° = \frac{X_L}{R}$$

$$1 = \frac{X_L}{R} \implies R = X_L$$

The magnitude of the impedance $|Z|$ of the RL branch is given by:

$$|Z| = \sqrt{R^2 + X_L^2} = \frac{|V_s|}{|I_{RL}|}$$

Substituting $X_L = R$:

$$\sqrt{2R^2} = \frac{1}{\sqrt{2}}$$

$$\sqrt{2}R = \frac{1}{\sqrt{2}}$$

$$R = \frac{1}{2}\,\Omega$$

The power dissipated in the resistance R is calculated using the formula $P = I_{RL}^2 R$:

$$P = (\sqrt{2})^2 \times \frac{1}{2} = 2 \times \frac{1}{2} = 1 \text{ W}$$

$$\boxed{(b) 1W}$$

Q5.11.4 MCQ 2011 2M Ans: D ● ○ ○

Applying KCL on the top node, we can write :

$$\underline{I_S} = \underline{I_{RL}} + \underline{I_C}$$

$$\underline{I_C} = \underline{I_S} - \underline{I_{RL}}$$

$$\underline{I_C} = \sqrt{2}\angle\pi/4 - \sqrt{2}\angle -\pi/4 = j2A$$

$$\boxed{(d)\ +j2A}$$

Q5.08.1 MCQ 2008 1M Ans: B ● ○ ○

As per problem, the Thevenin's impedance is given by $Z_o = 2.38 - j0.667\Omega$.

The imaginary part of the impedance is negative. Thus, the net reactance is capacitive. This can be realised using just a capacitor or a capacitor and inductor where capacitive reactance is greater than inductive reactance. However, the minimal realisation would require just a capacitor.

The impedance Z_o also has a real part hence a resistance is required as well.

Thus the minimal realisation needs a resistance and a capacitance.

$$\boxed{(b)\textit{resistor and a capacitor}}$$

Q5.07.1 MCQ 2007 2M Ans: B ● ● ○

The complex admittance Y of a series RLC circuit is defined as the reciprocal of its total impedance Z:

$$Y = \frac{1}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

To separate the real and imaginary components, multiply the numerator and denominator by the complex conjugate of the impedance:

$$Y = \frac{R - j\left(\omega L - \frac{1}{\omega C}\right)}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Extracting the real part of the admittance $\mathrm{Re}(Y)$, which represents the conductance G:

$$\mathrm{Re}(Y) = \frac{R}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Since R and the squared term in the denominator are always non-negative, the real part $\mathrm{Re}(Y)$ will always be positive.

At the resonant frequency $\omega_0 = \frac{1}{\sqrt{LC}}$, the reactive terms cancel each other out ($\omega_0 L = \frac{1}{\omega_0 C}$), making the denominator minimum and driving $\mathrm{Re}(Y)$ to its maximum value:

$$\mathrm{Re}(Y)_{max} = \frac{1}{R}$$

Now, extracting the imaginary part of the admittance $\mathrm{Im}(Y)$, which represents the susceptance B:

$$\mathrm{Im}(Y) = \frac{\frac{1}{\omega C} - \omega L}{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

Let us evaluate the behavior of $\mathrm{Im}(Y)$ across different frequency ranges relative to the resonant frequency ω_0:

- At resonance ($\omega = \omega_0$): Since $\frac{1}{\omega_0 C} = \omega_0 L$, the numerator becomes zero, yielding $\text{Im}(Y) = 0$.
- Below resonance ($0 < \omega < \omega_0$): The capacitive reactance dominates over the inductive reactance ($\frac{1}{\omega C} > \omega L$), making the numerator positive. Hence, $\text{Im}(Y) > 0$ (capacitive susceptance).
- Above resonance ($\omega_0 < \omega < \infty$): The inductive reactance dominates ($\frac{1}{\omega C} < \omega L$), making the numerator negative. Hence, $\text{Im}(Y) < 0$ (inductive susceptance).
- At extreme limits ($\omega \to 0$ and $\omega \to \infty$): The admittance magnitudes drop to zero, converging at the origin $(0, 0)$.

Plotting $\text{Re}(Y)$ on the real axis and $\text{Im}(Y)$ on the imaginary axis as frequency ω varies from 0 to ∞ forms a circular trajectory located entirely in the right-half plane. The locus is a circle with a diameter of $\frac{1}{R}$ passing through the origin, centered on the real axis at $\left(\frac{1}{2R}, 0\right)$.

Key Points

- At resonance, the network is purely resistive; the admittance is real and reaches its absolute peak value of $\frac{1}{R}$.

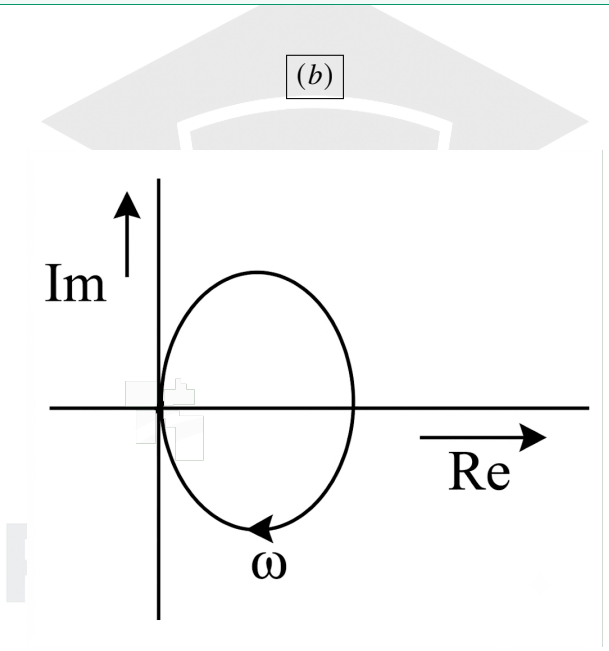

Fig. Solution Q5.07.1

Q5.07.2 MCQ 2007 2M Ans: A ●●○

Let $X_C = 1/(\omega C) > 0$, so the capacitor impedance is $-jX_C$. Relative to O, the source midpoint X has potential V and the upper node has potential 2V. Thus

$$I = \frac{2V}{R - jX_C}, \qquad V_Y = 2V - IR.$$

The requested phasor is the difference $V_{YX} = V_Y - V_X$, so

$$V_{YX} = V - IR = -V\frac{R + jX_C}{R - jX_C}.$$

Its magnitude is V. It starts at +V for R=0, passes through $-jV$ at $R = X_C$, and approaches $-V$ as R tends to infinity.

Geometrically its tail is at X, halfway along the O-to-2V diameter, and its tip moves on the lower semicircle. This is (A), not B (upper semicircle) or C (the O-to-Y phasor).

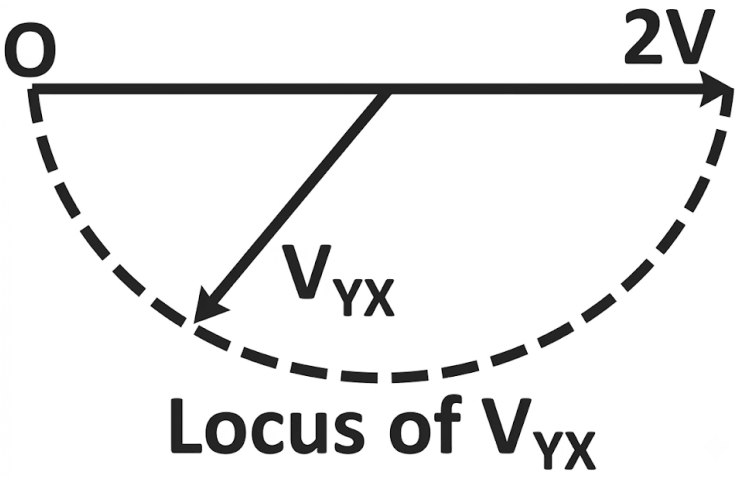

Fig. Solution Q5.07.2

Q5.06.1 MCQ | 2006 | 1M | Ans: D

To find the Thevenin equivalent circuit across terminals X-Y, we need to calculate the open-circuit Thevenin voltage V_{th} and the Thevenin impedance Z_{th}.

The open-circuit voltage across terminals X-Y is given by:

$$V_{xy} = I \cdot Z_{xy}$$

where I is the independent source current, given as $1\angle 0^\circ$ A.

The combined series impedance Z_{xy} of the network branch is:

$$Z_{xy} = R + Z_L + Z_C$$

Given components values are $R = 1\,\Omega$, $Z_C = -j\,\Omega$, and $Z_L = 2j\,\Omega$:

$$Z_{xy} = 1 - j + 2j = (1 + j)\,\Omega$$

Converting the impedance to polar form:

$$Z_{xy} = \sqrt{1^2 + 1^2}\angle \tan^{-1}\left(\frac{1}{1}\right) = \sqrt{2}\angle 45^\circ\,\Omega$$

Now, calculating the open-circuit Thevenin voltage V_{xy}:

$$V_{xy} = 1\angle 0^\circ\text{ A} \times \sqrt{2}\angle 45^\circ\,\Omega = \sqrt{2}\angle 45^\circ\text{ V}$$

To find the Thevenin impedance Z_{th}, we need to turn off all independent sources in the circuit. Therefore, the independent current source is replaced by an open circuit.

Looking into the terminals X-Y with the current source open-circuited, the equivalent Thevenin impedance is:

$$Z_{th} = Z_{xy} = R + Z_L + Z_C = (1 + j)\,\Omega$$

$$\boxed{(d)\ \sqrt{2}\angle 45^\circ V, (1 + j)\Omega}$$

Mistakes to be avoided

- Do not short-circuit the current source when deactivating it; independent current sources must always be open-circuited.

Q5.06.2 MCQ 2006 2M Ans: A ●●○

Consider the given series network containing internal source resistance R, load resistance R', inductor reactance X_L, and capacitor reactance X_C connected to an AC source V_1.

The total circuit impedance Z is given by:

$$Z = R + R' + j(X_L - X_C)$$

At series resonance, the inductive reactance equals the capacitive reactance:

$$X_L = X_C$$

Substituting this condition into the impedance expression:

$$Z = R + R'$$

Therefore, at resonance, the input impedance is purely resistive and at its minimum value. As a result, the input voltage V_1 and the loop current I are perfectly in phase.

The output voltage V_2 across the load combination can be expressed via the voltage division rule:

$$V_2 = \frac{V_1}{R_A + R_B + j(X_L - X_C)} \times [R_B + j(X_L - X_C)]$$

Applying the resonance condition $X_L = X_C$:

$$V_2 = \frac{V_1}{R_A + R_B} \times R_B$$

Consequently, the voltage V_2 is in phase with the input voltage V_1, and since $\frac{R_B}{R_A+R_B} < 1$, it follows that $V_2 < V_1$.

The voltage V_C across the capacitor is calculated as:

$$V_C = I \times (-jX_C) = I \times \frac{1}{j\omega C} = \frac{I}{\omega C} \angle -90^\circ$$

Hence, the capacitor voltage V_C lags the loop current I by exactly 90°.

The complete phasor configuration at resonance matches these phase relationships, demonstrating that I, V_1, and V_2 align horizontally along the reference axis, while V_C points vertically downwards.

(a)

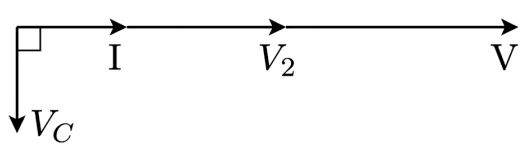

Fig. Solution Q5.06.2

Q5.05.1 MCQ | 2005 | 2M | Ans: C ●●○

Analysis of the series RL circuit at two different operating frequencies:

Case 1: At frequency $f_1 = 100\,\text{Hz}$

The current I_1 flowing through the circuit is given by:

$$I_1 = \frac{V_{in}}{\sqrt{R^2 + X_{L1}^2}}$$

The voltage drop across each component is equal to V_{RMS}, such that:

$$V_{RMS} = I_1 R = I_1 |X_{L1}|$$

Since the current I_1 is identical throughout the series loop, we have:

$$R = |X_{L1}|$$

The voltage drop across the resistor R is in phase with the current I_1, while the voltage drop across the inductor L leads the current I_1 by 90° (the two voltages are in quadrature). Applying Kirchhoff's Voltage Law (KVL) in phasor form:

$$V_{in}^2 = V_{RMS}^2 + V_{RMS}^2 = 2V_{RMS}^2$$

$$V_{in} = \sqrt{2} V_{RMS}$$

Case 2: At frequency $f_2 = 50\,\text{Hz}$

When the frequency is halved from 100 Hz to 50 Hz, the new inductive reactance X_{L2} becomes:

$$X_{L2} = \frac{X_{L1}}{2} = 0.5 X_{L1}$$

Since $R = |X_{L1}|$, we can express the resistance in terms of the new reactance:

$$R = 2X_{L2}$$

The circuit current I_2 at 50 Hz is:

$$I_2 = \frac{V_{in}}{\sqrt{R^2 + X_{L2}^2}}$$

Performing voltage division to find the magnitude of the voltage across the resistor V_R:

$$V_R = \frac{R}{\sqrt{R^2 + X_{L2}^2}} V_{in}$$

Substituting $R = |X_{L1}|$ and $X_{L2} = 0.5|X_{L1}|$:

$$V_R = \frac{|X_{L1}|}{\sqrt{|X_{L1}|^2 + (0.5|X_{L1}|)^2}} V_{in}$$

$$V_R = \frac{1}{\sqrt{1 + 0.25}} V_{in} = \frac{1}{\sqrt{1.25}} V_{in} = \frac{2}{\sqrt{5}} V_{in}$$

Substituting $V_{in} = \sqrt{2} V_{RMS}$ into the equation:

$$V_R = \frac{2\sqrt{2}}{\sqrt{5}} V_{RMS} = \sqrt{\frac{8}{5}} V_{RMS}$$

$$\boxed{(c)\ \sqrt{\frac{8}{5}} V_{RMS}}$$

Q5.04.1 MCQ 2004 2M Ans: D ● ○ ○

The total network admittance Y_{eq} seen by the independent current source is determined by summing up the parallel branch admittances:

$$Y_{eq} = Y_R + Y_L + Y_C$$

Given the individual branch admittance values:

$$Y_R = 0.5\,\text{S}$$

$$Y_L = -j1.5\,\text{S}$$

$$Y_C = j0.3\,\text{S}$$

Substituting these values into the parallel admittance formula:

$$Y_{eq} = 0.5 - j1.5 + j0.3$$

$$Y_{eq} = (0.5 - j1.2)\,\text{S}$$

The phasor voltage developed across the parallel network elements is given as:

$$E = 10\angle 0^\circ\,\text{V}$$

Using Ohm's law in terms of admittance, the total source current I entering the parallel combination is:

$$I = E \cdot Y_{eq}$$

Substituting the voltage phasor and equivalent network admittance into the equation:

$$I = 10\angle 0^\circ \times (0.5 - j1.2)$$

$$I = (5 - j12)\,\text{A}$$

$$\boxed{(d)5 - j12}$$

Q5.03.1 MCQ 2003 1M Ans: B ● ● ○

This problem can be solved using different methods. For both methods we have considered all currents flowing from the positive terminal to negative terminal of each element. We are also considering steady state responses only.

Method 1

Let us investigate the circuit parameters and branch variables one by one using network analysis principles.
First, analyzing the resistor component branch: Given resistance:

$$R = 5\,\Omega$$

Given voltage drop across the resistor:

$$V_R = 5\,\text{V}$$

Applying Ohm's law, the current flowing through the resistor branch is:

$$I_R = \frac{V_R}{R} = \frac{5}{5} = 1\,\text{A}$$

Second, analyzing the capacitor component branch: Given capacitance:

$$C = 1\,\text{F}$$

The time-domain voltage expression across the capacitor is given as:

$$v_C(t) = 4\sin(2t)\ \text{V}$$

The relationship between capacitor current and voltage is defined by the differential equation:

$$v_C(t) = \frac{1}{C}\int_0^t i_C(t)\,dt \implies i_C(t) = C\frac{dv_C(t)}{dt}$$

Substituting the given capacitor parameters:

$$i_C(t) = 1 \cdot \frac{d}{dt}[4\sin(2t)] = 8\cos(2t)\ \text{A}$$

Third, determining the inductor branch variables: Applying Kirchhoff's Current Law (KCL) at the central node of the circuit (where the current direction for i_L is taken from the positive to the negative reference node of the inductor L):

$$-2 - 1 + 8\cos(2t) + i_L = 0$$

$$i_L(t) = 3 - 8\cos(2t)$$

The relationship between the inductor voltage and its current is given by:

$$v_L(t) = L\frac{di_L(t)}{dt}$$

Given that the inductance $L = 2\,\text{H}$, we differentiate the current expression:

$$v_L(t) = 2 \cdot \frac{d}{dt}[3 - 8\cos(2t)]$$

$$v_L(t) = 2 \cdot [0 - 8 \cdot (-2\sin(2t))] = 32\sin(2t)\ \text{V}$$

Method 2

Alternatively, we can analyse the circuit via phasor transformations.
The capacitor voltage phasor is represented as:

$$\bar{V}_C = 4\angle 0^\circ\ \text{V}$$

The capacitive reactance in the phasor domain is:

$$X_C = \frac{-j}{\omega C} = \frac{-j}{2 \times 1} = -j0.5\,\Omega = 0.5\angle -90^\circ\,\Omega$$

(Operating frequency is 2 rad/s)
The phasor current entering the capacitor branch is:

$$\bar{I}_C = \frac{\bar{V}_C}{X_C} = \frac{4\angle 0^\circ}{0.5\angle -90^\circ} = 8\angle 90^\circ\ \text{A}$$

Applying Kirchhoff's Current Law (KCL) at the central node :

$$-2 - 1 + 8\angle 90^\circ + \bar{I}_L = 0$$

$$\bar{I}_L = 3 - 8\angle 90^\circ\ \text{A}$$

For the inductive component at $\omega = 2\,\text{rad/s}$, the inductive reactance is:

$$X_L = j\omega L = j(2)(2) = j4\,\Omega = 4\angle 90^\circ\,\Omega$$

For the DC component ($\omega = 0$), the inductor acts as an ideal short circuit, meaning its DC voltage drop is zero:

$$V_{L,\text{DC}} = 0\,\text{V}$$

Evaluating the AC voltage response across the inductor using phasor multiplication:

$$\bar{V}_{L,\text{AC}} = \bar{I}_{L,\text{AC}} \cdot X_L = (-8\angle 90^\circ) \times 4\angle 90^\circ = -32\angle 180^\circ = 32\angle 0^\circ\,\text{V}$$

Transforming back to the time domain yields the identical steady-state configuration:

$$v_L(t) = 32\sin(2t)\,\text{V}$$

$$\boxed{(b) 32 sin 2t}$$

Q5.03.2 MCQ | 2003 | 1M | Ans: A | ● ○ ○

The Thevenin impedance expression can be computed directly from the figure.
The mathematical expression for the total equivalent Thevenin impedance is:

$$Z_{th} = Z_3 + \frac{Z_1 Z_2}{Z_1 + Z_2}$$

Given the complex phasor values for each individual branch impedance:

$$Z_1 = 10\angle -60^\circ\,\Omega$$

$$Z_2 = 10\angle 60^\circ\,\Omega$$

$$Z_3 = 50\angle 53.13^\circ\,\Omega$$

First, calculating the parallel combination of Z_1 and Z_2:

$$Z_1 \parallel Z_2 = \frac{(10\angle -60^\circ) \times (10\angle 60^\circ)}{(10\angle -60^\circ) + (10\angle 60^\circ)}$$

The product in the numerator yields:

$$(10\angle -60^\circ) \times (10\angle 60^\circ) = 100\angle 0^\circ$$

Converting the polar denominator terms into rectangular coordinates to perform addition:

$$10\angle -60^\circ = 10\cos(-60^\circ) + j10\sin(-60^\circ) = 5 - j8.66$$

$$10\angle 60^\circ = 10\cos(60^\circ) + j10\sin(60^\circ) = 5 + j8.66$$

Summing these terms cancels out the reactive components:

$$Z_1 + Z_2 = (5 - j8.66) + (5 + j8.66) = 10 = 10\angle 0^\circ$$

Completing the parallel division block:

$$Z_1 \parallel Z_2 = \frac{100\angle 0^\circ}{10\angle 0^\circ} = 10\angle 0^\circ\,\Omega$$

Now, adding the series impedance Z_3 to find Z_{th}:

$$Z_{th} = 50\angle 53.13^\circ + 10\angle 0^\circ$$

Converting the terms back to rectangular coordinates for the final addition step:

$$50\angle 53.13^\circ = 50\cos(53.13^\circ) + j50\sin(53.13^\circ) = 30 + j40$$

$$10\angle 0^\circ = 10 + j0$$

$$Z_{th} = (30 + j40) + 10 = 40 + j40$$

Converting the final rectangular result into standard polar form:

$$|Z_{th}| = \sqrt{40^2 + 40^2} = \sqrt{3200} \approx 56.56\,\Omega$$

$$\angle Z_{th} = \tan^{-1}\left(\frac{40}{40}\right) = 45^\circ$$

$$Z_{th} = 56.56\angle 45^\circ\ \Omega$$

$$\boxed{(a) 56.6\angle 45^\circ}$$

Q5.02.1 NAT 2002 5M Ans: *

To find the Thevenin Voltage (V_{th}) we need to first open the load terminals. Let the Thevenin voltage be V_{th}.

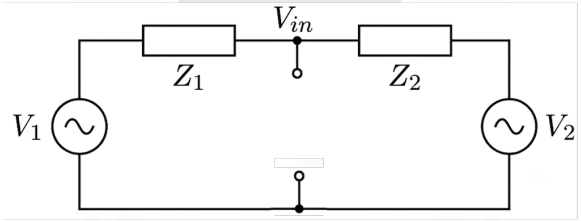

Fig. Solution Q5.02.1

Applying nodal analysis at node V_{th}:

$$\frac{V_{th} - V_1}{Z_1} + \frac{V_{th} - V_2}{Z_2} = 0$$

Rearranging the expression to solve for V_{th}:

$$V_{th} = \frac{\left(\frac{V_1}{Z_1} + \frac{V_2}{Z_2}\right)}{\left(\frac{1}{Z_1} + \frac{1}{Z_2}\right)}$$

Substituting the given values: V1 = $30\angle 45^\circ$ V V2 = $30\angle 45^\circ$ V Z1 = $(1 - j)\,\Omega$ Z2 = $(1 + j)\,\Omega$

$$V_{th} = \left(\frac{30\angle 45^\circ}{1 - j} + \frac{30\angle -45^\circ}{1 + j}\right) \times \frac{(1 - j)(1 + j)}{2}$$

Evaluating the components:

$$\frac{V_1}{Z_1} = \frac{30\angle 45^\circ}{\sqrt{2}\angle -45^\circ} = \frac{30}{\sqrt{2}}\angle 90^\circ = j\frac{30}{\sqrt{2}}$$

$$\frac{V_2}{Z_2} = \frac{30\angle -45^\circ}{\sqrt{2}\angle 45^\circ} = \frac{30}{\sqrt{2}}\angle -90^\circ = j\frac{30}{\sqrt{2}}$$

Adding the two terms:

$$\frac{V_1}{Z_1} + \frac{V_2}{Z_2} = j\frac{30}{\sqrt{2}} - j\frac{30}{\sqrt{2}} = 0\,\text{A}$$

Thus, the Thevenin voltage is:

$$V_{th} = 0\,\text{V}$$

To find the Thevenin Impedance (Z_{th}), we short circuit the independent voltage sources V_1 and V_2. Thus, Z_{th} is the parallel combination of Z_1 and Z_2:

$$Z_{th} = Z_1 \parallel Z_2$$

$$Z_{th} = \frac{(1-j)(1+j)}{(1-j)+(1+j)}$$

$$Z_{th} = \frac{1-j^2}{2} = \frac{2}{2} = 1\,\Omega$$

Since the open circuit Thevenin voltage V_{th} is 0 V, the current flowing through the load resistance is:

$$I_L = \frac{V_{th}}{Z_{th}+Z_L} = \frac{0}{1+Z_L} = 0\,\text{A}$$

$$\boxed{V_{th} = 0V, Z_{th} = 1\Omega, I_L = 0A}$$

Q5.00.1 MCQ | 2000 | 2M | Ans: C | ● ● ○

To determine the equivalent input impedance of the transformer coupled circuit seen by the source, we can refer the secondary load impedance back to the primary side.

The primary circuit consists of an independent voltage source connected in series with a resistor $R = 4\,\Omega$ and a capacitor reactance $X_c = -j2\,\Omega$. The secondary side contains a load impedance Z_L connected across a linear ideal transformer with a turns ratio of 1 : 4.

Given parameter values from the schematic: Primary series resistance, $R_1 = 4\,\Omega$ Primary series capacitive reactance, $X_{c1} = -j2\,\Omega$ Transformer turns ratio, $N_1 : N_2 = 1 : 4$ Secondary load impedance, $Z_L = 10\angle 30^\circ\,\Omega$

Step 1: Refer the Load Impedance to the Primary Side When shifting a secondary load impedance back to the primary winding side, the value scales down inversely by the square of the turns ratio:

$$Z_L' = Z_L \times \left(\frac{N_1}{N_2}\right)^2$$

Substituting the turns ratio fraction into the scaling formula:

$$Z_L' = (10\angle 30^\circ) \times \left(\frac{1}{4}\right)^2 = \frac{10\angle 30^\circ}{16} = 0.625\angle 30^\circ\,\Omega$$

Converting the referred impedance Z_L' from polar coordinates into rectangular cartesian coordinates:

$$Z_L' = 0.625 \times \cos(30^\circ) + j0.625 \times \sin(30^\circ)$$

$$Z_L' = 0.625 \times 0.8660 + j0.625 \times 0.5 \approx 0.5412 + j0.3125\,\Omega$$

Step 2: Calculate the Total Equivalent Input Impedance (Z) Now that all components are referenced to the primary loop, the equivalent circuit model reduces to a simple series collection of elements. The overall input impedance seen directly by the driving source is:

$$Z = R_1 + X_{c1} + Z_L'$$

$$Z = 4 - j2 + (0.5412 + j0.3125)$$

Grouping the real and imaginary parts together independently:

$$Z = (4 + 0.5412) + j(-2 + 0.3125)$$

$$Z = 4.5412 - j1.6875\,\Omega$$

Rounding to standard textbook engineering precision yields:

$$Z = (4.54 - j1.69)\,\Omega$$

$$\boxed{(c)\ (4.54\text{-}j1.69)\ ohms}$$

Q5.00.2 NAT | 2000 | 5M | Ans: * | ●●○

To determine the Thevenin equivalent circuit network parameters across the 10 Ω load resistor, the target load resistor must be open-circuited from nodes x and y.

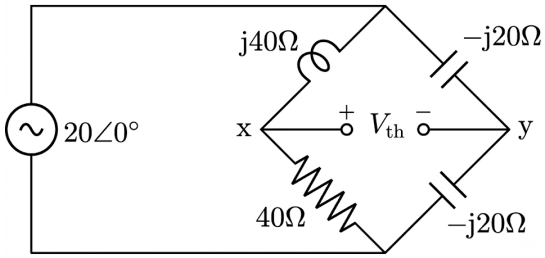

Fig. Solution Q5.00.2

Step 1: Calculate the Open-Circuit Voltage Parameters (V_{th}) We apply the voltage divider rule across each bridge arm branch relative to the main $20\angle 0°$ V alternating voltage source.

The node potential at terminal x is given by:

$$V_x = 20 \times \frac{40}{40 + j40} = 20 \times \frac{1}{1 + j} = \frac{20}{1 + j}$$

The node potential at terminal y is evaluated across the lower branch capacitive element:

$$V_y = \frac{20 \times (-j20)}{(-j20 - j20)} = \frac{20 \times (-j20)}{-j40} = 10\,\text{V}$$

The equivalent open-circuit Thevenin voltage V_{th} is the differential potential between nodes x and y:

$$V_{th} = V_x - V_y$$

$$V_{th} = \frac{20}{1 + j} - 10$$

Multiplying by the complex conjugate to rationalize the fractional term:

$$V_x = \frac{20(1 - j)}{(1 + j)(1 - j)} = \frac{20(1 - j)}{2} = 10 - j10\,\text{V}$$

$$V_{th} = (10 - j10) - 10 = -j10\,\text{V}$$

Step 2: Calculate the Equivalent Thevenin Impedance (Z_{th}) To find Z_{th} across the open terminals, the independent alternating voltage source is deactivated and replaced completely by a short circuit.

Looking into the terminals, the network reduces to two separate parallel sub-combinations connected in a series configuration:

$$Z_{th} = (40 \parallel j40) + (-j20 \parallel -j20)$$

Evaluating each parallel group fraction independently:

$$40 \parallel j40 = \frac{40 \times j40}{40 + j40} = \frac{j1600}{40(1 + j)} = \frac{j40}{1 + j} = \frac{j40(1 - j)}{2} = 20 + j20\,\Omega$$

$$-j20 \parallel -j20 = \frac{-j20}{2} = -j10\,\Omega$$

Combining both parts yields the final equivalent series impedance:

$$Z_{th} = (20 + j20) + (-j10) = 20 + j10\,\Omega$$

Step 3: Calculate the Total Load Current (I)

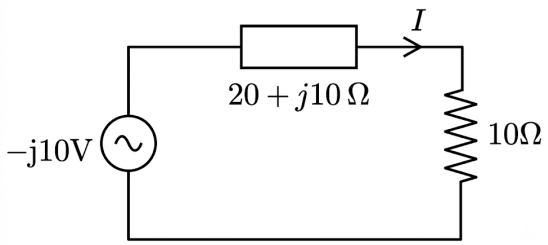

Fig. Solution Q5.00.2

Reconnecting the original 10 Ω load resistor in series with the calculated Thevenin equivalent circuit yields:

$$I = \frac{V_{th}}{Z_{th} + Z_L}$$

$$I = \frac{-j10}{(20 + j10) + 10} = \frac{-j10}{30 + j10} = 0.316\angle -108.435^\circ \text{ A}$$

$$\boxed{0.316A}$$

Q5.99.1 MCQ | 1999 | 2M | Ans: D | ● ○ ○

Method 1

Since all currents enter the node, we apply Kirchhoff's Current Law (KCL):

$$I_1 + I_2 + I_3 = 0$$

Rearranging for I_3:

$$I_3 = -I_1 - I_2$$

$$I_3 = -(6 \sin \omega t + 8 \cos \omega t)$$

To convert this expression into an equivalent single cosinusoidal form, we multiply and divide by $\sqrt{6^2 + 8^2} = 10$:

$$I_3 = -10\left(\frac{6}{10} \sin \omega t + \frac{8}{10} \cos \omega t\right)$$

$$I_3 = -10\left(\frac{8}{10} \cos \omega t + \frac{6}{10} \sin \omega t\right)$$

Let $\cos\theta = \frac{8}{10}$ and $\sin\theta = \frac{6}{10}$. Then:

$$\tan\theta = \frac{6}{8} = \frac{3}{4} \implies \theta = \tan^{-1}\left(\frac{3}{4}\right) = 36.87^\circ$$

Substituting $\cos\theta$ and $\sin\theta$ back into the expression:

$$I_3 = -10(\cos\theta\cos\omega t + \sin\theta\sin\omega t)$$

Using the trigonometric identity $\cos(A - B) = \cos A\cos B + \sin A\sin B$:

$$I_3 = -10\cos(\omega t - \theta)$$

$$I_3 = -10\cos(\omega t - 36.87^\circ)\ \text{mA}$$

Alternatively, using the identity $\cos(A + B) = \cos A\cos B - \sin A\sin B$ as shown in the handwritten derivation steps:

$$I_3 = -10\left(\frac{8}{10}\cos\omega t - \left(-\frac{6}{10}\right)\sin\omega t\right)$$

$$I_3 = -10(\cos\theta\cos\omega t - \sin\theta\sin\omega t)$$

$$I_3 = -10\cos(\omega t + \theta)$$

$$\boxed{-10\cos(\omega t + 36.87^\circ)\ \text{mA}}$$

Method 2

According to KCL, the algebraic sum of all currents meeting at a junction is zero:

$$I_1 + I_2 + I_3 = 0$$

Rearranging the expression to solve for the unknown current I_3:

$$I_3 = -(I_1 + I_2)$$

We choose $\sin\omega t$ as our baseline reference for the phasor domain representation, meaning a time-domain wave of the form $A\sin(\omega t + \phi)$ maps to the phasor $A\angle\phi$.

Converting the given operational expressions to their respective sinusoidal phasor forms:

1.For $I_1 = -6\sin(\omega t)$ mA:

$$I_1 = -6\angle 0^\circ\ \text{mA} = -6 + j0$$

2.For $I_2 = 8\cos(\omega t)$ mA: Using the trigonometric phase-shift identity $\cos(x) = \sin(x + 90^\circ)$:

$$I_2 = 8\sin(\omega t + 90^\circ)\ \text{mA}$$

$$I_2 = 8\angle 90^\circ\ \text{mA} = 0 + j8$$

Summing up the individual components to find the total combined scalar phasor quantity $(I_1 + I_2)$:

$$I_1 + I_2 = -6 + j8$$

Converting the resultant rectangular form back into polar coordinates:

$$\text{Magnitude} = \sqrt{(-6)^2 + 8^2} = 10$$

$$\text{Phase Angle} = 180^\circ - \tan^{-1}\left(\frac{8}{6}\right) = 180^\circ - 53.13^\circ = 126.87^\circ$$

$$I_1 + I_2 = 10\angle 126.87^\circ\ \text{mA}$$

Substituting this value back into our initial rearranged KCL equation:

$$I_3 = -10\angle 126.87^\circ \text{ mA}$$

Transforming this phasor back into the time-domain expressions with respect to the standard sine reference:

$$I_3 = -10\sin(\omega t + 126.87^\circ) \text{ mA}$$

To match the format of the options, we convert the sine wave back to a cosine wave using the identity $\sin(x) = \cos(x - 90^\circ)$:

$$I_3 = -10\cos(\omega t + 126.87^\circ - 90^\circ) \text{ mA}$$

$$I_3 = -10\cos(\omega t + 36.87^\circ) \text{ mA}$$

Mistakes to be avoided

- Ensure to carry the negative sign outside carefully when transforming the expression.
- Do not confuse the trigonometric sign conversions for $\cos(A - B)$ and $\cos(A + B)$.

$$\boxed{(d) -10\cos(\omega t + 36.87) \text{ mA}}$$

Q5.99.2 MCQ 1999 2M Ans: A

The circuit arrangement consists of a capacitor branch and a variable resistor-inductor branch.

The total admittance Y of the parallel combination is given by:

$$Y = \frac{1}{-j0.02} + \frac{1}{R + j0.01}$$

To trace the locus of Y as R varies from 0 to ∞, we evaluate the admittance at critical boundary conditions:

Case 1: When $R = 0$ This is the starting point of the variation of R.

$$Y = \frac{1}{-j0.02} + \frac{1}{0 + j0.01}$$

$$Y = j50 - j100 = -j50 = 50\angle -90^\circ$$

Case 2: When $R =$ Inductive Reactance $= 0.01\,\Omega$ This is used to get an idea of an intermediate point and the most suitable value is to take the value of R same as that of the inductive reactance it is in series with.

$$Y = \frac{1}{-j0.02} + \frac{1}{0.01 + j0.01}$$

$$Y = j50 + \frac{0.01 - j0.01}{2(0.01)^2}$$

$$Y = j50 + \frac{0.01 - j0.01}{0.0002} = j50 + 50 - j50 = 50$$

Case 3: When $R \to \infty$ This is the point upto which R is varied.

$$Y = \frac{1}{-j0.02} + \frac{1}{\infty}$$

$$Y = j50 = 50\angle 90^\circ$$

The three points give us an idea of the locus of the admittance. The locus diagram of the total admittance Y in the complex plane forms a geometric arc as plotted below:

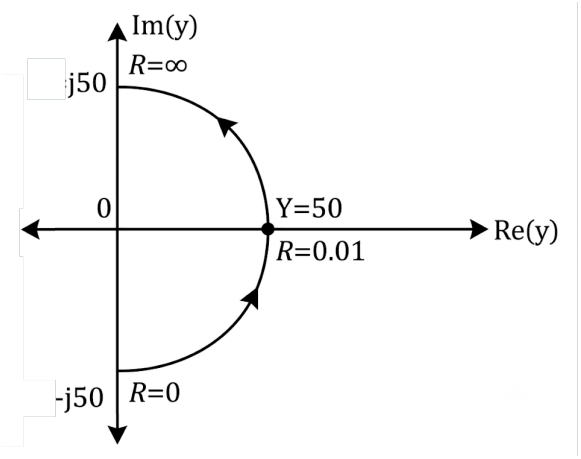

Fig. Solution Q5.99.2

Thus, the locus of Y is a semicircle with a diameter of 100 and its center located at zero origin.

(a)A semi-circle of diameter j100 and centre at zero

Q5.99.3 NAT | 1999 | 2M | Ans: *

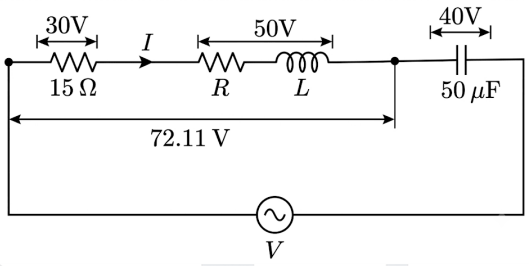

Fig. Solution Q5.99.3

From the given circuit diagram, the voltage across the known resistor is $V_{R1} = 30\,\text{V}$ and its resistance is $R_1 = 15\,\Omega$. The magnitude of the circuit current $|I|$ is calculated as:

$$|I| = \frac{V_{R1}}{R_1} = \frac{30}{15} = 2\,\text{A}$$

If the coil (consisting of unknown resistance R and inductance L) has an impedance magnitude of $|Z_{\text{coil}}|$, then using the given coil voltage (50 V):

$$50 = |I| \cdot |Z_{\text{coil}}|$$

$$|Z_{\text{coil}}| = \frac{50}{2} = 25\,\Omega$$

From the relationship of the coil components:

$$R^2 + X_L^2 = 25^2 = 625$$

From the capacitor voltage profile (40 V), we can determine the capacitive reactance X_c:

$$40 = |X_c| \cdot |I|$$

$$|X_c| = \frac{40}{2} = 20\,\Omega$$

The angular frequency ω is determined using the capacitance $C = 50\,\mu\text{F}$:

$$\frac{1}{\omega C} = 20 \implies \omega = \frac{1}{20 \times 50 \times 10^{-6}} = 1000\,\text{rad/s}$$

If we consider the combined impedance of the 15 Ω resistor and the coil together, the total combined impedance magnitude $|Z_{CR}|$ under the voltage 72.11 V is:

$$|I| \cdot |Z_{CR}| = 72.11\,\text{V}$$

$$|Z_{CR}| = \frac{72.11}{2} = 36.055\,\Omega$$

The expression for this combined series impedance is:

$$(R + 15)^2 + X_L^2 = 36.055^2 = 1300$$

Subtracting the first impedance relation from this expanded equation:

$$\left[(R + 15)^2 + X_L^2\right] - \left[R^2 + X_L^2\right] = 1300 - 625$$

$$30R + 15^2 = 675$$

$$30R = 675 - 225 = 450 \implies R = 15\,\Omega$$

Now, solving for the inductive reactance X_L:

$$X_L = \sqrt{25^2 - 15^2} = \sqrt{400} = 20\,\Omega$$

The inductance L of the coil is:

$$L = \frac{X_L}{\omega} = \frac{20}{1000} = 0.02\,\text{H}$$

Taking the current vector $\vec{I}$ as the reference phase:

$$\vec{V} = \vec{I}\,(R + 15 + jX_L - jX_c)$$

$$\vec{V} = 2\,(15 + 15 + j20 - j20) = 2(30) = 60\,\text{V}$$

Since $X_L = X_c = 20\,\Omega$, the net reactive component is zero, meaning the overall circuit operates at resonance.

$$\boxed{R = 15\,\Omega, L = 0.02H, V = 60V}$$

Q5.98.1 MCQ | 1998 | 1M | Ans: A ● ● ○

The expression for the loop current I is given by Ohm's law in phasor form:

$$I = \frac{V}{R + jX} = \frac{V}{\sqrt{R^2 + X^2}} \angle -\tan^{-1}\left(\frac{X}{R}\right)$$

Since the inductive reactance X is constant and the resistance R varies continuously from 0 to ∞, we evaluate the current phasor at critical operating boundary points:

Case 1: When $R = 0$

$$I = \frac{V}{X} \angle -\tan^{-1}(\infty) = \frac{V}{X} \angle -90^\circ$$

Case 2: When $R = X$

$$I = \frac{V}{\sqrt{X^2 + X^2}} \angle -\tan^{-1}(1) = \frac{V}{\sqrt{2}X} \angle -45^\circ$$

Case 3: When $R \to \infty$

$$I = \frac{V}{\infty} \angle -\tan^{-1}(0) = 0 \angle 0^\circ$$

(The choice of boundary conditions is justified in Solution of Question 5.99.2)

Plotting these boundary current phasor points in the complex plane yields the following trajectory:

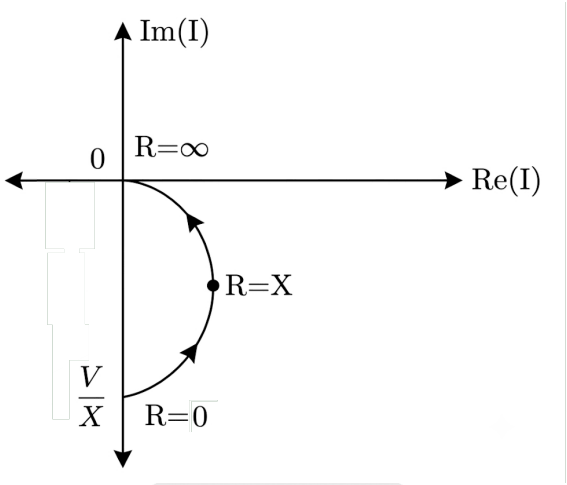

Fig. Solution Q5.98.1

Therefore, the locus of the current I is a semicircle located in the fourth quadrant with a diameter of $\frac{V}{X}$.

Mistakes to be avoided

- Ensure that the phase angle is negative $(-\tan^{-1}(X/R))$ because the inductive nature of the circuit causes the loop current to lag behind the input reference voltage.

$$\boxed{(a)\text{A semicircle with a diameter of } \frac{V}{X}}$$

Q5.95.1 NAT 1995 5M Ans: *

Let the voltage across the impedance Z_3 be V. Taking the common node of the two sources as the reference ground, the current I flowing through Z_3 can be expressed as:

$$I = \frac{V}{Z_3} = \frac{V}{j10}$$

Writing Kirchhoff's Current Law (KCL) at the essential node whose voltage is V:

$$\frac{V - 100\angle 0^\circ}{6 + j8} + \frac{V}{j10} + \frac{V - 100\angle -60^\circ}{6 - j8} = 0$$

Grouping the coefficients of the node voltage variable V:

$$V\left(\frac{1}{6 + j8} + \frac{1}{j10} + \frac{1}{6 - j8}\right) = \frac{100\angle 0^\circ}{6 + j8} + \frac{100\angle -60^\circ}{6 - j8}$$

Simplifying the admittances inside the parenthesis by rationalizing or using common denominators:

$$V\left(\frac{6 - j8}{100} - \frac{j10}{100} + \frac{6 + j8}{100}\right) = 18.39\angle -30^\circ$$

$$V\left(\frac{12 - j10}{100}\right) = 18.39\angle -30^\circ$$

Solving for the complex node voltage value V:

$$V = 117.74\angle 9.805^\circ \text{ V}$$

Now, calculating the branch current I passing through the inductive element branch:

$$I = \frac{V}{j10} = \frac{117.74\angle 9.805^\circ}{10\angle 90^\circ}$$

$$I = 11.77\angle -80.19^\circ \text{ A}$$

$$\boxed{11.77\angle -80.19^\circ \text{ A}}$$

Q5.95.2 NAT 1995 1M Ans: 15 ● ○ ○

Considering ideal ammeters where the ammeter resistance is $0\,\Omega$ (i.e., there is no voltage drop across them), the network reduces to a parallel combination of an inductor and a resistor.

Because the branches are connected in parallel, the same reference voltage appears across both elements. The current through the resistive branch, I_{A2}, remains perfectly in phase with the voltage, while the current through the inductive branch, I_{A3}, lags behind the voltage by 90°. Therefore, the two branch currents I_{A2} and I_{A3} are in quadrature phase relationship.

Applying Kirchhoff's Current Law (KCL) in phasor form:

$$\vec{I}_{A1} = \vec{I}_{A2} + \vec{I}_{A3}$$

Since the two current vectors are perpendicular to each other (90° out of phase), the magnitude of the total entering current $|I_{A1}|$ is calculated using the orthogonal vector addition formula:

$$|I_{A1}| = \sqrt{|I_{A2}|^2 + |I_{A3}|^2}$$

Substituting the given values ($I_{A2} = 9$ A and $I_{A3} = 12$ A) into the equation:

$$|I_{A1}| = \sqrt{9^2 + 12^2}$$

$$|I_{A1}| = \sqrt{81 + 144} = \sqrt{225} = 15 \text{ A}$$

$$\boxed{15}$$

Q5.94.1 MCQ 1994 1M Ans: D ● ○ ○

The voltage across the inductive component can be determined directly by applying the complex potential divider rule across the series combination:

$$V_L = \frac{V_s}{(17.32 + j10)} \times j10$$

To simplify this expression, we convert both the numerator reactance and the denominator impedance terms from rectangular coordinates into polar coordinates:

* For the numerator:

$$j10 = 10\angle 90^\circ$$

* For the denominator:

$$\text{Magnitude} = \sqrt{17.32^2 + 10^2} = \sqrt{300 + 100} = \sqrt{400} = 20$$

$$\text{Phase Angle} = \tan^{-1}\left(\frac{10}{17.32}\right) = \tan^{-1}(0.577) = 30^\circ$$

$$17.32 + j10 = 20\angle 30^\circ$$

Substituting these polar representations back into the potential divider equation:

$$V_L = \frac{V_s}{20\angle 30^\circ} \times 10\angle 90^\circ$$

$$V_L = \frac{V_s}{2}\angle(90^\circ - 30^\circ)$$

$$V_L = \frac{V_s}{2}\angle 60^\circ$$

The resulting phase angle indicates that the inductor voltage phasor V_L leads the input source voltage phasor V_s by an angle of 60°.

Therefore, the correct choice is D.

D

Q5.93.1 MCQ | 1993 | 1M | Ans: D ● ● ○

To analyze the network containing multiple electrical excitation frequencies, we apply the superposition theorem. This allows us to compute the individual steady-state response contributions by considering one source active at a time.

1: Response due to the 5 V DC source For a pure DC excitation, the angular frequency component is:

$$\omega = 0\,\text{rad/s}$$

Since inductive reactance $X_L \propto \omega$ and capacitive reactance $X_c \propto \frac{1}{\omega}$, at zero frequency:

a. The inductor behaves as an ideal short circuit ($X_L = 0\,\Omega$).

b. The capacitor behaves as an ideal open circuit ($X_c \to \infty\,\Omega$).

Consequently, no alternating or direct continuous loop current flows through this reactive series path due to the isolating break introduced by the capacitor:

$$i_1(t) = 0\,\text{A}$$

2: Response due to the $10\sin t$ AC voltage source For this harmonic source configuration, the operating frequency parameter is:

$$\omega = 1\,\text{rad/s}$$

The individual branch reactive elements are converted to their respective frequency-domain expressions:

$$X_L = j\omega L = j2\,\Omega$$

$$X_c = \frac{-j}{\omega C} = \frac{-j}{(1)(1)} = -j\,\Omega$$

The total series input impedance Z presentation encountered by the AC source is:

$$Z = 1 + j2 - j = 1 + j1\,\Omega = \sqrt{2}\angle 45^\circ\,\Omega$$

Expressing the AC input voltage source excitation in phasor form using sine phase tracking:

$$\vec{V} = 10\angle 0^\circ\,\text{V}$$

The current phasor $\vec{I}$ driven through the loop is calculated via Ohm's law:

$$\vec{I} = \frac{\vec{V}}{Z} = \frac{10\angle 0^\circ}{\sqrt{2}\angle 45^\circ} = 7.07\angle -45^\circ\,\text{A}$$

Transforming this phasor back into its corresponding time-domain formulation yields:

$$i_2(t) = 7.07\sin(t - 45^\circ)\,\text{A}$$

The net time-varying current profile $i(t)$ is the algebraic sum of the separate contributions:

$$i(t) = i_1(t) + i_2(t) = 0 + 7.07\sin(t - 45^\circ)\,\text{A}$$

$$i(t) = 7.07\sin(t - 45^\circ)\,\text{A}$$

Mistakes to be avoided

- Never compute a combined equivalent impedance by mixing or adding different operational frequencies (ω) directly.
- Ensure that the time-domain reference phase matches the initial function profile ($\sin t$) used during the polar phasor mapping stage.

$$\boxed{(d) 7.07 \sin(t - 45^\circ)}$$

Q5.91.1 Descriptive | 1991 | 2M | Ans: * | ●●○

The plot of admittance magnitude versus frequency on a logarithmic scale is shown in the figure in the problem.

The magnitude of admittance (Y) for individual ideal circuit elements is given by:

$$Y = \begin{cases} \frac{1}{R}, & \text{Resistor} \\ \frac{1}{\omega L}, & \text{Inductor} \\ \omega C, & \text{Capacitor} \end{cases}$$

Expressed in decibels (dB), the admittance relations become:

$$Y_{\text{dB}} = \begin{cases} -20 \log_{10}(R), & \text{Resistor} \\ -20 \log_{10}(\omega L), & \text{Inductor} \\ 20 \log_{10}(\omega C), & \text{Capacitor} \end{cases}$$

From the given asymptotic Bode plot of admittance:

- For $1 \leq \omega \leq 10$ rad/sec (corresponding to $\log_{10} \omega$ between 0 and 1), the admittance magnitude is constant (20 dB), indicating a purely resistive behavior.(R can be less that 0, making Y positive).
- For $10 \leq \omega \leq 1000$ rad/sec (corresponding to $\log_{10} \omega$ between 1 and 3), the curve exhibits a negative slope of -20 dB/decade, which signifies that the inductive admittance dominates.
- For $\omega \geq 1000$ rad/sec (corresponding to $\log_{10} \omega \geq 3$), the curve exhibits a positive slope of +20 dB/decade passing through 0 dB at $\log_{10} \omega = 4$, indicating capacitive admittance dominance.

To find the inductance (L), evaluate the inductive branch admittance at the corner frequency where it begins, $\omega = 10$ rad/sec:

$$-20 \log_{10}(10L) = 20 \text{ dB}$$
$$\log_{10}(10L) = -1$$
$$10L = 10^{-1} = 0.1$$
$$L = 0.01 \text{ H}$$

To find the capacitance (C), evaluate the capacitive branch admittance at its zero-crossing frequency, $\omega = 10^4$ rad/sec (where $\log_{10} \omega = 4$):

$$20 \log_{10}(10^4 C) = 0 \text{ dB}$$
$$\log_{10}(10^4 C) = 0$$
$$10^4 C = 10^0 = 1$$
$$C = \frac{1}{10^4} = 100 \, \mu\text{F}$$

Frequency Range	Type of Impedance	Value (Ω/H/F)
10000 rad/sec $< \omega$	Capacitive	100 μF
10 rad/sec $< \omega <$ 1000 rad/sec	Inductive	0.01 H

Q5.91.2 MCQ | 1991 | 2M | Ans: B

Considering ideal ammeters where the ammeter resistance is $0\,\Omega$ (i.e., there is no voltage drop across them), the network reduces to a parallel combination of an capacitor and a resistor.

Because the branches are connected in parallel, the same reference voltage appears across both elements. The current through the resistive branch, I_{A1}, remains perfectly in phase with the voltage, while the current through the capacitive branch, I_{A2}, leads the voltage by 90°. Therefore, the two branch currents I_{A1} and I_{A2} are in quadrature phase relationship.

Applying Kirchhoff's Current Law (KCL) in phasor form:

$$\vec{I}_{A3} = \vec{I}_{A1} + \vec{I}_{A2}$$

Since the two current vectors are perpendicular to each other (90° out of phase), the magnitude of the total entering current $|I_{A1}|$ is calculated using the orthogonal vector addition formula:

$$|I_{A3}|^2 = |I_{A1}|^2 + |I_{A2}|^2$$

$$|I_{A2}| = \sqrt{|I_{A3}|^2 - |I_{A1}|^2}$$

Substituting the given values ($I_{A1} = 5$ A and $I_{A3} = 13$ A) into the equation:

$$|I_{A2}| = \sqrt{13^2 - 5^2}$$

$$|I_{A2}| = 12\,\text{A}$$

$$\boxed{(b) 12A}$$

PrepFusion

SOLUTIONS

VI

Resonance

Topics

1. Series RLC Resonance Circuit
2. Quality Factor and Stored Energy in Series Resonance
3. Parallel RLC Resonance Circuit

Unit Snapshot*

Stream: EE	**Questions:** 18	**MCQ:** 14	**MSQ:** 0	**NAT:** 4
1M: 11	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–2M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q6.25.1 MCQ 2025 1M Ans: B ● ○ ○

Given the parameters of the practical inductor coil: Inductance of the coil: $L = 100\,\text{mH} = 100 \times 10^{-3}\,\text{H} = 0.1\,\text{H}$, series resistance of the coil: $R = 100\,\Omega$,supply voltage: $V = 20\,\text{V}$ and operating frequency: $f = 1\,\text{kHz} = 1000\,\text{Hz}$

The quality factor (Q-factor) of an inductive coil is defined as the ratio of its inductive reactance to its series resistance at a given operating frequency:

$$Q = \frac{X_L}{R}$$

First, let us calculate the inductive reactance X_L of the coil:

$$X_L = \omega L = 2\pi f L$$

Substituting the given values into the reactance formula:

$$X_L = 2\pi \times 1000 \times 0.1 = 200\pi\,\Omega$$

Now, calculating the quality factor Q:

$$Q = \frac{200\pi}{100} = 2\pi$$

Hence, the quality factor of the coil at 1 kHz is 2π.

$$\boxed{\text{(b)}2\pi}$$

Q6.22.1 NAT 2022 1M Ans: 48 to 48 ● ○ ○

When a practical coil and a practical capacitor are connected in series, each component contributes its own internal series resistance to the total network configuration.

Let Q_1 be the quality factor of the coil, Q_2 be the quality factor of the capacitor and Q be the overall quality factor of the series resonant circuit.

The relationship between individual quality factors and the overall quality factor for a series combination is given by the reciprocal addition formula:

$$\frac{1}{Q} = \frac{1}{Q_1} + \frac{1}{Q_2}$$

Given the individual quality factors from the problem data: $Q_1 = 60$ and $Q_2 = 240$

Substituting these values into the expression to find the overall system Q:

$$\frac{1}{Q} = \frac{1}{60} + \frac{1}{240}$$

Finding a common denominator (240):

$$\frac{1}{Q} = \frac{4}{240} + \frac{1}{240} = \frac{5}{240}$$

Taking the reciprocal to isolate Q:

$$Q = \frac{240}{5} = 48$$

$$\boxed{48}$$

Key Points

- Overall Q-factor: For series components, the total equivalent loss resistance is the sum of individual resistances ($R_{\text{total}} = R_1 + R_2$). Since Q is inversely proportional to resistance in a series circuit ($Q = \frac{X}{R}$), their quality factors combine reciprocally: $\frac{1}{Q} = \frac{1}{Q_1} + \frac{1}{Q_2}$.
- System Performance: The overall quality factor of a series system will always be lower than the smallest individual

component's quality factor due to accumulated series resistive losses.

Q6.22.2 NAT | 2022 | 1M | Ans: 250 to 250 ● ○ ○

The Quality Factor (Q-factor) of a resonant circuit can be defined in terms of its frequency selective bandwidth characteristics. It is the ratio of the resonant frequency to the frequency bandwidth of the circuit:

$$Q = \frac{f_r}{\text{BW}}$$

Given parameters from the problem data: Resonant frequency: $f_r = 150 \times 10^3\,\text{Hz} = 150\,\text{kHz}$ and bandwidth of the circuit: $\text{BW} = 600\,\text{Hz}$

Substituting these parameters directly into the definition equation to find the circuit selectivity profile:

$$Q = \frac{150 \times 10^3}{600}$$

$$Q = \frac{150000}{600} = 250$$

$$\boxed{250}$$

Q6.16.1 NAT | 2016 | 2M | Ans: 14.0 to 14.2 ● ● ○

The given parallel network consists of two branches connected across an AC voltage source. The first branch contains a series combination of a capacitor $C = 100\,\mu\text{F}$ and a resistor R. The second branch contains a series combination of an inductor L=0.02 H and a resistor R.

The admittance of the capacitive branch (Y_1) is given by:

$$Y_1 = \frac{1}{R - \frac{j}{\omega C}} = \frac{\omega C}{\omega CR - j} = \frac{\omega C(\omega CR + j)}{1 + (\omega CR)^2}$$

$$Y_1 = \frac{\omega^2 C^2 R}{1 + (\omega CR)^2} + j\frac{\omega C}{1 + (\omega CR)^2}$$

The admittance of the inductive branch (Y_2) is given by:

$$Y_2 = \frac{1}{R + j\omega L} = \frac{R - j\omega L}{R^2 + (\omega L)^2}$$

$$Y_2 = \frac{R}{R^2 + (\omega L)^2} - j\frac{\omega L}{R^2 + (\omega L)^2}$$

The total equivalent admittance of the parallel combination is $Y = Y_1 + Y_2$. For the network response to have only a real component (meaning the total current is in phase with the source voltage), the net imaginary part of the total admittance must equal zero:

$$\text{Im}\{Y\} = \text{Im}\{Y_1\} + \text{Im}\{Y_2\} = 0$$

$$\frac{\omega C}{1 + (\omega CR)^2} - \frac{\omega L}{R^2 + (\omega L)^2} = 0$$

$$\frac{C}{1 + (\omega CR)^2} = \frac{L}{R^2 + (\omega L)^2}$$

Cross-multiplying to solve the algebraic equation:

$$C\left[R^2 + (\omega L)^2\right] = L\left[1 + (\omega CR)^2\right]$$

$$CR^2 + \omega^2 L^2 C = L + \omega^2 L C^2 R^2$$
$$CR^2 - L + \omega^2 L^2 C - \omega^2 L C^2 R^2 = 0$$
$$(CR^2 - L) + \omega^2 LC(L - CR^2) = 0$$
$$(CR^2 - L)(1 - \omega^2 LC) = 0$$

This expression yields two valid operational conditions under which the parallel network can achieve zero net imaginary admittance:

Condition 1: Frequency-independent resonance

$$CR^2 - L = 0 \implies R^2 = \frac{L}{C} \implies R = \sqrt{\frac{L}{C}}$$

Condition 2: Frequency-dependent resonance

$$1 - \omega^2 LC = 0 \implies \omega = \frac{1}{\sqrt{LC}}$$

Given that the problem defines the operational state as working at all frequencies, Condition 1 must be satisfied. Substituting the given network values ($C = 100\,\mu$F and $L = 0.02$ H):

$$R = \sqrt{\frac{0.02}{100 \times 10^{-6}}} = 10\sqrt{2}\Omega$$
$$\therefore, R = 14.1\Omega$$

$$\boxed{14.0 \textit{ to } 14.2}$$

Key Points

- **Frequency-Independent Resonance:** A parallel network consisting of branches with balanced RC and RL structures can exhibit a unity power factor at all operational frequencies if the condition $R = \sqrt{L/C}$ is strictly met. The time constant of each branch must be equal.

Q6.13.1 MCQ | 2013 | 2M | Ans: C | ● ○ ○

For Coil 1, the quality factor is given by:

$$q_1 = \frac{\omega L_1}{R_1}$$

Rearranging for the self-inductance of Coil 1:

$$L_1 = \frac{q_1 R_1}{\omega}$$

Similarly, for Coil 2, the quality factor is:

$$q_2 = \frac{\omega L_2}{R_2}$$

Rearranging for the self-inductance of Coil 2:

$$L_2 = \frac{q_2 R_2}{\omega}$$

When the two coils are connected in series, the equivalent parameters are:

- Equivalent inductance: $L_{eq} = L_1 + L_2$
- Equivalent resistance: $R_{eq} = R_1 + R_2$

The overall quality factor Q of the series combination is defined as:

$$Q = \frac{\omega(L_1 + L_2)}{R_1 + R_2}$$

Substituting the expressions for L_1 and L_2 into the equation:

$$Q = \frac{\omega\left[\frac{q_1 R_1}{\omega} + \frac{q_2 R_2}{\omega}\right]}{R_1 + R_2}$$

Simplifying the expression by cancelling ω:

$$\boxed{Q = \frac{q_1 R_1 + q_2 R_2}{R_1 + R_2}}$$

$$\boxed{(c)(q_1 R_1 + q_2 R_2)/(R_1 + R_2)}$$

Q6.08.1 MCQ 2008 1M Ans: C ●○○

Let the operating angular frequency be ω_r, which denotes the resonant frequency of the circuit.

The complex reactances of the individual elements are given by:

- For the 0.1 H inductor:

$$Z_L = j\omega_r(0.1) = \frac{j\omega_r}{10}$$

- For the 1 F capacitor:

$$Z_C = \frac{-j}{\omega_r(1)} = \frac{-j}{\omega_r}$$

The total input impedance Z_{in} is calculated from the series-parallel combination matching the network structure:

$$Z_{in} = \frac{j\omega_r}{10} + \left[1 \parallel \left(\frac{-j}{\omega_r}\right)\right]$$

Evaluating the parallel combination block:

$$1 \parallel \left(\frac{-j}{\omega_r}\right) = \frac{1 \times \left(-\frac{j}{\omega_r}\right)}{1 - \frac{j}{\omega_r}} = \frac{-j}{\omega_r - j}$$

To rationalize the denominator, multiply the numerator and the denominator by the complex conjugate $(\omega_r + j)$:

$$\frac{-j(\omega_r + j)}{(\omega_r - j)(\omega_r + j)} = \frac{-j\omega_r + 1}{\omega_r^2 + 1} = \frac{1}{\omega_r^2 + 1} - \frac{j\omega_r}{\omega_r^2 + 1}$$

Substituting this back into the total input impedance expression:

$$Z_{in} = \frac{j\omega_r}{10} + \frac{1}{\omega_r^2 + 1} - \frac{j\omega_r}{\omega_r^2 + 1}$$

Grouping the real and imaginary parts together:

$$Z_{in} = \frac{1}{\omega_r^2 + 1} + j\left(\frac{\omega_r}{10} - \frac{\omega_r}{\omega_r^2 + 1}\right)$$

At resonance, the imaginary part of the input impedance must equal zero ($\text{Im}(Z_{in}) = 0$):

$$\frac{\omega_r}{10} - \frac{\omega_r}{\omega_r^2 + 1} = 0$$

$$\frac{\omega_r}{10} = \frac{\omega_r}{\omega_r^2 + 1}$$

This equation yields two possible mathematical roots:

- $\omega_r = 0$, which is practically invalid for an active alternating system.
- $10 = \omega_r^2 + 1$

Solving for the non-zero resonant frequency root:

$$\omega_r^2 = 9 \implies \omega_r = \pm 3\,\text{rad/s}$$

Since frequency must be a positive quantity:

$$\omega_r = 3$$

$$\boxed{(c) 3 rad/s}$$

Q6.04.1 MCQ | 2004 | 2M | Ans: D | ● ○ ○

For resonance to happen the imaginary part of the equivalent impedance or admittance must be 0. Clearly the impedance Z must be capacitive in nature in order to cancel out the effect of the inductance.

The total admittance Z of the network is given by:

$$Z = R + \frac{1}{j\omega C} || j\omega L$$

The admittance of the parallel part is :

$$Y = \frac{1}{j\omega L} + j\omega C$$

For the imaginary part of the admittance to equal zero:

$$\omega C - \frac{1}{\omega L} = 0$$

Solving for capacitance C:

$$C = \frac{1}{\omega^2 L}$$

Given parameters from the circuit problem: Frequency, $f = 500\,\text{Hz}$ Inductance, $L = 2\,\text{H}$ Angular frequency, $\omega = 2\pi f = 2\pi \times 500\,\text{rad/s}$

Substituting these values into the capacitance equation:

$$C = \frac{1}{(2\pi \times 500)^2 \times 2}$$

$$C = 5.066 \times 10^{-8}\,\text{F}$$

$$C = 0.05\,\mu\text{F}$$

$$\boxed{(d) 0.05\,\mu\text{F}}$$

Q6.03.1 MCQ 2003 2M Ans: * ● ○ ○

The series current is common to R, L and C. The given magnitudes imply $\omega L = 1/(\omega C)$ (resonance) and

$$\omega L = 2R = 10\ \Omega.$$

Hence $L = 10/\omega$ H. The supply phasor $5\angle 0°$ specifies its magnitude and reference phase, not frequency. Neither the frequency nor C is supplied, so L cannot be uniquely determined.

For example, *if* f were 50 Hz, L would be approximately 31.83 mH. This is only a conditional example; silently assuming mains frequency does not establish an answer to the printed question.

Missing-data qualification: A frequency or capacitance is needed to determine a unique numerical resonant inductance.

Q6.02.1 MCQ 2002 2M Ans: B ● ● ○

For a series R L C circuit, the alternating current magnitude is defined by:

$$I = \frac{V}{R + j(X_L - X_C)} = \frac{V}{R + j\left(\omega L - \frac{1}{\omega C}\right)}$$

Power dissipated in the circuit is proportional to the square of the current amplitude:

$$P \propto I^2$$

A half power frequency implies a frequency stage at which the current drops to $1/\sqrt{2}$ of its maximum possible value. Maximum current occurs exactly at the circuit resonance frequency:

$$I_{max} = \frac{V}{R}$$

Given parameters from the problem system:

$$\text{Resonant frequency, } \omega_o = \frac{1}{\sqrt{LC}} = 10^5 \text{ rad/s}$$

$$\text{Bandwidth, } B = \frac{R}{L} = 50 \times 10^3 \text{ rad/s}$$

Equating the current magnitude equation to the half power condition:

$$\left|\frac{V}{R + j\left(\omega L - \frac{1}{\omega C}\right)}\right| = \frac{V}{\sqrt{2}R}$$

$$R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2 = 2R^2$$

$$\left(\omega L - \frac{1}{\omega C}\right)^2 = R^2$$

$$\omega L - \frac{1}{\omega C} = \pm R$$

Case 1: Considering the upper frequency branch where $\omega L - \frac{1}{\omega C} = R$

$$\omega^2 LC - \omega RC - 1 = 0$$

Dividing the expressions throughout by LC:

$$\omega^2 - \frac{\omega R}{L} - \frac{1}{LC} = 0$$

$$\omega^2 - \omega B - \omega_o^2 = 0$$

Solving this quadratic expression for the upper root ω_1:

$$\omega_1 = \frac{B \pm \sqrt{B^2 + 4\omega_o^2}}{2} = \frac{B}{2} + \sqrt{\left(\frac{B}{2}\right)^2 + \omega_o^2}$$

The second root is disregarded since a negative frequency $\omega < 0$ is physically impossible.

Case 2: Considering the lower frequency branch where $\omega L - \frac{1}{\omega C} = -R$

$$\omega^2 LC + \omega RC - 1 = 0$$

$$\omega^2 + \omega B - \omega_o^2 = 0$$

Solving this quadratic expression for the lower root ω_2:

$$\omega_2 = \frac{-B + \sqrt{B^2 + 4\omega_o^2}}{2} = \sqrt{\left(\frac{B}{2}\right)^2 + \omega_o^2} - \frac{B}{2}$$

The alternate solution here is also ignored because $\omega < 0$ is not possible.

Comparing both derivations clearly indicates that $\omega_2 < \omega_1$. Substituting the operational data into the lower boundary frequency equation:

$$\omega_2 = 19.258\,\text{krad/s}$$

Converting the angular frequency down to cyclic linear frequency format:

$$f_{low} = \frac{\omega_2}{2\pi} = 3.055\,\text{kHz}$$

$$\boxed{(b) 3.055 kHz}$$

Q6.02.2 MCQ 2002 2M Ans: A ●○○

Unity power factor implies that the overall network voltage and current are in phase, which corresponds to the condition of resonance.

Thus, we need to make the imaginary part of the total impedance (or admittance) equal to zero. For this problem, since we have a parallel combination of elements, we calculate the equivalent admittance Y and set $\text{Im}(Y) = 0$.

The operational frequency parameters are:

$$\omega = 2\pi \times 50 = 314\,\text{rad/s}$$

The admittance of the capacitive branch Y_c is:

$$Y_c = +j\omega C = +j314C$$

The admittance of the inductive branch Y_l is derived from its branch impedance Z_l:

$$Y_l = \frac{1}{Z_l} = \frac{1}{30\angle 40^\circ} = \frac{1}{30}\angle -40^\circ$$

Converting Y_l into rectangular form:

$$Y_l = \frac{1}{30}\left(\cos(-40^\circ) + j\sin(-40^\circ)\right) \approx 0.0255 - j0.0214$$

The total equivalent network admittance Y is the sum of the parallel branch admittances:

$$Y = Y_c + Y_l$$

$$Y = j314C + 0.0255 - j0.0214$$

$$Y = 0.0255 + j(314C - 0.0214)$$

To satisfy the unity power factor resonance condition, set the imaginary component to zero:

$$\text{Im}(Y) = 314C - 0.0214 = 0$$

$$314C = 0.0214$$

Solving for the required capacitance C:

$$C = \frac{0.0214}{314}$$

$$C = 6.81 \times 10^{-5}\,\text{F}$$

$$C = 68.1\,\mu\text{F}$$

Wrapping the final calculated value to match standard rounding presentation:

$$\boxed{(a)68.1\,\mu\text{F}}$$

Q6.01.1 MCQ | 2001 | 1M | Ans: C | ● ○ ○

In a series RLC circuit we know that at resonance,'

$$|VL| = |V_C| = Q|V_R|$$

where Q is the quality factor which can also be greater than 1.

Also at resonance,

$$V_R = V_{in}$$

Hence we can conclude that it is possible for $|V_C|$ to be greater than the input voltage.

In a series RLC circuit current flowing through all 3 network elements are equal, say I. The current I is equal to V_{in}/R at resonance and is in phase with input voltage. Thus we can write

$$V_C = I \times -jX_C$$

Since V_C lags I by 90°, we can say that it is 90° out of phase with the input voltage as well.

$$\boxed{(c)\text{can be greater than the input voltage, however it is } 90^\circ \text{ out of phase with the input voltage}}$$

Q6.99.1 MCQ | 1999 | 1M | Ans: D | ● ○ ○

Given parameters from the problem: Resonant frequency:

$$f_0 = 100\,\text{Hz}$$

Bandwidth:

$$B = 5\,\text{Hz}$$

At resonance, the inductive reactance and capacitive reactance cancel each other out ($X_L = X_C$). Consequently, the net impedance of the circuit is purely resistive, and the entire source voltage appears across the resistor:

$$|V_R| = 10\,\text{V}$$

Let the voltage across the inductor be V_L. The magnitude of the voltage across the inductor at resonance is amplified by the quality factor (Q) of the circuit:

$$|V_L| = Q|V_R|$$

The quality factor Q in terms of angular resonant frequency ω_0 and circuit parameters is given by:

$$Q = \frac{\omega_0 L}{R}$$

We know that the relation between angular frequency and cyclic frequency is $\omega_0 = 2\pi f_0$, and the bandwidth in radians per second is given by $\mathcal{B} = \frac{R}{L}$ rad/s (or $B = \frac{R}{2\pi L}$ in Hz). Therefore, the quality factor can be rewritten in terms of frequency and bandwidth as:

$$Q = \frac{2\pi f_0}{2\pi B} = \frac{f_0}{B}$$

$$Q = \frac{100}{5} = 20$$

Substituting the value of Q back into the inductor voltage equation:

$$|V_L| = 20 \times 10\,\text{V} = 200\,\text{V}$$

The voltage across the inductor remains sinusoidal. Depending on whether the given source voltage is specified as a peak or root-mean-square (RMS) value, this calculated value represents the corresponding peak or RMS magnitude of the inductor voltage.

Key Points

- Remember that the voltage across the inductor at resonance can be significantly higher than the source voltage, a phenomenon known as voltage magnification.

Mistakes to be avoided

- Ensure that both the resonant frequency and the bandwidth are in the same units (either both in Hz or both in rad/s) before evaluating their ratio for Q.

$\boxed{(d) 200V}$

Q6.99.2 MCQ 1999 1M Ans: A ●○○

Since the magnitude of the voltage across the inductor equals the magnitude of the voltage across the capacitor:

$$|V_L| = |V_C| = 200\,\text{V}$$

This condition specifies that the circuit is operating at electrical resonance. Consequently, the net reactive components cancel each other out, and the network exhibits a purely resistive behavior.
The circuit current I can be calculated using Ohm's law with the total supply voltage V and the circuit resistance R:

$$I = \frac{V}{R} = \frac{100}{20} = 5\,\text{A}$$

Since the alternating voltage and current waveforms are sinusoidal, this obtained current value represents either the peak or root-mean-square (RMS) magnitude, depending directly on the definition of the given source parameters.

$\boxed{(a) 5A}$

Q6.98.1 MCQ 1998 1M Ans: D ● ○ ○

Let the initial values of the circuit elements in the resonant circuit be R, L, and C.
The initial resonant frequency f_0 is given by the formula:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Now, the values of the components are changed to $2R$, $2L$, and $2C$. The new resonant frequency f_{01} is calculated by substituting the new parameters into the frequency formula:

$$f_{01} = \frac{1}{2\pi\sqrt{(2L)\cdot(2C)}}$$

$$f_{01} = \frac{1}{2\pi\sqrt{4LC}}$$

$$f_{01} = \frac{1}{2\cdot(2\pi\sqrt{LC})}$$

Substituting the expression for the initial resonant frequency (f_0) into the equation:

$$f_{01} = \frac{f_0}{2}$$

Key Points

- Variations in the circuit resistance R do not alter the ideal resonant frequency value, though it affects the quality factor and damping ratio.

$$\boxed{(d)\ \frac{f_0}{2}}$$

Q6.96.1 MCQ 1996 2M Ans: D ● ● ○

The Quality factor of a series RL circuit is defined as:

$$Q = \frac{\omega L}{R}$$

When the operating frequency is doubled, the new angular frequency becomes $\omega_1 = 2\omega$. The modified quality factor Q_1 is given by:

$$Q_1 = \frac{\omega_1 L}{R} = \frac{2\omega L}{R} = 2Q$$

Hence, the quality factor Q is doubled.
The active power P dissipated by the series circuit is given by:

$$P = I^2 R$$

The magnitude of the current I in a series RL circuit connected to a voltage source V is expressed as:

$$I = \frac{V}{\sqrt{R^2 + (\omega L)^2}}$$

Substituting the expression for current I into the active power formula:

$$P = \left[\frac{V}{\sqrt{R^2 + (\omega L)^2}}\right]^2 \times R = \frac{V^2 \cdot R}{R^2 + (\omega L)^2}$$

Factoring out R^2 from the denominator:

$$P = \frac{V^2 \cdot R}{R^2\left[1 + \left(\frac{\omega L}{R}\right)^2\right]} = \frac{V^2}{R\left[1 + \left(\frac{\omega L}{R}\right)^2\right]}$$

Since $Q = \frac{\omega L}{R}$, the equation simplifies to:

$$P = \frac{V^2}{R(1 + Q^2)}$$

Given that the coil has a high quality factor ($Q^2 \gg 1$), we can approximate $1 + Q^2 \approx Q^2$:

$$P \approx \frac{V^2}{R \cdot Q^2}$$

When the frequency is doubled, the new active power P_1 becomes:

$$P_1 = \frac{V^2}{R \cdot (Q_1)^2}$$

Substituting $Q_1 = 2Q$ into the relation:

$$P_1 = \frac{V^2}{R \cdot (2Q)^2} = \frac{V^2}{4R \cdot Q^2}$$

$$P_1 = \frac{P}{4}$$

Therefore, the active power P decreases by 4 times.

(d)P is decreased 4 times, Q is doubled

Key Points

- For a high-Q coil ($Q \gg 1$), the real power consumption is inversely proportional to the square of the quality factor ($P \propto \frac{1}{Q^2}$).

Mistakes to be avoided

- Do not assume active power is independent of frequency. Even though $P = I^2R$, the current I depends heavily on the frequency-dependent inductive reactance $X_L = \omega L$.

Q6.95.1 NAT | 1995 | 1M | Ans: 0.03162 | ● ○ ○

At electrical resonance, the resonant angular frequency ω_0 is defined as:

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

The Quality factor (Q-factor) for a series RLC network is expressed as:

$$Q = \frac{\omega_0 L}{R}$$

Substituting the expression for ω_0 into the Q-factor equation:

$$Q = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R}$$

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}}$$

Substituting the given component parameters ($R = 10\,\Omega$, $L = 0.01$ H, and $C = 100 \times 10^{-3}$ F):

$$Q = \frac{1}{10}\sqrt{\frac{0.01}{100 \times 10^{-3}}} = 0.03162$$

$$\boxed{0.03162}$$

Q6.94.1 MCQ | 1994 | 1M | Ans: D | ● ○ ○

The total admittance (Y) of the parallel resonant circuit is given by the sum of the admittances of the inductive branch and the capacitive branch:

$$Y = \frac{1}{R + j\omega L} + j\omega C$$

$$\text{or, } Y = \frac{1}{R + j\omega L} \times \frac{R - j\omega L}{R - j\omega L} + j\omega C$$

Rationalizing the first term :

$$Y = \frac{R - j\omega L}{R^2 + (\omega L)^2} + j\omega C$$

Separating the real and imaginary parts:

$$Y = \frac{R}{R^2 + (\omega L)^2} + j\left(\omega C - \frac{\omega L}{R^2 + (\omega L)^2}\right)$$

$$\frac{\omega L}{R^2 + (\omega L)^2} = \omega C$$

From this condition, we can solve for the resonant frequency ω_0.
At resonance ($\omega = \omega_0$), the admittance becomes purely real:

$$Y = \frac{R}{R^2 + \omega_0^2 L^2}$$

The equivalent impedance (Z) at resonance is the reciprocal of the admittance (Y):

$$Z = \frac{1}{Y} = \frac{R^2 + \omega_0^2 L^2}{R}$$

Therefore, at parallel resonance, the impedance is maximum and is significantly greater than the resistance:

$$Z \gg R$$

$$\boxed{(d) \textit{A pure resistor of value much higher than } R}$$

Q6.93.1 MCQ | 1993 | 1M | Ans: B | ● ○ ○

The expression for the equivalent input impedance Z of the given network is:

$$Z = 10 + \left(j4\omega - \frac{j}{\omega}\right) \parallel \left(-\frac{j}{\omega}\right)$$

Simplifying the parallel reactive combination:

$$\left(j4\omega - \frac{j}{\omega}\right) \parallel \left(-\frac{j}{\omega}\right) = \frac{\left(j4\omega - \frac{j}{\omega}\right)\left(-\frac{j}{\omega}\right)}{\left(j4\omega - \frac{j}{\omega}\right) + \left(-\frac{j}{\omega}\right)}$$

Evaluating the numerator product:

$$\left(j4\omega - \frac{j}{\omega}\right)\left(-\frac{j}{\omega}\right) = \frac{4\omega}{\omega} - \frac{1}{\omega^2} = 4 - \frac{1}{\omega^2}$$

Evaluating the denominator sum:

$$j4\omega - \frac{j}{\omega} - \frac{j}{\omega} = j\left(4\omega - \frac{2}{\omega}\right)$$

Substituting these components back into the parallel impedance equation yields:

$$Z = 10 + \frac{4 - \frac{1}{\omega^2}}{j\left(4\omega - \frac{2}{\omega}\right)}$$

Bringing the complex imaginary unit j from the denominator to the numerator changes its sign:

$$Z = 10 - j\left[\frac{4 - \frac{1}{\omega^2}}{4\omega - \frac{2}{\omega}}\right]$$

For the circuit to be operating at electrical resonance, the net reactive component or the imaginary part of the total network impedance Z must be identically equal to zero:

$$\text{Im}(Z) = 0$$

Setting the numerator of the imaginary term equal to zero:

$$4 - \frac{1}{\omega_{\text{res}}^2} = 0$$

$$\frac{1}{\omega_{\text{res}}^2} = 4$$

$$\omega_{\text{res}}^2 = \frac{1}{4} = 0.25 \implies \omega_{\text{res}} = 0.5\,\text{rad/sec}$$

$$\boxed{(b) 0.5 rad/s}$$

SOLUTIONS

VII

Complex Power

Topics

1. Introduction to Power in AC Circuits
2. Active, Reactive and Apparent Power in Complex Networks

Unit Snapshot*

Stream: EE	**Questions:** 20	**MCQ:** 14	**MSQ:** 0	**NAT:** 6
1M: 10	**2M:** 10	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–7M(2012)

**See preface for how Avg Weightage and Range are calculated.*

PrepFusion

Q7.26.1 MCQ 2026 1M Ans: B ● ○ ○

To find the load power factor in a non-sinusoidal system containing harmonics, we calculate the ratio of total average power (P) to total apparent power (S).

Given the instantaneous voltage and current expressions:

$$v(t) = 325\sin(2\pi \cdot 50\,t)\ \text{V}$$

$$i(t) = 12\sin(2\pi \cdot 50\,t) + 9\sin(2\pi \cdot 150\,t)\ \text{A}$$

1. Calculation of average Power (P): Average power is contributed only by the voltage and current components of matching harmonic frequencies. Here, the voltage only contains the fundamental frequency component ($n = 1$ at 50 Hz).

$$P = \frac{V_{m1}I_{m1}}{2}\cos(\theta_1)$$

Since the phase difference θ_1 between the fundamental voltage and fundamental current is 0°:

$$P = \frac{325 \times 12}{2}\cos(0^\circ) = 1950\ \text{W}$$

2. Calculation of Apparent Power (S): Apparent power is defined as the product of the total RMS voltage and total RMS current of the system:

$$S = V_{\text{rms}}I_{\text{rms}}$$

The RMS value of the single-frequency sinusoidal voltage is:

$$V_{\text{rms}} = \frac{325}{\sqrt{2}}\ \text{V}$$

The total RMS value of the non-sinusoidal periodic current containing fundamental and third harmonic (150 Hz) components is:

$$I_{\text{rms}} = \sqrt{I_{\text{rms1}}^2 + I_{\text{rms3}}^2} = \sqrt{\left(\frac{12}{\sqrt{2}}\right)^2 + \left(\frac{9}{\sqrt{2}}\right)^2} = \sqrt{\frac{12^2 + 9^2}{2}}$$

Calculating the total apparent power (S):

$$S = \frac{325}{\sqrt{2}} \times \sqrt{\frac{144 + 81}{2}} = \frac{325}{\sqrt{2}} \times \frac{\sqrt{225}}{\sqrt{2}} = \frac{325 \times 15}{2} = 2437.5\ \text{VA}$$

3. Calculation of Load Power Factor (pf):**

$$\text{pf} = \frac{P}{S} = \frac{1950}{2437.5} = 0.8$$

$$\boxed{0.8}$$

Q7.26.2 MCQ 2026 2M Ans: C ● ○ ○

The mathematical derivation for the average active power (P_{avg}) in an AC circuit with a phase shift θ between the voltage and current waveforms is verified over a periodic cycle interval.

Given the alternating instantaneous system quantities:

$$V = V_m\sin(\omega t)$$

$$i = I_m\sin(\omega t - \theta)$$

The instantaneous power P expression is the direct product of the time-varying voltage and current profiles:

$$P = Vi = V_m I_m \sin(\omega t) \sin(\omega t - \theta)$$

Applying the product-to-sum trigonometric identity $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$, let $A = \omega t$ and $B = \omega t - \theta$:

$$P = \frac{V_m I_m}{2} [\cos(\omega t - (\omega t - \theta)) - \cos(\omega t + (\omega t - \theta))]$$

$$P = \frac{V_m I_m}{2} [\cos\theta - \cos(2\omega t - \theta)]$$

To find the average active power, we integrate the instantaneous power function over a half-cycle sequence interval $(0 \rightarrow \pi)$: The integration of the time-dependent harmonic component $\cos(2\omega t - \theta)$ over a symmetric periodic boundary interval equals zero:

$$\frac{1}{\pi} \int_0^{\pi} \cos(2\omega t - \theta)\, d(\omega t) = 0$$

Thus, only the time-invariant constant component remains after the averaging integration loop:

$$P_{\text{avg}} = \frac{V_m I_m}{2} \cos\theta$$

$$\boxed{(c)\ \frac{V_m I_m}{2} \cos\theta}$$

Q7.25.1 NAT | 2025 | 1M | Ans: 0.707 | ● ○ ○

From the information given:
The voltage phasor angle is oriented at: $\theta_V = -10°$ with an amplitude of 141.4 V
The current phasor angle is oriented at: $\theta_I = -55°$ with an amplitude of 0.7 A
The total phase angle difference ϕ between the two alternating quantities is:

$$\phi = |\theta_V - \theta_I|$$

$$\phi = |-10° - (-55°)| = 45°$$

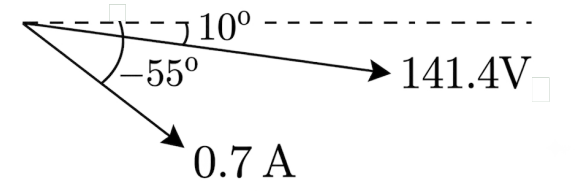

Fig. Solution Q7.25.1

Since the current phasor angle $(-55°)$ is more negative than the voltage phasor angle $(-10°)$, the circuit current lags behind the terminal voltage.

Calculating the power factor:

$$\text{pf} = \cos(\phi) = \cos(45°) \approx 0.707$$

Therefore, the power factor is 0.707 lagging.

$$\boxed{0.707}$$

Key Points

- **Power Factor Definition:** Power factor represents the scaling ratio of real power to apparent power, and is calculated directly via the cosine of the angular displacement: $\text{pf} = \cos(\theta_V - \theta_I)$.
- Lead and Lag: A lagging power factor explicitly indicates an inductive network state whereas a leading power factor explicitly indicates a capacitive network state.

Q7.19.1 NAT 2019 2M Ans: 585.0 to 590.0 ●●○

To find the average active power P delivered to the load, both the instantaneous current $i(t)$ and voltage $v(t)$ equations must be expressed in identical trigonometric base forms (such as sine waves) before extracting their relative phase angles.

Given the initial expressions:

$$i(t) = 8\cos\left(377t + \frac{\pi}{6}\right)\text{ A}$$

$$v(t) = -170\sin\left(377t - \frac{\pi}{6}\right)\text{ V}$$

First, we convert the current expression to a sine representation, using the identity $\cos(\theta) = \sin\left(\theta + \frac{\pi}{2}\right)$:

$$i(t) = 8\cos\left(377t + \frac{\pi}{6}\right) = 8\sin\left(377t + \frac{\pi}{6} + \frac{\pi}{2}\right)$$

$$i(t) = 8\sin\left(3\pi t + \frac{4\pi}{6}\right) = 8\sin\left(3\pi t + \frac{2\pi}{3}\right)\text{ A}$$

Then we absorb the negative sign of the voltage expression in the angle information :

$$v(t) = -170\sin\left(377t - \frac{\pi}{6}\right) = 170\sin\left(377t + \pi - \frac{\pi}{6}\right) = 170\sin\left(377t + \frac{5\pi}{6}\right)$$

In degrees, the phase angles of the voltage and current waveforms are:

Voltage phase angle: $\theta_V = \frac{5\pi}{6}\text{ rad} = 150^\circ$

Current phase angle: $\theta_I = \frac{2\pi}{3}\text{ rad} = 120^\circ$

The phase difference is :

$$\phi = \theta_V - \theta_I = 150^\circ - 120^\circ = 30^\circ$$

The average power consumed by an AC network branch is given by the formula:

$$P = V_{\text{rms}} I_{\text{rms}} \cos\phi$$

Substituting the maximum amplitudes ($V_m = 170\text{ V}$ and $I_m = 8\text{ A}$) transformed into root-mean-square values:

$$P = \frac{170}{\sqrt{2}} \times \frac{8}{\sqrt{2}} \times \cos(30^\circ)$$

$$P = \frac{170 \times 8}{2} \times \frac{\sqrt{3}}{2}$$

$$P = 170 \times 4 \times \frac{\sqrt{3}}{2} = 340\sqrt{3}\text{ W}$$

$$P \approx 340 \times 1.73205 = 588.9\text{ W}$$

$$\boxed{585.0 \textit{ to } 590.0}$$

Q7.18.1 MCQ | 2018 | 1M | Ans: C | ● ● ○

The direction of real power flow between two active AC voltage sources connected through an inductive line is primarily determined by the phase angle difference between the sources. Power flows from the leading voltage source to the lagging voltage source. If a source supplies power to the network, its active power P is considered positive ($P > 0$). Conversely, if a source absorbs power from the network, its active power P is considered negative ($P < 0$).

To determine the nature of the power (P_1, P_2, and P_3) for each voltage source, we need to analyze their respective phase angles.

Given the voltage expressions:

$$V_1 = 100\cos(\omega t)$$

$$V_2 = 100\cos\left(\omega t + \frac{\pi}{18}\right)$$

$$V_3 = 100\cos\left(\omega t + \frac{\pi}{36}\right)$$

Converting the phase angles from radians to degrees to easily compare them:

$$\theta_1 = 0^\circ$$

$$\theta_2 = \frac{\pi}{18} \times \frac{180^\circ}{\pi} = 10^\circ$$

$$\theta_3 = \frac{\pi}{36} \times \frac{180^\circ}{\pi} = 5^\circ$$

Comparing the phase angles shows that:

$$\theta_2 > \theta_3 > \theta_1$$

This phase relationship means that V_2 leads both V_1 and V_3. In an inductive transmission network, current and real power naturally flow away from the node with the leading phase angle toward the nodes with lagging phase angles.

Because V_2 possesses the highest phase angle, it acts as the primary generator delivering power to the rest of the circuit. Therefore, source V_2 delivers power, making its power positive:

$$P_2 > 0$$

Since V_1 and V_3 have lagging phase angles relative to V_2, they receive power from the network rather than delivering it. Consequently, their power values are negative:

$$P_1 < 0$$

$$P_3 < 0$$

Thus, the final power flow characteristics match the conditions:

Mistakes to be avoided

- Do not confuse the voltage magnitude with the phase angle when determining real power flow direction. Even though all three sources have identical peak magnitudes of 100 V, the power distribution is dictated entirely by their phase displacements.
- Ensure standard sign conventions are strictly followed. A negative power calculation signifies that the source is absorbing active power from the system rather than generating it.

$$\boxed{(c)\, P_1 < 0, P_2 > 0, P_3 < 0}$$

Q7.18.2 NAT 2018 2M Ans: 10.0 to 10.0 ● ○ ○

For the calculation of average power, only the voltage and current terms of the identical frequency contribute to the net power.

Pertaining to this problem, thus, the expression for average power is given by:

$$P = V_o I_o + \frac{V_1 I_1}{2} \cos(\theta_v - \theta_i)$$

Given the fundamental voltage component:

$$V_1 = -10\cos(\omega t + 60^\circ) = 10\cos(\omega t + 60^\circ + 180^\circ) = 10\cos(\omega t + 240^\circ)$$

The phase difference $\theta_v - \theta_i$ is thus 240°.

Given that the total average power $P = 0\,\text{W}$, and the DC components are $V_o = 5\,\text{V}$ and $I_o = 5\,\text{A}$:

$$P = V_o I_o + \frac{V_1 I_1}{2} \cos(\theta_v - \theta_i) = 0$$

$$5 \times 5 + \frac{10 \times X}{2} \cos(240^\circ) = 0$$

$$25 = 0$$

$$25 + 5X\cos(240^\circ) = 0$$

$$25 - \frac{5X}{2} = 0$$

$$25 = \frac{5X}{2}$$

$$X = 10$$

$$\boxed{10}$$

Q7.17.1 MCQ 2017 1M Ans: A ● ○ ○

The total instantaneous power in a multi-phase system is the sum of the instantaneous powers of the individual phases.

The equation for the total instantaneous power is given by:

$$P = V_{an} i_a + V_{bn} i_b$$

Substituting the given expressions for the phase voltages and phase currents into the equation:

$$P = 220\sin(100\pi t) \cdot 10\sin(100\pi t) + 220\cos(100\pi t) \cdot 10\cos(100\pi t)$$

Factoring out the common constant coefficient $220 \times 10 = 2200$:

$$P = 2200\left(\sin^2(100\pi t) + \cos^2(100\pi t)\right)$$

Using the fundamental trigonometric identity $\sin^2\theta + \cos^2\theta = 1$:

$$P = 2200 \times 1 = 2200\,\text{W}$$

Thus, the total instantaneous power is constant and independent of time.

$$\boxed{(a) 2200W}$$

Key Points

- In balanced multi-phase systems with appropriate phase shifts (such as the orthogonal sine and cosine relations shown here), the time-dependent fluctuating components cancel out, resulting in a constant total instantaneous power.

Q7.16.1 NAT 2016 1M Ans: 249.0 to 251.0 ●○○

For the calculation of average power, only the voltage and current terms of the identical frequency must be considered at a time. Since the voltage has only a single frequency component, we consider the component of that frequency in current (fundamental). The third and fifth harmonic components do not contribute to the average power.

The phase difference between the fundamental components of the voltage and current is as per the given expressions $\pi/6$ (current lags voltage).

$$P_{avg} = \frac{V_m I_m}{2} cos(\theta_v - \theta_i) = \frac{100 \times 10}{2} cos60^\circ = 250W$$

249.0 *to* 251.0

Q7.14.1 NAT 2014 2M Ans: 17.3 to 17.4 ●●○

An AC source supplies power to a network containing a combination of capacitive branches and a series-connected load containing an inductive reactance X_L and a resistor R.

The problem states that the total real power supplied by the AC source is $P_{total} = 1\,\text{kW} = 1000\,\text{W}$. Real power is consumed exclusively by the resistive elements of the network. The network includes a known resistor $R_1 = 1\,\Omega$ carrying a current of $I_1 = 2\,\text{A}$. The real power dissipation in this $1\,\Omega$ resistor is:

$$P_1 = I_1^2 \times R_1 = (2)^2 \times 1 = 4\,\text{W}$$

The remaining real power supplied by the source must be completely consumed by the load resistance R:

$$P_R = P_{total} - P_1 = 1000 - 4 = 996\,\text{W}$$

Assuming an ideal voltmeter (∞ resistance), no current flows through the voltmeter branch. Therefore, by applying Kirchhoff's Current Law (KCL) at the input parallel node, the entire current leaving the source branch (10 A) that does not trace through the parallel capacitive paths passes directly into the series load structure. Thus, the load current is:

$$I_L = 10\,\text{A}$$

The power dissipated inside the load resistor R is expressed as:

$$P_R = I_L^2 \times R$$

$$996 = (10)^2 \times R \implies 996 = 100 \times R$$

$$R = \frac{996}{100} = 9.96\,\Omega$$

The ideal voltmeter directly records the voltage magnitude drop across the load combination:

$$V_L = 200\,\text{V}$$

The total magnitude of the load impedance Z_L is:

$$|Z_L| = \frac{V_L}{I_L} = \frac{200}{10} = 20\,\Omega$$

Since the load structure consists of a series configuration of R and X_L, its total impedance magnitude satisfies the relationship:

$$|Z_L|^2 = R^2 + X_L^2$$

$$X_L = \sqrt{|Z_L|^2 - R^2}$$

Substituting the derived numerical metrics into the expression:

$$X_L = \sqrt{20^2 - (9.96)^2} = \sqrt{400 - 99.2016} = \sqrt{300.7984} \approx 17.34\,\Omega$$

$$\boxed{17.34}$$

Q7.13.1 MCQ 2013 1M Ans: B ● ● ○

We know that complex power S is given by :

$$S = VI^*$$

Here $V = 100\angle 60°V$ and $I = 10\angle -150°A$. Therefore complex power $S = 100\angle 60° \times 10\angle 150° = 1000\angle 210°$ VA Now, $S = P + jQ$ Thus, $1000\angle 210° = 1000\cos 210° + j1000\sin 210°$ From this we can conclude, $P = 1000\cos 210° = -866.025$ W and $Q = 1000\sin 210° = -500$ VAR According to convention when **current flows from load to source when the terminal where current enters load is taken as positive**, power > 0 indicates power is absorbed by load and power < 0 indicates power is delivered by load.

Here **current flows from load to source**. Since only one of the conditions is altered, the criteria to determine whether power is absorbed or delivered is reversed, i.e.,here power < 0 indicates power is absorbed by load and power > 0 indicates power is delivered by load.

Since both active and reactive powers are negative, both active and reactive powers are absorbed by load.

$$\boxed{(b)\textit{load absorbs real power and absorbs reactive power.}}$$

Q7.13.2 MCQ 2013 1M Ans: C ● ○ ○

According to maximum power transfer theorem, the purely resistive load should have a value equal to the magnitude of the internal impedance of the source in order to have maximum power transferred to it.

$$R_L = |Z_{in}| \implies R_L = \sqrt{4^2 + 3^2} = 5\Omega$$

$$\boxed{(c)5}$$

Q7.12.1 MCQ 2012 1M Ans: B ● ○ ○

It is given that $i(t) = 5cos(100\pi t + 100)$ A.

$$\therefore I_{rms} = \frac{5}{\sqrt{2}}\,\text{A}$$

Average power delivered $P = I_{rms}^2 R$, where R = 4Ω Substituting, we get $P = 50W$

$$\boxed{(b)50W}$$

Q7.12.2 MCQ 2012 2M Ans: A ● ● ○

First, we need to convert circuit B to its Thevenin equivalent form.

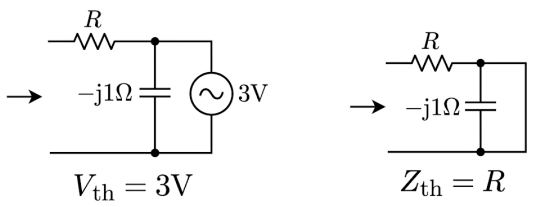

Fig. Solution Q7.12.2

The simplified circuit now looks like :

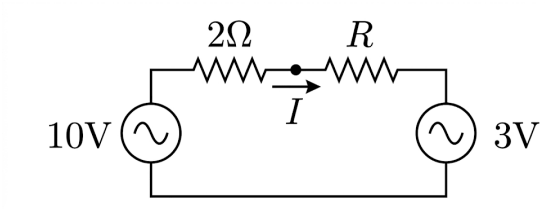

Fig. Solution Q7.12.2

The circuit current I as a function of load resistance R is expressed as:

$$I = \frac{10 - 3}{2 + R} = \frac{7}{2 + R}$$

The power transferred from circuit A to circuit B is given by the relation:

$$P = I^2R + 3I$$

Substituting the expression for current I:

$$P = I(IR + 3)$$

$$P = \left(\frac{7}{2 + R}\right)\left[\frac{7R}{2 + R} + 3\right]$$

$$P = \left(\frac{7}{2 + R}\right)\left[\frac{7R + 3(2 + R)}{2 + R}\right]$$

$$P = \left(\frac{7}{2 + R}\right)\left[\frac{7R + 6 + 3R}{2 + R}\right]$$

$$P = \frac{7(10R + 6)}{(2 + R)^2} = \frac{70R + 42}{(2 + R)^2}$$

To find the condition for maximum power transfer, we set the first derivative of power with respect to R to zero ($\frac{dP}{dR} = 0$):

$$\frac{dP}{dR} = \frac{(2 + R)^2 \cdot (70) - (70R + 42) \cdot 2(2 + R)}{(2 + R)^4} = 0$$

Factoring out $(2 + R)$ from the numerator (where $2 + R \neq 0$):

$$(2 + R)\left[70(2 + R) - 2(70R + 42)\right] = 0$$

$$140 + 70R - 140R - 84 = 0$$

$$56 - 70R = 0$$

$$R = \frac{56}{70} = 0.8\,\Omega$$

Since a negative solution ($R = -2\,\Omega$) is physically non-realizable for a passive resistor, it is ignored.

Key Points

- **Maximum Power Transfer Theorem (Modified):** When a circuit contains independent or dependent sources within the load network, the traditional condition $R_L = R_{th}$ changes. Maximization must be derived from fundamental calculus principles using $\frac{dP}{dR_L} = 0$.

Mistakes to be avoided

- Do not apply $R = R_{th} = 2\,\Omega$ here, because circuit B contains an active terminal characteristic ($3I$ component) rather than a purely passive linear resistor.

$$\boxed{(a) 0.8\Omega}$$

Q7.12.3 MCQ 2012 2M Ans: A ●●○

As per figure, V_2 is the voltage across a resistor, and thus is in the same phase as the circuit current I. However, the voltage V_3 across the R_L load (which contains an inductive component) will lead the current vector by the power factor angle ϕ (inductor has lagging power factor and current will lag voltage).

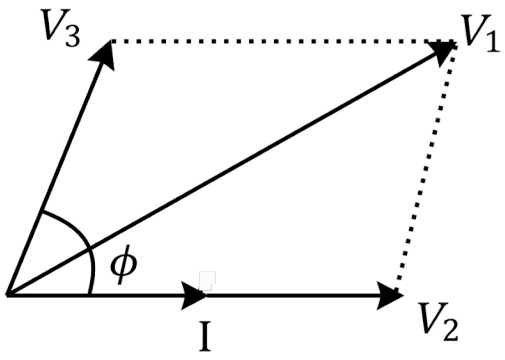

Fig. Solution Q7.12.3

From the phasor diagram, the total supply voltage V_1 is the phasor sum of V_2 and V_3. By applying the law of cosines for vector addition:

$$V_1^2 = V_2^2 + V_3^2 + 2V_2V_3\cos\phi$$

Substituting the given values from the problem ($V_1 = 220\,\text{V}$, $V_2 = 122\,\text{V}$, and $V_3 = 136\,\text{V}$):

$$(220)^2 = (122)^2 + (136)^2 + 2(122)(136)\cos\phi$$

Evaluating the squares:

$$48400 = 14884 + 18496 + 33184\cos\phi$$

$$48400 = 33380 + 33184\cos\phi$$

Solving for $\cos\phi$:

$$33184\cos\phi = 48400 - 33380$$

$$33184\cos\phi = 15020$$

$$\cos\phi = \frac{15020}{33184} \approx 0.45$$

Key Points

- For an inductive load (R_L load), the voltage leads the current by an angle ϕ, where $0° < \phi < 90°$.
- The total voltage in an AC series circuit must be calculated using phasor (vector) addition, not algebraic addition.

$\boxed{(a)0.45}$

Q7.12.4 MCQ 2012 2M Ans: B ● ○ ○

Given the load resistance:

$$R_L = 5\,\Omega$$

The power factor ($\cos\phi$) of the load impedance is defined as:

$$\cos\phi = \frac{R_L}{\sqrt{R_L^2 + X^2}} = \frac{R_L}{|Z|}$$

From the previously calculated power factor ($\cos\phi = 0.45$), the magnitude of the load impedance $|Z|$ can be determined as:

$$|Z| = \frac{5}{0.45} = 11.11\,\Omega$$

The RMS current I flowing through the load impedance is:

$$I = \frac{V_3}{|Z|} = \frac{136}{11.11}$$

The real power consumption in the load impedance is dissipated entirely across its resistive part, given by I^2R:

$$P = \left(\frac{136}{11.11}\right)^2 \times 5 = 749.08\,\text{W}$$

Rounding to the nearest whole number:

$$P \simeq 750\,\text{W}$$

Key Points

- We need to use the formula $P = I^2R$ as here we don't have enough information to calculate the voltage across the resistance or the phase angles of voltage and current. The voltage V_3 is the voltage across the entire RL load, and not just the resistance.

$\boxed{750W}$

Q7.04.1 MCQ 2004 1M Ans: B ● ○ ○

To find the total instantaneous power supplied by a balanced 3-phase AC supply to a balanced R-L load, we derive the sum of the instantaneous power of all three phases.

Let the balanced three-phase source voltages be represented as:

$$v_A(t) = V_m \sin(\omega t)$$

$$v_B(t) = V_m \sin(\omega t - 120^\circ)$$

$$v_C(t) = V_m \sin(\omega t - 240^\circ)$$

Since the load is a balanced R-L load, the currents will lag the corresponding phase voltages by an impedance phase angle ϕ:

$$i_A(t) = I_m \sin(\omega t - \phi)$$

$$i_B(t) = I_m \sin(\omega t - 120^\circ - \phi)$$

$$i_C(t) = I_m \sin(\omega t - 240^\circ - \phi)$$

The total instantaneous power $p(t)$ is the sum of the individual phase powers:

$$p(t) = v_A(t)i_A(t) + v_B(t)i_B(t) + v_C(t)i_C(t)$$

Applying the trigonometric identity $\sin(X)\sin(Y) = \frac{1}{2}[\cos(X-Y) - \cos(X+Y)]$ to each phase:
For Phase A:

$$v_A(t)i_A(t) = \frac{V_m I_m}{2}[\cos(\phi) - \cos(2\omega t - \phi)]$$

For Phase B:

$$v_B(t)i_B(t) = \frac{V_m I_m}{2}[\cos(\phi) - \cos(2\omega t - 240^\circ - \phi)]$$

For Phase C:

$$v_C(t)i_C(t) = \frac{V_m I_m}{2}[\cos(\phi) - \cos(2\omega t - 480^\circ - \phi)]$$

Summing these three equations together:

$$p(t) = \frac{V_m I_m}{2}[3\cos(\phi)] - \frac{V_m I_m}{2}[\cos(2\omega t - \phi) + \cos(2\omega t - 240^\circ - \phi) + \cos(2\omega t - 480^\circ - \phi)]$$

The three double-frequency cosine terms inside the brackets represent a balanced three-phase set of sinusoidal quantities separated by 120° (480° is co-terminal with 120°). The sum of any balanced three-phase sinusoidal set at any instant is identically zero:

$$\cos(2\omega t - \phi) + \cos(2\omega t - 120^\circ - \phi) + \cos(2\omega t - 240^\circ - \phi) = 0$$

Therefore, the time-dependent pulsating components completely cancel out, leaving:

$$p(t) = \frac{3}{2}V_m I_m \cos(\phi) = 3V_{rms}I_{rms}\cos(\phi)$$

Since V_{rms}, I_{rms}, and $\cos(\phi)$ are time-invariant constants for a steady balanced system, the total instantaneous power remains absolutely constant with respect to time.

$\boxed{(b)\,Constant}$

Key Points

- The total constant instantaneous power of a balanced three-phase system equals three times the average active power per phase.

Mistakes to be avoided

- Do not confuse single-phase power traits with three-phase power traits; while single-phase real power is pulsating, balanced three-phase real power is invariant.

Q7.01.1 MCQ | 2001 | 2M | Ans: B | ●○○

Real power will only be absorbed by the load resistance and not by pure reactive components such as an ideal capacitor.
First, we determine the load current I_L flowing through the branch impedance by applying Ohm's law:

$$I_L = \frac{V}{Z_L}$$

Given parameters from the circuit setup: System voltage, $V = 240\angle 0^\circ$ V Load branch impedance, $Z_L = 10\angle 60^\circ\ \Omega$
Substituting these values into the current expression:

$$I_L = \frac{240\angle 0^\circ}{10\angle 60^\circ} = 24\angle -60^\circ \text{ A}$$

The formula for active real power P delivered to a circuit branch is given by:

$$P = V_L I_L \cos \phi_L$$

Where V_L represents the voltage magnitude across the load branch, I_L represents the current magnitude, and ϕ_L represents the power factor phase angle between the voltage and current.

Extracting the parameters for power evaluation: Voltage magnitude, $V_L = 240\,\text{V}$ Current magnitude, $I_L = 24\,\text{A}$ Phase angle, $\phi_L = 60^\circ$

Substituting these terms to calculate the total absorbed real power:

$$P = 240 \times 24 \times \cos(60^\circ)$$

$$P = 5760 \times 0.5 = 2880\,\text{W}$$

$$\boxed{(b) 2880W}$$

Q7.99.1 MCQ | 1999 | 2M | Ans: A | ● ○ ○

Given values from the problem statement: Voltage source:

$$V = 10\angle 15^\circ\,\text{V}$$

Supply current:

$$I = 2\angle -45^\circ\,\text{A}$$

The complex power S is expressed as:

$$S = P + jQ = V \cdot I^*$$

Substituting the given phasor values into the complex power formula:

$$P + jQ = (10\angle 15^\circ) \cdot (2\angle -45^\circ)^*$$

$$P + jQ = (10\angle 15^\circ) \cdot (2\angle 45^\circ)$$

$$P + jQ = 20\angle 60^\circ$$

Converting the polar form into rectangular form using Euler's identity:

$$P + jQ = 10 + j17.32$$

Equating the real and imaginary parts: Active Power:

$$P = 10\,\text{W}$$

Reactive Power:

$$Q = 17.32\,\text{VAr}$$

$$\boxed{(a)\ 10\ W \text{ and } 17.32\ VAr}$$

Q7.96.1 MCQ 1996 1M Ans: C ● ○ ○

In the given problem, the supply is a single phase supply (household supply).

We know that for a single phase system, the instantaneous power has double the frequency of the supply.

$$p(t) = v(t)i(t)$$

A boiler can be modelled as a resistance, hence $i(t) = v(t)/R$ Thus

$$p(t) = v(t)^2/R$$

Given that $v(t) = 230\sqrt{2}\sin(2\pi.50t)V$ Thus,

$$p(t) = \frac{230^2 \times 2}{R} sin^2(2\pi.50t)$$

$$\text{or,} \frac{230^2 \times 2}{2R}(1 - cos(2 \times 2\pi.50t))$$

$$\text{or,} \frac{230^2 \times 2}{2R}(1 - cos(2\pi.100t))$$

Thus we can see that the frequency of instantaneous power (100Hz) is double that of the supply frequency (50 Hz).

$$\boxed{(c)100Hz}$$

Q7.91.1 NAT 1991 2M Ans: * ● ● ○

Let V_1 and V_2 be the voltages across the LT and HT windings of the transformer respectively, and I_1 and I_2 be the currents in the corresponding coils.Thus the following relations hold:

$$\frac{V_1}{V_2} = \frac{1}{2} \quad \text{and} \quad \frac{I_1}{I_2} = 2$$

Which gives:

$$V_2 = 2V_1 \quad \text{and} \quad I_1 = 2I_2$$

Applying KVL in the left loop:

$$-5\angle 0^\circ + 6I_1 + V_1 = 0$$

$$V_1 + 6I_1 = 5 \quad \text{— (1)}$$

Applying KVL in the right loop:

$$-V_2 + 16I_2 + 20\angle -60^\circ = 0 \quad \text{— (2)}$$

Putting $V_2 = 2V_1$ and $I_1 = 2I_2$ in (2):

$$-2V_1 + 16\left(\frac{I_1}{2}\right) = -20\angle -60^\circ$$

$$-2V_1 + 8I_1 = -20\angle -60^\circ$$

Dividing by 2:

$$-V_1 + 4I_1 = -10\angle -60^\circ \quad \text{— (3)}$$

Adding equation (1) and equation (3):

$$10I_1 = 5 - 10\angle -60^\circ$$

$$10I_1 = 5 - 10(\cos(-60^\circ) + j\sin(-60^\circ))$$

$$10I_1 = 5 - 10(0.5 - j0.866) = 5 - 5 + j8.66$$

$$I_1 = \frac{5\sqrt{3}}{10} j = 0.866j \text{ A}$$

Using the transformer relationships:

$$I_2 = \frac{I_1}{2} = 0.433j \text{ A}$$

(a) Power delivered by the sources: The complex power delivered by source 1 is:

$$S_1 = V_{S1} I_1^* = (5\angle 0^\circ)(-0.866j) = -j4.33 \text{ VA}$$

Thus, Active Power $P_1 = 0$ W and Reactive Power $Q_1 = -4.33$ VAR.

The complex power delivered by source 2 is:

$$S_2 = V_{S2} \times (-I_2)^* = (20\angle -60^\circ)(-0.433j)^*$$

$$S_2 = (10 - j17.32)(j0.433) = 7.5 + j4.33 \text{ VA}$$

(Note: We take $-I_2$ as the current I_2 goes into the positive terminal of the source. When using $S = VI^*$ to calculate complex power the current I must come out of the positive terminal of the source.)

Thus, Active Power $P_2 = 7.5$ W and Reactive Power $Q_2 = 4.33$ VAR.

(b) Power dissipated in the resistors: Power dissipated in the 6 Ω resistor:

$$P_{6\Omega} = |I_1|^2 \times 6 = (0.866)^2 \times 6 = 4.5 \text{ W}$$

Power dissipated in the 16 Ω resistor:

$$P_{16\Omega} = |I_2|^2 \times 16 = (0.433)^2 \times 16 = 3 \text{ W}$$

(a)Source 1: $-j4.33$, Source 2:$7.5 + j4.33$, (b)6 Ω resistor:4.5, 16 Ω resistor:3

PrepFusion

SOLUTIONS

VIII

Two Port Parameters

Topics

1. Z-Parameters
2. Y-Parameters, Reciprocity and Symmetry
3. H, G and ABCD Parameters
4. Series and Parallel Connection of Two Port Networks

Unit Snapshot*

Stream: EE	**Questions:** 19	**MCQ:** 12	**MSQ:** 0	**NAT:** 7
1M: 6	**2M:** 13	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2018)–2M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q8.26.1 NAT | 2026 | 2M | Ans: 2.81 to 2.91 | ● ● ○

From the given circuit figure, the input voltage is:

$$V_1 = 10\,\text{V}$$

We can convert the 2-port network from π-network to T-network using delta-to-star conversion: Given parameters: $R_1 = 5\,\Omega$, $R_2 = 10\,\Omega$, $R_3 = 4\,\Omega$

The T-network resistances are:

$$R_a = \frac{10 \times 5}{10 + 5 + 4} = \frac{50}{19}\,\Omega$$

$$R_b = \frac{10 \times 4}{10 + 5 + 4} = \frac{40}{19}\,\Omega$$

$$R_c = \frac{5 \times 4}{10 + 5 + 4} = \frac{20}{19}\,\Omega$$

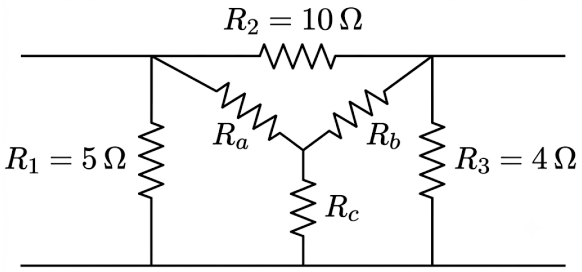

Fig. Solution Q8.26.1

The simplified network looks like :

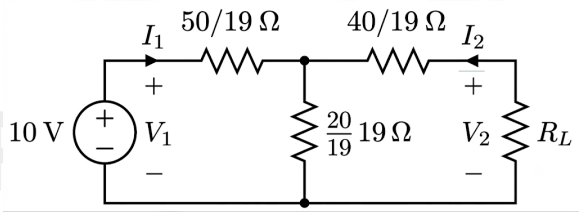

Fig. Solution Q8.26.1

Let us find out the general z parameters for a T network. Consider the given T-network consisting of three impedances Z_a, Z_b, and Z_c. The port voltages are V_1 and V_2, and the port currents entering the network are I_1 and I_2.

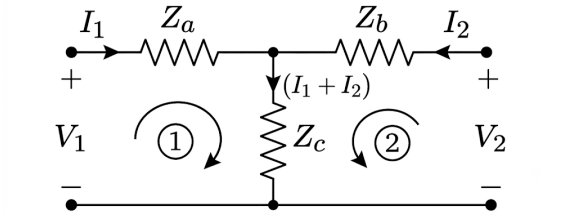

Fig. Solution Q8.26.1

Applying Kirchhoff's Voltage Law (KVL) in loop 1:

$$V_1 = Z_a I_1 + Z_c (I_1 + I_2)$$

$$V_1 = I_1(Z_a + Z_c) + I_2(Z_c)$$

Applying Kirchhoff's Voltage Law (KVL) in loop 2:

$$V_2 = Z_b I_2 + Z_c(I_1 + I_2)$$

$$V_2 = I_1(Z_c) + I_2(Z_b + Z_c)$$

The standard open-circuit impedance parameter (Z-parameter) matrix form is given by:

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

By comparing the KVL loop equations with the standard Z-parameter matrix equations, we obtain the parameter terms:

$$\begin{bmatrix} Z_a + Z_c & Z_c \\ Z_c & Z_b + Z_c \end{bmatrix}$$

The new network parameters of the standard T-form (Z-parameters) are written as:

$$Z_{11} = R_a + R_c = \frac{50}{19} + \frac{20}{19} = \frac{70}{19}\,\Omega$$

$$Z_{22} = R_b + R_c = \frac{40}{19} + \frac{20}{19} = \frac{60}{19}\,\Omega$$

$$Z_{12} = Z_{21} = R_c = \frac{20}{19}\,\Omega$$

The network equations are:

$$V_1 = \frac{70}{19} I_1 + \frac{20}{19} I_2$$

$$V_2 = \frac{20}{19} I_1 + \frac{60}{19} I_2$$

According to Maximum Power Transfer Theorem, the load resistance R_L should be equal to the Thevenin resistance R_{th} for maximum power to be transferred to the load.

$$R_{th} = \frac{V_{th}}{I_{sc}}$$

(where V_{th} is the open circuit Thevenin voltage and I_{sc} is the short circuit current)

Step 1: Find Thevenin voltage (V_{th}) Open-circuit the load resistance R_L, which implies $I_2 = 0$.

$$V_2 = \frac{20}{19} I_1$$

$$V_1 = \frac{70}{19} I_1$$

Given $V_1 = 10\,\text{V}$, we have:

$$10 = \frac{70}{19} I_1 \implies I_1 = \frac{190}{70}\,\text{A} = \frac{19}{7}\,\text{A}$$

Substituting I_1 into the expression for V_2:

$$V_{th} = V_2 = \frac{20}{19} \times \frac{190}{70} = \frac{20}{7}\,\text{V}$$

Step 2: Find Short-circuit current (I_{sc}) Replace the load resistance R_L by a short-circuit. Here, $I_{sc} = -I_2$ (as per standard convention). Since the output is shorted, $V_2 = 0$:

$$\frac{20}{19} I_1 + \frac{60}{19} I_2 = 0 \implies I_1 = -3I_2$$

Using the input loop equation where $V_1 = 10\,\text{V}$:

$$10 = \frac{70}{19}I_1 + \frac{20}{19}I_2$$

Substituting $I_1 = -3I_2$:

$$10 = \frac{70}{19}(-3I_2) + \frac{20}{19}I_2$$

$$10 = -\frac{210}{19}I_2 + \frac{20}{19}I_2 = -\frac{190}{19}I_2$$

$$10 = -10I_2 \implies I_2 = -1\,\text{A}$$

Thus, the short-circuit current is:

$$I_{sc} = -I_2 = 1\,\text{A}$$

Step 3: Calculate Thevenin Resistance (R_{th})

$$R_{th} = \frac{V_{th}}{I_{sc}} = \frac{\frac{20}{7}}{1} = \frac{20}{7}\,\Omega \approx 2.86\,\Omega$$

Therefore, for maximum power transfer, $R_L = R_{th} = 2.86\,\Omega$.

Mistakes to be avoided

- Pay careful attention to sign conventions when relating the standard 2-port network current I_2 to the short-circuit current I_{sc}. By convention, I_2 flows into the port, so $I_{sc} = -I_2$.
- Ensure that when computing I_{sc}, the input source voltage V_1 remains connected and active in the system equations.

$$\boxed{2.81 \textit{ to } 2.91}$$

Q8.24.1 NAT | 2024 | 2M | Ans: -19.90 to -19.70 | ● ● ○

Use the convention that both port currents enter each network:

$$V_1 = AV_2 - BI_2, \qquad I_1 = CV_2 - DI_2.$$

For P, $V_{P2} = 80I_{P1} + 100I_{P2}$ gives $I_{P1} = V_{P2}/80 - 1.25I_{P2}$. Therefore D is *positive* 1.25. Substituting into $V_{P1} = 40I_{P1} + 60I_{P2}$ gives $V_{P1} = 0.5V_{P2} + 10I_{P2}$, hence

$$T_P = \begin{bmatrix} 0.5 & -10 \\ 0.0125 & 1.25 \end{bmatrix}.$$

For Q the actual off-diagonal Y value is -2.5 S, rather than -25. From $I_{Q2} = -2.5V_{Q1} + V_{Q2}$,

$$V_{Q1} = 0.4V_{Q2} - 0.4I_{Q2}.$$

Then $I_{Q1} = 5V_{Q1} - 2.5V_{Q2} = -0.5V_{Q2} - 2I_{Q2}$, so

$$T_Q = \begin{bmatrix} 0.4 & 0.4 \\ -0.5 & 2 \end{bmatrix}.$$

Cascading in the shown P-then-Q order gives

$$T_R = T_P T_Q = \begin{bmatrix} 5.2 & -19.8 \\ -0.62 & 2.505 \end{bmatrix}.$$

Thus $\boxed{\beta = -19.80\,\Omega}$, within the official accepted range -19.90 to -19.70. The requested beta is the upper-right matrix element, -19.8.

Q8.23.1 NAT 2023 2M Ans: 237 to 239 ●●○

The input figure specifies $V_1 = 20$ V, not 10 V. The bridging 3 Ohm resistor adds a two-port admittance with diagonal entries 1/3 S and off-diagonal entries -1/3 S. Hence

$$Y_{eq} = \begin{bmatrix} 16/3 & -17/6 \\ -17/6 & 4/3 \end{bmatrix} \text{ S.}$$

With current I2 defined into port 2 and load 6 Ohm, $V_2 = -6I_2$. The second port equation gives

$$I_2 = -\frac{17}{6}(20) + \frac{4}{3}(-6I_2).$$

Thus $9I_2 = -170/3$, or $I_2 = -170/27$ A. The load receives

$$P_L = 6(170/27)^2 = 237.86008\ldots \text{ W.}$$

Rounded to two decimal places, $\boxed{P_L = 237.86 \text{ W}}$. The official 2023 EE Q52 accepted interval is 237 to 239 W.

Q8.18.1 NAT 2018 1M Ans: 0.5 ●○○

As required in the problem, we can short-circuit terminal 2 by setting $V_2 = 0$.

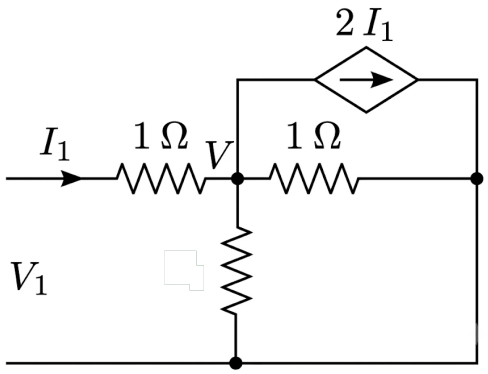

Fig. Solution Q8.18.1

Let the internal node voltage be V. Applying KCL at node V:

$$\frac{V}{1} + \frac{V}{1} + 2I_1 - I_1 = 0$$

$$2V + I_1 = 0 \implies I_1 = -2V$$

The input current I_1 flowing through the input 1 Ω resistor is expressed as:

$$I_1 = \frac{V_1 - V}{1} = V_1 - V$$

Rearranging for V:

$$V = V_1 - I_1$$

Substituting the expression for V back into the KCL relation $I_1 = -2V$:

$$I_1 = -2(V_1 - I_1)$$

$$I_1 = -2V_1 + 2I_1$$

$$I_1 = 2V_1$$

The hybrid parameter h_{11} is defined under short-circuit output conditions ($V_2 = 0$) as:

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0} = \frac{1}{2} = 0.5\,\Omega$$

$$\boxed{0.5}$$

Q8.17.1 NAT | 2017 | 1M | Ans: 3 | ● ● ○

Method 1

We can analyze the network by converting the lower T (or Y) sub-network into a Δ (or π) network to easily combine it with the parallel elements. Given parameters of the lower Y-network: $R_1 = 4\,\Omega$, $R_2 = 2\,\Omega$, $R_3 = 2\,\Omega$

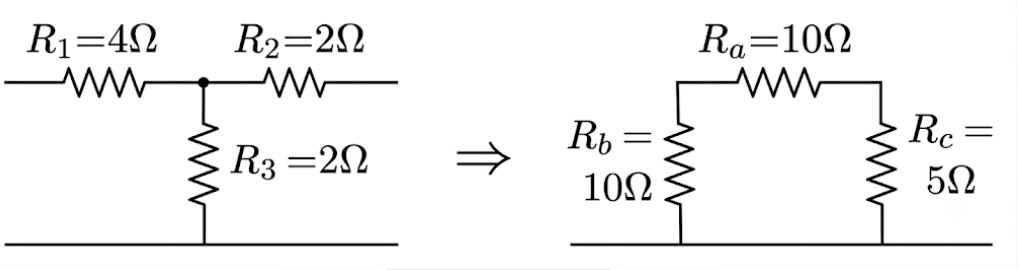

Fig. Solution Q8.17.1

Using the star-to-delta conversion formulas:

$$R_a = R_1 + R_2 + \frac{R_1 R_2}{R_3} = 4 + 2 + \frac{4 \times 2}{2} = 10\,\Omega$$

$$R_b = R_1 + R_3 + \frac{R_1 R_3}{R_2} = 4 + 2 + \frac{4 \times 2}{2} = 10\,\Omega$$

$$R_c = R_2 + R_3 + \frac{R_2 R_3}{R_1} = 2 + 2 + \frac{2 \times 2}{4} = 5\,\Omega$$

Let us find out the generalised Y parameters for a pi network. Consider the given π-network containing three admittance branches Y_a, Y_b, and Y_c. The port voltages are V_1 and V_2, and the port currents entering the network at node a and node b are I_1 and I_2, respectively.

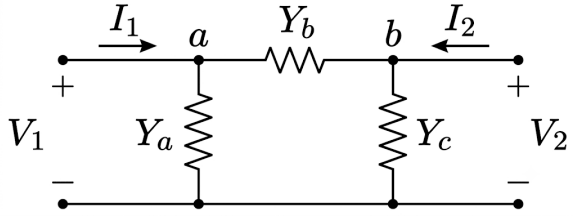

Fig. Solution Q8.17.1

Applying Kirchhoff's Current Law (KCL) at node a:

$$I_1 = V_1 Y_a + (V_1 - V_2) Y_b$$

$$I_1 = V_1 (Y_a + Y_b) + V_2 (-Y_b)$$

Applying Kirchhoff's Current Law (KCL) at node b:

$$I_2 = V_2Y_c + (V_2 - V_1)Y_b$$

$$I_2 = V_1(-Y_b) + V_2(Y_c + Y_b)$$

The standard short-circuit admittance parameter (Y-parameter) matrix relations are expressed as:

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

By comparing the nodal equations derived above with the standard Y-parameter matrix framework, the parameter elements are structured as:

$$\begin{bmatrix} Y_a + Y_b & -Y_b \\ -Y_b & Y_c + Y_b \end{bmatrix}$$

The admittance parameters (Y-parameters) for a standard Δ-form network are defined as:

$$Y_{11} = \frac{1}{R_b} + \frac{1}{R_a} = \frac{1}{10} + \frac{1}{10} = \frac{1}{5}$$

$$Y_{12} = Y_{21} = -\frac{1}{R_a} = -\frac{1}{10}$$

$$Y_{22} = \frac{1}{R_c} + \frac{1}{R_a} = \frac{1}{5} + \frac{1}{10} = \frac{3}{10}$$

Thus, the admittance matrix for this sub-network is:

$$[Y]_1 = \begin{bmatrix} \frac{1}{5} & -\frac{1}{10} \\ -\frac{1}{10} & \frac{3}{10} \end{bmatrix}$$

The top 2 Ω resistor is connected in parallel with this sub-network. Its corresponding standard admittance matrix is:

$$[Y]_2 = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

(Refer to Solution of Q8.23.1) Since the two sub-networks are in parallel, their admittance parameters are directly added:

$$[Y]_{eq} = [Y]_1 + [Y]_2 = \begin{bmatrix} \frac{1}{5} + \frac{1}{2} & -\frac{1}{10} - \frac{1}{2} \\ -\frac{1}{10} - \frac{1}{2} & \frac{3}{10} + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{7}{10} & -\frac{3}{5} \\ -\frac{3}{5} & \frac{4}{5} \end{bmatrix}$$

The impedance parameters (Z-parameters) can be obtained by finding the inverse of the equivalent admittance matrix:

$$[Z] = [Y]_{eq}^{-1}$$

The determinant of the matrix $[Y]_{eq}$ is calculated as:

$$\Delta Y = \left(\frac{7}{10}\right)\left(\frac{4}{5}\right) - \left(-\frac{3}{5}\right)\left(-\frac{3}{5}\right) = \frac{28}{50} - \frac{9}{25} = \frac{14}{25} - \frac{9}{25} = \frac{5}{25} = \frac{1}{5}$$

Using the matrix inversion formula:

$$[Z] = \frac{adj(Y)}{\Delta Y} = \frac{1}{\Delta Y}\begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{7}{10} \end{bmatrix} = 5\begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{7}{10} \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 3 & 3.5 \end{bmatrix}$$

From the resulting impedance matrix, the open-circuit reverse transfer impedance parameter is:

$$Z_{21} = 3\,\Omega$$

Method 2

The Z-parameter equations are given as :

$$V_1 = I_1 Z_{11} + I_2 Z_{12}$$

$$V_2 = I_1 Z_{21} + I_2 Z_{22}$$

Thus, the impedance parameter Z_{21} is defined as:

$$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$$

This implies that port 2 must be open-circuited ($i_2 = 0$).

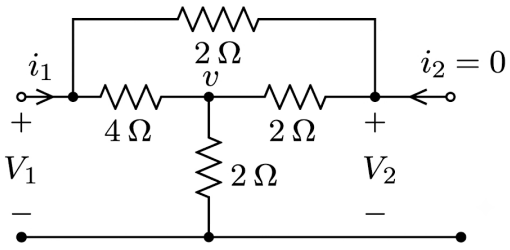

Fig. Solution Q8.17.1

Applying KCL at port 1:

$$i_1 = \frac{V_1 - V}{4} + \frac{V_1 - V_2}{2}$$

$$4i_1 = V_1 - V + 2V_1 - 2V_2$$

$$4i_1 = 3V_1 - 2V_2 - V \quad \text{— (1)}$$

Applying KCL at port 2:

$$\frac{V_2 - V_1}{2} + \frac{V_2 - V}{2} = 0$$

$$2V_2 = V_1 + V \quad \text{— (2)}$$

Applying KCL at node V:

$$\frac{V - V_1}{4} + \frac{V}{2} + \frac{V - V_2}{2} = 0$$

$$V - V_1 + 2V + 2V - 2V_2 = 0$$

$$5V = V_1 + 2V_2 \quad \text{— (3)}$$

Substituting equation (3) into equation (2):

$$2V_2 = V_1 + \left(\frac{V_1 + 2V_2}{5}\right)$$

$$10V_2 = 5V_1 + V_1 + 2V_2$$

$$8V_2 = 6V_1$$

$$V_1 = \frac{4}{3}V_2 \quad \text{— (4)}$$

Substituting equations (3) and (4) into equation (1):

$$4i_1 = 3\left(\frac{4}{3}V_2\right) - 2V_2 - \left(\frac{V_1 + 2V_2}{5}\right)$$

$$20i_1 = 20V_2 - 10V_2 - V_1 - 2V_2$$

$$20i_1 = 8V_2 - \frac{4}{3}V_2 = \frac{20}{3}V_2$$

$$3i_1 = V_2$$

$$Z_{21} = \frac{V_2}{i_1} = 3\,\Omega$$

Key Points

- When we have a 2 port network consisting a single impedance Z, it assumes the standard form : $\begin{bmatrix} 1/Z & -1/Z \\ -1/Z & 1/Z \end{bmatrix}$
- Parallel combination of two-port sub-networks is straightforward when using admittance (Y) matrices, as individual parameter elements simply sum up: $[Y]_{eq} = [Y]_1 + [Y]_2$.

Mistakes to be avoided

- Be careful to include the negative sign for transfer terms (Y_{12} and Y_{21}) in the admittance matrix for passive π-networks.

3

Q8.17.2 MCQ | 2017 | 2M | Ans: D | ●●○

When two two-port networks are connected in a cascaded arrangement, the overall transmission ($ABCD$) matrix is determined by multiplying the individual transmission matrices of each network stage.

1. Matrix Multiplication for Cascaded Stages Let the individual transmission parameters of the two networks be denoted with subscripts 1 and 2. The overall transmission matrix equation is defined as:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} \begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} = \begin{bmatrix} A_1A_2 + B_1C_2 & A_1B_2 + B_1D_2 \\ C_1A_2 + D_1C_2 & C_1B_2 + D_1D_2 \end{bmatrix}$$

This yields the following overall network terminal relationships:

$$V_1 = (A_1A_2 + B_1C_2)V_3 - (A_1B_2 + B_1D_2)I_3$$

$$I_1 = (C_1A_2 + D_1C_2)V_3 - (C_1B_2 + D_1D_2)I_3$$

2. Evaluation of Parameters under Ideal Voltage Transfer Conditions For the Thevenin voltage calculation we inspect the network under an open-circuit output condition ($I_2 = 0$):

$$V_1 = (A_1A_2 + B_1C_2)V_3 \implies V_T = V_3 = \frac{V_1}{A_1A_2 + B_1C_2}$$

We know that Thevenin impedance looking into the network is calculated by turning off all sources ($V_1 = 0$) and can be formulated from the matrix equations. However Z_T is also calculated as the ratio of the Thevenin equivalent voltage and the short circuit current. The short circuit current is conventionally defined as leaving the output port, hence $I_{sc} = -I_3$.
First we find I_3 when the output port is shorted $V_3 = 0$.

$$V_1 = -(A_1B_2 + B_1D_2)I_3 \implies I_3 = -\frac{V_1}{A_1B_2 + B_1D_2}$$

Next finding

$$Z_T = \frac{V_T}{I_{sc}} = \frac{V_3}{-I_3} = \frac{\frac{V_1}{A_1A_2+B_1C_2}}{\frac{V_1}{A_1B_2+B_1D_2}} = \frac{A_1B_2 + B_1D_2}{A_1A_2 + B_1C_2}$$

Key Points

- Due to sequential current matching, the equivalent matrix of cascaded systems must maintain precise matrix multiplication order: $[T_{\text{total}}] = [T_1][T_2]$.

$$(d)V_T = \frac{V_1}{A_1A_2 + B_1C_2}, Z_T = \frac{A_1B_2 + B_1D_2}{A_1A_2 + B_1C_2}$$

Q8.16.1 NAT | 2016 | 2M | Ans: 35.55 ●●○

Given a two-port network specified by its open-circuit impedance (Z) parameters connected to a real voltage source at the input port and a load resistor R_l at the output port, we determine the power delivered to the load resistor.

1. Two-Port Network Parameter Equations The given impedance matrix is:

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 40 & 60 \\ 80 & 100 \end{bmatrix}$$

The standard defining matrix equation $[V] = [Z][I]$ yields:

$$V_1 = 40I_1 + 60I_2 \quad \text{— (1)}$$

$$V_2 = 80I_1 + 100I_2 \quad \text{— (2)}$$

2. Input and Output Port Boundary Conditions From the input side of the circuit, applying KVL:

$$I_1 = \frac{20 - V_1}{10} \implies V_1 = 20 - 10I_1 \quad \text{— (3)}$$

From the output side of the circuit, the load resistor determines the terminal relationship:

$$V_2 = -I_2R_l$$

The final calculation shows $R_l = 20\,\Omega$. Substituting $R_l = 20\,\Omega$ into the relation:

$$V_2 = -20I_2 \quad \text{— (4)}$$

3. Solving for Port Currents Substitute equation (3) into equation (1):

$$20 - 10I_1 = 40I_1 + 60I_2$$

$$50I_1 + 60I_2 = 20 \quad \text{— (5)}$$

Substitute equation (4) into equation (2):

$$-20I_2 = 80I_1 + 100I_2$$

$$80I_1 + 120I_2 = 0 \implies I_1 = -1.5I_2 \quad \text{— (6)}$$

Now, substitute equation (6) into equation (5):

$$50(-1.5I_2) + 60I_2 = 20$$

$$-75I_2 + 60I_2 = 20 \implies -15I_2 = 20$$

$$I_2 = -\frac{20}{15} = -\frac{4}{3}\,\text{A}$$

4. Calculation of Power Delivered to the Load The power P_{R_l} delivered to the load resistor R_l is calculated using the square of the load current:

$$P_{R_l} = I_2^2R_l = \left(-\frac{4}{3}\right)^2 \times 20 = \frac{16}{9} \times 20 = \frac{320}{9} \approx 35.55\,\text{W}$$

$$\boxed{35.55}$$

Q8.15.1 NAT | 2015 | 2M | Ans: 0.4 to 0.6 | ● ● ○

Define the load current I_L as flowing out of port 2. The transmission relation is $V_1 = AV_2 + BI_L$.

With port 2 shorted, $V_2 = 0$ and $10 = 4B$, so $B = 2.5\ \Omega$. With a 1 Ohm load and $I_L = 1.25$ A, the second condition gives $5 = (A + 2.5)(1.25)$, hence $A = 1.5$.

For the requested 2 Ohm load at $V_1 = 3$ V,

$$3 = (2A + B)I_L = 5.5I_L.$$

Therefore $I_L = 6/11 = 0.5454545\ldots$ A, or $\boxed{0.545\text{ A}}$ to three decimal places. The original question does not specify precision; the authenticated official archive Session 1 Q56 accepts 0.4 to 0.6 A. Both 0.545 and 0.5454 are valid within that published interval.

Q8.10.1 MCQ | 2010 | 2M | Ans: C | ● ○ ○

Let the internal two-port network terminal voltage variables be denoted as V_1 at port ab and V_2 at port cd. The port currents are $I_a = I_1 = I_e$ and $I_c = I_2$. Let the total input voltage across terminal ef be V_1'.

The standard open-circuit impedance parameter (Z-parameter) relations for the core network are:

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Now, analyzing the series connection at the input side, a $1\ \Omega$ resistor is connected in series with terminal a. The relationship between the total input voltage V_1' and the core network input voltage V_1 is given by:

$$V_1' - I_1(1) = V_1$$

$$V_1' = I_1 + V_1$$

Substituting the expression for V_1 into the equation for V_1':

$$V_1' = (Z_{11} + 1)I_1 + Z_{12}I_2$$

The output loop voltage expression remains unchanged because no additional element modifies the cd port connections directly:

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Therefore, the modified open-circuit impedance matrix (Z-parameters) for the combined network configuration is written as:

$$\begin{bmatrix} Z_{11} + 1 & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

$$\boxed{(c)\ \begin{bmatrix} Z_{11} + 1 & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}}$$

Q8.06.1 MCQ | 2006 | 2M | Ans: * | ● ○ ○

We analyze the given criteria from the circuit configuration parameters step by step to determine which two-port parameters are valid or zero.

From the network configuration given in the problem statement:

$$I_1 = 0 \quad \text{and} \quad V_2 = 0$$

Let us verify the options using various two-port parameters:

1. Checking Z-parameters: The standard Z-parameter equations are defined as:

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

Since $I_1 = 0$, port 1 behaves as an open circuit. The driving-point input impedance Z_{11} becomes infinitely large:

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} = \infty$$

The parameter Z_{21} is expressed as:

$$Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$$

Since $V_2 = 0$ and $I_1 = 0$, Z_{21} turns out to be indeterminate $(\frac{0}{0})$. There is no indication of valid finite Z-parameters. Thus, choices involving Z-parameters are incorrect.

2. Checking h-parameters: The standard hybrid h-parameter equations are:

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

Since $I_1 = 0$ implies an open-circuit at port 1, the short-circuit input impedance h_{11} would evaluate to infinity (∞). Since no such infinite option exists for hybrid equations, standard h-parameters are invalid here.

We will also check the g parameters. The inverse hybrid g-parameter equations relate the network variables as follows:

$$I_1 = g_{11}V_1 + g_{12}I_2$$

$$V_2 = g_{21}V_1 + g_{22}I_2$$

Given the terminal conditions from the figure where $I_1 = 0$ and $V_2 = 0$ independently of the values of the independent variables V_1 and I_2:

$$0 = g_{11}V_1 + g_{12}I_2$$

$$0 = g_{21}V_1 + g_{22}I_2$$

For these equations to hold true identically for any arbitrary applied voltage V_1 and current I_2, all the individual coefficients must be zero:

$$g_{11} = 0, \quad g_{12} = 0, \quad g_{21} = 0, \quad g_{22} = 0$$

Q8.05.1 MCQ | 2005 | 2M | Ans: B | ● ○ ○

Method 1

Based on the provided circuit analysis of cascaded two-port networks, we determine the equivalent ABCD parameters by multiplying the individual transmission matrices.

1. Series Impedance Element (Z_1)

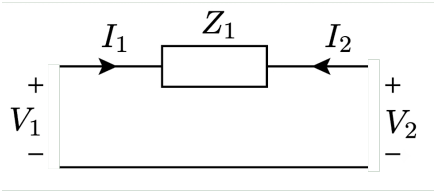

Fig. Solution Q8.05.1

For a series impedance Z_1, the transmission (ABCD) matrix is given by:

$$\begin{bmatrix} A_1 & B_1 \\ C_1 & D_1 \end{bmatrix} = \begin{bmatrix} 1 & Z_1 \\ 0 & 1 \end{bmatrix}$$

2. Shunt Admittance Element (Z_2)

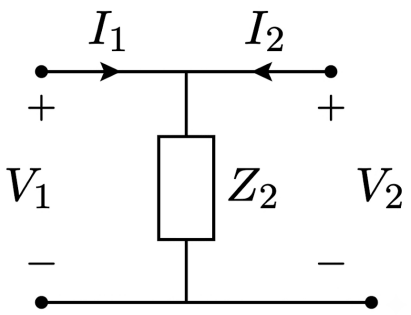

Fig. Solution Q8.05.1

For a shunt impedance Z_2, the transmission (ABCD) matrix is given by:

$$\begin{bmatrix} A_2 & B_2 \\ C_2 & D_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{Z_2} & 1 \end{bmatrix}$$

3. Cascaded Network Parameters The equivalent ABCD parameters of the cascaded connection are found by taking the product of the individual matrices:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}_{eq} = \begin{bmatrix} 1 & Z_1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{Z_2} & 1 \end{bmatrix} = \begin{bmatrix} 1 + \frac{Z_1}{Z_2} & Z_1 \\ \frac{1}{Z_2} & 1 \end{bmatrix}$$

From the equivalent matrix, the parameter C is defined as:

$$C = \frac{1}{Z_2}$$

Given that:

$$C = 0.025\angle 45^\circ \text{ S}$$

Therefore, the shunt impedance Z_2 is calculated as:

$$Z_2 = \frac{1}{C} = \frac{1}{0.025\angle 45^\circ} = 40\angle -45^\circ\ \Omega$$

Method 2

Alternatively, we can calculate Z_2 by directly computing the coefficient C without using the concept of cascade connection. The transmission parameters ($ABCD$ parameters) equations are defined as:

$$V_1 = AV_2 - BI_2$$

$$I_1 = CV_2 - DI_2$$

The coefficient C is given by:

$$C = \left.\frac{I_1}{V_2}\right|_{I_2=0} \quad \text{(port 2 must be open)}$$

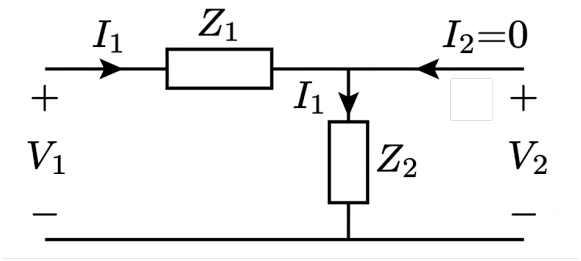

Fig. Solution Q8.05.1

From the circuit diagram, when port 2 is open ($I_2 = 0$), the current I_1 flows entirely through the impedance Z_2. Therefore:

$$V_2 = I_1 Z_2$$

Thus, the parameter C is:

$$C = \frac{I_1}{V_2} = \frac{1}{Z_2}$$

Given that:

$$C = 0.025\angle 45^\circ$$

We can solve for Z_2:

$$Z_2 = \frac{1}{C} = \frac{1}{0.025\angle 45^\circ} = 40\angle -45^\circ\ \Omega$$

$$\boxed{(b) 40\angle -45^\circ\ \Omega}$$

Q8.05.2 MCQ 2005 1M Ans: D ●○○

Based on the loop analysis of the given two-port network, we determine the impedance (Z) parameters by applying KVL to the left and right loops.

The standard defining equations for a two-port Z-parameter network are:

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

1. KVL in Left Loop Applying KVL to the left loop, the voltage V_1 is expressed as:

$$V_1 = (I_1 + I_2) Z_1$$

$$V_1 = Z_1 I_1 + Z_1 I_2$$

Comparing this with the standard equation, we get:

$$Z_{11} = Z_1, \quad Z_{12} = Z_1$$

2. KVL in Right Loop Applying KVL to the right loop, the voltage V_2 is expressed as:

$$V_2 = I_2 Z_2 + (I_1 + I_2) Z_1$$

$$V_2 = Z_1 I_1 + (Z_1 + Z_2) I_2$$

Comparing this with the standard equation, we get:

$$Z_{21} = Z_1, \quad Z_{22} = Z_1 + Z_2$$

Combining the coefficients into the matrix form gives:

$$\begin{bmatrix} Z_1 & Z_1 \\ Z_1 & Z_1 + Z_2 \end{bmatrix}$$

$$\boxed{(d)\ \begin{bmatrix} Z_1 & Z_1 \\ Z_1 & Z_1 + Z_2 \end{bmatrix}}$$

Q8.03.1 MCQ | 2003 | 2M | Ans: D | ● ● ○

Method 1

We can carry out the following steps in order to find the parameter h_{21}.

1. Delta-to-Wye (Δ-Y) Transformation

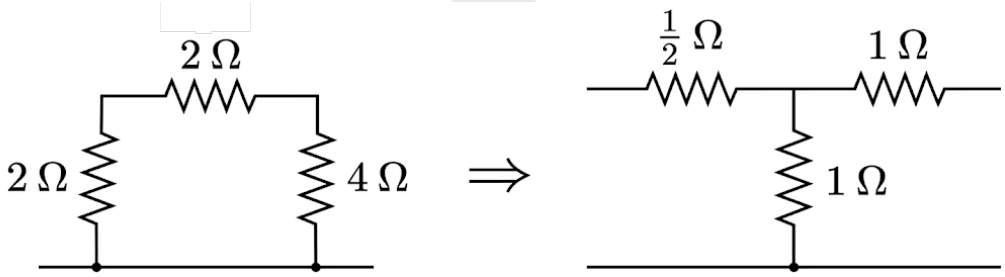

Fig. Solution Q8.03.1

The middle delta network consists of three resistors: 2 Ω, 2 Ω, and 4 Ω. Converting this Δ-network into an equivalent Y-network yields the following branch resistances:

$$R_A = \frac{2 \times 2}{2 + 2 + 4} = \frac{4}{8} = 0.5\,\Omega = \frac{1}{2}\,\Omega$$

$$R_B = \frac{2 \times 4}{2 + 2 + 4} = \frac{8}{8} = 1\,\Omega$$

$$R_C = \frac{2 \times 4}{2 + 2 + 4} = \frac{8}{8} = 1\,\Omega$$

2. Network Simplification Combining the series elements with the external series resistors (4 Ω on the left and 2 Ω on the right), the total series arm resistances become:

$$R_{\text{left}} = 4\,\Omega + 0.5\,\Omega = 4.5\,\Omega$$

$$R_{\text{right}} = 2\,\Omega + 1\,\Omega = 3\,\Omega$$

The shunt arm resistance remains 1 Ω.

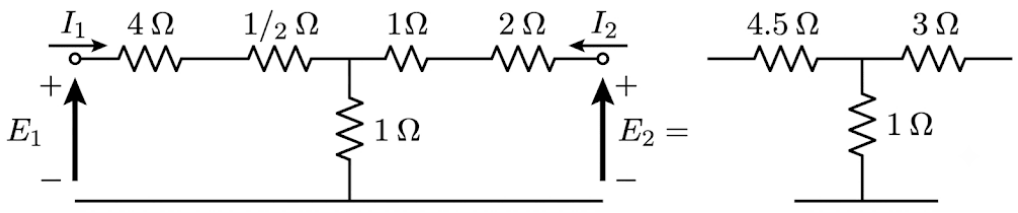

Fig. Solution Q8.03.1

For this simplified T-network, the equivalent Z-parameter matrix is written by inspection (Refer to Solution of Question 8.26.1):

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 4.5+1 & 1 \\ 1 & 3+1 \end{bmatrix} = \begin{bmatrix} 5.5 & 1 \\ 1 & 4 \end{bmatrix}$$

The corresponding two-port network equations are:

$$E_1 = 5.5I_1 + I_2$$

$$E_2 = I_1 + 4I_2$$

The standard hybrid parameter equations are:

$$E_1 = h_{11}I_1 + h_{12}E_2$$

$$I_2 = h_{21}I_1 + h_{22}E_2$$

From the second Z-parameter equation, express I_2 in terms of I_1 and E_2:

$$4I_2 = E_2 - I_1 \implies I_2 = -0.25I_1 + 0.25E_2$$

Substituting this expression for I_2 into the first Z-parameter equation:

$$E_1 = 5.5I_1 + (-0.25I_1 + 0.25E_2)$$

$$E_1 = 5.25I_1 + 0.25E_2$$

Comparing these rewritten equations with the standard h-parameter form gives the matrix:

$$[h] = \begin{bmatrix} 5.25 & 0.25 \\ -0.25 & 0.25 \end{bmatrix}$$

Thus, the reverse voltage gain parameter h_{12} is 0.25.

Method 2

The hybrid parameter (h-parameter) equations are defined as:

$$E_1 = h_{11}I_1 + h_{12}E_2$$

$$I_2 = h_{21}I_1 + h_{22}E_2$$

The reverse voltage gain parameter h_{12} is determined as:

$$h_{12} = \left.\frac{E_1}{E_2}\right|_{I_1=0} \quad \text{(port 1 must be open)}$$

When port 1 is open-circuited ($I_1 = 0$), there is no voltage drop across the input series resistor (the 4 Ω resistor). By applying the voltage division rule to find the voltage at the node and further scaling, we get:

$$E_1 = \frac{(2+2) \parallel 4}{(2+2) \parallel 4 + 2} \cdot E_2 \times \frac{1}{2}$$

Therefore, the expression for h_{12} simplifies to:

$$h_{12} = \frac{(2+2) \parallel 4}{(2+2) \parallel 4 + 2} \times \frac{1}{2}$$

Calculating the parallel combination:

$$(2+2) \parallel 4 = 4 \parallel 4 = 2\,\Omega$$

Substituting this value back into the expression for h_{12}:

$$h_{12} = \frac{2}{2+2} \times \frac{1}{2} = 0.5 \times \frac{1}{2} = 0.25$$

$$\boxed{(d)0.25}$$

Q8.02.1 MCQ 2002 2M Ans: B ●○○

Based on the provided circuit calculations, the admittance (Y) parameters of the given T-network (Y-form) can be found using two different methods.

Method 1

The given network is of the T-form consisting of three resistors of $2\,\Omega$ each.
First, we write down the open-circuit impedance (Z) parameters by inspection (Refer to Solution of Question 8.26.1):

$$Z = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} 2+2 & 2 \\ 2 & 2+2 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

The short-circuit admittance (Y) parameters are obtained by taking the inverse of the Z-matrix ($Y = Z^{-1}$):

$$Y = Z^{-1} = \frac{\text{adj}(Z)}{|Z|}$$

The determinant of the Z-matrix is calculated as:

$$|Z| = (4 \times 4) - (2 \times 2) = 16 - 4 = 12$$

The adjoint of the Z-matrix is:

$$\text{adj}(Z) = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$$

Therefore, the admittance parameters matrix is:

$$Y = \frac{1}{12}\begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix} \text{S}$$

Thus, the individual admittance parameters are:

$$Y_{11} = \frac{1}{3}\,\text{S}, \quad Y_{12} = -\frac{1}{6}\,\text{S}, \quad Y_{21} = -\frac{1}{6}\,\text{S}, \quad Y_{22} = \frac{1}{3}\,\text{S}$$

Method 2

Alternatively, we can directly convert the T-form (Y-network) into an equivalent π-form (Δ-network).
For a symmetrical T-network with equal resistors ($R = 2\,\Omega$), each resistor in the equivalent Δ-network is calculated as:

$$R_\Delta = \frac{(2 \times 2) + (2 \times 2) + (2 \times 2)}{2} = \frac{4+4+4}{2} = \frac{12}{2} = 6\,\Omega$$

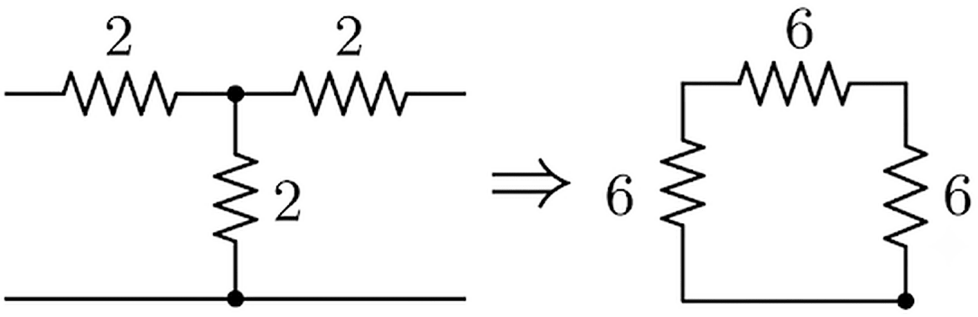

Fig. Solution Q8.02.1

The resulting π-network has a series branch resistor of $6\,\Omega$ and two shunt branch resistors of $6\,\Omega$ each.

By standard definition for a π-network with admittance elements Y_A, Y_B (shunts) and Y_C (series) (Refer to Solution of Question 8.17.1):

$$Y_{11} = Y_A + Y_C = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}\,\text{S}$$

$$Y_{22} = Y_B + Y_C = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}\,\text{S}$$

$$Y_{12} = Y_{21} = -Y_C = -\frac{1}{6}\,\text{S}$$

Writing the final Y-parameter matrix form:

$$[Y] = \begin{bmatrix} \frac{1}{3} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{3} \end{bmatrix}\,\text{S}$$

Method 3

We can also find the y-parameters by shorting each port one by one.

The y-parameter equations are :

$$I_1 = y_{11}V_1 + y_{12}V_2$$

$$I_2 = y_{21}V_1 + y_{22}V_2$$

The y-parameters are defined as:

$$y_{11} = \left.\frac{I_1}{E_1}\right|_{E_2=0}, \quad y_{12} = \left.\frac{I_1}{E_2}\right|_{E_1=0}, \quad y_{21} = \left.\frac{I_2}{E_1}\right|_{E_2=0}, \quad y_{22} = \left.\frac{I_2}{E_2}\right|_{E_1=0}$$

Thus we need to investigate under two conditions :

Case 1: Port 1 shorted ($E_1 = 0$)

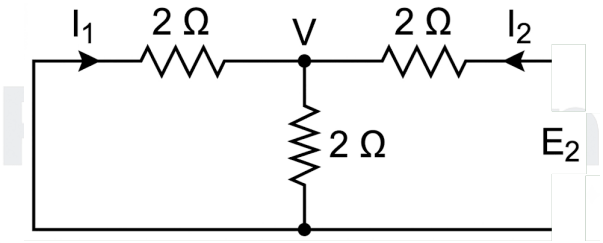

Fig. Solution Q8.02.1

By applying the voltage division rule to find node voltage V:

$$V = \frac{2 \parallel 2}{(2 \parallel 2) + 2}E_2 = \frac{1}{1+2}E_2 = \frac{1}{3}E_2$$

The current I_1 is given by:

$$I_1 = -\frac{V}{2} = -\frac{E_2/3}{2} = -\frac{E_2}{6}$$

Therefore, the parameter y_{12} is:

$$y_{12} = \left.\frac{I_1}{E_2}\right|_{E_1=0} = -\frac{1}{6}\,\text{S}$$

The current I_2 is given by:

$$I_2 = \frac{E_2 - V}{2} = \frac{E_2 - E_2/3}{2} = \frac{2E_2}{6} = \frac{E_2}{3}$$

Therefore, the parameter y_{22} is:

$$y_{22} = \left.\frac{I_2}{E_2}\right|_{E_1=0} = \frac{1}{3}\,\text{S}$$

Case 2: Port 2 shorted ($E_2 = 0$)

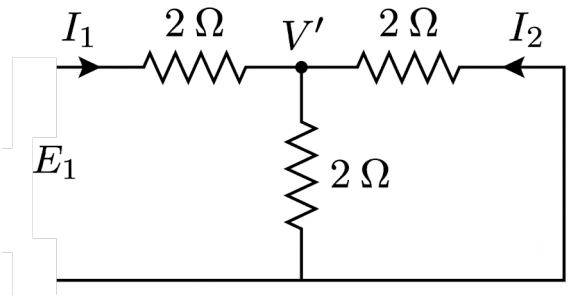

Fig. Solution Q8.02.1

By applying the voltage division rule to find node voltage V':

$$V' = \frac{2 \parallel 2}{(2 \parallel 2) + 2} E_1 = \frac{E_1}{3}$$

The current I_1 is given by:

$$I_1 = \frac{E_1 - V'}{2} = \frac{2E_1}{6} = \frac{E_1}{3}$$

Therefore, the parameter y_{11} is:

$$y_{11} = \left.\frac{I_1}{E_1}\right|_{E_2=0} = \frac{1}{3}\,\text{S}$$

The current I_2 is given by:

$$I_2 = -\frac{V'}{2} = -\frac{E_1}{6}$$

Therefore, the parameter y_{21} is:

$$y_{21} = \left.\frac{I_2}{E_1}\right|_{E_2=0} = -\frac{1}{6}\,\text{S}$$

Thus, the calculated short-circuit admittance parameters are:

$$y_{11} = \frac{1}{3}\,\text{S}, \quad y_{12} = -\frac{1}{6}\,\text{S}, \quad y_{21} = -\frac{1}{6}\,\text{S}, \quad y_{22} = \frac{1}{3}\,\text{S}$$

Mistakes to be avoided

- When finding the parameters from a π-network, do not forget the negative sign on the off-diagonal terms (Y_{12} and Y_{21}).
- Ensure to convert resistance values (Ω) to admittances (S or $\mho$) before computing matrix sum relations in the standard π-form.

$$(b)\ \frac{1}{3}\ mho,\ -\frac{1}{6}\ mho,\ -\frac{1}{6}\ mho\ and\ \frac{1}{3}\ mho\ respectively$$

Q8.01.1 MCQ 2001 1M Ans: C ● ○ ○

A network is classified as passive if it contains no internal independent energy sources or amplification mechanisms (such as operational amplifiers or biased transistors). It can only consist of passive circuit elements: resistors (R), inductors (L), and capacitors (C).

Let us analyze each of the parameters given in the options:

Option A: A passive network can amplify voltage amplitudes under specific steady-state alternating current (AC) conditions. For example, a resonant series RLC tank circuit or a step-up transformer can provide a steady-state output voltage magnitude that is significantly higher than the input voltage magnitude ($V_{out} > V_{in}$).

Option B: The output impedance or equivalent impedance seen looking into a port can easily be altered, transformed, or made lower than the source impedance by utilizing matching networks or passive transformations.

Option C: By the law of conservation of energy (specifically Tellegen's Theorem applied to passive systems), the total real power consumed by the passive components must be greater than or equal to zero ($P_{\text{consumed}} \geq 0$). Therefore, the steady-state real power delivered at the output port (P_{out}) can never exceed the real power injected into the input port (P_{in}):

$$P_{out} \leq P_{in}$$

Hence, a passive two-port network can never offer greater power at its output compared to its input.

Option D: Voltage regulation is a measure of a network's ability to maintain a constant output voltage under varying load conditions. Passive networks can alter regulation attributes depending on their structural impedances.

Thus, the strict physical limitation imposed by passivity ensures that the network can never amplify power.

$$\boxed{(c) \textit{Greater power}}$$

Q8.00.1 MCQ 2000 2M Ans: B ● ○ ○

Based on the provided mathematical derivation for a two-port network, we determine the impedance (Z) parameters by finding the inverse of the admittance (Y) parameter matrix.

1. Extraction of Admittance (Y) Parameters From the given terminal equations of the two-port network:

$$I_1 = 2V_1 + V_2$$

$$I_2 = V_1 + V_2$$

The standard defining equations for Y-parameters are:

$$I_1 = Y_{11}V_1 + Y_{12}V_2$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2$$

By comparing the coefficients, the admittance matrix $[Y]$ is constructed as:

$$[Y] = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

2. Matrix Inversion to Find Impedance (Z) Parameters The relationship between the impedance matrix $[Z]$ and the admittance matrix $[Y]$ is given by the matrix inverse:

$$[Z] = [Y]^{-1} = \frac{\text{adj}(Y)}{|Y|}$$

First, find the determinant of the matrix $[Y]$:

$$|Y| = (2 \times 1) - (1 \times 1) = 2 - 1 = 1$$

Next, determine the adjoint matrix of $[Y]$ by swapping the diagonal elements and changing the signs of the off-diagonal elements:

$$\text{adj}(Y) = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Now, calculate $[Z]$:

$$[Z] = \frac{1}{1}\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Thus, the individual open-circuit impedance parameters are:

$$Z_{11} = 1, \quad Z_{12} = -1, \quad Z_{21} = -1, \quad Z_{22} = 2$$

$$\boxed{(b)(1, -1, -1, 2)}$$

Q8.96.1 MCQ | 1996 | 2M | Ans: * | ● ● ○

To determine the correct matching between the network properties in Table A and the conditions in Table B, let us analyze each constraint individually:

1. Reciprocity at the ports : A two-port network is reciprocal if the transmission of a signal from port 1 to port 2 is the same as from port 2 to port 1. In terms of open-circuit impedance parameters (z-parameters), the condition for reciprocity is defined as:

$$z_{12} = z_{21}$$

Looking at the choices provided in Table B, option (Q) shows $z_{12} = -z_{21}$, which is the condition for an anti-reciprocal network, not a reciprocal one. Let us check the reciprocity condition in terms of hybrid parameters (h-parameters):

$$h_{12} = -h_{21}$$

Since this is also not listed, let's examine the provided solution context which indicates a typo in the original question's options or matching pairs, where (a) matches to an condition defining equal transfer impedances ($z_{12} = z_{21}$). Given the specific provided key where (a) → (Q), it implies a direct matching based on the question text provided in the official key.

2. Passivity at the ports A passive network contains only energy-absorbing or energy-storing elements and cannot deliver net energy. For a passive network, there is an absence of negative resistance elements inside the two-port network because a negative resistance acts as an active element supplying power. Thus:

(b) → (T)

3. Electrical symmetry at the ports A two-port network is electrically symmetrical if its ports can be interchanged without changing the port voltages and currents. In terms of h-parameters, the condition for symmetry is that the determinant of the h-parameter matrix must be equal to unity:

$$\Delta h = h_{11}h_{22} - h_{12}h_{21} = 1$$

Thus:

(c) → (P)

Key Points

- **Reciprocity Conditions:** $z_{12} = z_{21}$, $y_{12} = y_{21}$, $h_{12} = -h_{21}$, $AD - BC = 1$.
- **Symmetry Conditions:** $z_{11} = z_{22}$, $y_{11} = y_{22}$, $h_{11}h_{22} - h_{12}h_{21} = 1$, $A = D$.
- **Passivity:** Requires that the total real power absorbed by the network is non-negative for all excitation conditions, implying no internal active elements or negative resistances.

Q8.94.1 MCQ 1994 1M Ans: B ● ○ ○

For a general linear two-port network, the hybrid parameters (h-parameters) express the relationships between the port variables as:

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

If the network consists entirely of passive, linear, and bilateral components (such as standard resistors, inductors, and capacitors), it is fundamentally a reciprocal network.

The condition for reciprocity in terms of impedance parameters (z-parameters) is:

$$z_{12} = z_{21}$$

To find the corresponding condition in terms of h-parameters, we can express the z-parameters using h-parameters:

$$z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0} = \frac{h_{12}V_2}{h_{22}V_2} = \frac{h_{12}}{h_{22}}$$

$$z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$$

Setting $I_2 = 0$ in the second hybrid equation:

$$0 = h_{21}I_1 + h_{22}V_2 \implies V_2 = -\frac{h_{21}}{h_{22}}I_1$$

$$z_{21} = -\frac{h_{21}}{h_{22}}$$

Equating z_{12} and z_{21} for reciprocity:

$$\frac{h_{12}}{h_{22}} = -\frac{h_{21}}{h_{22}} \implies h_{12} = -h_{21}$$

Therefore, if a two-port network is passive (and thus reciprocal), the required relationship is $h_{12} = -h_{21}$.

$$\boxed{(b) h_{12} = -h_{21}}$$

Q8.91.1 MCQ 1991 1M Ans: C ● ○ ○

For a two-port network to be reciprocal, the transmission of a signal from port 1 to port 2 must be identical to the transmission from port 2 to port 1 under identical termination conditions. Let us analyze the reciprocity requirements across different parameter sets:

1. Hybrid Parameters (h-parameters) The hybrid equations relating the port variables are given by:

$$V_1 = h_{11}I_1 + h_{12}V_2$$

$$I_2 = h_{21}I_1 + h_{22}V_2$$

The condition for reciprocity in terms of h-parameters is:

$$h_{12} = -h_{21}$$

This perfectly matches option (C).

2. Verification of Other Options

Option A: The relationship between z and y parameters involves the determinant of the z-matrix ($\Delta z = z_{11}z_{22} - z_{12}z_{21}$). The entry Y_{12} is given by $-\frac{z_{12}}{\Delta z}$. Thus, $\frac{z_{12}}{Y_{12}} = -\Delta z = z_{12}^2 - z_{11}z_{22}$. This is a general property of matrix inversion and does not uniquely represent the condition for reciprocity.

Option B: $Z_{12} = \frac{1}{Y_{22}}$ is generally false; $z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0}$, whereas $y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$.
Option D: For transmission parameters ($ABCD$ parameters), the condition for reciprocity is:

$$AD - BC = 1$$

Option D incorrectly states $AD - BC = 0$ (which is the condition for a singular transmission matrix).

Thus, the only correct statement defining a reciprocal network among the given options is $h_{12} = -h_{21}$.

Key Points

- **Reciprocity Conditions Summary:**
  - Impedance parameters: $z_{12} = z_{21}$
  - Admittance parameters: $y_{12} = y_{21}$
  - Transmission parameters: $AD - BC = 1$
  - Hybrid parameters: $h_{12} = -h_{21}$
  - Inverse hybrid parameters: $g_{12} = -g_{21}$

Mistakes to be avoided

- Be careful with transmission parameters; the reciprocity condition is $AD - BC = 1$, not $AD - BC = 0$.
- Remember the unique negative sign associated with the reciprocity conditions for hybrid (h) and inverse hybrid (g) parameter networks.

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